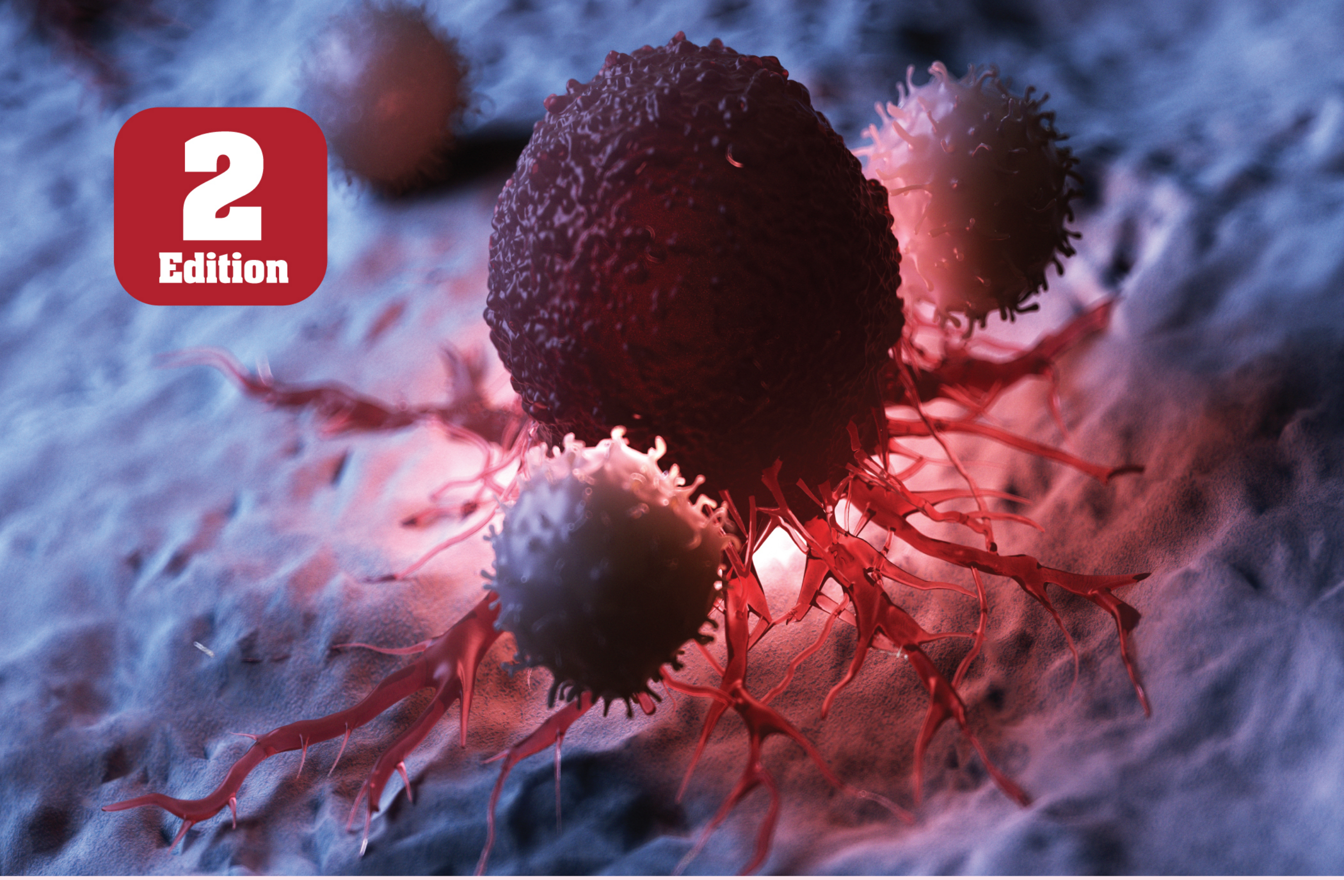

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Preface

Immunology plays a vital role in protecting our bodies against diseases and maintaining overall health. It is the branch of biomedical science that focuses on the study of the immune system, which is a complex network of cells, tissues, and organs that work together to defend the body against harmful pathogens, such as bacteria, viruses, fungi, and parasites.

The primary role of immunology is to understand how the immune system functions and how it can be manipulated to prevent or treat diseases. It involves studying the different components of the immune system, including white blood cells (such as T cells and B cells), antibodies, cytokines, and immune receptors. Immunology helps us understand how our immune system recognizes and responds to foreign substances, known as antigens. It investigates the mechanisms by which the immune system distinguishes between self and non-self, ensuring that it targets only harmful pathogens while sparing our own healthy cells and tissues.

Furthermore, immunology plays a crucial role in vaccine development by studying how vaccines activate the immune response and provide long-lasting protection against specific diseases. This field also investigates autoimmune disorders, allergies, and immunodeficiencies, aiming to develop effective therapies and treatments. Ultimately, the knowledge gained through immunology research helps advance medicine, enabling the development of new diagnostic tools, therapies, and preventive measures to combat infectious diseases, cancer, and other immune-related disorders. Immunology is at the forefront of medical innovation, paving the way for a healthier and more resilient future.

Immunology is a fascinating field of study that delves into the intricate workings of our immune system. This book, Core Concepts in Immunology, explores the core concept of immunology, unraveling the mysteries of how our body defends itself against diseases and infections. Through detailed explanations and comprehensive research, readers will gain a deeper understanding of the immune system's complex mechanisms. Immunology is not just about fighting off pathogens, but also about the delicate balance of immune responses and the potential for immune-related disorders. From the basics to the latest advancements, this book provides a comprehensive overview of immunology, making it an essential read for students and anyone intrigued by the fascinating world of our body's defense system.

CHAPTER 1

INTRODUCTION TO IMMUNE SYSTEM

The immune system is a complex network of cells, tissues, and organs in the human body that works together to defend against harmful pathogens, such as bacteria, viruses, fungi, and other foreign invaders. Its primary function is to recognize and eliminate these pathogens while distinguishing them from the body's own cells and tissues.

After studying this chapter, you will understand the core concept of:

- Functional Components of the Immune System
- Innate Immune System
- Adaptive Immune System
- Evolution of the Immune System

INTRODUCTION

The immune system is made up of a complex network of organs, cells and proteins that fight infection (microbes).

The immune system is a complex network of cells, tissues, and organs that work collaboratively to defend the body against harmful invaders, such as bacteria, viruses, fungi, and other pathogens. Its primary function is to identify and eliminate these foreign substances while maintaining tolerance to the body's own tissues and cells. The immune system plays a crucial role in maintaining overall health and preventing infections and diseases.

The immune system keeps a record of every microbe it has ever defeated, in types of white blood cells (B-lymphocytes and T-lymphocytes) known as memory cells. This means it can recognise and destroy the microbe quickly if it enters the body again, before it can multiply and make you feel sick.

Some infections, like the flu and the common cold, have to be fought many times because so many different viruses or strains of the same type of virus can cause these illnesses. Catching a cold or flu from one virus does not give you immunity against the others.

CORE CONCEPT: FUNCTIONAL COMPONENTS OF THE IMMUNE SYSTEM

The innate immune system is always working to protect the body and does not require any special preparation to stop infection. The immune system is precisely that a system, not a single entity. To function well, it requires balance and harmony.

Each of the cells in the immune response has a particular role to play.

Macrophages

These cells may be divided into two main groups: the dendritic cell and the mature macrophage. The dendritic cell's major function is to present antigen to the lymphocyte, and it is the earliest cell to recognize foreign antigen.

> Once it has been activated a lymphocyte can become an effector cell, which is usually short-lived, or a long-lived memory cell capable of being reactivated should an infection be reoccur.

There are two forms of dendritic cells: immature and mature. The induction of an adaptive immune response begins when a pathogen is ingested by an immature dendritic cell. These cells reside in most tissues and are relatively long-lived. As seen in Figure 1, they are derived from the same cell myeloid precursor as the macrophage. This immature cell carries receptors on its surface that recognize common features of many pathogens such as cell wall carbohydrates of bacteria. Once the bacterium is in contact with these receptors, the dendritic cell is stimulated to engulf the pathogen and degrade it intracellularly. These cells also continue engulfing extracellular material (both viruses and bacteria) by a receptor-independent mechanism of macropinocytosis. Once accomplished, the main function of the "activated" dendritic cell is to carry pathogenic antigens to the peripheral lymphoid organs to present them to T lymphocytes. Once arrived, the dendritic cell matures into an APC, which now permits it to activate pathogen-specific lymphocytes. Another function of activated dendritic cells is to secrete cytokines that influence both the innate and adaptive immune responses.

Engage in regular physical activity to improve circulation, reduce inflammation, and support overall immune system health.

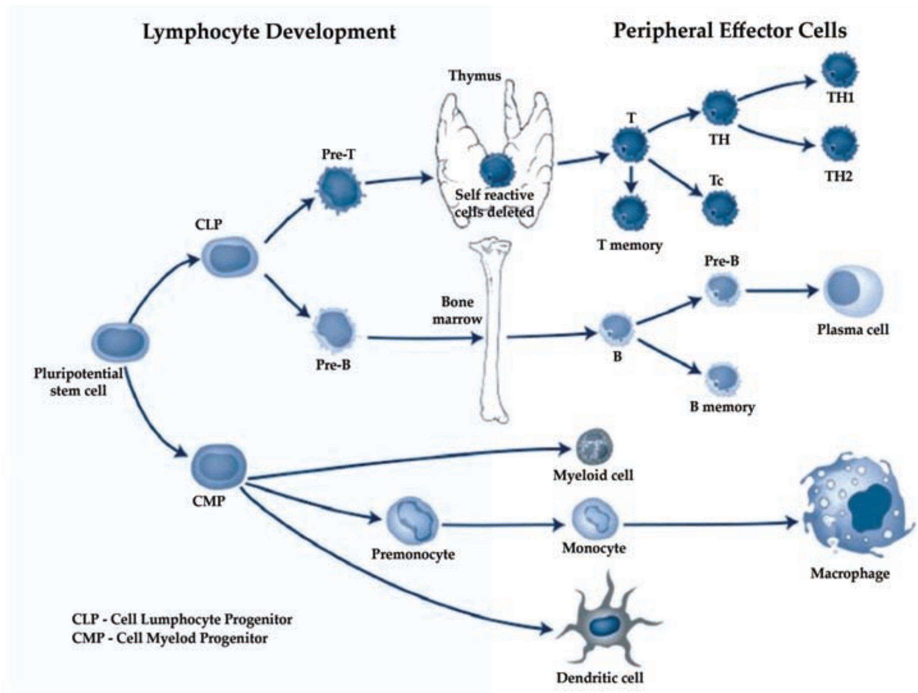


Figure 1: Development and differentiation of lymphocytes from pluripotential stem cells.

The mature macrophage also derives from primitive stem cells in the bone marrow, but unlike the lymphocyte, it matures in the tissues. Thus monocytes, the precursors of mature macrophages, circulate for only a few hours before entering the tissues where they live for months as mature macrophages. There is great variety in the tissue macrophages; they are heterogeneous in appearance and metabolism. They include mobile alveolar and peritoneal macrophages. There are also fixed cells in the liver called **Kupffer cells** and skin macrophages called Langerhans cells.

Kupffer cells are the specialized macrophages located in the liver, particularly at the lining of liver sinusoids.

The primary function of these mononuclear cells is to phagocytose invading organisms, dead cells, immune complexes, and antigens. To do this, these cells are equipped with powerful lyososomal granules containing acid hydrolases and other degrading enzymes. Macrophages need activation to carry out these functions. These include cytokines, which can bind to IgG: Fc receptors or most importantly (as we shall see later) receptors for bacterial polysaccharides. In addition, they can be activated by soluble inflammatory products such as C5a. In turn, the macrophages can release monokines, such as TNF or IL-1, which increase the inflammation in inflamed tissues.

Neutrophils

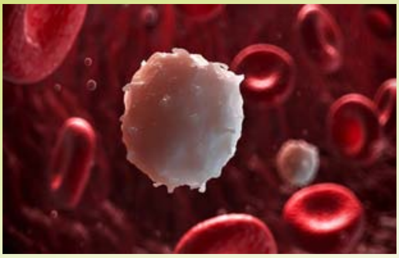
These circulating cells also play an important role in the body's defense against infection. These cells produce molecule receptors, permitting them to adhere to and migrate from the blood vessels to the site of infection. They are attracted to the site by IL-8, C3a, and C3b, cytokines released by T_H1 cells, and finally factors produced by mast cells. These cells are also phagocytic cells, and the process of phagocytosis is similar to that seen in macrophages. They are particularly effective when the invading organism becomes coated with antigen-specific antibodies (often called opsonins) along with activated complement components.

Other Functional Cells

Natural killer (NK) cells also can kill target cells in the absence of either antigen or antibody stimulation. Their lineage is not known, but they are probably in some manner related to T cells. Unlike other cells, they can be nonspecifically activated by mitogens, interferon, and IL-12. These cells are particularly useful in the early response to viral infection. As in other cells, they have receptors on their surface that recognize particular ligands. For example, NKR-PI is a lectin-like receptor that recognizes carbohydrate moieties on target cells, which initiates killing. As in other cell systems, there is also an inhibiting receptor called KIR. This molecule binds to ligands on MHC class I ligands, and this prevents killing of the target cell.

NK cells are not immune cells, and they have a broad range of specificity and no real memory. Studies of animals with NK deficiencies indicate that they have a greater incidence of viral infections and malignancies. This suggests that they have broad "immunological surveillance" properties but the exact mechanisms whereby they exert those properties are not known.

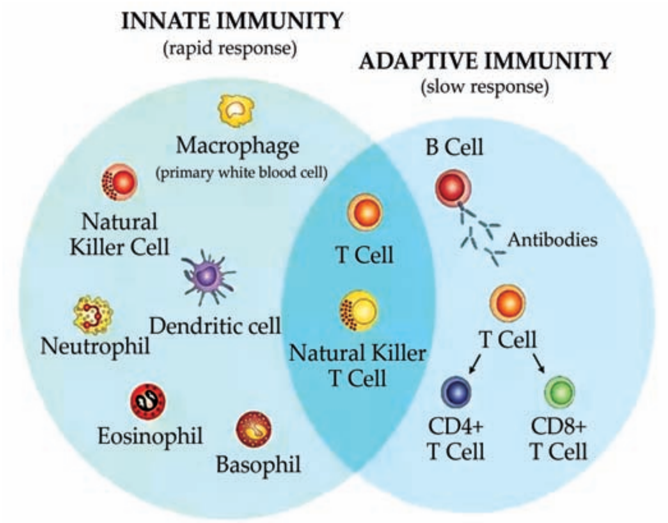
The use of an antibody-coated target to destroy foreign target cells is called antibody-dependent cell-mediated cytotoxicity, or ADCC. This killing is dependent on the recognition by cells bearing Fc receptors and includes monocytes, neutrophils, and NK cells. These cells do not need simultaneous recognition by MHC molecules. The mechanisms of killing most likely involve the release of cytoplasmic components of the target cells and perforin, but additional factors are also probably involved.



CORE CONCEPT: INNATE IMMUNE SYSTEM

Innate immunity refers to nonspecific defense mechanisms that come into play immediately or within hours of an antigen's appearance in the body. These mechanisms include physical barriers such as skin, chemicals in the blood, and immune system cells that attack foreign cells in the body.

The body is sometimes described as constantly being 'at war'. Everywhere around us there are dangerous bacteria, viruses, parasites and toxins that can cause the human body harm. To combat this, the body uses the immune system to fight away anything that may cause damage. The innate immune system are those parts of the **immune system** that work no matter what the damage is caused by, and are all aimed at protecting the body without the need for a lot of preparation. They are always at work and do not need to have seen the offending invader before to be able to start attacking it.



Physical Barriers to Infection

The major defences of the innate immune system are those that stop the infection from getting into the body in the first place. If they can do this, then it makes killing the few that get through a much easier task for the body.

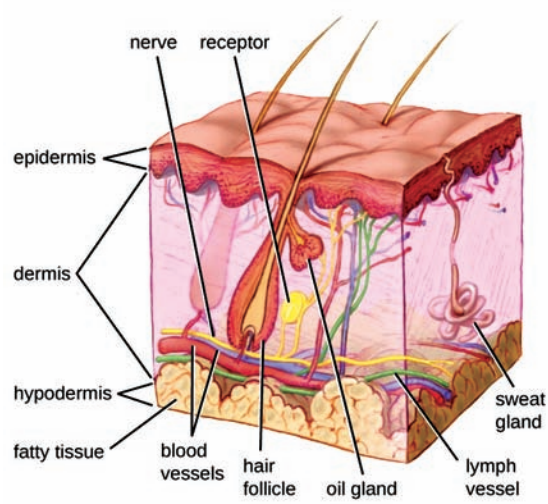
The **immune system** defends our body against invaders, such as viruses, bacteria, and foreign bodies.

Skin

The skin is a thick barrier to infection and protects from bad bacteria in a few ways.

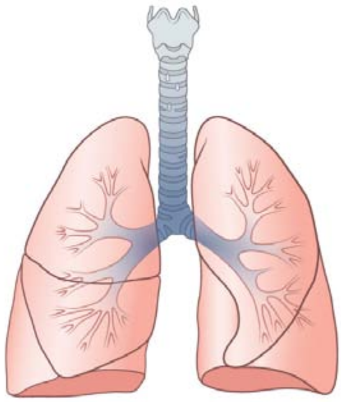
- It has a lot of dead skin cells that block the entry of bacteria
- The skin has its own, perfectly healthy, population of bacteria. These compete with the bad bacteria to stop them from multiplying.
- Sweat glands and sebaceous glands (that secrete a thick, oily substance called sebum) both release a slightly acidic substance onto the skin, slowing the growth of bacteria.

If the skin is constantly damp, such as the skin between toes, under arms or even under rolls of fat, it weakens the barrier. This wet skin can be more easily broken by the fungus that usually live on the skin and allows them to grow, causing a fungal *infection*. If the skin is broken by a cut or graze then it allows a far more easy access to the body than otherwise and these areas can get infected more easily. During surgery, even though every attempt to stop infection is made, sometimes the body's own normal bacteria can cause an infection because of the cut.



Lungs

The lining of the airways of the respiratory system are also coated in a thick, mucous-like substance that acts as a protective barrier. These airways are basically like big pipes, with lots of air flowing through all the time. With this air comes many possibly damaging things so the lungs must be protected. In the lungs, as air rushes past the walls, bacteria adhere to sticky mucus. In this mucus is an enzyme, which is the name for a substance that causes a chemical reaction. Here, the enzyme, which is called **lysozyme,** causes the bacteria to be broken down into its basic parts. It is being treated much like a food, and is being digested by this enzyme. After it has been digested, the remains of the dead bacteria as well as other substances are swept out of the air ways by small hair-like projections called 'cilia' which line the walls.



Eyes

Tears act to stop infection in two ways. Firstly, the tears physically wash away a lot of bacteria. Secondly, tears also contain the same enzyme that was mentioned before, lysozyme, which protects the eye from bacteria.

Mouth

The mouth is full of highly dangerous bacteria that can easily destroy tissue and cause tooth decay. Saliva helps prevent the destructive processes in several ways:

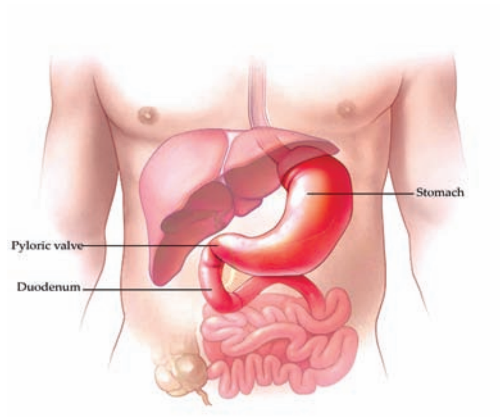
- The flow of saliva helps wash away bacteria, as well as the food particles that the bacteria eat.
- The saliva contains substances, including lysozyme and others, that work to destroy bacteria.
- Lysozyme also destroys the bits of food that the bacteria use for food.

> Remember that a balanced approach to overall health is key to supporting your immune system. It's also important to consult with a healthcare professional before making significant changes to your lifestyle, especially if you have underlying health conditions or concerns.

- A large amount of protective antibodies (which are substances that recognise infective particles) are found in the saliva.

Stomach

The stomach is a highly acidic environment, which is very destructive to most types of bacteria. Also, if a bacteria or other dangerous substance gets into the stomach, then they are treated just like any type of food and often broken down and destroyed by digestive enzymes before they can cause any damage.

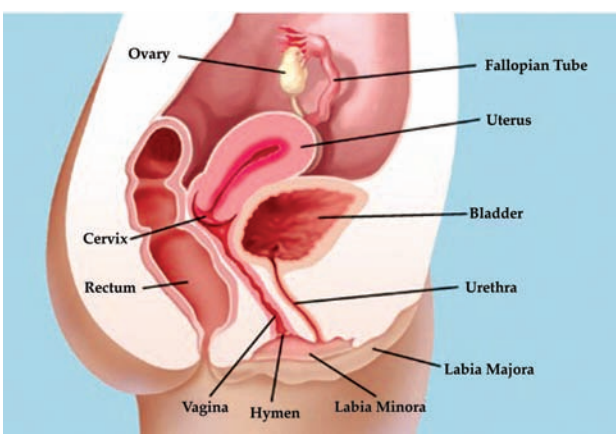


Urinary Tract

The urine is slightly acidic, meaning bacteria do not grow very well. The frequent flushing action of urination also helps to lower the amount of urinary infections.

Vagina

In a similar way to the skin, mucous linings of the vagina help form physical barriers to the entry of infective organisms. Also like the skin, the vagina is home to a lot of native bacteria that compete with the dangerous types and stop them from growing. Also in the vagina, there is a population of a type of bacteria called *Lactobacillus* that produces an acidic substance called lactate, helping to keep the environment slightly acidic and slowing bacterial growth.



Innate Immune Responses in the Blood

If infective agents manage to get through the boundaries described above and enter the body, there are several other immune responses that take place. One defensive process involves the white **blood cells** (also called leukocytes). Types of white blood cells are the neutrophils, eosinophils and basophils can respond to foreign substances that have never been seen by the immune system before. Another immune defence involves the presence within the blood of specific chemical compounds that attach to foreign organisms and destroy them including:

A **blood cell** is a cell produced through hematopoiesis and found mainly in the blood.

- Lysozyme: Attacks bacteria, causing them to be 'digested'
- Basic polypeptides: A substance that deactivates some types of bacteria
- Complement cascade
- Natural killer lymphocytes

Complement cascade

The complement cascade is a system of about 20 proteins. All the complement proteins are formed by the liver and are normally among the plasma proteins in the blood as well as among the proteins that leak into the intercellular spaces. The complement cascade helps provide an innate immunity through the activation of the alternative pathway. This is a complicated system but basically involves these complement proteins recognizing some substances that only occur in bacteria. When they are detected, they attach to the bacteria and punch holes in the walls of it, causing the bacteria to burst. The complement cascade also signals for phagocytes to come and attack the bacteria.

Phagocytes

1. **Neutrophils:** A neutrophil is a very common type of phagocyte. On approach to a target, the neutrophil first attaches itself to the particle and then captures it in little cellular 'arms'. These 'arms' (also called pseudopodia) join together around the other side of the target, forming an enclosed chamber which holds the particle. This chamber detaches from the cell wall and creates a small cavity called a phagocytic vesicle. Once this has happened, the cell can inject all sort of

digestive enzymes into the chamber destroying the bacteria. A neutrophil can eat between 3 and 20 bacteria before it stops working.

2. **Macrophages:** A macrophage is a much more powerful eater than a neutrophil. A macrophage can digest up to 100 bacteria before becoming deactivated, and can even engulf entire red blood cells. Apart from being much bigger, they have special digestive substances that break down the cell walls of some particularly strong bacteria.
3. **Basophils:** Basophils secrete a substance that keeps inflammation (detailed below) going after it has begun.
4. **Eosinophils:** Eosinophils are immune cells which play a role in the destruction of parasitic worms. While eosinophils are far too small to 'eat' the worms, they attach to the worms and punch small holes in their side, causing them to die. Eosinophils are also often found in tissues where an allergic reaction has taken place such as the lungs of people with asthma. Their purpose there may be to detoxify some of the inflammation-inducing substances.
5. **Bactericidal agents:** Even if this digestion does not destroy a bacterium, these cells are able to neutralise many more through the use of bactericidal agents. Much of this is done by oxidizing agents which can kill a bacteria at even low quantities. Oxidants are usually very dangerous, even to normal body tissue and so are under careful control by the phagocytes. Some bacteria, such as tuberculosis, can avoid even these mechanisms. These are often responsible for chronic infections.

Natural killer cells

Natural killer (NK) cells arise from the bone marrow. Bone marrow is in the centre of some types of bone, and is the manufacturing area for lots of blood cells. Unlike other cells, NK cells attack the body's own cells that have been infected by microbes, rather than the microbes themselves. They scout the body for anything that appears to be wrong with the body's own cells such as happens in tumours and viruses. NK cells, when the detect a problem, use some specialised enzymes to punch holes in these cells and inject a chemical that causes the cell to die in a special type of cell death called apoptosis. Apoptosis involves the cell being dismantled rather than just exploding which keeps the virus contained.

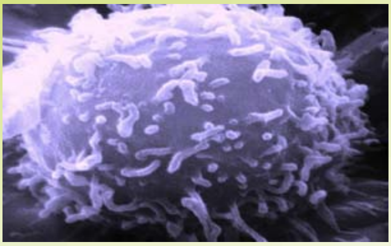
> The immune system has evolved to have a defensive role against any infectious microbe, but should not cause harm to the body.

Inflammation

When any type of tissue injury occurs, substances are released from the body tissues causing inflammation. The signs of inflammation are redness, warmth, swelling and pain. These signs are caused by:

- Increases local blood flow.
- Leakage of water and protein from blood vessels.
- Movement of lots of immune cells into the area.
- Swelling of some of the cells.
- Stimulation of pain receptors.

The substances that cause inflammation are mostly those released from cells called mast cells, which are located mostly in connective tissues. This helps to repel invaders because it absolutely saturates the area with all the substances that have been described above. Complement comes out of the leaky vessels, as do the phagocytes giving them the most possible access to the damaged area.



CORE CONCEPT: ADAPTIVE IMMUNE SYSTEM

The adaptive immune system, also known as the acquired immune system or, more rarely, as the specific immune system, is a subsystem of the overall immune system that is composed of highly specialized, systemic cells and processes that eliminate pathogens or prevent their growth.

The adaptive immune system, also called *acquired immunity*, uses specific antigens to strategically mount an immune response. Unlike the innate immune system, which attacks only based on the identification of general threats, the adaptive immunity is activated by exposure to pathogens, and uses an immunological memory to learn about the threat and enhance the immune response accordingly.

The adaptive immune response is much slower to respond to threats and infections than the innate immune response, which is primed and ready to fight at all times.

The adaptive, or acquired, immune response takes days or even weeks to become established—much longer than the innate response; however, adaptive immunity is more specific to pathogens and has memory.

Adaptive immunity is an immunity that occurs after exposure to an antigen either from a pathogen or a vaccination. This part of the immune system is activated when the innate immune response is insufficient to control an infection.

In fact, without information from the innate immune system, the adaptive response could not be mobilized. There are two types of adaptive responses: the cell-mediated immune response, which is carried out by T cells, and the humoral immune response, which is controlled by activated B cells and antibodies.

Activated T cells and B cells that are specific to molecular structures on the pathogen proliferate and attack the invading pathogen. Their attack can kill pathogens

Phagocytosis is a major mechanism used to remove pathogens and cell debris.

directly or secrete antibodies that enhance the **phagocytosis** of pathogens and disrupt the infection. Adaptive immunity also involves a memory to provide the host with long-term protection from reinfection with the same type of pathogen; on re-exposure, this memory will facilitate an efficient and quick response.

Antigen-presenting Cells

Unlike NK cells of the innate immune system, B cells (B lymphocytes) are a type of white blood cell that gives rise to antibodies, whereas T cells (T lymphocytes) are a type of white blood cell that plays an important role in the immune response. T cells are a key component in the cell-mediated response—the specific immune response that utilizes T cells to neutralize cells that have been infected with viruses and certain bacteria. There are three types of T cells: cytotoxic, helper, and suppressor T cells. Cytotoxic T cells destroy virus-infected cells in the cell-mediated immune response, and helper T cells play a part in activating both the antibody and the cell-mediated immune responses. Suppressor T cells deactivate T cells and B cells when needed, and thus prevent the immune response from becoming too intense.

An antigen is a foreign or "non-self" macromolecule that reacts with cells of the immune system. Not all antigens will provoke a response. For instance, individuals produce innumerable "self" antigens and are constantly exposed to harmless foreign antigens, such as food proteins, pollen, or dust components. The suppression of immune responses to harmless macromolecules is highly regulated and typically prevents processes that could be damaging to the host, known as tolerance. The innate immune system contains cells that detect potentially harmful antigens, and then inform the adaptive immune response about the presence of these antigens. An **antigen-presenting cell (APC)** is an immune cell that detects, engulfs, and informs the adaptive immune response about an infection. When a pathogen is detected, these APCs will phagocytose the pathogen and digest it to form many different fragments of the antigen. Antigen fragments will then be transported to the surface of the APC, where they will serve as an indicator to other immune cells. Dendritic cells are immune cells that process antigen material; they are present in the skin (Langerhans cells) and the lining of the nose, lungs, stomach, and intestines. Sometimes a dendritic cell presents

An **antigen-presenting cell (APC)** or accessory cell is a cell that displays antigen complexed with major histocompatibility complexes (MHCs) on their surfaces; this process is known as antigen presentation.

on the surface of other cells to induce an immune response, thus functioning as an antigen-presenting cell. Macrophages also function as APCs. Before activation and differentiation, B cells can also function as APCs.

After phagocytosis by APCs, the phagocytic vesicle fuses with an intracellular lysosome forming phagolysosome. Within the phagolysosome, the components are broken down into fragments; the fragments are then loaded onto MHC class I or MHC class II molecules and are transported to the cell surface for antigen presentation, as illustrated in Figure 2. Note that T lymphocytes cannot properly respond to the antigen unless it is processed and embedded in an MHC II molecule. APCs express MHC on their surfaces, and when combined with a foreign antigen, these complexes signal a "non-self" invader. Once the fragment of antigen is embedded in the MHC II molecule, the immune cell can respond. Helper T- cells are one of the main lymphocytes that respond to antigen-presenting cells. Recall that all other nucleated cells of the body expressed MHC I molecules, which signal "healthy" or "normal."

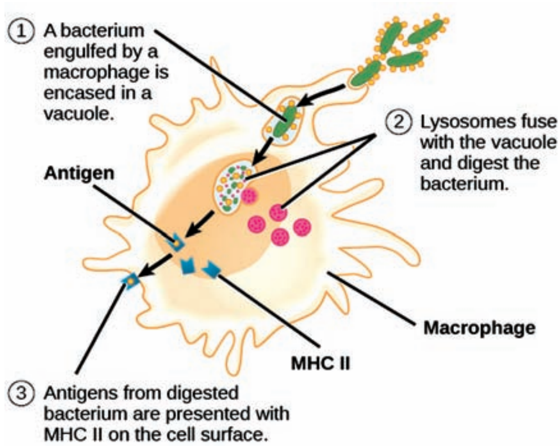


Figure 2: An APC, such as a macrophage, engulfs and digests a foreign bacterium. An antigen from the bacterium is presented on the cell surface in conjunction with an MHC II molecule Lymphocytes of the adaptive immune response interact with antigen-embedded MHC II molecules to mature into functional immune cells.

T and B Lymphocytes

Lymphocytes in human circulating blood are approximately 80 to 90% T cells, shown in Figure 3, and 10 to 20 percent B cells. Recall that the T cells are involved in the cell-mediated immune response, whereas B cells are part of the humoral immune response.

T cells encompass a heterogeneous population of cells with extremely diverse functions. Some T cells respond to APCs of the innate immune system, and indirectly induce immune responses by releasing cytokines. Other T cells stimulate B cells to prepare their own response. Another population of T cells detects APC signals and directly kills the infected cells. Other T cells are involved in suppressing inappropriate immune reactions to harmless or "self" antigens.

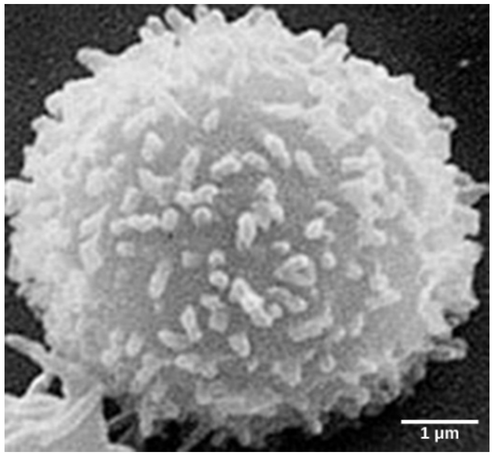


Figure 3: This scanning electron micrograph shows a T lymphocyte, which is responsible for the cell-mediated immune response. T cells are able to recognize antigens.

T and B cells exhibit a common theme of recognition/binding of specific antigens via a complementary receptor, followed by activation and self-amplification/maturation to specifically bind to the particular antigen of the infecting pathogen. T and B lymphocytes are also similar in that each cell only expresses one type of antigen receptor. Any individual may possess a population of T and B cells that together express a near limitless variety of antigen receptors that are capable of recognizing virtually any infecting pathogen. T and B cells are activated when they recognize small components of antigens, called epitopes, presented by APCs, illustrated in Figure 4.

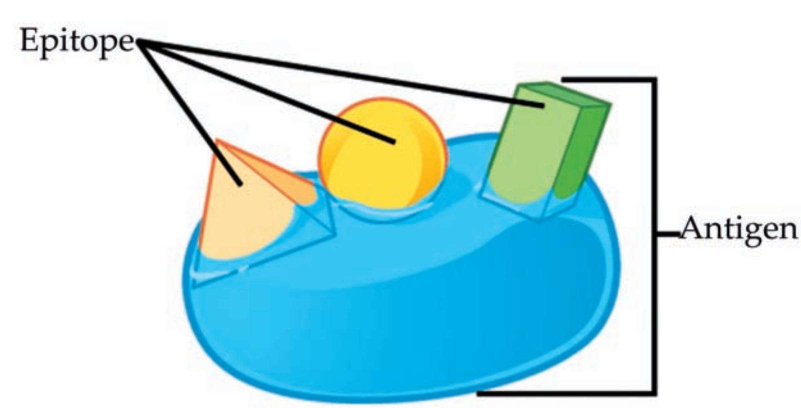


Figure 4: An antigen is a macromolecule that reacts with components of the immune system. A given antigen may contain several motifs that are recognized by immune cells. Each motif is an epitope.

Note that recognition occurs at a specific epitope rather than on the entire antigen; for this reason, epitopes are known as "antigenic determinants." In the absence of information from APCs, T and B cells remain inactive, or naïve, and are unable to prepare an immune response. The requirement for information from the APCs of innate immunity to trigger B cell or T cell activation illustrates the essential nature of the innate immune response to the functioning of the entire immune system.

Naïve T cells can express one of two different molecules, CD4 or CD8, on their surface, as shown in Figure 5, and are accordingly classified as $CD4^+$or $CD8^+$ cells. These molecules are important because they regulate how a T cell will interact with and respond to an APC. Naïve $CD4^+$ cells bind APCs via their antigen-embedded MHC II molecules and are stimulated to become helper T (T_H) lymphocytes, cells that go on to stimulate B cells (or cytotoxic T cells) directly or secrete cytokines to inform more and various target cells about the pathogenic threat. In contrast, $CD8^+$cells engage antigen-embedded MHC I molecules on APCs and are stimulated to become cytotoxic T lymphocytes (CTLs), which directly kill infected cells by apoptosis and emit cytokines to amplify the immune response. The two populations of T cells have different mechanisms of immune protection, but both bind MHC molecules via their antigen receptors called T cell receptors (TCRs). The CD4 or CD8 surface molecules differentiate whether the TCR will engage an MHC II or an MHC I molecule. Because they assist in binding specificity, the CD4 and CD8 molecules are described as coreceptors.

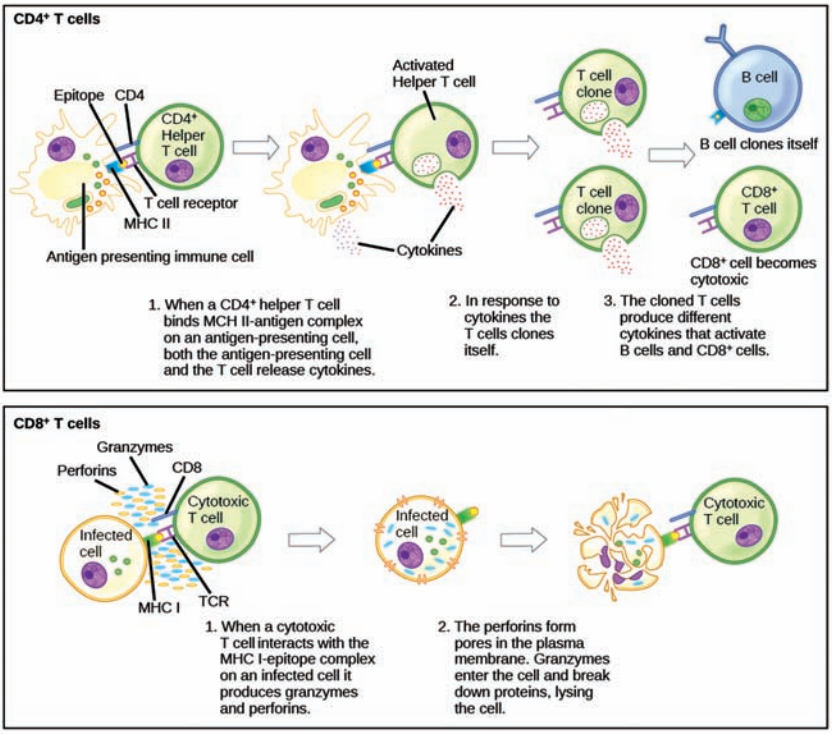


Figure 5: Naïve $CD4^+$ T cells engage MHC II molecules on antigen-presenting cells (APCs) and become activated. Clones of the activated helper T cell, in turn, activate B cells and $CD8^+$ T cells, which become cytotoxic T cells. Cytotoxic T cells kill infected cells.

Consider the innumerable possible antigens that an individual will be exposed to during a lifetime. The mammalian adaptive immune system is adept in responding appropriately to each antigen. Mammals have an enormous diversity of T cell populations, resulting from the diversity of TCRs. Each TCR consists of two polypeptide chains that span the T cell membrane, as illustrated in Figure 6; the chains are linked by a disulfide bridge. Each polypeptide chain is comprised of a constant domain and a variable domain: a domain, in this sense, is a specific region of a protein that may be regulatory or structural. The intracellular domain is involved in intracellular signaling. A single T cell will express thousands of identical copies of one specific TCR variant on its cell surface. The specificity of the adaptive immune system occurs because it synthesizes millions of different T cell populations, each expressing a TCR that differs in its variable domain. This TCR diversity is achieved by the mutation and recombination of genes that encode these receptors in stem cell precursors of T cells. The binding between an antigen-displaying MHC molecule and a complementary TCR "match" indicates that the adaptive immune system needs to activate and produce that specific T cell because its structure is appropriate to recognize and destroy the invading **pathogen**.

A **pathogen** or a germ in the oldest and broadest sense is anything that can produce disease; the term came into use in the 1880s.

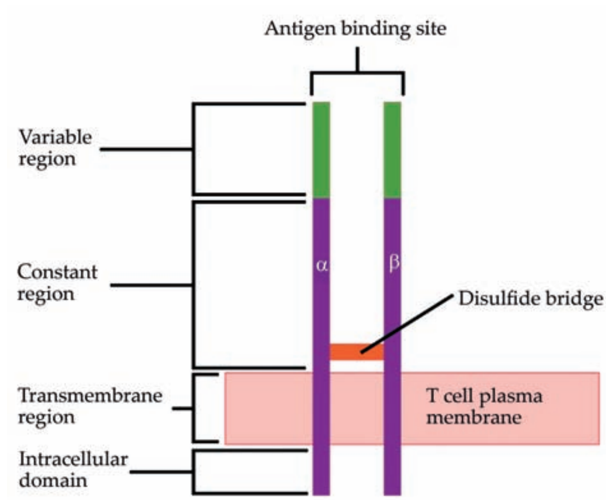


Figure 6: A T cell receptor spans the membrane and projects variable binding regions into the extracellular space to bind processed antigens via MHC molecules on APCs.

Helper T Lymphocytes

The T_H lymphocytes function indirectly to identify potential pathogens for other cells of the immune system. These cells are important for extracellular infections, such as those caused by certain bacteria, helminths, and protozoa. T_H lymphocytes recognize specific antigens displayed in the MHC II complexes of APCs. There are two major populations of T_H cells: T_H1 and T_H2. T_H1 cells secrete cytokines to enhance the activities of macrophages and other T cells. T_H1 cells activate the action of cyotoxic T cells, as well as macrophages. T_H2 cells stimulate naïve B cells to destroy foreign invaders via antibody secretion. Whether a T_H1 or a T_H2 immune response develops depends on the specific types of cytokines secreted by cells of the innate immune system, which in turn depends on the nature of the invading pathogen.

The T_H1-mediated response involves macrophages and is associated with inflammation. Recall the frontline defenses of macrophages involved in the innate immune response. Some intracellular bacteria, such as *Mycobacterium tuberculosis*, have evolved to multiply in macrophages after they have been engulfed. These pathogens evade attempts by macrophages to destroy and digest the pathogen. When *M. tuberculosis* infection occurs, macrophages can stimulate naïve T cells to become T_H1 cells. These stimulated T cells secrete specific cytokines that send feedback to the macrophage to stimulate its digestive capabilities and allow it to destroy the colonizing *M. tuberculosis*. In the same manner, T_H1-activated macrophages also become better suited to ingest and kill tumor cells. In summary; T_H1 responses are directed toward intracellular invaders while T_H2 responses are aimed at those that are extracellular.

B Lymphocytes

When stimulated by the T_H2 pathway, naïve B cells differentiate into antibody-secreting plasma cells. A **plasma cell** is an immune cell that secrets antibodies; these cells arise from B cells that were stimulated by antigens. Similar to T cells, naïve B cells initially are coated in thousands of B cell receptors (BCRs), which are membrane-bound forms of Ig (immunoglobulin, or an antibody). The B cell receptor has two heavy chains and two light chains connected by disulfide linkages. Each chain has a constant and a variable region; the latter is involved in antigen binding. Two other membrane proteins, Ig alpha and Ig beta, are involved in signaling. The receptors of any particular B cell, as shown in Figure 7 are all the same, but the hundreds of millions of different B cells in an individual have distinct recognition domains that contribute to extensive diversity in the types of molecular structures to which they can bind. In this state, B cells function as APCs. They bind and engulf foreign antigens via their BCRs and then display processed antigens in the context of MHC II molecules to T_H2 cells. When a T_H2 cell detects that a B cell is bound to a relevant antigen, it secretes specific cytokines that induce the

Plasma cells are white blood cells that secrete large volumes of antibodies. They are transported by the blood plasma and the lymphatic system.

B cell to proliferate rapidly, which makes thousands of identical (clonal) copies of it, and then it synthesizes and secretes antibodies with the same antigen recognition pattern as the BCRs. The activation of B cells corresponding to one specific BCR variant and the dramatic proliferation of that variant is known as **clonal selection**. This phenomenon drastically, but briefly, changes the proportions of BCR variants expressed by the immune system, and shifts the balance toward BCRs specific to the infecting pathogen.

T and B cells differ in one fundamental way: whereas T cells bind antigens that have been digested and embedded in MHC molecules by APCs, B cells function as APCs that bind intact antigens that have not been processed. Although T and B cells both react with molecules that are termed "antigens," these lymphocytes actually respond to very different types of molecules. B cells must be able to bind intact antigens because they secrete antibodies that must recognize the pathogen directly, rather than digested remnants of the pathogen. Bacterial carbohydrate and lipid molecules can activate B cells independently from the T cells.

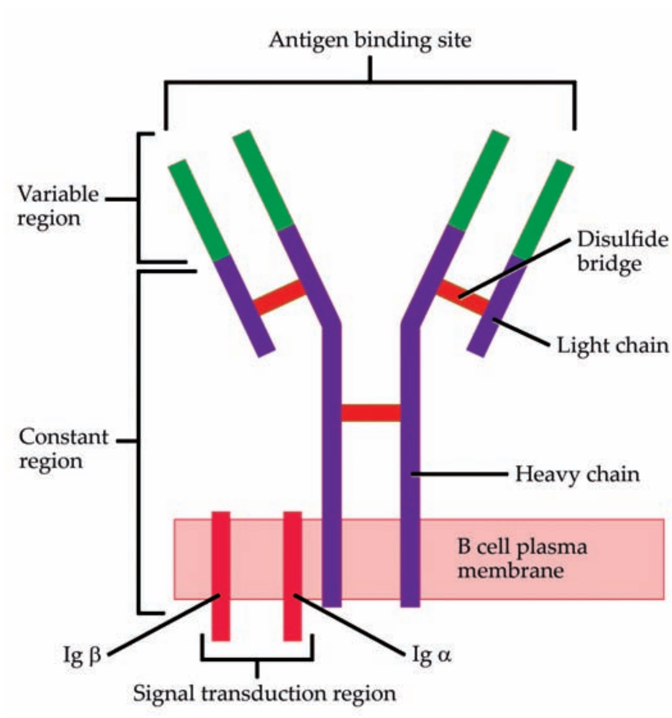


Figure 7: B cell receptors are embedded in the membranes of B cells and bind a variety of antigens through their variable regions. The signal transduction region transfers the signal into the cell.

Cytotoxic T Lymphocytes

CTLs, a subclass of T cells, function to clear infections directly. The cell-mediated part of the adaptive immune system consists of CTLs that attack and destroy infected cells. CTLs are particularly important in protecting against viral infections; this is because viruses replicate within cells where they are shielded from extracellular contact with circulating antibodies. When APCs phagocytize pathogens and present MHC I-embedded antigens to naïve $CD8^+$ T cells that express complementary TCRs, the $CD8^+$ T cells become activated to proliferate according to clonal selection. These resulting CTLs then identify non-APCs displaying the same MHC I-embedded antigens (for example, viral proteins)—for example, the CTLs identify infected host cells.

Macrophages are specialised cells involved in the detection, phagocytosis and destruction of bacteria and other harmful organisms.

Intracellularly, infected cells typically die after the infecting pathogen replicates to a sufficient concentration and lyses the cell, as many viruses do. CTLs attempt to identify and destroy infected cells before the pathogen can replicate and escape, thereby halting the progression of intracellular infections. CTLs also support NK lymphocytes to destroy early cancers. Cytokines secreted by the T_H1 response that stimulates **macrophages** also stimulate CTLs and enhance their ability to identify and destroy infected cells and tumors.

CTLs sense MHC I-embedded antigens by directly interacting with infected cells via their TCRs. Binding of TCRs with antigens activates CTLs to release perforin and granzyme, degradative enzymes that will induce apoptosis of the infected cell. Recall that this is a similar destruction mechanism to that used by NK cells. In this process, the CTL does not become infected and is not harmed by the secretion of perforin and granzymes. In fact, the functions of NK cells and CTLs are complementary and maximize the removal of infected cells, as illustrated in Figure 8. If the NK cell cannot identify the “missing self” pattern of down-regulated MHC I molecules, then the CTL can identify it by the complex of MHC I with foreign antigens, which signals “altered self.” Similarly, if the CTL cannot detect antigen-embedded MHC I because the receptors are depleted from the cell surface, NK cells will destroy the cell instead. CTLs also emit cytokines, such as interferons, that alter surface protein expression in other infected cells, such that the infected cells can be easily identified and destroyed. Moreover, these interferons can also prevent virally infected cells from releasing virus particles.

Plasma cells and CTLs are collectively called effector cells: they represent differentiated versions of their naïve counterparts, and they are involved in bringing about the immune defense of killing pathogens and infected host cells.

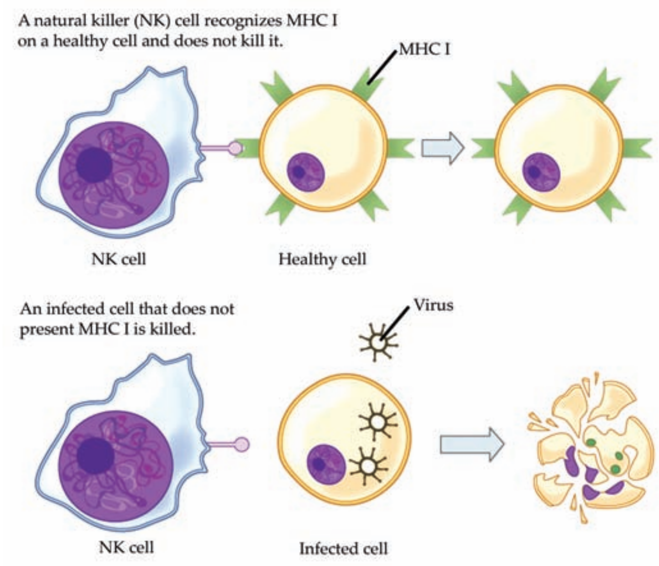


Figure 8: Natural killer (NK) cells recognize the MHC I receptor on healthy cells. If MHC I is absent, the cell is lysed.

Mucosal Surfaces and Immune Tolerance

Mucosa-associated lymphoid tissue (MALT) is scattered along mucosal linings in the human body and constitutes the most extensive component of human lymphoid tissue.

The innate and adaptive immune responses discussed thus far comprise the systemic immune system (affecting the whole body), which is distinct from the mucosal immune system. Mucosal immunity is formed by mucosa-associated lymphoid tissue, which functions independently of the systemic immune system, and which has its own innate and adaptive components. **Mucosa-associated lymphoid tissue (MALT)**, illustrated in Figure 9, is a collection of lymphatic tissue that combines with epithelial tissue lining the mucosa throughout the body. This tissue functions as the immune barrier and response in areas of the body with direct contact to the external environment. The systemic and mucosal immune systems use many of the same cell types. Foreign particles that make their way to MALT are taken up by absorptive epithelial cells called M cells and delivered to APCs located directly below the mucosal tissue. M cells function in the transport described, and are located in the Peyer’s patch, a lymphoid nodule. APCs of the mucosal immune system are primarily dendritic cells, with B cells and macrophages having minor roles. Processed antigens displayed on APCs are detected by T cells in the MALT and at various mucosal induction sites, such as the tonsils, adenoids, appendix, or the mesenteric lymph nodes of the intestine. Activated T cells then migrate through the lymphatic system and into the circulatory system to mucosal sites of infection.

MALT is a crucial component of a functional immune system because mucosal surfaces, such as the nasal passages, are the first tissues onto which inhaled or ingested pathogens are deposited. The mucosal tissue includes the mouth, pharynx, and esophagus, and the gastrointestinal, respiratory, and urogenital tracts.

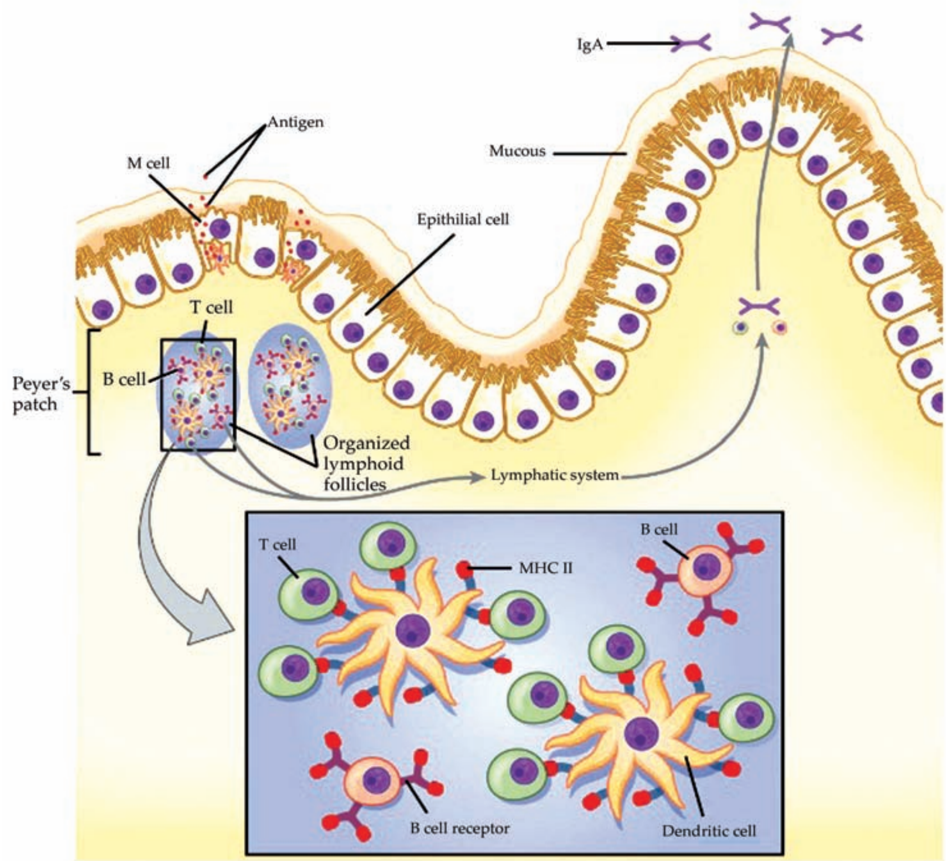


Figure 9: The topology and function of intestinal MALT is shown.

The immune system has to be regulated to prevent wasteful, unnecessary responses to harmless substances, and more importantly so that it does not attack "self." The acquired ability to prevent an unnecessary or harmful immune response to a detected foreign substance known not to cause disease is described as immune tolerance. Immune tolerance is crucial for maintaining mucosal homeostasis given the tremendous number of foreign substances (such as food proteins) that APCs of the oral cavity, pharynx, and gastrointestinal mucosa encounter. Immune tolerance is brought about by specialized APCs in the liver, lymph nodes, small intestine, and lung that present harmless antigens to an exceptionally diverse population of regulatory T (T_{reg}) cells, specialized lymphocytes that suppress local inflammation and inhibit the secretion of stimulatory immune factors. The combined result of T_{reg} cells is to prevent immunologic activation and inflammation in undesired tissue compartments and to allow the immune system to focus on pathogens instead. In addition to promoting immune tolerance of harmless antigens, other subsets of T_{reg} cells are involved in the prevention of the autoimmune response, which is an inappropriate immune response to host cells or self-antigens. Another T_{reg} class suppresses immune responses to harmful pathogens after the infection has cleared to minimize host cell damage induced by **inflammation** and cell lysis.

Immunological Memory

Inflammation is part of the body's defense mechanism. It is the process by which the immune system recognizes and removes harmful and foreign stimuli and begins the healing process.

The adaptive immune system possesses a memory component that allows for an efficient and dramatic response upon reinvasion of the same pathogen. Memory is handled by the adaptive immune system with little reliance on cues from the innate response. During the adaptive immune response to a pathogen that has not been encountered before, called a primary response, plasma cells secreting antibodies and differentiated T cells increase, then plateau over time. As B and T cells mature into effector cells, a subset of the naïve populations differentiates into B and T memory cells with the same antigen specificities, as illustrated in Figure 10.

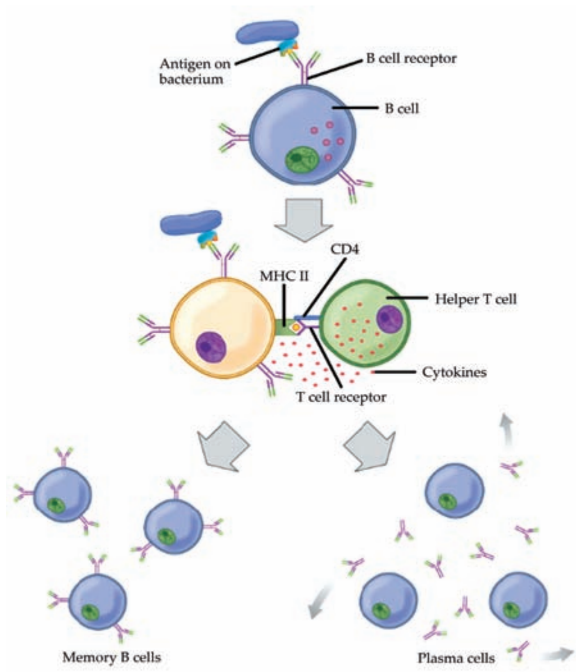


Figure 10: After initially binding an antigen to the B cell receptor (BCR), a B cell internalizes the antigen and presents it on MHC II. A helper T cell recognizes the MHC II–antigen complex and activates the B cell. As a result, memory B cells and plasma cells are made.

A memory cell is an antigen-specific B or T lymphocyte that does not differentiate into effector cells during the primary immune response, but that can immediately become effector cells upon re-exposure to the same pathogen. During the primary immune response, memory cells do not respond to antigens and do not contribute to host defenses. As the infection is cleared and pathogenic stimuli subside, the effectors are no longer needed, and they undergo apoptosis. In contrast, the memory cells persist in the circulation.

If the pathogen is never encountered again during the individual's lifetime, B and T memory cells will circulate for a few years or even several decades and will gradually die off, having never functioned as effector cells. However, if the host is re-exposed to the same pathogen type, circulating memory cells will immediately differentiate into plasma cells and CTLs without input from APCs or T_H cells. One reason the adaptive immune response is delayed is because it takes time for naïve B and T cells with the appropriate antigen specificities to be identified and activated. Upon reinfection, this step is skipped, and the result is a more rapid production of immune defenses. Memory B cells that differentiate into plasma cells output tens to hundreds-fold greater antibody amounts than were secreted during the primary response, as the graph in Figure 11 illustrates. This rapid and dramatic antibody response may stop the infection before it can even become established, and the individual may not realize they had been exposed.

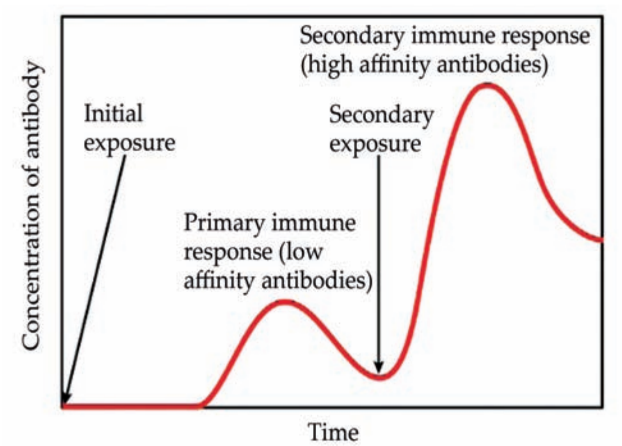


Figure 11: In the primary response to infection, antibodies are secreted first from plasma cells. Upon re-exposure to the same pathogen, memory cells differentiate into antibody-secreting plasma cells that output a greater amount of antibody for a longer period of time.

Vaccination is based on the knowledge that exposure to noninfectious antigens, derived from known pathogens, generates a mild primary immune response. The immune response to vaccination may not be perceived by the host as illness but still confers immune memory. When exposed to the corresponding pathogen to which an individual was vaccinated, the reaction is similar to a secondary exposure. Because each reinfection generates more memory cells and increased resistance to the pathogen, and because some memory cells die, certain vaccine courses involve one or more booster vaccinations to mimic repeat exposures: for instance, tetanus boosters are necessary every ten years because the memory cells only live that long.

Mucosal Immune Memory

A subset of T and B cells of the mucosal immune system differentiates into memory cells just as in the systemic immune system. Upon reinvasion of the same pathogen type, a pronounced immune response occurs at the mucosal site where the original pathogen deposited, but a collective defense is also organized within interconnected or adjacent mucosal tissue. For instance, the immune memory of an infection in the oral cavity would also elicit a response in the pharynx if the oral cavity was exposed to the same pathogen.

Primary Centers of the Immune System

Although the immune system is characterized by circulating cells throughout the body, the regulation, maturation, and intercommunication of immune factors occur at specific sites. The blood circulates immune cells, proteins, and other factors through the body. Approximately 0.1 percent of all cells in the blood are leukocytes, which encompass monocytes (the precursor of macrophages) and lymphocytes. The majority of cells in the blood are erythrocytes (red blood cells). Lymph is a watery fluid that bathes tissues and organs with protective white blood cells and does not contain erythrocytes. Cells of the immune system can travel between the distinct lymphatic and blood circulatory systems, which are separated by interstitial space, by a process called extravasation (passing through to surrounding tissue).

The cells of the immune system originate from hematopoietic stem cells in the bone marrow. Cytokines stimulate these stem cells to differentiate into immune cells. B cell maturation occurs in the bone marrow, whereas naïve T cells transit from the bone marrow to the thymus for maturation. In the thymus, immature T cells that express TCRs complementary to self-antigens are destroyed. This process helps prevent autoimmune responses.

The **macrophage** is a large white blood cell that is an integral part of our immune system. Its job is to locate microscopic foreign bodies and 'eat' them.

On maturation, T and B lymphocytes circulate to various destinations. Lymph nodes scattered throughout the body, as illustrated in Figure 12, house large populations of T and B cells, dendritic cells, and **macrophages**. Lymph gathers antigens as it drains from tissues. These antigens then are filtered through lymph nodes before the lymph is returned to circulation. APCs in the lymph nodes capture and process antigens and inform nearby lymphocytes about potential pathogens.

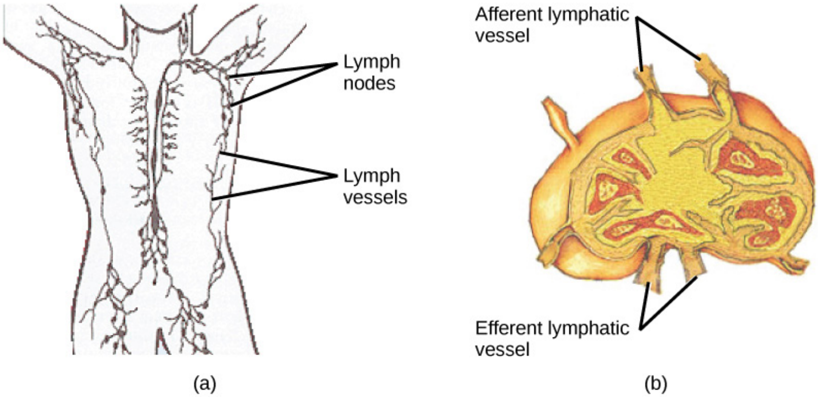


Figure 12: (a) Lymphatic vessels carry a clear fluid called lymph throughout the body. The liquid enters (b) lymph nodes through afferent vessels. Lymph nodes are filled with lymphocytes that purge infecting cells. The lymph then exits through efferent vessels.

The spleen houses B and T cells, macrophages, dendritic cells, and NK cells. The spleen, shown in Figure 13, is the site where APCs that have trapped foreign particles in the blood can communicate with lymphocytes. Antibodies are synthesized and secreted by activated plasma cells in the spleen, and the spleen filters foreign substances and antibody-complexed pathogens from the blood. Functionally, the spleen is to the blood as lymph nodes are to the lymph.

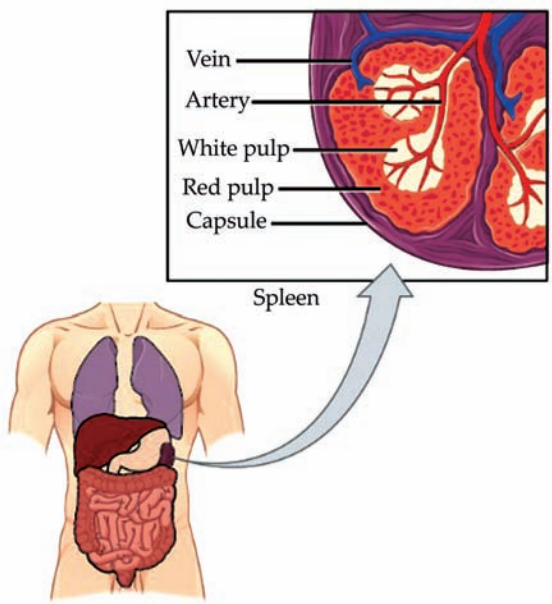
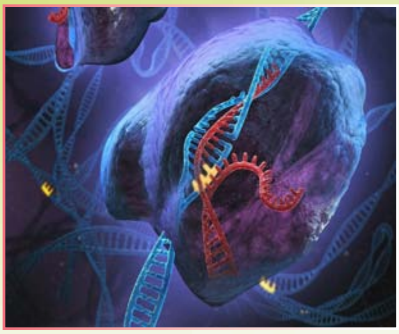


Figure 13: The spleen is similar to a lymph node but is much larger and filters blood instead of lymph. Blood enters the spleen through arteries and exits through veins. The spleen contains two types of tissue: red pulp and white pulp. Red pulp consists of cavities that store blood. Within the red pulp, damaged red blood cells are removed and replaced by new ones. White pulp is rich in lymphocytes that remove antigen-coated bacteria from the blood.



CORE CONCEPT:EVOLUTION OF THE IMMUNE SYSTEM

The immune system is spread throughout the body and involves many types of cells, organs, proteins, and tissues. Crucially, it can distinguish our tissue from foreign tissue — self from non-self. Dead and faulty cells are also recognized and cleared away by the immune system.

Virtually all organisms have at least one form of defense that helps repel disease-causing organisms. Advanced vertebrate animals, a group that includes humans, defend themselves against such microorganisms by means of a complex group of defense responses collectively called the immune system. This protective system evolved from simpler defense mechanisms, but the evolutionary twists and turns that led to its development are not entirely clear. To unravel the path that the vertebrate immune system followed in its evolution, investigators have studied the defense responses of various living organisms. They also have examined the genes of immune system proteins for clues to the genetic origins of immunity.

The Development of Immunity in Major Animal Groups

Because the immune system is composed of cells and tissues that do not lend themselves to fossilization, it is impossible to trace the evolution of immunity from the paleontological record. But, because all animals exhibit some general ability to recognize self and to repel foreign substances, it is possible to study the immune capacity of living animals and, based on the relative positions of these animals in the evolutionary tree, to extrapolate a reasonable evolutionary history of the immune system.

Immune Capacity among Invertebrates

From the lowliest protozoans to the higher marine tunicates, invertebrates have means of distinguishing self components from nonself components. Sponges from one colony will reject tissue grafts from a different colony but will accept grafts from their own. When tissue grafts are made in animals higher up the evolutionary tree—between individual annelid worms or starfish, for example—the foreign tissue is commonly invaded by phagocytic cells (cells that engulf and destroy foreign material) and cells resembling lymphocytes (white bloodcells of the immune system), and it is destroyed.

Yet tissues grafted from one part of the body to another on the same individual adhere and heal readily and remain healthy. So it seems that something akin to cellular immunity is present at this level of evolution. Insects engulf and eliminate foreign invaders through the process of phagocytosis (“cellular eating”). They have factors present in their circulatory fluids that can bind to foreign cells and cause clumping, or agglutination, of a number of these cells, an event that facilitatesphagocytosis. Insects also seem to acquire immunity to infectious agents.

Immune Capacity among Vertebrates

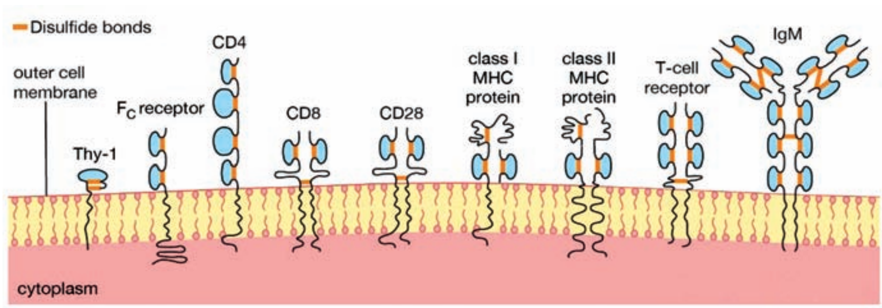
The most sophisticated immune systems are those of the vertebrates. Recognizable lymphocytes and immunoglobulins (Ig; also called antibodies) appear only in these organisms. The most primitive living vertebrates—the jawless fishes (hagfish and lampreys)—do not have lymphoid tissues corresponding to a spleen or a thymus, and their immune responses, although demonstrable, are very weak and sluggish. Farther up the evolutionary tree, at the level of the cartilaginous fishes (sharks and rays) and the bony fishes, a thymus and a spleen are present, as are immunoglobulins, although only those immunoglobulins of the IgM class are detectable. Fish lack specialized lymph nodes, but they do have clusters of lymphocytes in the gut that may serve an analogouspurpose.

It is not until the level of the terrestrial vertebrates—amphibians, reptiles, birds, and mammals—that a complete immune system with thymus, spleen, bone marrow, and lymph nodes is present and IgM and IgG antibodies are made. Antibodies of the IgA class are found only in birds and mammals, and IgE antibodies are confined to mammals. So it appears that the most primitive devices for producing specific, acquired immunity gradually diversified to meet the new environmental hazards as animals moved out of the sea onto the land.

The evolution of the complement system (a group of proteins involved in immune responses) may have occurred faster than that of the immunoglobulin system. The jawless fishes have complement components corresponding only to the later-acting (i.e., cytolytic, or cell-killing) aspects of complement function, but all higher vertebrates have components similar to the complete complement system of mammals. The fact that the complement system has been so well conserved during evolution implies not only that it has been of great biological value but also that complement and immunoglobulins have interacted throughout the evolution of the immune system in higher vertebrates. (For more information on the complement system, see Antibody-mediated immune mechanisms.)

Genetic Origins of the Immune System

Researchers have found many similarities between the structures of proteins involved in antigen recognition and those in cell-to-cell recognition in the immune system. (Antigens are the foreign proteins that antibodies recognize and bind to.) These proteins include the antigen receptors of lymphocytes, the (MHC) proteins, the coreceptors involved in cell-to-cell recognition in immune reactions (such as the receptors named CD4, CD8, and CD28), and the F_C receptor that binds to the stem of the Y-shaped immunoglobulin molecule. A number of proteins not involved in the immune system also share structural features with these proteins. The main feature similarity is a structure called the immunoglobulin domain. Each protein is composed of one or more Ig domains of nearly identical size. The domains are formed into a loop by bonds between sulfur atoms on the amino acids at the ends. Although each domain is different and serves a different function in the molecule as a whole, the number and order of the amino acids forming each domain are far more similar than would be expected if each had arisen independently in the course of evolution. Equally remarkable is the fact that nerve cells, thymus cells, and T lymphocytes (T cells) in mice and rats carry a surface protein called Thy-1 (thymus-1 antigen) that also has this same basic structure and a similar arrangement of amino acids. The similarities suggest that the genes for all these molecules originated from some primitive gene involved in the recognition of one cell by another, which is required for orderly development of a complex, multicellular organism, and that during evolution they had acquired different functions. Researchers named this group of genes and their protein products the immunoglobulin superfamily.



Not surprisingly, molecules that have a similar function in different species (e.g., immunoglobulins or MHC components) show an even closer resemblance. By analyzing the number of differences in the amino acids and their position in the polypeptide chains that make up IgM and IgG molecules in, for example, humans, mice, and rabbits and by making reasonable assumptions about mutation rates, scientists can estimate roughly how many generations would have had to elapse—and hence how much time—for the present immunoglobulins of these species to have evolved from a common ancestral IgM-like molecule. Such calculations suggest that divergence from the ancestral immunoglobulin took place some 200 million years ago. That was about the same time amphibians are thought to have diverged from the main vertebrate line. So one may conclude that a functional immune system arose even earlier and has continued to provide a defense against foreign agents ever since.

A CONCEPT OF IMMUNE TOLERANCE

Immune tolerance refers to the body's ability to prevent an immune response against its own tissues, thereby avoiding autoimmune reactions. It is a crucial concept in immunology and plays a fundamental role in maintaining a balanced and functional immune system. Immune tolerance mechanisms ensure that the immune system can distinguish between self and non-self antigens, preventing the harmful attack on the body's own cells and tissues.

There are two main types of immune tolerance:

1. Central Tolerance: This occurs during the development of immune cells in the thymus (for T cells) and bone marrow (for B cells). Immature immune cells that recognize self-antigens with high affinity are eliminated or undergo changes to become non-reactive to self-antigens. This process helps ensure that only immune cells capable of recognizing foreign antigens with appropriate affinity are allowed to mature.
2. Peripheral Tolerance: This mechanism prevents potentially self-reactive immune cells that escape central tolerance from causing harm. Peripheral tolerance includes various regulatory mechanisms that suppress or modulate immune responses to self-antigens. Examples include regulatory T cells (Tregs) that inhibit the activation of other immune cells and mechanisms such as anergy (functional inactivation of immune cells) and deletion (elimination) of self-reactive immune cells.

Immune tolerance is essential for preventing autoimmune diseases, which occur when the immune system mistakenly attacks the body's own tissues. Disorders like rheumatoid arthritis, lupus, multiple sclerosis, and type 1 diabetes are examples of autoimmune diseases that result from a breakdown in immune tolerance.

SUMMARY

- The immune system is a complex network of cells, tissues, and organs that work collaboratively to defend the body against harmful invaders, such as bacteria, viruses, fungi, and other pathogens.
- Natural killer (NK) cells also can kill target cells in the absence of either antigen or antibody stimulation. Their lineage is not known, but they are probably in some manner related to T cells.
- Sweat glands and sebaceous glands (that secrete a thick, oily substance called sebum) both release a slightly acidic substance onto the skin, slowing the growth of bacteria.
- If infective agents manage to get through the boundaries described above and enter the body, there are several other immune responses that take place.
- The adaptive immune system, also called acquired immunity, uses specific antigens to strategically mount an immune response. Unlike the innate immune system, which attacks only based on the identification of general threats, the adaptive immunity is activated by exposure to pathogens, and uses an immunological memory to learn about the threat and enhance the immune response accordingly.
- Some T cells respond to APCs of the innate immune system, and indirectly induce immune responses by releasing cytokines.
- The adaptive immune system possesses a memory component that allows for an efficient and dramatic response upon reinvasion of the same pathogen.

MULTIPLE CHOICE QUESTIONS

1. **What is the primary function of the immune system?**
   a. Regulation of body temperature
   b. Digestion of food
   c. Defense against foreign invaders
   d. Production of hormones
2. **Which of the following is an example of an innate immune response?**
   a. Antibody production
   b. Delayed hypersensitivity
   c. Memory cell formation
   d. Inflammation at a wound site
3. **Which cells are responsible for antibody production in the immune system?**
   a. T cells
   b. B cells
   c. Natural killer cells
   d. Macrophages
4. **The immune system's ability to differentiate between self and non-self molecules is known as:**
   a. Immune memory
   b. Immunization
   c. Autoimmunity
   d. Immune tolerance
5. **Which type of immunity is acquired through vaccination or exposure to a pathogen?**
   a. Passive immunity
   b. Innate immunity
   c. Adaptive immunity
   d. Natural immunity

Answers

1. (c) 2. (d) 3. (b) 4. (d) 5. (c)

REVIEW QUESTIONS

1. Describe the components of the immune system.
2. What is the major function of the B-Cells?
3. What is a pathogen?
4. Discuss about the innate immune system.
5. What is immunological memory?
6. How to evolution of the immune system? Explain.
7. Explain the antigen-presenting cells

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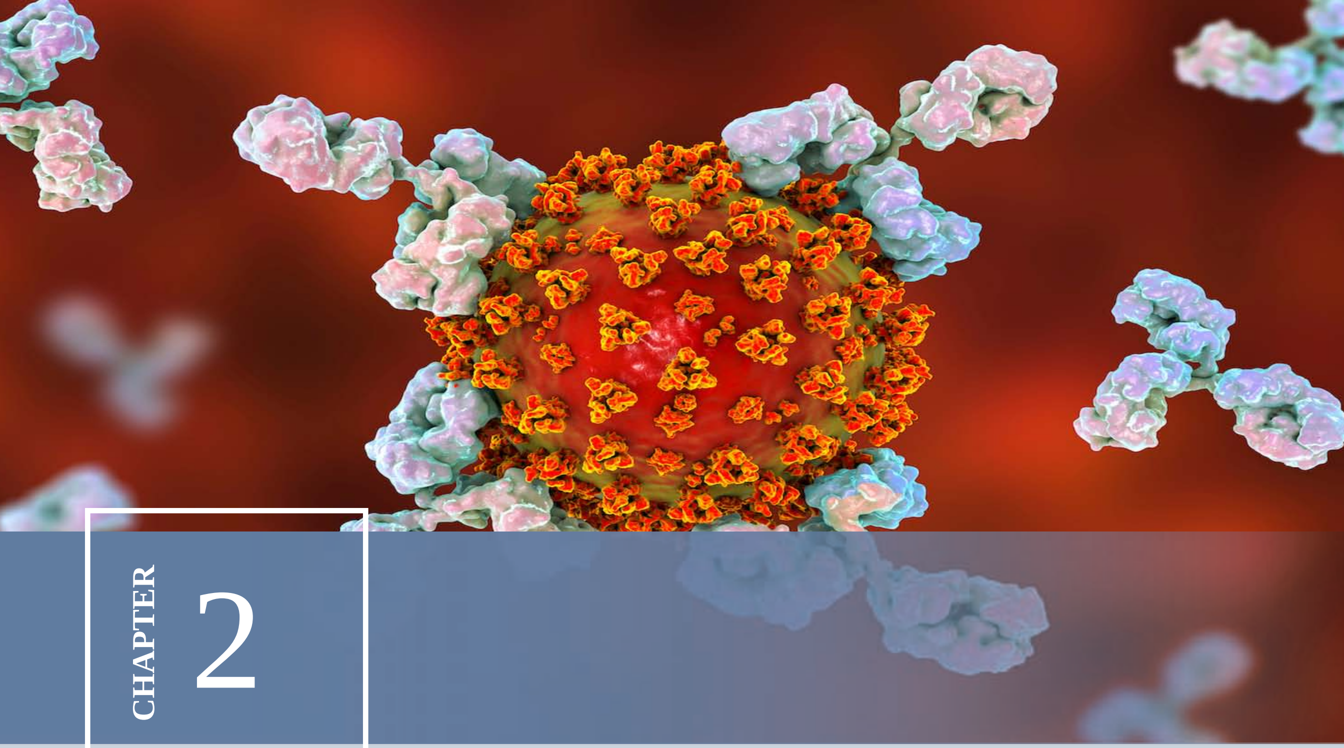
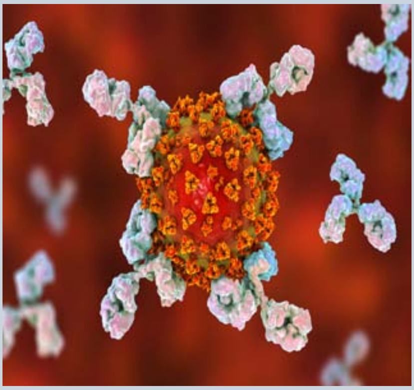


CHAPTER 2

ANTIBODIES AND ANTIGENS



Antigen and antibodies are two very different entities. In a nutshell, an antibody is a glycoprotein which is produced in response to and counteract a particular antigen. On the other hand, an antigen is a foreign substance (usually harmful) that induces an immune response, thereby stimulating the production of antibodies.

After studying this chapter, you will understand the core concept of:

- Motion of Antibodies and Antigens
- Major Histocompatibility Complex
- Antigen Receptors and Accessory Molecules of T lymphocytes

INTRODUCTION

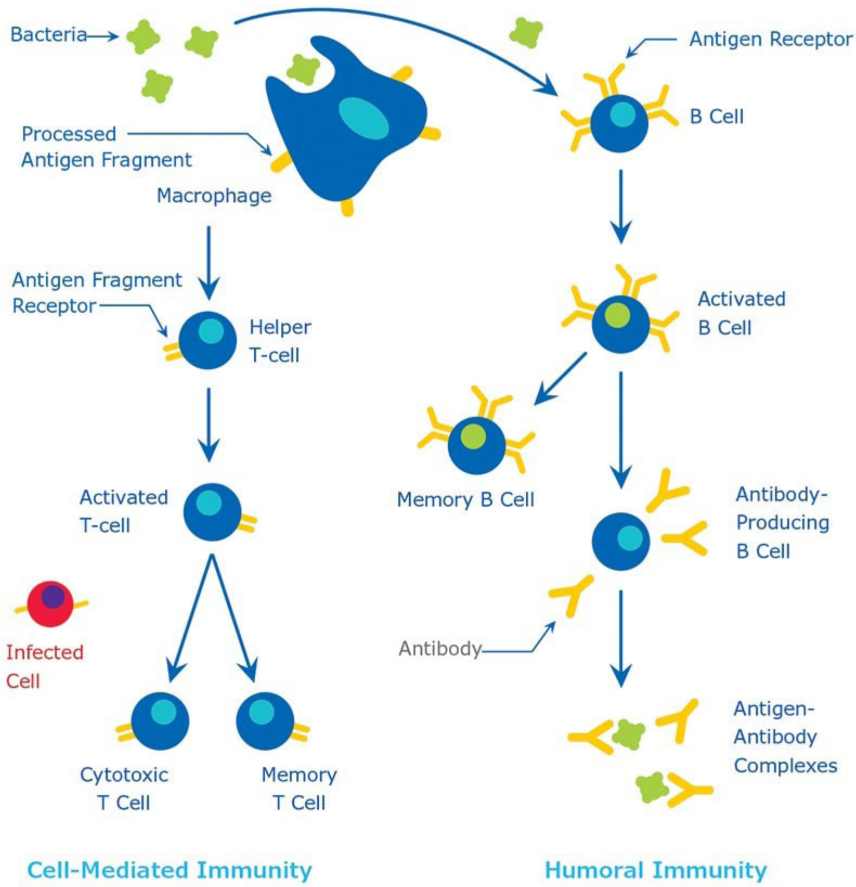
Antigens are molecules that can trigger an immune response in the body. They are usually proteins or large polysaccharides found on the surface of pathogens like bacteria, viruses, fungi, and parasites, as well as on the surface of foreign substances like pollen or toxins. Antigens are recognized by the immune system as foreign invaders, and the immune system mounts a response to eliminate them.

The term antigen is derived from antibody generation, referring to any substance that is capable of eliciting an immune response (e.g., the production of specific antibody molecules). By definition, an antigen (Ag) is capable of combining with the specific antibodies formed by its presence.

Generally, antigens are foreign proteins or their fragments that enter host body via an infection. However, in some cases, the body's own proteins may act as antigens and induce an autoimmune response. Bacteria and viruses contain antigens, either on their surface, or inside. These antigens can be isolated and used to develop vaccines.

Antigens are generally of high molecular weight, and commonly are proteins or polysaccharides. Polypeptides, lipids, nucleic acids, and many other materials can also function as antigens. Immune responses may also be generated against smaller substances, called haptens, if these are chemically coupled to a larger carrier protein, such as bovine serum albumin, keyhole limpet hemocyanin (KLH), or other synthetic matrices. A variety of molecules such as drugs, simple sugars, amino acids, small peptides, phospholipids, or triglycerides may function as haptens. Thus, given enough time, just about any foreign substance will be identified by the immune system and evoke specific antibody production. However, this specific immune response is highly variable and depends much in part on the size, structure, and composition of antigens. Proteins or glycoproteins are considered as the most suitable antigens due to their ability to generate a strong immune response; in other words, they are strongly immunogenic. Antigens are recognized by the host body by two distinct processes (1) by B cells and their surface antibodies (sIgM) and (2) by the T cell receptor on T cells. Although both B and T cells respond to the same antigen, they respond to different parts of the same molecule. Antibodies on the surface of B cells can recognize the tertiary structure of proteins. On the other hand, T cells require antigens that have been ingested and degraded into recognizable fragments by the antigen-presenting cells. Commonly employed antigen-presenting cells are macrophages and dendritic cells.

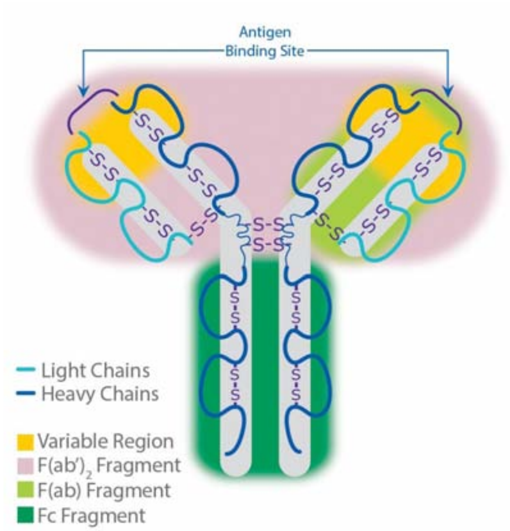
An antibody is defined as "an immunoglobulin capable of specific combination with the antigen that caused its production in a susceptible animal." Antibodies are produced in response to the invasion of foreign molecules in the body. An antibody, abbreviated as Ab, is commonly referred to as an immunoglobulin or Ig. Human immunoglobulins are a group of structurally and functionally similar glycoproteins (82-96% protein and 4-18% carbohydrate) that confer humoral immunity.



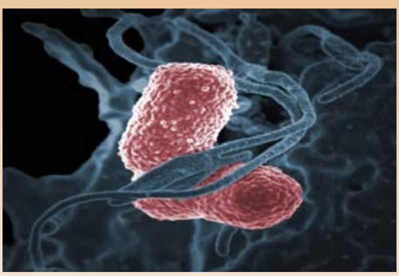
Antibodies, also known as immunoglobulins (Ig), are proteins produced by specialized immune cells called B cells. These proteins are specifically designed to bind to antigens with high specificity. When antibodies encounter antigens, they can neutralize them directly by preventing them from causing harm, or they can tag them for destruction by other immune cells. Antibodies are a crucial component of the adaptive immune response, providing targeted defense against a wide range of pathogens.

Antibodies, also known as immunoglobulins, are ~150 kDa, Y-shaped proteins that are both a natural part of the immune system and a tool that can be used for a variety of research applications. Within the immune system, antibodies are produced by B cells. They bind to proteins on the surface of extracellular pathogens such as parasites or microbes, or to proteins expressed on the surface of cells that have been infected with a microbe, to trigger immune cascades that clear these infections. Anything that generates an antibody response in the immune system is referred to as an antigen.

Antibodies exist as one or more copies of a Y-shaped unit, composed of four polypeptide chains. Each Y contains two identical copies of a heavy chain and two identical copies of a light chain, named as such by their relative molecular weights. This Y-shaped unit is composed of the two variable, antigen-specific F(ab) arms, which are critical for actual antigen binding, and the constant Fc "tail" that binds immune cell Fc receptors and also serves as a useful "handle" for manipulating the antibody during most immunochemical procedures. The number of F(ab) regions on the antibody corresponds with its subclass (see below), and determines the valency of the antibody (loosely stated, the number of "arms" with which the antibody may bind its antigen).



The ability of antibodies to bind proteins is useful for research applications as well because they allow scientists to target specific proteins they're interested in. Once a protein is targeted with an antibody, you can visualize the protein via fluorescence or chemiluminescence, precipitate the protein out of solution, or isolate cells expressing this protein.



CORE CONCEPT: MOTION OF ANTIBODIES AND ANTIGENS

Antibodies are small proteins that circulate in the bloodstream. They are part of the body's defense (immune) system and are sometimes called immunoglobulins. They are made by a type of white blood cell (a B lymphocyte). Antibodies attach to proteins and other chemicals in the body, which they recognize to be not normally found in the body ('foreign'). The foreign proteins and chemicals that antibodies attach to are called antigens.

Antibodies may be defined as the proteins that recognize and neutralize any microbial toxin or foreign substance such as bacteria and viruses. The only cells that make antibodies are **B lymphocytes**. Mainly two forms of antibodies exist. One those that are membrane-bound and act as receptor for antigens on the surface of B lymphocytes and the other that are involved in inhibition of entry and spread of pathogens and are found in blood circulation and connective tissues. The substance or molecule identified by antibodies or that can evoke antibody response is called an antigen.

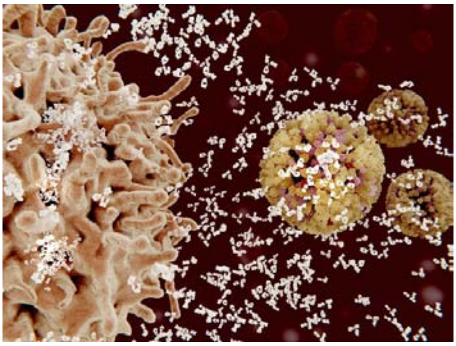
B lymphocytes are a type of white blood cell of the lymphocyte subtype. They function in the humoral immunity component of the adaptive immune system by secreting antibodies.

Some commonly used terminologies:

Serum: Clot formation in the blood leaves the residual fluid that contains antibodies. These antibodies in the residue form the serum.

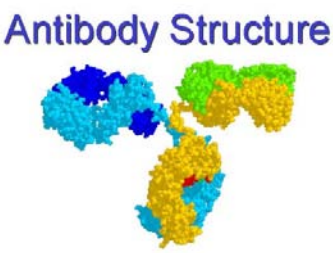
Antiserum: Serum contains a bunch of antibodies and when these antibodies show specificity to a particular antigen by binding to it, those antibodies are known as antiserum.

Serology: Serology may be defined as the study of blood serum or antibodies and their reactions with particular antigens.



Antibody Structure

Antibodies are also called as immunoglobulins and are Y-shaped protein structures. Antibodies consist of two identical light and heavy chains. Amino terminal variable (V) regions are found in both heavy and light chains and they take part in antigen recognition. Effector functions are directed by carboxy – terminal constant (C) regions of the heavy chains but C regions are also found in both the chains. Both the heavy and light chains are composed of Immunoglobulin (Ig) domain. Ig domain is a protein domain that consists of folded repeating units of 110 amino acids in length sandwiched between two layers of β-pleated sheet.



The two layers of β–pleated sheet are held together by a disulfide bridge and there are short loops that connect the adjoining strands of each β sheet. Amino acids in some of these loops are most crucial for antigen recognition. Light and heavy chain structure is almost similar. In light chain there is one V region Ig domain and one C region Ig domain whereas in heavy chain V region comprises of one Ig domain and the C region comprises of three or four Ig domains.

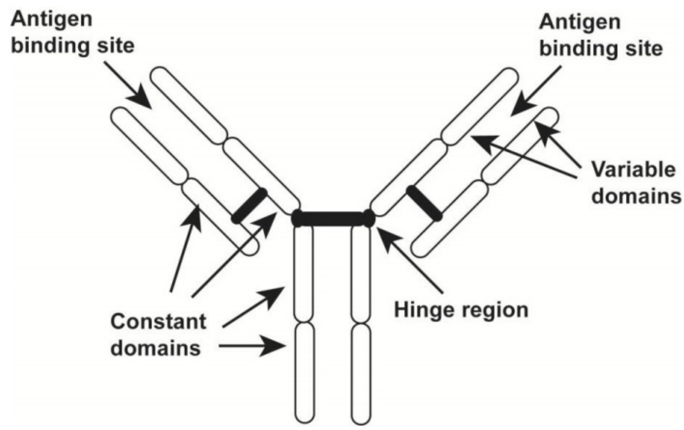


Figure 1: Immunoglobulin-G (IgG) molecule.

Fragment antigen-binding is a region on an antibody that binds to antigens. It is composed of one constant and one variable domain of each of the heavy and the light chain.

Antigen-binding site is formed by the V region of one heavy chain and the adjacent V region of one light chain. Disulfide bonds formed between cysteine residues connect the light and heavy chains in the carboxyl terminus of the light chain and the CH-1 domain of the heavy chain. Association of heavy and light chains occurs partly due to the non-covalent interactions between the VL and VH domains and between the CL and CH1 domains. Two heavy chains of each antibody entity are connected covalently by disulfide bonds. In IgG antibodies disulfide bonds are formed between cysteine residues in the CH2 regions which are near to a region known as hinge. This hinge region is more likely to undergo proteolytic cleavage. **Fragment antigen binding** (Fab fragment) is a portion on antibody that has the capability to bind to antigen and consists of one variable and one constant domain of each of the heavy and the light chain. Fragment crystallizable region (Fc region) is the distal region of an antibody that is composed of two identical, disulfide linked peptides containing the heavy chain CH2 and CH3 domains. Fc region communicates with some cell surface receptors called Fc receptors and this feature of Fc region helps antibodies to stimulate the immune system.

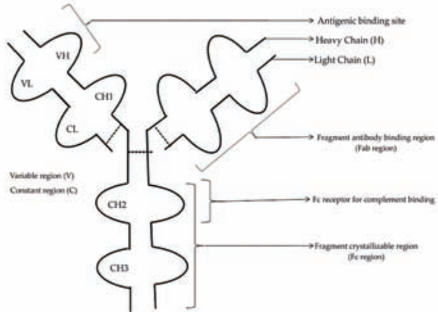


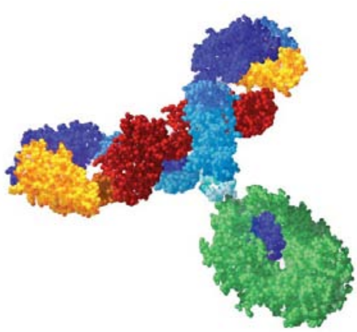
Figure 2: Schematic representation of immunoglobulin domains.

Monoclonal Antibodies

The concept of monoclonal antibodies was given for the first time by Georges Kohler and Cesar Milstein in the year 1975. Monoclonal antibodies are the antibodies that are specific to one particular antigen as they are made by identical immune cells that are several copies of a same parent cell e.g. any tumor cell of a specific region say plasma cells, are monoclonal and thus have the ability to produce antibodies of same specificity. The basic technique involved in making of monoclonal antibodies relies on fusion of B cells from an immunized mouse with a myeloma (tumor cell line) cell line and let the cells grow in a condition where unfused normal and tumor cells cannot survive. The cells that are fused and able to grow through this procedure are called as **hybridomas**.

> Monoclonal antibodies are likely to be less efficient at immunoprecipitation than polyclonal antibodies.

Hybridomas are produced by injecting a specific antigen into a mouse, collecting an antibody-producing cell from the mouse's spleen, and fusing it with a tumor cell called a myeloma cell.



Uses of Monoclonal Antibodies

1. Monoclonal antibodies help in immunodiagnosis by detection of a particular antigen or antibody.

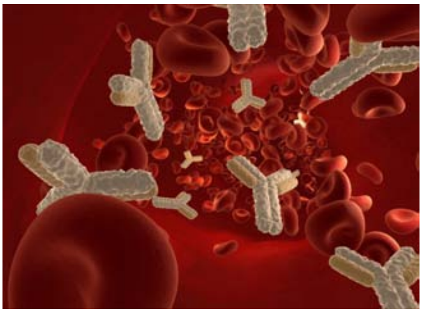
2. Many tumor-specific antibodies help in tumor detection.
3. Some of the monoclonal antibodies have therapeutic uses. E.g. cytokine tumor necrosis factor (TNF) is used to treat many inflammatory conditions.
4. Monoclonal antibodies help in identification of individual cell populations e.g. lymphocyte and leukocyte differentiation has become possible now.
5. They help in the purification of cells in order to generate the info about their features and functions.

Genesis of Immunoglobulin (Ig) Molecules

Like most of the proteins, immunoglobulin heavy and light chains are formed in the rough endoplasmic reticulum. Chaperones are the proteins that are required for proper folding or unfolding of Ig heavy chains and also are needed during the assembly of heavy chain with light chain. Assembly process includes stabilizing of both the heavy and light chains by disulfide linkage and mutual association of heavy and light chains and the whole process occurs in endoplasmic reticulum.

Carbohydrate is a biomolecule consisting of carbon (C), hydrogen (H) and oxygen (O) atoms, usually with a hydrogen–oxygen atom ratio of 2:1 (as in water); in other words, with the empirical formula $C_m(H_2O)_n$ (where m may be different from n).

This is followed by **carbohydrate** modification which is required at the end of assembly process. At the end of this process Ig molecules get separated from chaperones and are shifted to cisternae of Golgi complex for carbohydrate modification, and finally find the way into the plasma membrane in vesicles. Membrane bound Ig molecules lie within the plasma membrane and the secreted form find its way out of the cell. Membrane form of the μ heavy chain is synthesized by a prototype called the pre-B cell, which synthesizes the Ig polypeptides. Pre- B cell receptor expression on cell surface requires the association of μ heavy chain with surrogate light chains. Further maturation of B- cells is associated with modification in Ig gene expression leading to the generation of Ig molecules in different forms. The mature B lymphocytes differentiate into the antibody-secreting cells only when stimulated by foreign object or any antigen.



Half- life of Antibodies

Half-life of antibodies varies in circulation. IgG molecules have a half life of 21 to 28 days while IgE has the shortest half-life of about 2 days. Long half-life of IgG is assigned to its ability to bind to Fc receptor called the neonatal Fc receptor. It is neonatal Fc receptor that is responsible for transfer of maternal IgG across the placental barrier.

Table 1: Biological properties of different Ig molecules

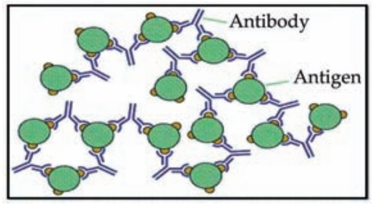
Immunoglobulin class/Property	Molecular weight (Daltons)	Subunits	Constant heavy region (CH)	Heavy chain	Synthesis area
IgM	900,000	5	4	μ	Spleen and lymph nodes
IgG	180,000	1	3	γ	Spleen and lymph nodes
IgA	360,000	2	4	α	Intestinal and respiratory tracts
IgE	200,000	1	4	ε	Intestinal and respiratory tracts
IgD	180,000	1	3	δ	Spleen and lymph nodes

Antigen-Antibody Reactions

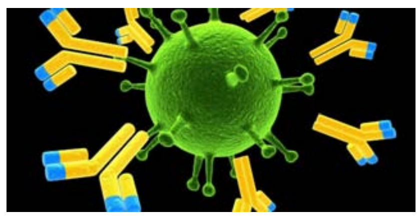
The interactions between antigens and antibodies are known as antigen–antibody reactions. The reactions are highly specific, and an antigen reacts only with antibodies produced by itself or with closely related antigens. Antibodies recognize molecular shapes (epitopes) on antigens. Generally, the better the fit of the epitope (in terms of geometry and chemical character) to the **antibody** combining site, the more favorable the interactions that will be formed between the antibody and antigen and the higher the affinity of the antibody for antigen. The affinity of the antibody for the antigen is one of the most important factors in determining antibody efficacy in vivo.

Antibody is a large, Y-shaped protein produced mainly by plasma cells that is used by the immune system to neutralize pathogens such as pathogenic bacteria and viruses.

The antigen- antibody interaction is bimolecular irreversible association between antigen and antibody. The association between antigen and antibody includes various non-covalent interactions between epitope (antigenic determinant) and variable region (V_H/V_L) domain of antibody.

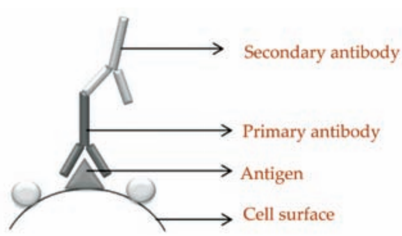


Antigen-antibody reactions can be seen as if they were equations with one unknown variable. In other words, based on the known variable (antigen or antibody) and the final result of the reaction (presence or absence of antigen-antibody complex), it is possible to determine the presence and the quantity of the other component in reaction (antibody or antigen, respectively). Thus, in order to detect an antigen of interest in a given sample, an antibody specific for that antigen should be used. However, if the goal is to detect antibodies in the sample, in addition to a known antigen, another antibody must be added as a reagent in the reaction.



This antibody is specific for the epitopes of Fc fragment (constant region of heavy immunoglobulin chain) of an antibody of a particular isotype (class), such as γ-chain in the case of IgG detection, and, therefore, it is called anti-immunoglobulin antibody or antiantibody (because they are specific for other antibodies). Anti-immunoglobulin antibodies are sometimes also referred to as the secondary antibodies, considering the fact that the antibodies that bind directly to an antigen are, so called, **primary antibodies**. These anti-immunoglobulin antibodies are usually obtained from animals that are immunized with immunoglobulins of another species (typically human) and they not only enable detection of specific antibodies in a sample, but also allow determining their class, which can help in diagnosis of many diseases.

Primary antibodies are produced as monoclonal (mAbs) or polyclonal antibodies (pAbs) using animal species such as the rat, mouse, goat, and rabbit as hosts.

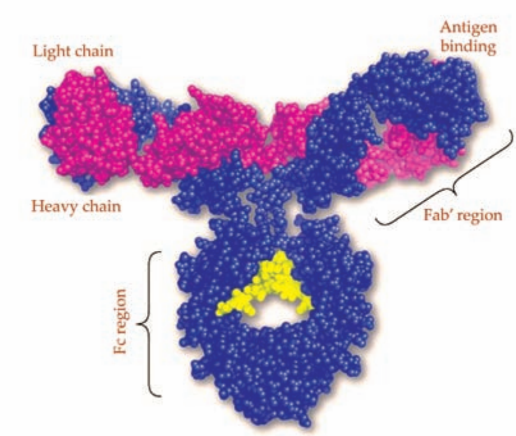


Chemical Bonds Responsible for the Antigen Antibody Reaction

The interaction between the Ab-binding site and the epitope involves exclusively non-covalent bonds, in a similar manner to that in which proteins bind to their cellular receptors, or enzymes bind to their substrates. The binding is reversible and can be prevented or dissociated by high ionic strength or extreme pH. The following intermolecular forces are involved in Ag–Ab binding:

1. ***Electrostatic bonds*****:** This result from the attraction between oppositely charged ionic groups of two protein side chains; for example, an ionized amino group (NH_4^+) on a lysine in the Ab, and an ionized carboxyl group (COO^-) on an aspartate residue in the Ag.
2. ***Hydrogen bonding***: When the Ag and Ab are in very close proximity, relatively weak hydrogen bonds can be formed between hydrophilic groups (e.g., OH and C=O, NH and C=O, and NH and OH groups).
3. ***Hydrophobic interactions***: Hydrophobic groups, such as the side chains of valine, leucine, and phenylalanine, tend to associate due to Van der Waals bonding and coalesce in an aqueous environment, excluding water molecules from their surroundings. As a consequence, the distance between them decreases, enhancing the energies of attraction involved. This type of interaction is estimated to contribute up to 50% of the total strength of the Ag–Ab bond.
4. ***Van der Waals bonds*****:** These forces depend upon interactions between the "electron clouds" that surround the Ag and Ab molecules. The interaction has been compared to that which might exist between alternating dipoles in two molecules, alternating in such a way that, at any given moment, oppositely oriented dipoles will be present in closely apposed areas of the Ag and Ab molecules.

Each of these non-covalent interactions operates over very short distance (generally about 1 Å) so, Ag-Ab interactions depends on very close fit between antigen and antibody.



Strength of Ag-Ab interactions

Affinity

- Combined strength of totalnon-covalent interactions between single Ag-binding site of Ab and single epitope is affinity of Ab for that epitope.
- Low affinity Ab: Bind Ag weakly and dissociates readily.
- High affinity Ab: Bind Ag tightly and remain bound longer.

Relies on the ability to co-valently conjugate chemi-cals to the Fc region of Ig without interfering with antigen binding ("enzyme-linked") and the ability of plastic to nonspecifically bind proteins (immunosor-bent).

Avidity

Strength of multiple interactions between multivalent Ab and Ag is avidity. Avidity is better measure of binding capacity of antibody than affinity. High avidity can compensate low affinity.

Cross Reactivity

Antibody elicited by one Ag can cross react with unrelated Ag if they share identical epitopeor have similar chemical properties.

Characteristics of Biologic Antigens

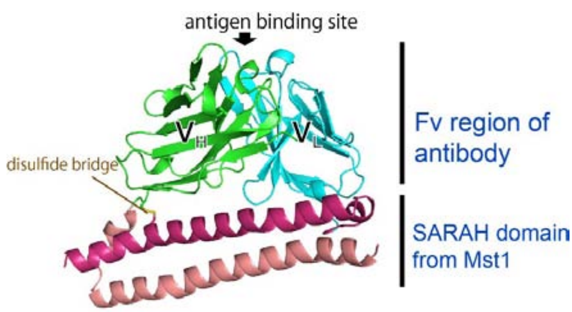
Epitope is the part of an antigen that is recognized by the immune system, specifically by antibodies, B cells, or T cells.

1. One of the most important characters of antigen is to bind specifically to an antibody.
2. Almost all the antigens are identified by specific antibodies but very few have the ability to stimulate the antibodies. Sometimes in order to provoke an immune response, immunologists adjoin several copies of small molecules called hapten to a protein prior to immunization and the protein to which it is attached is known as carrier.
3. Foreign antigens are usually much bigger than the region where actual binding occurs between the antigen and the antibody and this region is known as antigen binding region. An antibody prefers to bind to this small region of the antigen known as **epitope**. Epitopes are hence also called as antigenic determinants.
4. Random structure on the antigenic molecule that is identified by the antibody as an antigenic binding site forms the epitope of that antigen.

5. Different epitopes are so organized on a single protein molecule that their spacing may affect the binding of antibody molecules in various ways

Chemistry of Antigen Binding

The interaction of an antigen antibody is a reversible binding process that requires several non-covalent interactions like hydrogen bonds, electrostatic forces and hydrophobic interactions. Affinity and Avidity between the antigen antibodies also play a major role in their interaction. The potency of the reaction between a specific antigenic determinant and its single combining site on the antibody determines its affinity. The overall potential of binding of an antigen with many antigenic determinants to its multivalent antibody determines its avidity. Normally antigen-antibody binding site on antibodies are more or less flat and hence spacious so that they can attach large complexes or structures.



Antigen Recognition

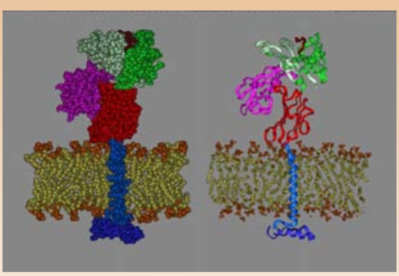
Specificity: Antibodies are very specific to an antigen and can even understand the minute difference between almost similar antigens. It may however happen that an antibody may bind to different but structurally similar antigen and this phenomenon is termed as a cross-reaction.

Diversity: Diversity determines the ability of antibody to bind specifically to a large number of different antigens. The pool of antibodies with different specificities describes the antibody repertoire.

Affinity maturation: The efficiency of antibody bonding to antigen is measured in terms of affinity and avidity. Some modification is required in structure of V region of antibodies during T cell dependent humoral immune response to antigens so that the antibodies having high affinity can be generated. B-cells that are responsible for generating high affinity antibodies preferentially bind to the antigen due to selection and become the prominent cells with each antigen antibody reaction. This mechanism is termed as affinity maturation, and it leads to an increase in binding affinity of antigen and antibody as antibody mediated response develops further.

Effector Functions of Antigen Antibody Reaction

- As two or more Fc portions are required to stimulate effector functions so effector functions are carried out only by molecules with bound antigens and not with free Ig.
- Fc region of the antibody molecules play a critical role in effector stimulation, so antibody isotypes varying in Fc region can be easily distinguished on the basis of interactions they carry.
- Distribution of antibody molecules through different tissues is decided solely by the constant region in the heavy chain of an antibody molecule. This directed distribution through constant region of heavy chain is the reason behind IgA presence in mucosal secretions or recruitment of other antibodies to a particular tissue.
- In antibody mediated immune response, variation in the isotypes of antibodies decides the ways to eliminate antigen from the body. In addition, isotype switching or class switching also has some role in it. e.g. antibody response to bacteria and viruses is carried out by IgG antibodies but switching to IgG isotype can also lengthen the humoral response because it has the longest half life period among all the antibodies.



CORE CONCEPT: MAJOR HISTOCOMPATIBILITY COMPLEX

Major histocompatibility complex (MHC), group of genes that code for proteins found on the surfaces of cells that help the immune system recognize foreign substances. MHC proteins are found in all higher vertebrates. In human beings the complex is also called the human leukocyte antigen (HLA) system.

The major histocompatibility complex (MHC) is an area of the genome which codes for a series of proteins expressed on the cells in the body. These proteins serve as flags for the immune system which allow the immune system to distinguish between “self” proteins which belong in the body, and "nonself" proteins which are foreign. The T cells of the immune system interface with the proteins produced by the major histocompatibility complex, using this information to determine whether or not material encountered in the body belongs there.

These proteins take the form of antigens. In humans, they are known as **human leukocyte antigens (HLA)**. Fixed on the outside of a cell, these antigens can be presented, much like credentials, to the immune system. If the immune system recognizes an antigen as harmful, it can take steps to kill the cell it is attached to. This is designed to allow the immune system to kill bacteria and other organisms which make their way into the body, and to allow the immune system to identify cells which have been infected by viruses so that the spread of the virus can be stopped. The MHC is extremely diverse. Some of the genes involved have hundreds of alleles, which is rather unusual; few genes have so many different ways in which they can express. The diversity of the major histocompatibility complex is both a blessing and a curse. Genetic diversity makes the human race stronger, especially when it comes to immune defense, but it also makes it difficult to transplant tissue between humans.

Human leukocyte antigens (HLA) system or complex is a gene complex encoding the major histocompatibility complex (MHC) proteins in humans. These cell-surface proteins are responsible for the regulation of the immune system in humans.

When blood, tissue, or other donated biological material is transplanted from one person into another, the HLA antigens may not match. As a result, the recipient's immune system will consider the donor material to be alien, and it will attack it. This causes transplant rejection with things like skin and organs, and with blood, it can cause severe reactions. In some cases, a major histocompatibility complex conflict can be deadly. There are a number of different tests which can be used to identify key areas of someone's major histocompatibility complex. These tests are run when people are evaluated for organ donation, so that a match which will fit as perfectly as possible can be found. While some people may assume that blood type is the only thing tested for, in fact, there are an assortment of antigens which can conflict and testing must be thorough to avoid wasting donor material on someone who cannot receive it.

The binding of antibodies to antigens can neutralize pathogens, tag them for destruction, or activate other immune responses.

Major Histocompatibility Complex (MHC) Gene

The MHC locus contains two types of MHC genes, class I and class II. MHC genes are codominantly expressed in an individual that means the alleles of the gene are inherited from both the parents. MHC class I molecules display peptides to the CD8+ lymphocytes to activate cell mediated immune response, and MHC class II molecules display the peptides to CD4+ lymphocytes to activate humoral mediated immune response. The diversity of the immune system has made MHC class I and II genes to be the most polymorphic genes present in the human genome. In humans, the gene responsible for encoding MHC molecule is located in the chromosome 6 (chromosome 17 in mice). The human MHC class I is encoded by three class of genes namely, HLA-A, HLA-B, and HLA-C. Similarly MHC class II is encoded by genes HLA-DP, HLA-DQ, and HLA-DR. In mice nomenclature for MHC changed to H-2K, H-2D, and H-2L for class I and I-A and I-E for class II (only 2 genes in mice). The set of MHC alleles present on each chromosome are called MHC haplotype.

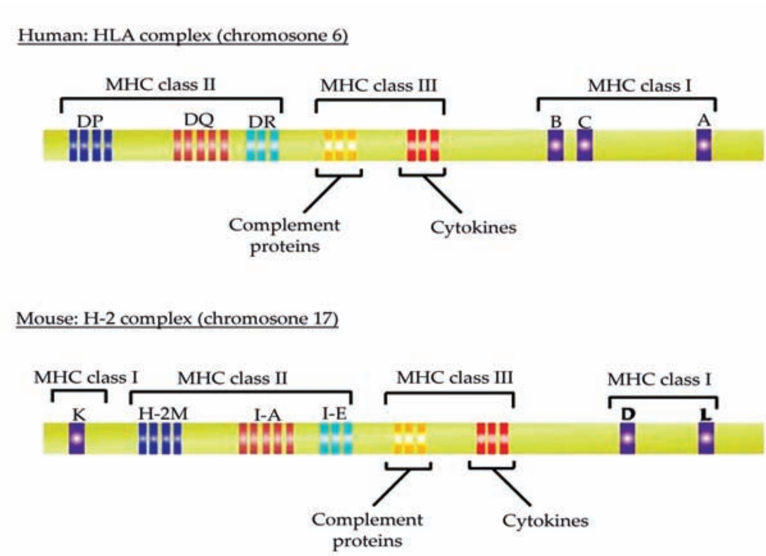
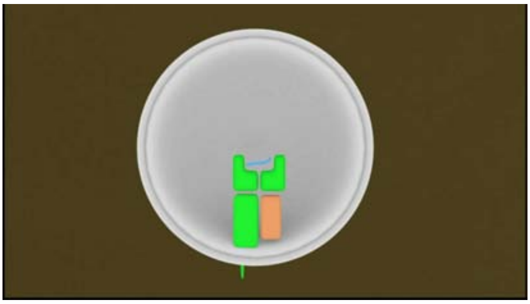


Figure 3: Map of human and mice MHC gene loci:

Bare lymphocyte syndrome is a condition caused by mutations in certain genes of the major histocompatibility complex or involved with the processing and presentation of MHC molecules. It is a form of severe combined immunodeficiency.

MHC Expression

MHC class I molecules are expressed on all the nucleated cells, while class II are expressed only in dendritic cells, B cells, macrophages and few other cells. Class I restricted CD8+ cells kill the virus infected cells, the cells containing intracellular antigens and tumor antigens. Class II restricted CD4+ cells kill the extracellular antigen presented by mostly dendritic cells. The expressions of MHC molecules are stimulated by the cytokines such as interferons (type-I and II). Interferon-γ secreted by natural killer cells during the early innate immune response is the major cytokine responsible for activating the expression of MHC class II molecules in dendritic cell and macrophages. The rate of transcription of MHC gene is the major determinant for the expression of MHC molecules. Any mutation in the transcription factor leads to many immunodeficiency diseases such as **bare lymphocyte syndrome**.



Properties of MHC Molecules

- MHC molecule consists of peptide binding groove, an immunoglobulin like domain, transmembrane domain, and a cytoplasmic domain. MHC class I molecule is made up of one MHC encoded and one non-MHC encoded chain. MHC class II molecule is made up of two MHC encoded chains.
- The peptide binding groove is located at the adjacent to polymorphic amino acid residue. Because of the variability in the region, different MHC molecule binds and displays different peptides and are recognized by different T cells.
- An immunoglobulin like domains contains the binding site for CD4 and CD8 cells.

Table 2: Features of MHC class I and II molecules

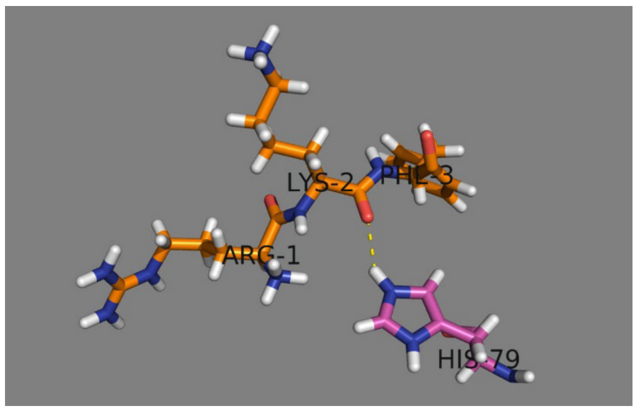
Characters	MHC class I	MHC class II
Polypeptide chains	α1, α2, α3 and β2 microglobulin	α1, α2, β1 and β2
Size of peptide	8-11 amino acid long	10-30 or more amino acid long
Peptide binding site	Between α1 and α2	Between α1 and β1
Binding site for T cell coreceptor	α3 region (CD8+ binding)	β2 region (CD4+ binding)

Peptide-MHC Interaction

There are some characteristic features of peptide-MHC interaction.

- MHC class I and II molecules have a single peptide binding cleft that accommodates one peptide at a time but can bind to different peptides.
- The processed peptide that binds to MHC shares structural compatibility that promotes their interaction.

- MHC acquires the peptide over their cleft during the processing of the antigen inside the cell.
- Only small populations of peptide loaded over the MHC molecules are capable of eliciting the immune responses.
- MHC molecules present both the self and non-self peptide to the T cells. Remarkably it is the T lymphocyte that decides to which the body should produce an immune response. Majority of the MHC present in the body are loaded with the self peptide and T cells activated against the self peptides are either killed or inactivated by the host immune surveillance system. Hence, T cells normally do not respond to a self antigen.



Amino acids are organic compounds containing amine and carboxyl functional groups, along with a side chain specific to each amino acid.

MHC Class I

MHC class I molecules are made up of two polypeptide chains, α and β2-microglobulin. The "α" chain is around 44 kD and "β2-microglobulin" chain is around 12kD in size. Each α chain is divided into three parts to accommodate extracellular α1, α2, and α3 domains, transmembrane domain and a cytoplasmic tail. The α1 and α2 is around 90 **amino acids** long and binds to only 8-11 amino acid long peptides (peptide binding cleft). The ends of MHC class I peptide binding cleft is closed and the larger peptide cannot be accommodated in the designated space. The accommodation length of peptide is highly conformational and globular proteins need to process into 8-11 amino acid length in order to load over the MHC class I cleft. The α1 and α2 contains the polymorphic residues which are responsible for the variation among the MHC I allele and their recognition by a specific T cell. The α3 segment of α chain contains the binding site for CD8+ cells.

The α3 segment extends to 25 amino acids residue towards its carboxy terminal covering the lipid bilayer and more 30 amino acids as a cytoplasmic tail. The β2-microglobulin noncovalently interacts with α3 chain. Binding of the peptide in the cleft between α1 and α2 strengthens the interaction between α and β2- microglobulin chain. The fully formed MHC class I molecule is a heterotrimer consists of α1, α2, α3 and β2-microglobulin chain.

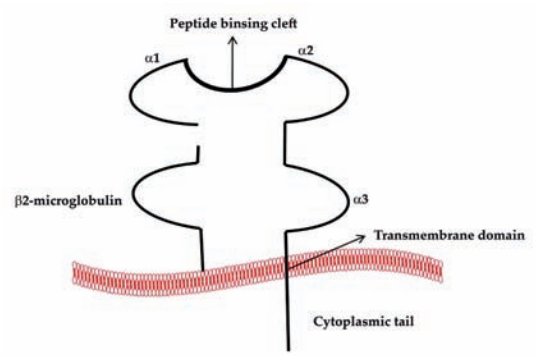


Figure 4: Schematic representation of a MHC class I molecule.

MHC Class II

MHC class II molecules are also made up of two polypeptide chains, α and β. The “α” chain is around 33 kD and “β” chain is around 31kD in size. The α1 and β1 chain interacts with the peptide which is longer than peptide binding to class I molecule. The peptide binding cleft at α1 and β1 are open and so can fit peptides of length 30 or more amino acids. The β2 segment contains the binding site for CD4+ cells. The fully formed MHC class II molecule is a heterotrimer consisting of α1, α2, β1 and β2 microglobulin chain.

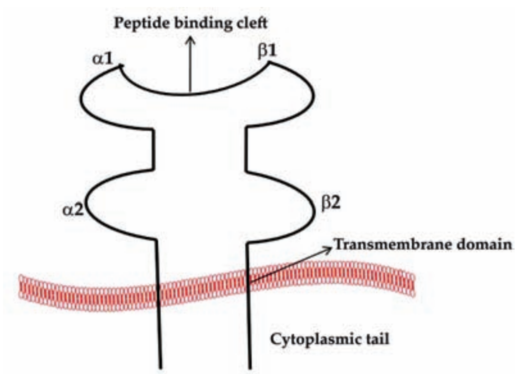


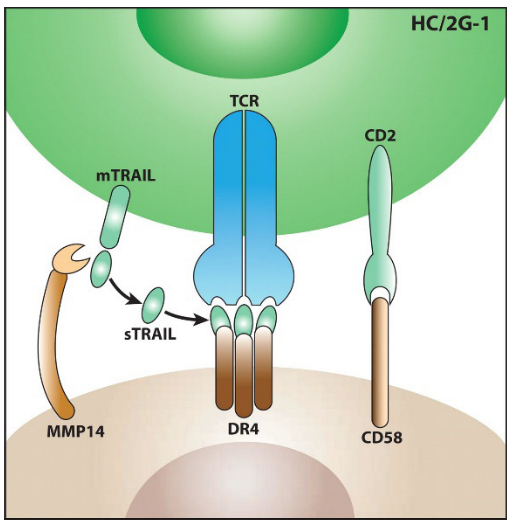
Figure 5: Schematic representation of a MHC class II molecule

Antigen recognition by T lymphocyte

In order to generate an acquired immune response an antigen molecule must be broken inside the cells and presented to the immune cells with the help of major histocompatibility complex (MHC) molecules. These are encoded by the genes of MHC complex and vary between different species. Antigens can trigger an immune response only after bounding to MHC molecules. All vertebrate animals contain the MHC encoding loci in their chromosomes.

Most of the T lymphocytes can recognize the small peptide fragments while the B cells can recognize the peptides, carbohydrates, lipid, nucleic acid and other chemicals. Because of different antigen specificity of T and B lymphocytes, cell mediated immune responses are usually activated by a protein antigen while humoral immune responses are activated by non-protein antigens. The T cells recognize only protein antigens displayed by MHC molecules because the MHC cannot bind to any other molecules. An individual T cell can recognize only one specific MHC molecules loaded with the peptide, the property is called as MHC restricted.

T cells can recognize only the linear peptides and not the conformational epitopes of an antigen because the conformations of the proteins are lost during the processing and loading into the peptide binding cleft of MHC molecules. T cells can recognize only the antigens that are associated with the antigen presenting cells and not to the soluble protein.



Antigen Presenting Cells

Many cell types function as antigen presenting cells to activate the naïve and effector T cells. Dendritic cells are the most common and effective antigen presenting cells in the body. Macrophages and B cells also act as an antigen presenting cells, but only to the previously activated T cells. All the above mentioned cells expresses the MHC type II molecule over their surface and hence also called professional antigen presenting cells. The antigen presenting cells displays the peptide MHC complex to T cells and also provides the additional stimuli to T cells for its proper functioning. These stimuli are sometimes called as co-stimulatory molecules because they function together with the antigen presenting cells.

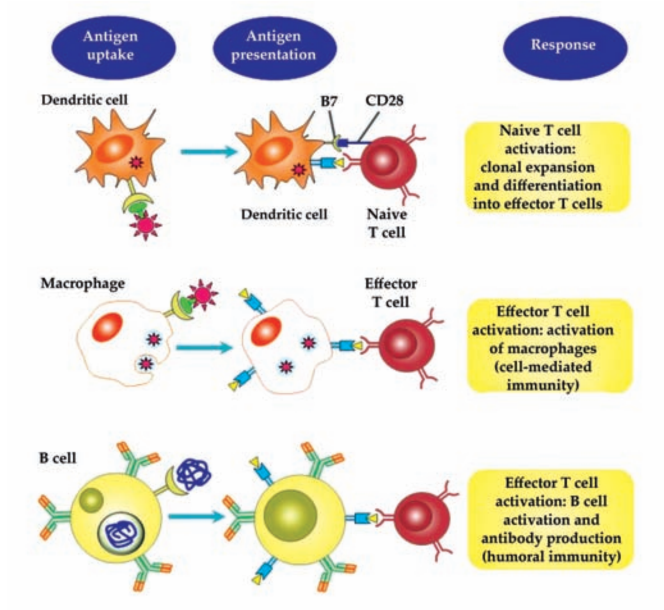


Figure 6: Different antigen presenting cells.

The antigen presenting function of the antigen presenting cells can be enhanced by microbial products. The induction of T cell response against an antigen is usually enhanced by the administration of purified protein products called as adjuvants. Adjuvants are derived from microbes such as killed mycobacterium which mimics the microbes and stimulate the production of immune response.

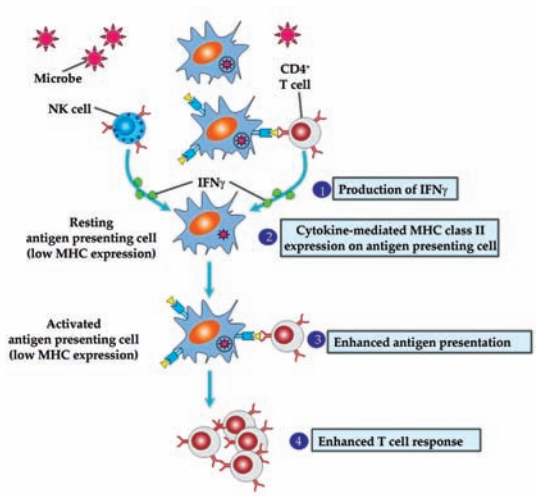


Figure 7: Enhancement of class II expression by interferon-γ.

Dendritic Cells

Dendritic cells are the major cells of **immune system** that act as an antigen presenting cell. Dendritic cells are present in the lymphoid organs and epithelial cells of gastrointestinal tract and respiratory tract. All dendritic cells are derived from the bone

Immune system is a host defense system comprising many biological structures and processes within an organism that protects against disease.

marrow precursor mononuclear phagocytic cells. Dendritic cells capture the antigen from the skin and epithelial lining of the tissues and enter the lymphatic vessels. Lymph acts as a reservoir of the cell associated as well as free antigen. There are mainly two subsets of dendritic cells.

Conventional Dendritic Cells

These are also called myeloid dendritic cells. They are the most abundant dendritic cells in the body and responsible for producing a strong T cell immune response. In tissues, they are called Langerhans cells because of its long cytoplasmic process that occupies a large surface area in the epithelial surface which make them highly accessible to the antigens. The surface marker molecules for conventional dendritic cell are CD11c and CD11b. They express high level of Toll like receptor 4, 5, and 8. In addition they secrete high level of tumor necrosis factor and interleukin-6.

Plasmacytoid Dendritic Cells

They are named so because of their morphology which is similar to plasma cells. The surface marker molecule for plasmacytoid dendritic cell is B220. They express high level of Toll like receptor 7 and 9. They are responsible for the secretion of large amount of type I interferon (α and β) following viral infection.

Processing of Antigen through MHC Class I Pathway

Usual antigens that are processed by MHC class I include intracellular bacteria, viruses, and tumor antigens. MHC class I peptides are processed in the cytosol by the proteolytic degradation of the protein. Occasionally the proteins are phagocytized and imported to the cytoplasm in order to load over the MHC class I molecules. The proteins are degraded by the proteasome in the cytoplasm which are the complex structure responsible for the degradation of unwanted or improperly folded proteins. After degradation the peptides are transferred to the endoplasmic reticulum with the help of transported proteins called transporter associated with antigen processing (TAP). TAP is associated with another protein called Tapasin, which acquires specific affinity with the new and empty MHC molecules. Tapasin brings the TAP-antigen complex towards the new MHC molecule. The proper folding and assembly of the MHC class I molecules inside the endoplasmic reticulum is modulated by the chaperons calnexin and calreticulin. The peptides from TAP-antigen-tapasin complex is processed and loaded over the MHC molecule with the help of endoplasmic reticulum associated peptidases (ERAP). The peptide transported to the endoplasmic reticulum are generally presented through the MHC class I pathway. The peptide bound to the MHC class I molecules are then transported to Golgi apparatus and then to cell surface with the help of exocytic vesicles. The MHC class I loaded peptides are recognized by CD8+ T lymphocyte to induce cell mediated immune response.

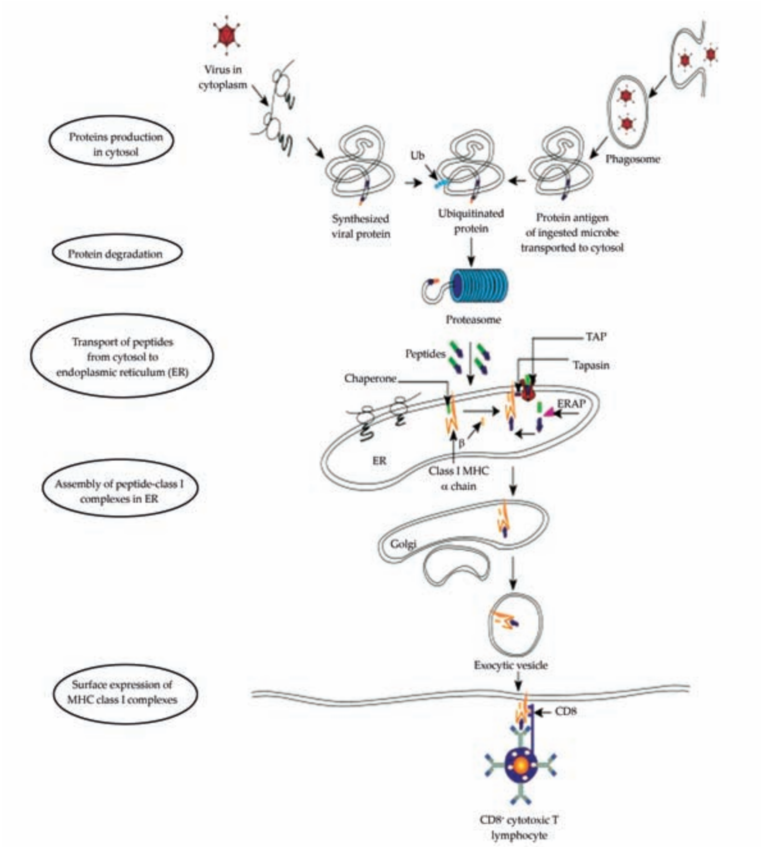


Figure 8: MHC class I antigen processing pathway.

Processing Of Antigen through MHC Class Ii Pathway

Majority of the peptides associated with the MHC class II are generated from extracellular antigens (protein) that are captured inside endosomes of the antigen presenting cells. The antigen containing endosomes are fused with the lysosome to form endolysosome, the acidic pH of the endolysosome helps in the degradation of the proteins into smaller peptides. MHC class II molecules are synthesized in the endoplasmic reticulum and transported to the endosomes with the help of invariant chain (Ii), which binds to the peptide binding cleft of a newly synthesized MHC class II molecule. Class II molecules with bound invariant chain (CLIP) are transported to endosomes and are degraded by proteolysis to release the invariant chain. The remaining part of CLIP is removed by HLA-DM present in the endosomes in order to create space for peptide. Once CLIP is removed the peptides are loaded over the MHC class II molecule. The MHC class II molecule bound to peptides is delivered to the surface for their recognition by the CD4+ T lymphocytes (humoral immune response).

Plasma membrane is a biological membrane that separates the interior of all cells from the outside environment. It consists of a lipid bilayer with embedded proteins.

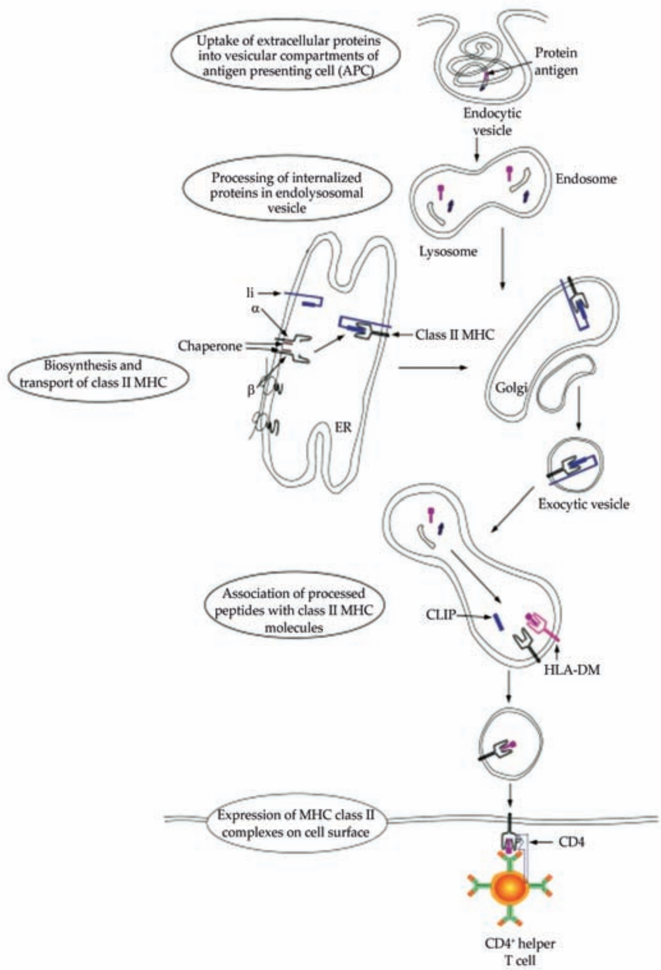
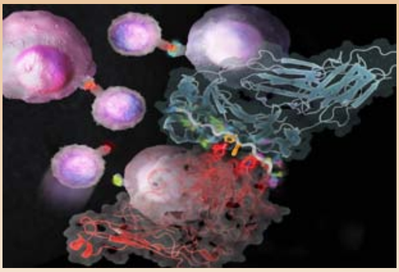


Figure 9: MHC class II antigen processing pathway.

Cross Presentation of Peptides

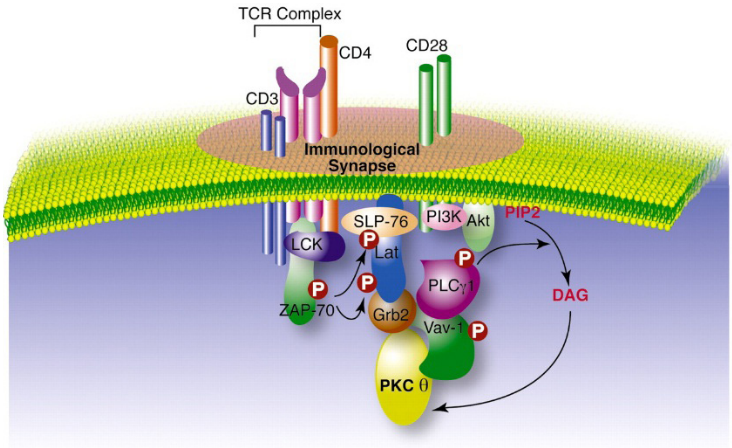
Occasionally dendritic cells capture and ingest the viruses or tumor antigens and present the antigens to the CD8+ T lymphocytes. As mentioned above, antigens captured into vesicles initiate the MHC class II pathway. The deviation of some dendritic cells to present the endocytic degraded peptides to MHC class I molecules and activate the CD8+ mediated immune response is called cross presentation.



CORE CONCEPT: ANTIGEN RECEPTORS AND ACCESSORY MOLECULES OF T LYMPHOCYTES

A lymphocyte that has gotten larger after being stimulated by an antigen. Lymphoblast also refers to an immature cell that can develop into a mature lymphocyte. Lymphocytes. A lymphocyte is a type of white blood cell that is part of the immune system. There are two main types of lymphocytes: B cells and T cells.

Receptors that initiate the signaling pathways are generally associated with the **plasma membrane**. The extracellular domain of the receptors recognizes the ligands present over the cell surface and this interaction may lead to conformational changes in the receptor. The conformational changes are associated with the recruitment of the phosphate group at its carboxy terminal on tyrosine, serine, or threonine residue. The enzyme that adds the phosphate group on amino acid residues are called protein kinases. The tyrosine is the major amino acid residue that takes part in this event (phosphorylation) hence the enzymes are referred as protein tyrosine kinases. Alternatively the enzymes which are responsible for the removal of phosphate group from amino acids are called phosphatase. In general the protein kinases can initiate while phosphatase can inhibit the signaling pathways. Several types of protein modification can also modulate the binding of an antigen to the receptor such as phosphorylation (addition of phosphate group), acetylation (addition of acetyl group), methylation (addition of methyl group), and ubiquitination (addition of ubiquitin). The ubiquitination is the event that takes place during the degradation of proteins through proteasome.



Types of Cellular Receptors

There are several types of cell receptors based on their signaling mechanism and biochemical pathways.

Receptor Tyrosine Kinases

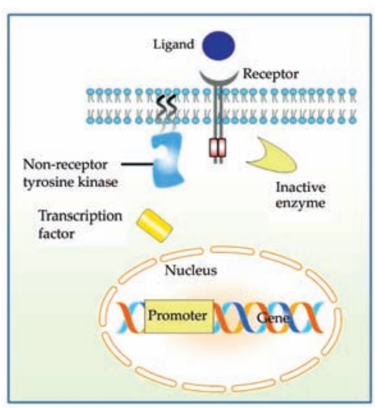
They are associated with the cell membrane and are involved in the phosphorylation of tyrosine residue located in their cytoplasmic tail. The pathway begins after binding with a suitable ligand over the receptor. e.g. Insulin receptor, epidermal growth factor receptor, platelet derived growth factor receptor, and receptor involved in the process of hematopoiesis

Non-receptor Tyrosine Kinases

They are associated with the cell membrane and are involved in the phosphorylation of proteins by a non-receptor tyrosine kinases following binding with a ligand. Immune receptors, cytokine receptors, and integrins are known to follow non-receptor tyrosine kinases signaling pathway.

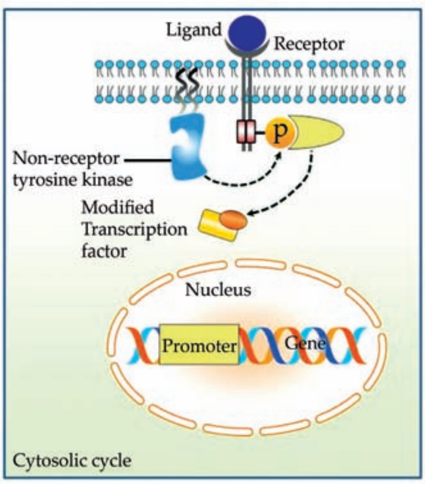
Seven Transmembrane Receptors

These are the polypeptide receptors that traverse seven times in the plasma membrane and hence also named as serpentine receptor. The receptor generally binds to GTP hence also called as G protein-coupled receptors (GPCR). Binding of the ligand to GPCR activates the hetrotrimeric G protein and initiates the downstream signaling pathway. Inflammatory cytokines and cAMP are activated following binding to GPCR.



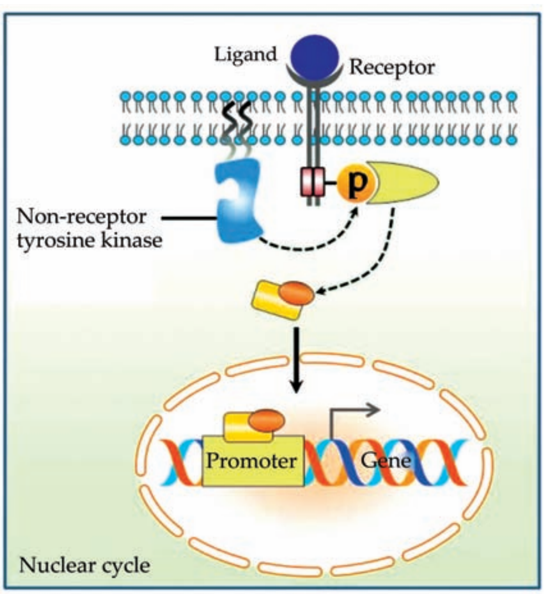
Nuclear Receptors

The modulation of transcription is usually done at the level of nuclear membrane. The receptors that use lipids as its ligand either increase or decrease the transcription of genes. Vitamin D receptor and glucocorticoid receptor are the examples of nuclear receptors.



Miscellaneous Receptors

Notch receptors are involved in the embryonic development and tissue maturation. The binding of the specific ligand with notch receptor leads to the proteolytic cleavage of the cytoplasmic tail of the receptor that can act as a transcription factor for different developmental pathways. A group of ligand called "Wnt" can modulate the level of β- catenin which contributes to B and T cell development.



Ligands are ions or neutral molecules that bond to a central metal atom or ion.

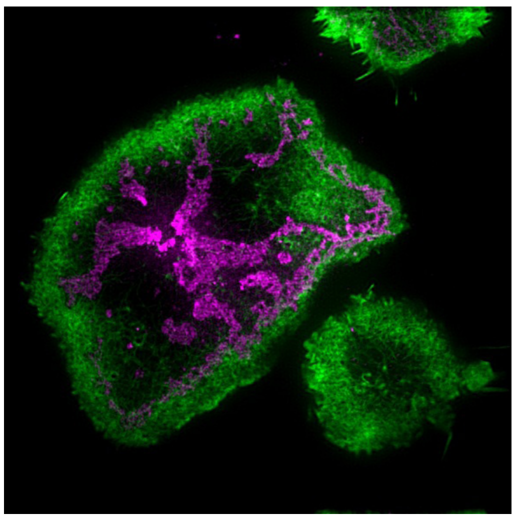
Immune Receptor Family

Immune receptors are made up of immunoglobulin superfamily which are involved in **ligands** recognition and contain tyrosine motif in their cytoplasmic tails. The cytoplasmic tail contains the immunoreceptor tyrosine-based activating motifs (ITAM) which are involved in the activation process. Phosphorylation of ITAM recruits the Syk/ZAP-70 tyrosine kinase which activates the immune cells. Contrary to ITAM some immune receptor contains immunoreceptor tyrosine-based inhibitory motifs (ITIM) which lead to inhibition of immune signaling. Immune receptors family includes B and T cell receptors, IgE receptor on mast cells, and activating and inhibitory receptor Fc receptors.

Remember, antibodies and antigens are fundamental to our immune system's ability to defend against infections and maintain overall health. Their intricate interactions form the basis of immunology and have significant implications for medical research and treatment.

Characteristics of Antigen Receptor Signaling

Signaling event in the T and B cells undergoes similar downstream pathway. Binding of ligands to the receptor activates the Src family kinase that phosphorylates the ITAM motif in the cytoplasmic tail of the receptor. The phosphorylated tyrosine in ITAM recruits the Syk tyrosine kinases which further activate the downstream signaling cascade.

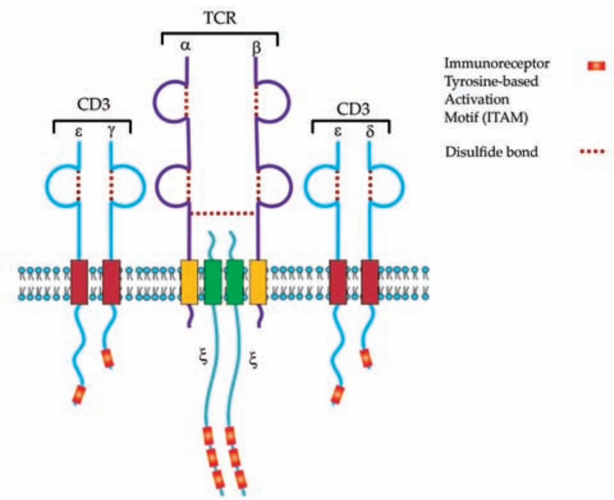


T cell Receptor Complex and T cell Signaling

The T cell receptors (TCR) are made up of heterodimer of two polypeptide chains, α and β that are covalently linked with each other by a disulfide linkage. The TCR containing these two chains are called as αβ T cells. Another type of TCR called γδ T cells contains γ and δ polypeptide chains. Each polypeptide chain in TCR is made up of amino terminus, variable and carboxy terminal, constant region (similar like immunoglobulin molecules). The variable regions are the complementarity determining regions in the TCR and are responsible for its polymorphism. The α and β chains contain 5-12 amino acids residue at their cytoplasmic tail to take part in signal transduction pathway. The other two structures that are associated with the TCR are CD3 and *z* proteins which are noncovalently associated with the αβ chain. TCR together with CD3 and δ proteins forms the TCR complex.

When the TCR engages with antigenic peptide and MHC (peptide/MHC), the T lymphocyte is activated through signal transduction, that is, a series of biochemical events mediated by associated enzymes, co-receptors, specialized adaptor molecules, and activated or released transcription factors.

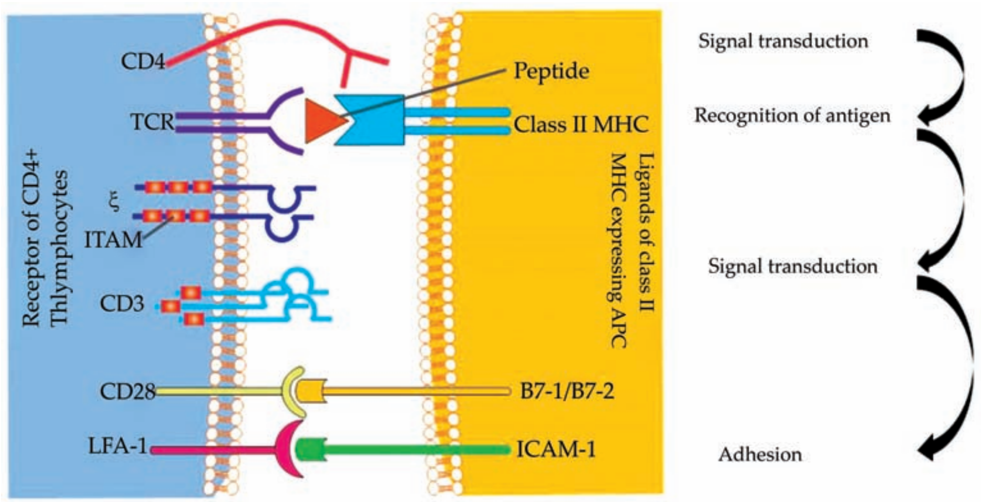
The CD3 and δ proteins are constant in all T cells regardless to its specificity towards any ligand. CD3 contains ε, γ, and δ polypeptide chains which resembles the immunoglobulin superfamily members. CD3 contains two heterodimer made up of εγ and εδ polypeptide chains. The ε, γ, and δ polypeptide chains of CD3 molecule is made up of 44-81 amino acids and contains one ITAM molecule for modulating the downstream signaling pathway. CD3 polypeptide chain contains a negatively charged aspartic acid residue that interacts with the positive charged residue present in the αβ chain. The *z* polypeptide chain contains a long cytoplasmic tail that contains three ITAM molecules and usually expressed as a homodimer in a TCR complex.



Signaling in T Cell Receptor

The ligation of TCR with a peptide loaded MHC molecule results in grouping of CD3, δ proteins and other coreceptors. CD4 and CD8 are T cell coreceptors that bind to the MHC molecule and facilitate the TCR signaling pathway. As we learned in MHC molecule CD4 recognizes class II while CD8 recognizes class I molecule, to maintain this equilibrium, T cells can express either CD4 or CD8 receptor but never both. The interaction of TCR with an MHC-peptide complex results in phosphorylation of the ITAM residue present over the TCR complex. Phosphorylation of the ITAM activates the tyrosine kinase which phosphorylates the tyrosine present over the other coreceptor molecules. The cytoplasmic tails of CD4 and CD8 recruits a Src family kinase Lck. Another Src family kinase associated with the TCR complex is Fyn. Lck phosphorylates the tyrosine in the ITAM present over the CD3 and δ chain.

The phosphorylated ITAM in the δ chain recruits the Syk family tyrosine kinase called δ associated protein of 70kD (ZAP70). ZAP70 in turn phosphorylates the adaptor proteins such as SLP-76 and LAT. Phosphorylated LAT then recruits several components of adaptor proteins including PLCγ1 (key enzyme involved in T cell activation). This entire event activates the Ras and mitogen activated protein kinase (MAPK) pathway which in turn activates the transcription factors and activated T cell response.



A CONCEPT OF ANTIGEN RECEPTOR SIGNALING

Antigen receptor signaling is a fundamental process in the immune system that involves the activation of immune cells in response to the recognition of antigens. Antigens are molecules, often proteins, that are recognized by the immune system as foreign or non-self. The immune system's ability to detect and respond to antigens is crucial for mounting effective immune responses against pathogens and other threats.

Here's an overview of antigen receptor signaling and its importance:

1. Antigen Recognition: Immune cells, particularly T cells and B cells, possess specialized receptors on their surfaces known as antigen receptors. T cells have T cell receptors (TCRs), and B cells have B cell receptors (BCRs), also known as antibodies when secreted. These receptors are highly diverse and can bind to specific antigens.
2. Activation of Immune Cells: When an immune cell's antigen receptor encounters its target antigen, it initiates a series of intracellular signaling events. These signaling pathways trigger the activation of the immune cell and lead to various responses, such as proliferation, differentiation, cytokine production, and antibody secretion.
3. Signal Transduction: Antigen receptor signaling involves the transmission of information from the extracellular antigen-binding domain of the receptor to the intracellular signaling domain. This often involves the recruitment and activation of specific kinases, phosphatases, and other signaling molecules.
4. Major Histocompatibility Complex (MHC): TCRs on T cells recognize antigens presented by major histocompatibility complex (MHC) molecules on the surface of antigen-presenting cells (APCs). This interaction is crucial for the activation of T cells and their subsequent functions, including cytotoxicity and cytokine secretion.
5. Co-Stimulation: Full activation of immune cells typically requires co-stimulatory signals provided by interactions between co-stimulatory molecules on antigen-presenting cells and their corresponding receptors on T cells. Co-stimulation helps prevent inappropriate immune responses.
6. Immune Memory: Antigen receptor signaling is also involved in the generation of immune memory. Upon encountering an antigen,

immune cells undergo clonal expansion and differentiation into effector cells. A subset of these cells becomes memory cells that can respond more rapidly and robustly to subsequent encounters with the same antigen.

7. Autoimmune Diseases: Dysregulation of antigen receptor signaling can lead to autoimmune diseases. In autoimmune disorders, immune cells mistakenly target and attack the body's own tissues due to aberrant antigen recognition and signaling.

8. Immunotherapy: Understanding antigen receptor signaling has led to the development of immunotherapies, such as monoclonal antibodies and immune checkpoint inhibitors, which modulate immune responses to treat cancer and other diseases.

SUMMARY

- Antibodies may be defined as the proteins that recognize and neutralize any microbial toxin or foreign substance such as bacteria and viruses.
- Antigen-binding site is formed by the V region of one heavy chain and the adjacent V region of one light chain.
- Monoclonal antibodies are the antibodies that are specific to one particular antigen as they are made by identical immune cells that are several copies of a same parent cell e.g. any tumor cell of a specific region say plasma cells, are monoclonal and thus have the ability to produce antibodies of same specificity.
- The interaction between the Ab-binding site and the epitope involves exclusively non-covalent bonds, in a similar manner to that in which proteins bind to their cellular receptors, or enzymes bind to their substrates.
- Foreign antigens are usually much bigger than the region where actual binding occurs between the antigen and the antibody and this region is known as antigen binding region.
- When blood, tissue, or other donated biological material is transplanted from one person into another, the HLA antigens may not match.
- MHC class I molecules are expressed on all the nucleated cells, while class II are expressed only in dendritic cells, B cells, macrophages and few other cells.
- MHC molecules present both the self and non-self peptide to the T cells. Remarkably it is the T lymphocyte that decides to which the body should produce an immune response.
- T cells can recognize only the linear peptides and not the conformational epitopes of an antigen because the conformations of the proteins are lost during the processing and loading into the peptide binding cleft of MHC molecules.
- Immune receptors are made up of immunoglobulin superfamily which are involved in ligands recognition and contain tyrosine motif in their cytoplasmic tails.

MULTIPLE CHOICE QUESTIONS

1. **What are antigens?**
   a. Proteins produced by B cells
   b. Molecules that trigger an immune response
   c. Antibodies found in bodily fluids
   d. Cells responsible for phagocytosis

2. **Which immune cells produce antibodies?**
   a. T cells
   b. Macrophages
   c. B cells
   d. Natural killer cells

3. **What is the primary function of antibodies in the immune system?**
   a. Directly attacking and killing pathogens
   b. Phagocytosis of antigens
   c. Initiating fever responses
   d. Binding to antigens and marking them for destruction

4. **What is the term for the specific region on an antibody that binds to an antigen?**
   a. Epitope
   b. Fc region
   c. Complement
   d. Hinge region

5. **Which antibody class is the most abundant in the bloodstream and is involved in opsonization and complement activation?**
   a. IgA
   b. IgD
   c. IgE
   d. IgG

Answers

1. (b) 2. (c) 3. (d) 4. (a) 5. (d)

REVIEW QUESTIONS

1. Describe the antibodies and antigens.
2. Define the antibody structure.
3. What are uses of monoclonal antibodies?
4. Briefly explain the genesis of immunoglobulin (ig) molecules.
5. Define the antigen-antibody reactions.
6. Discuss about chemical bonds responsible for the antigen antibody reaction.
7. Write the characteristics of biologic antigens.
8. What do you mean by major histocompatibility complex (MHC) gene?

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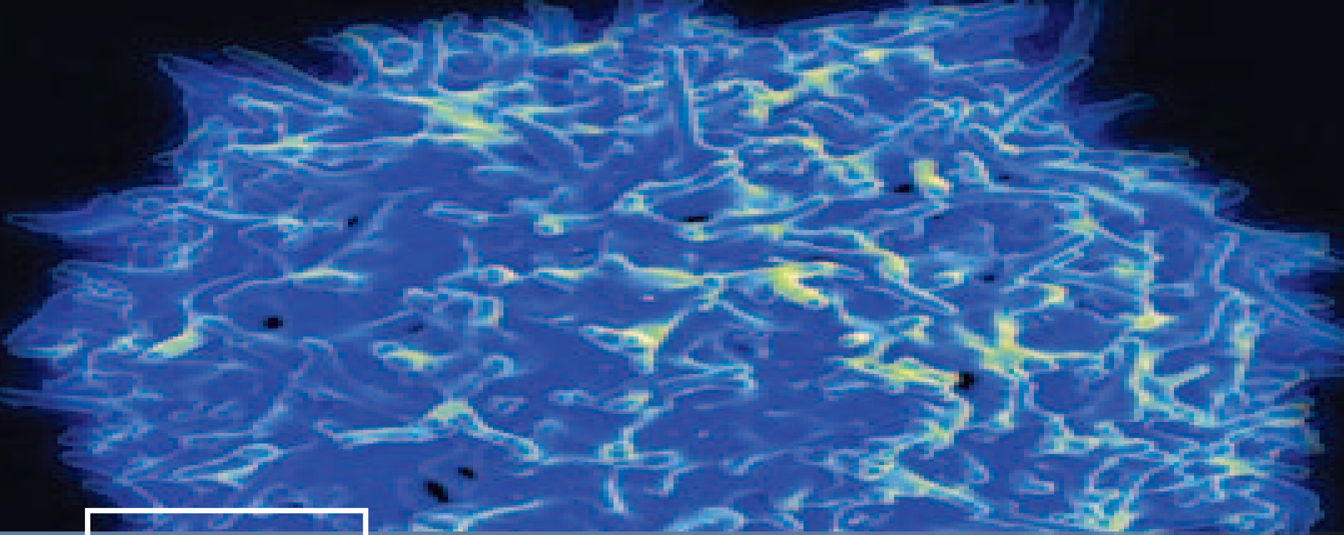
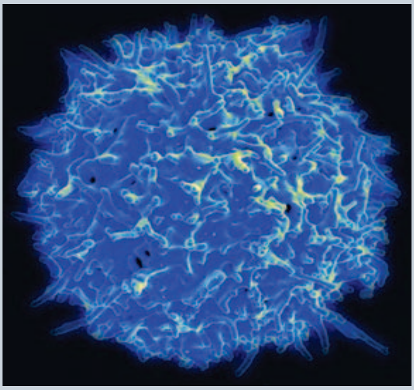


CHAPTER 3

DEVELOPMENT OF LYMPHOCYTES



Lymphocyte development involves a complex series of tightly choreographed events. Current models are based on studies of genetically engineered mice, in vitro culture systems that support lymphoid development, and rare patients with genetic forms of immunodeficiency. The process is orchestrated by genes that act at specific stages of B or T cell differentiation, which encode lineage-specific transcription factors, growth factors, and chemokines; DNA recombinases (RAG1 and RAG2) and terminal deoxytransferase (TdT), which direct the rearrangement and diversification of B and T cell antigen receptor genes, respectively; and the DNA modifying enzyme activation-induced cytosine deaminase (AID), which is needed for immunoglobulin class-switching and somatic hypermutation, a phenomenon that is required for the production of high affinity antibodies.

After studying this chapter, you will understand the core concept of:

- Lymphocyte
- Types of Lymphocytes
- Lymphocyte Functions

INTRODUCTION

Lymphocytes are a type of white blood cell that play a crucial role in the immune system. They are responsible for recognizing and attacking foreign substances, such as pathogens or antigens, and maintaining immune responses. The development of lymphocytes is a complex process that occurs mainly in the bone marrow and thymus.

The development of lymphocytes is a remarkable and intricate process that lies at the heart of our immune system's ability to safeguard our health. Lymphocytes, a type of white blood cell, serve as the body's defense force, tirelessly patrolling for invaders and orchestrating immune responses. From their origin as hematopoietic stem cells to their maturation into specialized B and T cells, this developmental journey is a symphony of genetic rearrangements, selections, and transformations.

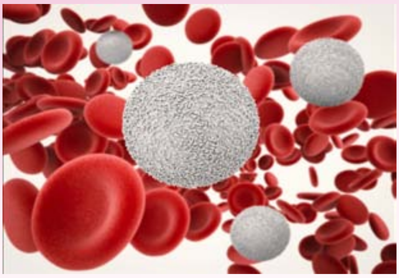
This journey unfolds primarily within the bone marrow and thymus, two vital organs where lymphocytes are sculpted into powerful agents of immunity. The orchestration of this process ensures a diverse and adaptable army of lymphocytes, each equipped with a unique set of antigen receptors, capable of recognizing an astonishing array of potential threats.

B-cell development is a captivating narrative of gene rearrangements and cellular refinement. As B cells mature, they acquire the ability to generate a diverse repertoire of antibodies, each with the potential to neutralize specific antigens. This dynamic process not only contributes to our immediate immune defenses but also provides the basis for immunological memory, allowing rapid and targeted responses upon re-encounter with familiar adversaries.

The narrative of T-cell development unfolds in the thymus, where young T-cell precursors undergo a rigorous training regimen. The process involves a delicate balance of positive and negative selections, ensuring that T cells are both capable of recognizing foreign invaders and refraining from attacking the body's own tissues. This careful curation transforms these early precursors into a potent force, ready to execute immune responses with precision.

However, this developmental journey is not without its challenges. Disruptions or abnormalities in lymphocyte development can lead to immune system deficiencies, leaving the body vulnerable to infections or autoimmune disorders. Scientists and clinicians continue to unravel the mysteries of lymphocyte development, seeking insights that could lead to new therapies and interventions to bolster immune function and restore health.

In this exploration of lymphocyte development, we delve into the captivating saga of cellular transformations, genetic rearrangements, and functional specialization that underpin our immune system's remarkable ability to protect us from harm.



CORE CONCEPT: LYMPHOCYTE

A small white blood cell (leukocyte) that plays a large role in defending the body against disease. In human adults lymphocytes make up roughly 20 to 40 percent of the total number of white blood cells

Lymphocyte type of white **blood cell** (leukocyte) that is of fundamental importance in the immune system because lymphocytes are the cells that determine the specificity of the immune response to infectious microorganisms and other foreign substances. In human adults lymphocytes make up roughly 20 to 40 % of the total number of white blood cells. They are found in the circulation and also are concentrated in central lymphoid organs and tissues, such as the spleen, tonsils, and lymph nodes, where the initial immune response is likely to occur.

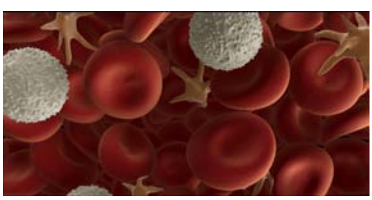
Blood cell is a cell produced through hematopoiesis and found mainly in the blood.

Some of the divided daughter cells become effector cells and some other cells become memory cells:

i. The activity of the effector lymphocytes lead to the elimination of the antigen. The effector lymphocytes die within short time.

ii. But the memory lymphocytes are not concerned with the immediate elimination of the antigen. The memory lymphocytes remain alive in the body for longer time (many months to years). The memory cells remain in a quiescent state until it comes in contact with the same antigen, which has induced its formation. Upon contact with the cognizant antigen, the memory cell is activated.

The activated memory cell divides many times. Some of the daughter cells become effector cells and others become memory cells. The functions of the effector cells lead to the removal of the antigen from the body. Whereas the memory cells remain in the body, waiting for the future arrival of the antigen which has induced its formation.



Activation of B Lymphocytes and Their Functions

B lymphocytes have immunoglobulin molecules anchored to their cell membrane. The immunoglobulin molecules on the B cell surface are called surface immunoglobulin's (sIgs) or membrane-bound immunoglobulin's (migs). The sIgs on B cell bind to antigens and hence act as antigen receptors of B cell.

> Mammalian stem cells differentiate into several kinds of blood cell within the bone marrow. This process is called haematopoiesis.

- Upon binding of antigen with the sIgs, the B cell is said to be activated.
- The antigen-sIg complex is internalized into the B cell.
- The antigen is processed and then presented to nearby helper T cell.
- The helper T cell helps the B cell in its activation process.
- The activated B cell divides many times to produce daughter cells.

Some of the daughter cells become effector cells and others become memory B cells. The effector B cells are called plasma cells. The plasma cells secrete large amounts of antibodies that bind with the antigen that induced its formation.

The binding of antibodies with antigens leads to the elimination of the antigen by different ways:

i. The antibodies bind to the antigen and leads to the lysis of the antigen-bearing cells through complement activation.

ii. The antibodies help in the phagocytosis of the antigen by phagocytes.

iii. The antibodies bind to toxins produced by bacteria and prevent the host from developing disease and death from the toxins. [For example, antibodies against the tetanus toxin (produced by the bacteria Clostridium tetani) bind to the tetanus toxin molecules and prevent the host from developing a deadly disease called tetanus].

Antibody-dependent cell-mediated cytotoxicity (ADCC) is a mechanism of cell-mediated immune defense whereby an effector cell of the immune system actively lyses a target cell, whose membrane-surface antigens have been bound by specific antibodies.

iv. Antibodies bind to antigens and lead to the antigens destruction through a mechanism called **antibody- dependent cell-mediated cytotoxicity (ADCC)**.

Activation of T Lymphocytes and Their Functions

In contrast to B cells, the T cells do not directly bind to antigens in the body fluid. The antigens should be presented by antigen presenting cells (APCs) to T cells. The T cells bind to the antigens presented by the APCs and become activated.

There are two subpopulations of T cells called helper T (T_H) cells and cytotoxic T (Tc) cells. The mechanisms of activation of T_H cells and Tc cells are different. Generally, T_H cells are activated by extracellular microbes, whereas Tc cells are activated by intracellular microbes.

Activation of T_H cells

The bacteria entering the host are engulfed by macrophages.

The bacteria are acted upon by the lysozymes in the macrophage and the microbial proteins are split into short antigenic peptides.

The short antigenic peptide is complexed to a molecule in APCs, called MHC class II molecule. The MHC class Il-antigen complex is then presented on the surface of the macrophage membrane.

The **T cell receptor (TCR)** of helper T cell bind with the MHC class Il-antigen complex on the macrophage surface.

Upon binding with the antigen, the helper T cell is activated.

The activated helper T cell secretes many cytokines and the cytokines, in turn, act upon other cell types and influence their activities.

i. Activated helper T cells help the B cell in B cell activation and consequent antibody production.

ii. Cytokines of helper T cell activate the macrophages. Activated macrophages exhibit enhanced phagocytosis and enhanced microbial killing powers.

iii. Cytokines of activated helper T cell help the cytotoxic T cells to kill virus-infected cells.

T-cell receptor is a molecule found on the surface of T cells, or T lymphocytes that is responsible for recognizing fragments of antigen as peptides bound to major histocompatibility complex (MHC) molecules.

Activation of Cytotoxic T (Tc) Cells

The second subpopulation of T lymphocytes called cytotoxic T cells play important roles in killing intracellular microbes, such as viruses and cancer cells.

The viruses live inside the host cells and multiply.

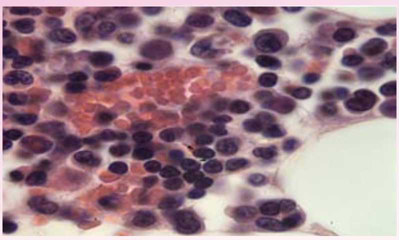
The viral antigen peptides produced during the viral multiplication are complexed to MHC class I molecules of the host cell.

The MHC class I-viral antigen complex is then transported to the surface of the host cell.

The MHC class I-viral antigen complex is presented by the host cell to the Tc cell.

The TCR of Tc cell binds to the MHC class I-viral antigen complex and the Tc cell becomes activated.

i. Activated Tc cell secretes cytokines. The cytokines act on the virus-infected cell and destroy the cell, resulting in the elimination of virus.



CORE CONCEPT: TYPES OF LYMPHOCYTE

Lymphocytes are responsible for immune responses. There are two main types of lymphocytes: B cells and T cells. The B cells make antibodies that attack bacteria and toxins while the T cells attack body cells themselves when they have been taken over by viruses or have become cancerous.

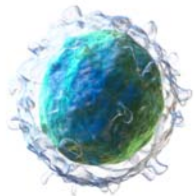
Antigen is a molecule, moiety, foreign particulate matter, or an allergen, such as pollen, that can bind to a specific antibody or T-cell receptor.

The two primary types of lymphocytes are B lymphocytes and T lymphocytes, or B cells and T cells. Both originate from stem cells in the bone marrow and are initially similar in appearance. Some lymphocytes migrate to the thymus, where they mature into T cells; others remain in the bone marrow, where—in humans—they develop into B cells. Most lymphocytes are short-lived, with an average life span of a week to a few months, but a few live for years, providing a pool of long-lived T and B cells. These cells account for immunologic “memory,” a more rapid, vigorous response to a second encounter with the same antigen. Through receptor molecules on their surfaces, lymphocytes are able to bind antigens (foreign substances or microorganisms that the host recognizes as “nonself”) and help remove them from the body. Each lymphocyte bears receptors that bind to a specific antigen. The ability to respond to virtually any **antigen** comes from the enormous variety of lymphocyte populations that the body contains, each of them with a receptor capable of recognizing a unique antigen.

B Lymphocytes

B lymphocyte derived its name from its site of maturation, the bursa of Fabricious, in the birds.

B lymphocytes are produced from the hematopoietic stem cells in the bone marrow of adult. The mature B lymphocytes released into circulation from the bone marrow are in a resting or virgin state. Resting B lymphocytes don’t secrete antibodies. Instead the resting B cells express surface immunoglobulin’s (sigs) on their membrane.



B cell Development

It is estimated that 5 x B cells/day are produced in the bone marrow. However, only 10 % of these B cells are released into the circulation. The remaining 90 % B cells produced each day die within the bone marrow. During their development in the bone marrow, enormous number of developing B cells is killed by a process known as negative selection of B cells. The developing B cells capable of reacting with self-antigens are killed in the bone marrow.

Humoral Immune Response

There are two arms of acquired immunity called humoral immunity and cell-mediated immunity. B cells are involved in the humoral immunity. Upon antigen binding, the B cell is activated. The activated B cell divides repeatedly to produce plasma cells and memory B cells. Plasma cells secrete antibodies and antibodies are the important molecules of the humoral immune response.

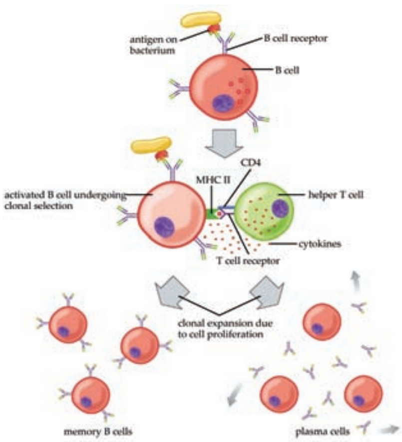


Figure 1: B cell activation.

Plasma cells are white blood cells that secrete large volumes of antibodies.

A resting B cell is activated by the binding of its surface immunoglobulin's (slg) with antigen. Activated B cell divides repeatedly. Some of the daughter cells become effector (plasma) cells and others become memory B cells. The plasma cell secretes antibodies.

The **plasma cell** has s short life and dies in few days. The memory B cells are in a resting state and they remain in the host for many months to years. When the memory B cell happens to contact the similar antigen (which induced the production of memory B cells from an activated B cell), the memory B cell gets activated. The activated B cell divides many times to produce effector (plasma) cells and memory B cells

B Lymphocyte Activation

The mature B lymphocytes released from bone marrow are in a resting state and they don't secrete antibodies. The resting B cell, which don't contact the antigen die within few days. Whereas, a resting B cell, which binds to the antigen through the B cell surface immunoglobulin (sigs) becomes activated. The activated B cell divides and produces plasma cells and memory B cells.

Activation of B cell requires two signals and cytokines from T_H cell:

i. The first signal is provided by the binding of antigen to surface immunoglobulin's [sIgs; also called membrane immunoglobulin's (migs)] on B cell surface.

ii. Cell-to-cell contact between B cell and T_H cell provides the second signal required for B cell activation.

Binding of antigen to the surface immunoglobulins provides the first signal and initiates the B cell activation. The antigen binds to sIgs on B cell and cross-links the sIgs.

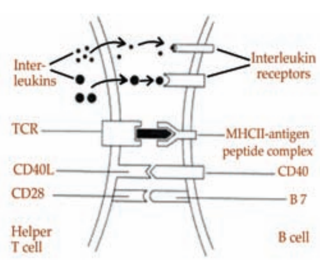


Figure 2: T Helper cell and B cell activation.

The sIg-antigen complex is internalized into the B cell by endocytosis (referred to as receptor-mediated or immunoglobulin-mediated endocytosis).

The sigs on a resting B cell bind to the antigen and the antigen is internalized. The lysosomal enzymes cleave the antigen into small peptides and complex them to the MHC class II antigen. The MHC class ll-antlgen peptide complex is expressed on the surface of B cell.

The following bindings occur between the molecules on B cell and T cell:

1. The T cell receptor (TCR) of helper T cell binds to the MHC class II-antigen peptide complex on the B cell.
2. The binding between CD40L molecules (on T cell) with CD40 (on B cell) delivers the second signal for B cell activation.
3. The CD28 molecule (on T cell) interacts with B7 (on B cell) and provides the necessary stimulatory signal to B cell. Furthermore, the cytokines secreted by T cell bind to B cell cytokine receptors on B cell and help in the B cell activation

The internalized antigen is processed into antigen peptides through the endocytic pathway. The lysosomal enzymes of the B cell cleave the antigen into short antigen peptides.

The short antigen peptide is complexed to MHC class II molecule. The MHC class Il-antigen peptide complex is transported to the B cell surface and expressed on the surface of B cell.

The MHC class II antigen peptide complex on B cell is presented to the T_H cell. It generally takes about 30-60 minutes for the B cell to process and present the antigen to T_H cell. (Antigen binding to sIgs on B cell also induces the B cell to express large number of class II molecules and B7 molecules on B cell surface. Induction of MHC class II molecule expression helps in the better presentation of MHC class Il-antigen peptide complex to T_H cell.)

Cell–cell interaction refers to the direct interactions between cell surfaces that play a crucial role in the development and function of multicellular organisms.

The T cell receptor (TCR) of T_H cell binds to the MHC class Il-antigen complex on B cell and forms a T-B conjugate.

The second signal for B cell activation is provided by the direct contact between some molecules on T_H cell and B cell. The interaction between CD40 molecules on B cell and CD40L

(CD40 ligand) on T_H cell delivers the second signal needed for B cell activation. In general, antigen binding to surface immunoglobulin and CD40L stimulation acts synergistically to trigger B cell activation.

Also, B7 molecules (on B cell) interact with CD28 molecules (on T_H cell). This **cell-to-cell interaction** provides a stimulatory signal required by the T_H cell for T cell activation. The activated T_H cell, in turn secretes many cytokines such as IL-2, IL-4, IL-5, and others.

The activated B cell expresses surface receptors for various cytokines, such as IL-2, IL-4, IL-5, and others. The cytokines secreted by T_H cell bind to their respective receptors on B cell and help in further proliferation and differentiation of B cell.

The activated B cell divides many times. Some of the daughter cells become plasma cells and others become memory B cells.

Apart from sIgs, some other molecules on B cell surface also play important roles in B cell activation. They are Ig-a/ Ig-P chains, B cell co-receptor complex, and CD22 molecules on the B cell surface.

B Cell Receptor and B Cell Activation

Each sig on B cell is associated with two signal- transducing Ig-α/Igβ heterodimer polypeptides to form B cell receptor (BCR). The Ig-α and Ig-β polypeptide chains have long cytoplasmic tails. The cytoplasmic tails of both Ig-α and Ig-β chains contain the 18-residue motif termed the immuno-receptor tyrosine- based activation motif (ITAM).

The antigen binding and cross-linking of sigs provides the initial stimulus for B cell activation. The stimulus produced by the cross- linking of sIgs is transduced into the B cell by the cytoplasmic tails of Ig-α/ Ig-β. The B cell activation signal is mediated by protein tyrosine kinases (PTKs). The signal-transduction processes leads to the generation of active transcription factors. The transcription factors stimulate the transcription of specific genes in the nucleus of B cell.

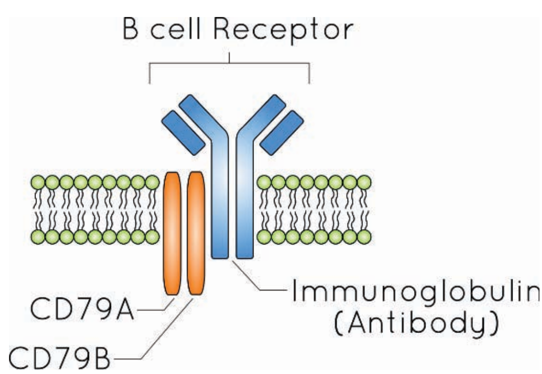


Figure 3: B cell receptor.

(A) The surface immunoglobulin (sIg) and two signal-transducing Ig-α/lg-β polypeptide chains form the B cell receptor on the B cell membrane. The Ig-α/lg-β chains have long cytoplasmic tails. The cytoplasmic tails contain the 18-residue motif called the immuno-receptor tyrosine-based activation motif (ITAM). (B) Initiation of B cell activation. The antigen binds and cross links the adjacent sIgs on B cell. The antigen binding with sIgs provides the initial signal for B cell activation. Upon cross linking of the sigs, the ITAMs interact with many members of the Src family of tyrosine kinases (Fyn, BIk, and Lck) and activate the kinases

Compare the development of B cells and T cells. Highlight the similarities and differences in terms of where they develop (bone marrow vs. thymus), their receptor types (BCRs vs. TCRs), and their functions.

B cell Coreceptor Complex and B cell Activation

The B cell coreceptor complex consists of three protein chains designated CD19, CR2 (CD21), and CD81 (TAPA- 1) (Figure 4). CD19 has a long cytoplasmic tail and three extra cellular Ig-fold domains. CR2 (complement receptor 2) acts as a receptor for C3d, a breakdown product formed during complement activation. CD81 is a membrane sparming polypeptide chain.

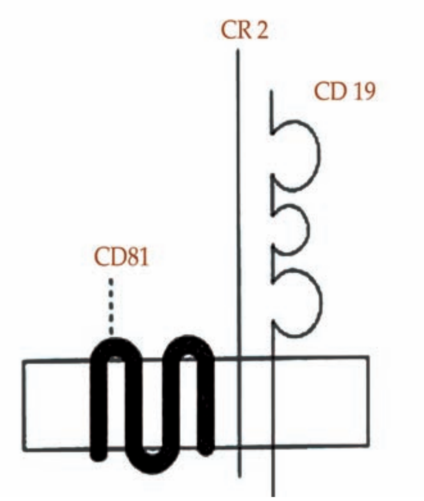


Figure 4: B cell coreceptor.

Three B cell membrane polypeptide chains together constitute the B cell coreceptor. GDI 9 has a long cytoplasmic tail and three extracellular Ig-fold domains. CR2 (complement receptor) has a short cytoplasmic tail. Extracellular portion of CR2 acts as a receptor for complement fragment C3d. CD81 is a **membrane** spanning polypeptide chain

Membrane is a selective barrier; it allows some things to pass through but stops others. Such things may be molecules, ions, or other small particles.

Antibodies formed against an antigen bind to the specific antigen.

The binding of antibody with antigen activates the classic complement pathway and leads to the deposition of C3d on the antigen.

When the antigen in the antigen-antibody complex binds to the sIgs of a B cell, the adjacent CR2 molecule (on B cell) binds to the C3d on the antigen (CR2 acts as a receptor for C3d). Thus, sIgs and B cell co-receptor are bridged to each other through the antigen-antibody complex (Figure 5).

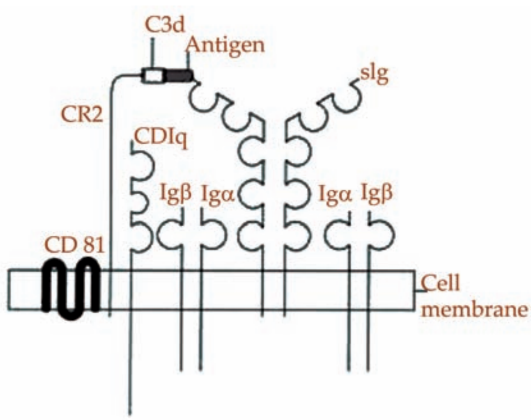
The cross-linking of sIgs with co-receptor enables the CD19 chain (of the B cell co-receptor) to interact with Ig- a / Ig-β chains of B cell receptor. The co-receptor complex serves to amplify the activating signals transmitted through BCR.

CD22 and Negative Signal for B cell Activation:

The B cells also express a molecule called CD22 on their surface. CD22 is constitutively associated with the B cell receptors in resting B cells. CD22 delivers a negative signal that makes the B cells more difficult to activate.

The antigen binds and cross links the sigs. The ITAMs of Ig-α/lg-β interact with members of the Src family of tyrosine kinases and activate the kinases. The CR2 chain of B cell co-receptor complex acts as a receptor for C3d and binds to C3d on the surface of antigen. Binding of CR2 with C3d leads to phosphorylation of CD19. The Src family of tyrosine kinase Lyn binds to phosphorylated CD19. The co-receptor complex amplifies the activating signals transmitted through **B cell receptor**

B cell receptor (BCR) is a transmembrane protein on the surface of a B cell.



Antibody (Ab) is a large, Y-shaped protein produced mainly by plasma cells that is used by the immune system to neutralize pathogens such as pathogenic bacteria and viruses.

Figure 5: B cell receptor and co-receptor during B cell activation.

Plasma Cells and Antibodies

When the B cell is activated, the activated B cell divides repeatedly. Some of the divided cells become plasma cells while others become memory B cells. Plasma cells secrete

antibodies. Plasma cells are spherical or elliptical. The cytoplasm is abundant and it may have a granular character. The nucleus is small in relation to the size of the cells. The nucleus is eccentrically placed and contains dense masses of chromatin, often arranged in a wheel-spoke fashion. Plasma cells don't replicate.

They live for few days only and later they die by a process called programmed cell death. A plasma cell may secrete thousands of antibody molecules per minute. The initial antibodies secreted by plasma cells in response to antigen always belong to IgM class. Usually the humoral responses wane after the subsidence of antigenic challenge because the antibody producing plasma cells don't live long.

The antibody secreted by plasma cell is a 'Y' shaped four-chain polypeptide molecule. The Fab region of antibody binds to its specific antigen. The antibody from a plasma cell binds only to the antigen that is responsible for its production (by activating a B cell and consequently, leading to the development of plasma cell). The antibody is specific to a particular antigen, because the antibody doesn't bind to other antigens.

Also the antibody is said to be a bi-functional molecule. The primary function of antibody is to bind to its specific antigen through its Fab regions. Binding of antibody to the antigen leads to the functioning of the Fc region of antibody (and the functions mediated by the Fc region are said to be secondary functions).

Memory B Lymphocytes

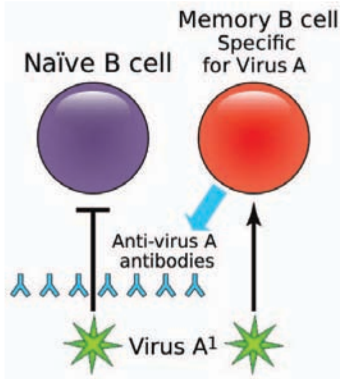
Upon activation of B cell, the activated B cell divides to produce two groups of cells called plasma cells and memory B cells. The memory B cells don't secrete antibodies immediately. They remain in a resting state for many months to years. In fact, in an adult, the lymph nodes are packed with memory B cells. The memory B cells in the lymph nodes are waiting for the contact with their specific antigens.

Upon its contact with the specific antigen the memory B cell becomes activated. The activated memory B cell divides and produces plasma cells and memory B cells (Figure 1). The plasma cells produce antibodies for immediate removal of antigen, while the memory B cells migrate to lymph nodes and wait for the future contact with antigen.

Naive B cells express only sIgM and sIgD on their cell surface. But memory B cells express sIgM/sIgG/sIgA/ sIgE/sIgD on their membranes (Table 1).

Table 1: Comparison of resting B cell and memory B cell

Properties	**Resting B cell**	**Memory B cell**
Surface	IgM and IgD	IgM/IgG/IgA/
immunoglobulin		IgE/IgD
Complement receptor	Low	High
Anatomic location	Spleen, lymph node	Bone marrow, lymph node, spleen
Life span	Short-lived	Long-lived



Inhibition of B cell Activation

Once the infecting agent is removed, there is no further necessity for the host to produce antibodies.

i. Plasma cells die shortly as they have a life span of few days only. As the plasma cells die, further production of antibodies is stopped.

ii. It seems that there is a negative feedback mechanism that regulates antibody production. Generation of new plasma cells is interfered by inhibition new B cell activation. The inhibition of B cell activation is probably mediated by the binding of antigen- antibody (especially IgG type) complexes to the B cell.

The Fc region of antibody combines with Fc receptor on B cell surface while the antigen (in an antigen-antibody complex) combines with sigs (Figure 6).

The simultaneous binding of antigen-antibody complex to the Fc receptor and surface immunoglobulin's on B cell may interfere with the signaling mechanism inside the B cell. Such interference may result in the suppression of B cell activation.

Consequently, new plasma cells production and antibody secretion do not occur.

iii. The antibody molecules are proteins and they are degraded over a period of time. As the antibodies in the circulation are degraded, the level of antibodies induced against an antigen decreases.

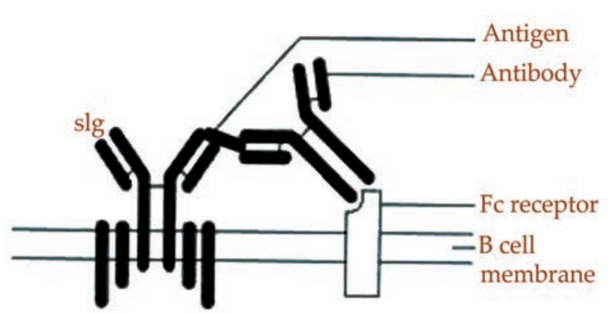


Figure 6: Inhibition of B cell activation.

B cell activation is probably inhibited by the binding of antigen-antibody complex to the sig and Fc receptor present on the B cell membrane. The antibody-bound antigen binds to the slg and the Fc region of antibody binds to the Fc receptor. These bindings interfere with the signaling mechanisms inside the B cell

Heavy-Chain Class Switch:

Always, IgM is the first antibody class produced by a plasma cell during primary **immune response** against antigen. But as the B cell clone proliferates, daughter cells capable of producing other classes of immunoglobulins (such as IgG or IgA or IgE or IgD) appear. This phenomenon is called class switching or isotype switching. The class switching occurs by rearrangement of genes coding the constant region of immunoglobulin.

Immune response is the body's response caused by its immune system being activated by antigens.

But there is no change in the gene coding for variable region. Hence, any class of immunoglobulin from a particular B cell clone will have the same antigen specificity, i.e. they will combine with the same antigen. The choice of a switch to a new immunoglobulin class (IgG/IgA/IgE/IgD) is influenced by many factors, like the tissue in which the B cell activation and proliferation occurs as well as the effect of some cytokines on B cells.

i. The microenvironment in Peyer's patches of intestine favors the switching to IgA class.

ii. IFNγ promotes class switching to IgG1.

iii. IL-4 promotes class switching to IgE.

Primary and Secondary Immune Responses

The immune responses induced at the time of first entry of antigen into the host are called primary immune responses. The immune responses induced during the second and subsequent entry of similar antigens into the host are called secondary immune responses.

Primary immune response

The primary immune response to an antigen is described in four phases.

1. Lag (latent) phase

Lag phase is the interval between the times of entry of antigen to the time of detection of antibodies against the antigen in blood. In human the lag phase is about one week. During the lag phase, the antigen is processed and presented to the T cells; the B cells are activated and the plasma cells begin to secrete antibodies.

2. Exponential phase

Exponential phase is the period during which the level of antibodies rapidly increases. This period reflects the large quantity of antibodies secreted by enormous number of plasma cells.

3. Steady-state (plateau) phase

During the plateau phase the antibody level remains at a constant level over a long period of time. The secretion of antibodies and degradation of antibodies occur approximately equal rates; and hence the antibody level remains at a steady state.

4. Declining phase

During declining phase, the level of antibodies slowly decreases. After the removal of antigen, new plasma cells are not produced; the already formed plasma cells die rapidly within few days after their production; and consequently, new antibodies are not secreted. The degradation of the already secreted antibody molecules leads to a decline in the antibody level.

Memory B cells are a B cell sub-type that are formed within germinal centers following primary infection and are important in generating an accelerated and more robust antibody-mediated immune response in the case of re-infection (also known as a secondary immune response).

Lag phase varies between different antigens and the period depends upon many factors. Depending upon the persistence of antigen, the period of primary response can be lost for various periods, ranging from few days to several weeks.

Secondary immune response

In contrast to the primary immune response, the peak concentration of antibodies is reached in 2-5 days during secondary immune response (Table 2). This is due to the greater number, speed, and intensity of **memory B cell** activity during the secondary immune response. The memory B cells respond to the antigen more rapidly than naive B cells. The number of naive B cells available for a primary immune response to an antigen is few.

Whereas, after a primary immune response, the number of memory B cells available to respond to the antigen are numerous. The number of memory B cells activated during a secondary response is more than the number of B cells activated during the primary response. Consequently the antibody level during secondary response is 100-1000 fold higher than that in primary response.

Table 2: Primary and secondary humor& immune responses

	Primary response	Secondary response
Antibody response	5-10 days	2-5 days
Lag phase	Relatively low	Relatively high
Peak concentration	1gM produced in early response and then lgG is produced	IgG predominates
lmmunoglobulin class produced		
Antigen	Thymus dependent antigen	Thymus dependent antigen
Average antibody affinity	Thymus independent anti-gen Relatively low	Relatively high
Responding cell	Resting B cell	Memory B cell

The first class of antibody produced against an antigen during a primary response (by the plasma cells derived from activated naive B cells) is always IgM. Afterwards other class/ classes of antibodies are produced against the antigen. Most of the plasma cells during secondary immune response secrete IgG, IgA, or IgE antibodies. Yet few plasma cells also secrete IgM antibodies during secondary immune response.

The sequence of events from antigen entry into the host to the time of maximal antibody secretion in a primary (For example, at the first time entry of antigen) immune response needs more time (about 5 to 10 days) when compared to the time required for secondary response.

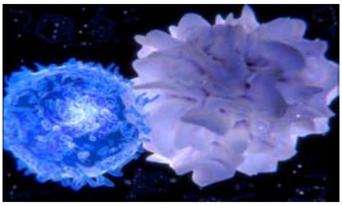
The reason for this longer period to achieve maximum antibody concentration during a primary response is due to the time needed for the following events:

a. Antigen specific T cells and B cells are few at the time of first entry of antigen into the host. The initial binding of antigen with specific T cells and B cells may take more time.

b. The antigen must be processed, and presented to antigen specific T_H cells.

c. The activated T_H cells should multiply and contact antigen specific B cells.

d. Then the activated B cells should proliferate and produce plasma cells to secrete antibodies.



B cell Activation by T-Independent Antigen

Generally antigen contact with surface immunoglobulins of B cell alone is insufficient to activate the B cell. Apart from antigen contact, the B cell also requires some help from the nearby antigen – specific helper ($CD4^+$) T cell. Such antigens, which require T cell help for activating the B cells, are called T-dependent antigens

However, there are some antigens, which can activate the B cells without the help from helper T cells. Such antigens are called T-independent antigens. There are two types of T-independent antigens (TI-1 and TI-2 antigens).

TI-1 antigens at high concentrations induce activation of antigen specific as well as antigen nonspecific B cells. Since many B cells are activated, these antigens are called polyclonal B cell activators (e.g. Lipopolysaccharide of gram-negative bacteria cell walls).

These antigens also stimulate macrophages to produce IL-1 and TNFa, which augment immune responses. On the other hand, TI-2 antigens do not have polyclonal activity nor do they activate macrophages (e.g. polysaccharides of bacterial cell walls)

T lymphocytes

T lymphocytes develop from the hematopoietic stem cells in the bone marrow. The progenitor T cells released from the bone marrow into blood circulation are immature T cells.

The progenitor cells then enter into an organ called thymus. Further maturation of T cells occurs in the thymus.

T cell Subpopulations (Helper T Cells and Cytotoxic T Cells):

Among the T cells, there are two functionally different sub- populations and each population has its own surface markers. These T cell subpopulations are also called as T cell subsets.

1. The T cells that express protein molecules called CD4 on their cell membranes are called helper T cells ($T_H$cells/ $CD4^+$T cells; $CD2^+CD3^+CD4^+CD8^-$ cells). T_H cells promote the immunologic functions of other cell types such as B cells, Tc cells, and macrophages.
2. The T cells which express CD8 protein molecules on their cell membranes are called cytotoxic T lymphocytes or cytolytic T lymphocytes (Tc cells or CTLs; $CD2^+CD3^+$ $CD8^+$ $CD4^-$cells). Tc cells play an important role in killing virus-infected cells, cancer cells, and cells of transplanted organs.

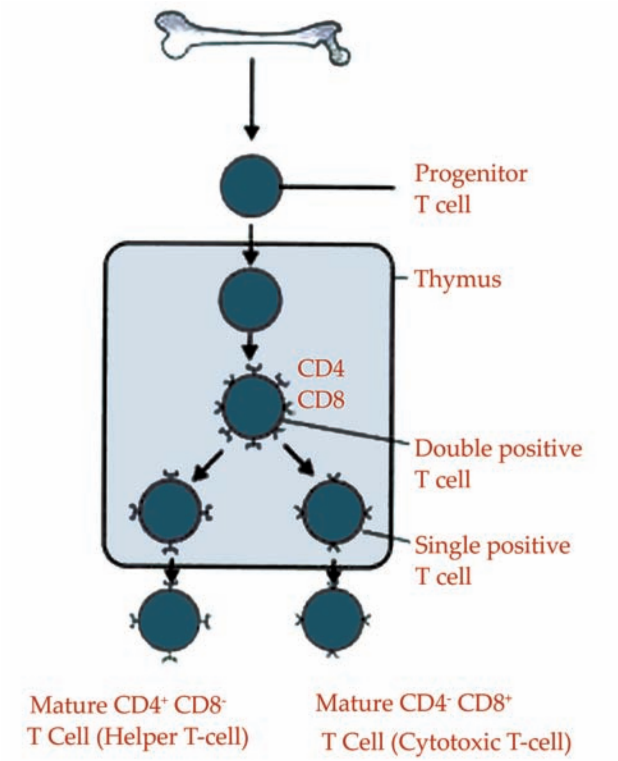


Figure 7:

Thymus is a specialized primary lymphoid organ of the immune system.

T lymphocytes are produced by hematopoietic stem cells in the bone marrow. T lymphocytes released from the bone marrow into circulation are not mature T lymphocytes and they are called progenitor T lymphocytes. The progenitor T lymphocytes enter the **thymus**, where T lymphocytes development is completed. The progenitor T cell entering the thymus does not express CD4 and CDS molecules on its cell surface (and hence called double-negative cells; $CD4^{-}CD8^{-}$).

As the cell develops, both CD4 and CDS molecules appear on its surface (and hence the cell is called double-positive cell; $CD4^{+}$ $CD8^{+}$). As the cell develops further, the cell shuts off either CD4 or CDS molecule expression and expresses any one of the molecules on the cell surface (and hence called single positive cell; $CD4^{+}$ $CD8^{-}$ or $CD4^{-}$ $CD8^{+}$). Mature, single positive T cells are released from thymus into blood circulation

About 70 % of T cells in human are helper T (also called $C04^{+}$ T) cells and 25 % are cytotoxic T (also called $CD8^{+}$ T) cells. About 4 % of T cells don’t express both CD4 and CDS molecules on their cell membranes. These $CD4^{+}CD8^{-}$ T cells are called as double negative T lymphocytes. They express a different form of T cell receptor composed of γ and δ

polypeptides. Remaining 1 % of T cells express both CD4 and CDS molecules and are called double positive T cells ($CD4^{+}CD8^{+}$). The functions of double positive T cells and double negative T cells are not known.

T Cell Receptors (TCRs)

The success of immune responses depends on the remarkable ability of the lymphocytes to recognize the antigens, which have entered into the host. The ways by which the T cells and B cells recognize antigens are different. T lymphocytes do not recognize antigens directly by themselves. T cell needs the help of another cell (called antigen presenting cell-APC) to present the antigen in a suitable form to the T cell.

(On the other hand, B cells don't require the antigen presenting cells to present the antigen to them. B cells directly bind to antigens through their surface immunoglobulin receptors. Incidentally, B cell itself acts as an APC to helper T cell).

The T cell receptor (TCR) on the cytoplasmic membrane of T cell is a complex of at least eight polypeptide chains (Figure 8). The α and β polypeptide chains of TCR bind to the antigenic peptide presented by APC. The other six polypeptide chains of TCR is called CD3 complex. CD3 complex is involved in signal transduction of TCR-antigen combination into the T cell. The intracellular signals lead to the activation of T cell.

The α and β chains of TCR are trans membrane polypeptide chains anchored to the T cell membrane. Each chain has three regions called extracellular region, trans membrane region, and intracellular region (or cytoplasmic tail). The extracellular portion of each chain is folded into two domains (similar to immunoglobulin domains) called variable domain and constant domain. Variable domain in a chain is called Vα domain and the variable domain in P chain is called Vβ domain.

The constant region domain of α chain is called Ca and the constant domain of P chain is called Cp. Similar to the variable region of immunoglobulin molecule, the variable region of TCR has three hyper variable regions (equivalent to the CDRs in antibody). The α and β chains are connected to each by disulfide bonds between their constant region sequences.

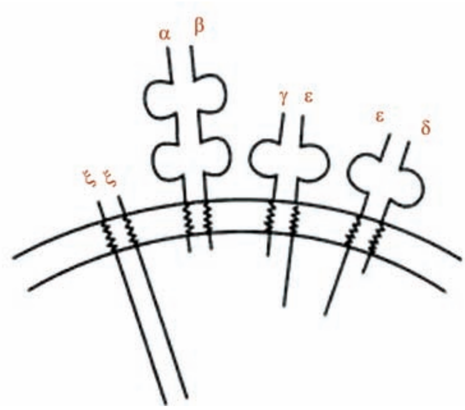


Figure 8

T cell receptor. T cell receptor on the T cell is a complex of eight polypeptide chains. The extracellular portions of α and β chains are folded into domains known as variable domains (Vα and Vβ) and constant domains (Cα and Cβ).

NK cells are a part of the innate immune system and play a major role in defending the host from both tumors and virally infected cells. NK cells distinguish infected cells and tumors from normal and uninfected cells by recognizing changes of a surface molecule called MHC (major histocompatibility complex) class I.

The variable domains in the α and β chains bind to the MHC class II-antigen peptide complex on antigen presenting cell. The remaining 3 sets of polypeptides together constitute the CD3 complex. There are two ξξ (zeta) chain homodimers, two γε (gamma and epsilon) chain heterodimers, and two e5 (epsilon and delta) chain heterodimers.

The cytoplasmic domains of CDS chains contain one or more immune receptor tyrosine- based activation motifs (ITAMs). The CDS complex converts the recognition of antigen by a and p chains into transmembrane signals

The amino terminal (i.e. the variable domain) of α and β chains of TCR, which binds to antigen is polymorphic. Therefore there are a large number of different forms of α and β chains. Again different combinations of α and β chains lead to the formation of different TCRs. Each TCR can bind to a particular antigen only. Since there are numerous forms of TCRs, the immune system has TCRs for a number of different antigens.

CDS complex consists of 3 pairs of polypeptide chains [ξξ (zeta) chain homodimers, γε (gamma and epsilon) chain heterodimes, and e6 (epsilon and delta) chain heterodimers]. The long cytoplasmic tails of the CDS chains contain a common sequence, the immunoreceptor tyrosine-based activation motif (IT AM). The IT AM site interacts with tyrosine residues and play important roles in signal transduction.

Antigen-presenting cell (APC) or accessory cell is a cell that displays antigen complexed with major histocompatibility complexes (MHCs) on their surfaces; this process is known as antigen presentation.

T cell Activation and T cell Functions:

Virtually any cell in the body can act as **antigen presenting cell (APC)** to T cell. Yet certain cell types (macrophages, dendritic cells, Langerhan's cells, and B cells) are specially adopted for this purpose and they are referred to as professional APCs.

The antigen peptide fragment of bacteria or virus is complexed to a protein molecule in APC called major histocompatibility complex (MHC) molecule. The MHC molecule-antigen peptide complex is transported to the cell membrane and expressed on the cell membrane of APC. The TCR (on the T cell) binds to the MHC-antigen peptide complex (on the surface of APC) and this binding activates the T cell.

i. Helper T cells are activated upon binding to MHC class II-antigen complex presented by professional APCs (such as macrophages, dendritic cells, and B cells).

ii. Cytotoxic T cells are activated upon binding to MHC class I-antigen complex presented by virus-infected cells or cancer cells.

Helper T_H Cell Activation:

Activation of helper T cell requires at least two signals (Figure 9):

a. The binding of T cell receptor (TCR) of T_H cell with MHC class Il-antigen complex (present on APC) provides the first signal:

i. The α and β chains of TCR (of T_H cell) bind to the antigen in the MHC class 11- antigen complex, and

ii. The CD4 molecule of T_H cell binds to the β_2 domain of MHC class II molecule.

b. The second signal (called co-stimulatory signal) is thought to be provided by the binding of a separate protein molecule on T_H cell with a protein molecule on APC. CD28 is a surface protein molecule on T_H cell. B7 is a surface protein molecule on APC. Binding between CD28 on T_H cell and B7 on APC delivers the second signal to T_H cell. Other surface proteins on T_H cell and APC may also mediate the co-stimulation of T_H cell.

Upon activation by the two signals, the T_H cell begins to secrete a cytokine called interleukin-2 (IL-2) and also expresses lL-2 receptors (1L-2R) on its surface. lL-2 and lL-2 receptors are essential for the proliferation and differentiation of activated T_H cell. lL-2 secreted by the T_H cell binds to the IL-2 receptor of the same T_H cell, which secreted it (a phenomenon known as autocrine effect). The activated T cell divides 2 to 3 times a day for about 4 to 5 days, resulting in the generation of a large number of cells; some of the daughter cells differentiate into effector T_H cells and others differentiate into memory T_H cells. Effector T_H cells have a short life span (few days to few weeks). Effector T_H cells also display several other surface molecules on their surfaces (such as CD25, CD28, CD29, CD40L, MHC class II molecules, and transferritin receptors). Memory T_H cells are generally thought to live for longer time.

(A) Binding between surface molecules on T_H cell and APC during T_H cell activation. The variable regions in the Vα and Vβ domains of α and β chains of TCR bind to the MHC class II-antigen peptide complex presented by the APC. The polypeptide chains of CD3 complex converts the recognition of antigen by α and β chains into trans membrane signals. The CD4 chain of T_H cell binds to the β_2 domain of MHC class II molecule. The co stimulation signal for T_H cell activation is provided by the binding of CD28 molecule on T_H cell with the B7 molecule on APC.

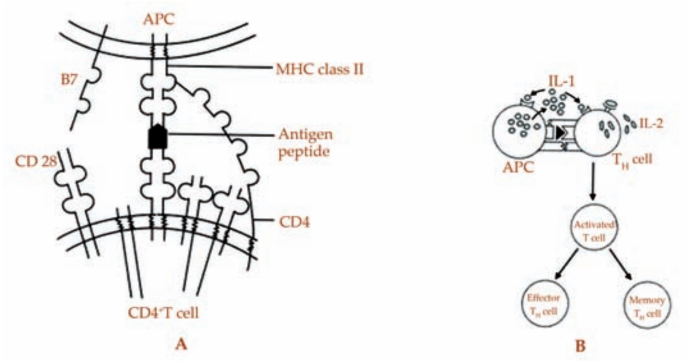


Figure 9: A and B: Helper T lymphocyte activation.

Apart from these bindings other surface molecules on T_H cell and APC may also participate in the T_H cell activation. (B) T_H cell activation and interleukin-1. The bindings between T_H cell and APC lead to the secretion of IL-1 by APC. IL-1 acts on the IL-1 secreting APC (known as autocrine effect) and on the nearby T_H cell (known as paracrine effect).

The autocrine effect of IL-1 leads to increased surface expression of MHC molecules and adhesion molecules on APC. The paracrine effect of IL-1 on T_H cell leads to increased IL-2 receptor expression on T_H cell and increased IL-2 secretion by T_H cell

Interleukin-I and T_H Cell Activation:

Cell-to-cell contact between T_H cell and APC leads to the activation of T_H cell. At the same time, the cell-to-cell contact also leads to the secretion of a cytokine called interleukin-1 (IL-1) by the APC. IL-1 seems to have autocrine (on IL-1 secreting APC) and paracrine (on the nearby T_H cells) effects.

Autocrine action of IL-1 increases the surface expression of MHC molecules and various adhesion molecules on the APC, which helps in a stronger cell-to-cell contact between APC and T_H cell. Thus IL-1 helps in a better antigen presentation to T_H cell. IL-1 also acts on the nearby T_H cell and promotes IL-2 secretion and IL-2 receptor expression by T_H cell. Thus IL-1 also helps in the proliferation of activated T_H cell (Figure 9).

Two other cytokines, **tumor necrosis factor** (TNF) and interleukin-6 (IL-6) secreted by APC also synergize with IL-1 and help in the T_H cell proliferation. [Thus the cell- to-cell contact between T_H cell and APC has bidirectional effects (i.e. T_H cell is activated by APC; at the same time APC is induced by T_H cell to secrete cytokines such as IL-1)].

Tumor necrosis factor (TNF) superfamily is a protein superfamily of type II transmembrane proteins containing TNF homology domain and forming trimers.

Functions of Activated T_H Cells

Effector T_H cells secrete many cytokines and the cytokines act over many cell types.

The cytokines of effector T_H cells perform the following main functions:

1. Activation and proliferation of T_C cells.
2. Help in the activation of B cells to produce plasma cells, which secrete antibodies.
3. Regulate the activities of monocyte-macrophages, and other cells of the immune system.

Virgin T_H lymphocytes are in a resting state and their ability to secrete cytokine is very limited. Binding of resting T_H cell to the MHC class Il-antigen complex on APC initiates the activation of T_H cell. The activated $T_H$cell divides many times to produce effector T_H cells and memory T_H cells. The effector T_H cells may fall into any one of the two subsets called T_H l subset or T_H 2 subset. The cytokines produced by the T_H l and T_H 2 subsets are different and consequently their immune functions are also different.

T_H1 Cells

T_H 1 cells produce IL-2, interferon-gamma (IFNγ) and tumor necrosis factor P (TNPP) (Table 3).

Helminths are large multicellular organisms, which can generally be seen with the naked eye when they are mature.

i. These lymphokines activate macrophages and other phagocytes leading to enhanced phagocytosis and intracellular killing of engulfed microbes.
ii. IFNγ induces immunoglobulin class switching of B cells to produce IgGl subclass of antibodies. The IgGl can strongly bind to Fc receptors (of IgG) on macrophages so that the opsonization and the subsequent intracellular killing of microbes by macrophages are enhanced.
iii. IL-2 secreted by T_H cells helps in the activation of cytotoxic T cells.
iv. Apart from IL-2, the T_H cell also secretes many other cytokines, which act upon B cells, macrophages, and other cell types.

T_H 2 Cells

T_H2 cells produce cytokines that are usually involved in actions against large, multicelluar parasites such as **helminths**, which are too large to be engulfed by macrophages. T_H l cells secrete interleukin-4 (IL-4), interleukin-5 (IL-5), interleukin-6 (IL-6), interleukin-10 (IL-10), and inteleukin-13 (IL-13) (Table 3).

i. The T_H2 cell derived cytokines chemo attract B cells, mast cells, basophils, and eosinophils and also promote the growth and differentiation of these cells at the site where parasite is present.

ii. lL-4 also promotes B cell class switching to IgE. IgE combines with the Fc receptors (of IgE) on mast cells and eosinophils and induces these cells to release their cellular contents. The released cellular contents of mast cells and eosinophils act against the parasites.

Table 3: Cytokines secretion by T_H1 and T_H2 cells

T_H subtype	*Cytokines secreted*	*Major immunological effects*
T_H1	IFNy	Activate macrophages. Promote B cell proliferation and class switching to IgGI. Inhibit T_H2 cell differentiation.
	IL-2 TNFI3	Promote activation of antigen specific T_H and T_c cells. Activate macrophages and neutrophils. Promote B cell growth and immunoglobulin production.
T_H2	1L4	Inhibit T_H1 cell differentiation. Chemo-attract lymphocytes, mast cells, and basophils. Enhances growth of mast cells and eosinophils. Promote B cell proliferation and class switching to IgE and IgC4. Inhibit cytokine production by macro-phages.
	IL-5	Enhance growth and development of eosinophils.
	IL-6	Promote B cell growth and immuno-globulin production.
	IL-10	Inhibit production of cytokines (including IFNy) by T_H1 cells, macrophages, and other APCs. Inhibit T_H1 cell differentiation. Promote B cell growth and immunoglobu-lin production.
	IL-13	Same as 114.

Cytotoxic Cell Activation

Cytotoxic T (T_c) cells or cytolytic T lymphocytes (CTLs) are $CD8^+$ T cells and they play a major role in defense against viral infections. Viral infected cells present the viral antigens in association with MHC class I molecules on the infected cell surface.

Binding of T_C cells to the MHC class I-viral antigen complex on the cell membrane of the AFC initiates the T_H activation. Activation of T_C cell requires two important signals (Figure 10).

The binding of TCR of T_C cell with the MHC class I- viral antigen complex on the viral infected cell surface provides the first signal.

i. The variable regions of α and β chains (Vα and Vβ) of TCR of Tc cell bind to the viral antigen in the MHC class 1-viral antigen complex, and

ii. The CD8 molecule on T_C cell binds to the α 3 domain of MHC class 1 molecule.

The first signal induces the expression of lL-2 receptors on the T_C cell surface.

The second signal is provided by the cytokine IL-2 secreted by the nearby-activated T_C cell. (T_C cells generally do not produce enough IL-2 to stimulate their own proliferation). The IL-2 produced by activated T_C cell binds to the IL-2 receptors on T_C cell and helps in the activation and proliferation of T_C cell.

A third signal for T_C cell activation may be provided by the interaction of CD28 (on T_C cell) with B7 molecule (on virus infected cell).

Cytotoxic T Cell Functions

1. Destruction of virus infected cell, leading to the elimination of virus from the host.

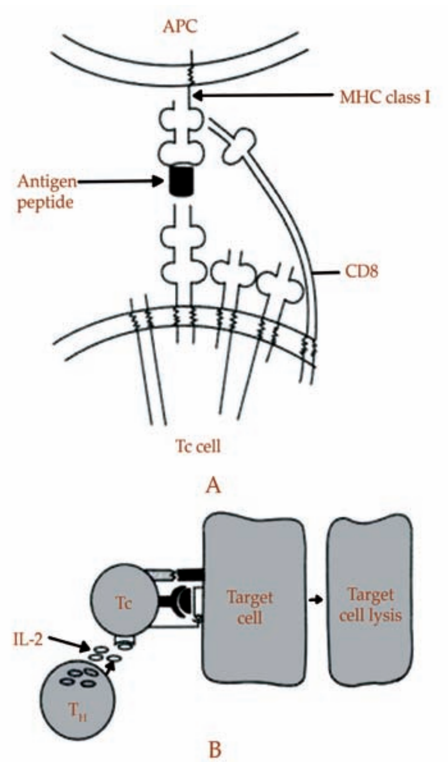


Figure 10: A and B: Cytotoxic T cell activation.

(A) Binding between TCR and MHC class l-viral antigen peptide complex on APC.

The variable regions in the Vα and Vβ chains of TCR bind to the MHO class l-viral antigen peptide complex on target cell (which acts as the APC). Upon this binding, the CD3 complex sends transmembrane signal into the T_c cell leading to the activation of T_c cell. The CDS polypeptide on T_c binds to the α3 domain of MHC class I molecule, and (B) IL-2 secreted by T_H cell helps the activation of T_H cell. Activated T_C cell secretes IL-2. The IL-2 binds to IL-2 receptors on T_c cell and helps in the activation of T_c cell. The activated T_C cell lyses the target cell, which presented the antigen to the T_c cell

2. Destruction of cancer cells, which may express tumor- specific antigens on their cell surface.
3. Destruction of cells of the transplanted organ from HLA- unrelated donors.

How T Cells (CTLs) destroy target cells?

The following sequence of events results in the destruction of target cells (such as virus infected cell, cancer cell, and transplanted organ cell) by CTLs.

Binding of TCR (of CTL cell) with the MHC class I- antigen peptide complex (on target cell) provides the signal needed for the initiation of CTL's action against target cell.

Integrin receptor molecule LFA-1 (on CTL cell) binds to intercellular cell adhesion molecule (ICAM) on the target cell; and forms a CTL-target cell conjugate.

Granzymes are serine proteases that are released by cytoplasmic granules within cytotoxic T cells and natural killer (NK) cells.

The CTL releases its granules over the target cell. The granules contain enzymes perforin and granzymes.

1. Perforin is a 534-amino acid protein. Perforin shows limited sequence homology with the pore forming complement proteins C6, C7, C8, and C9. Perforin molecules insert and polymerize in the target cell membrane by a mechanism similar to that of C9. Approximately, 20 perforin molecules polymerize to form a tubular hole (about 16 nm width) in the target cell membrane. Through the pores intracellular proteins and ions of target cell leak out. Ultimately, the target lyses by osmotic effects.
2. The granules of the CTLs also contain a family of serine proteases known as **granzymes**. As

explained above the perforins punch holes in the target cell membrane. Subsequently, the granzyme B enter into the target cell through the perforin pores. Within the target cell, granzyme B activate caspases in the target cell. Caspases, in turn, cause nuclear damage and leads to apoptotic death of the cell (Figure 11).

3. Apart from the perforin and granzyme mediated destruction of target cell, the CTL also kills the target cell by another mechanism. Activation of CTL leads to the expression of protein molecules called Fas ligands (FasL) on the surface of CTL. Fas protein is a transmembrane protein on the cell membrane of target cell.

Binding of FasL (on CTL) with Fas (on target cell) delivers a death signal into the target cell; and triggers apoptosis of the target cell, resulting in death of the target cell (Figure 11). Both granzyme and FAS pathway initiate a caspase cascade of apoptotic death of target cell.

Apart from target cell DNA, the viral DNA inside the target cell is also fragmented during apoptic death of target cells, resulting in viral elimination. After delivering a lethal hit, the CTL moves away from the attacked target cell and searches another target cell.

Accessory Molecules that Strengthen the Cell-to-Cell Contact between T cell and APC:

The interaction of TCR on T cell with MHC-antigen peptide on APC is usually weak. Therefore the cell-to- cell contact between T cell and APC has to be strengthened. Cell-adhesion molecules on both T cell and APC strengthen the cell-to-cell contact between T cell and APC (Figure 12).

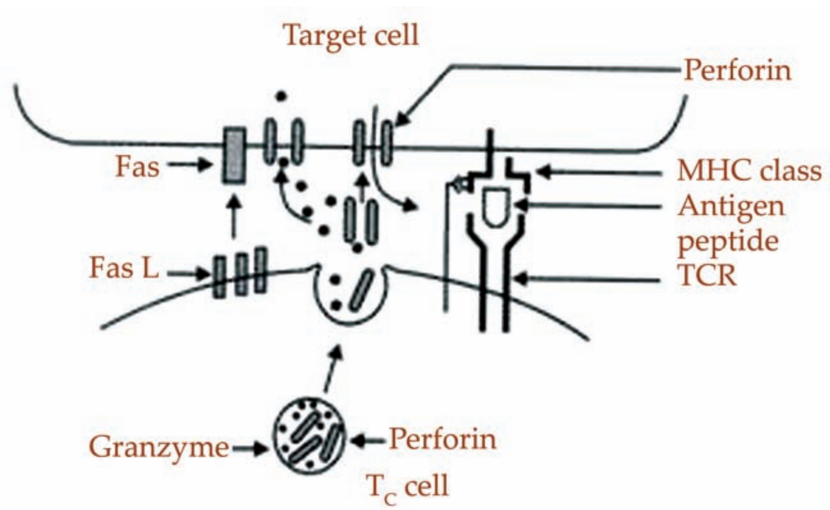


Figure 11: Different mechanisms by which the cytotoxic cell attacks the target cell.

The binding of MHC class l-antigen peptide complex on the target cell with the TCR of T_c cell activates the T_C cell. The activated T_c cell secretes enzymes perforin and granzyme. Mechanism 1. **Perforin** inserts itself into the target cell membrane. Polymerization of many perforin molecules on the target cell membrane leads to the formation of small pore in the target cell membrane. The target cell contents leak out through the pores and consequently, the target cell dies. Mechanism 2.

Perforin-1 is a protein that in humans is encoded by the PRF1 gene and the Prf1 gene in mice.

The granzyme molecules enter into the target cell through the pores created by the perforins and activates the caspases in the target cell. The activated caspases, in turn, lead to apoptotic death of the target cell. Mechanism 3. The activated T_c ceil expresses FasL (Fas ligand) on its cell membrane. If the target cell membrane expresses Fas molecules, the FasL on T_c cell bind to Fas on target cell and such binding leads to the apoptic death of target cell

T cells express a number of adhesion molecules such as leukocyte functional antigen-1 (LFA-1; also called CD11a/CD18) and CD2. These adhesion molecules on T cell bind to molecules on APC and promote cell-to-cell contact. The binding of adhesion molecules probably initiates the interaction between T cell and APC. Subsequently, the TCR binds to the MHC -antigen complex on APC leading to signal transduction into T cell. Consequently, the T cell is activated. During activation of T cell there is a transient increase in the expression of accessory molecules. The transient expression of accessory molecules helps in the interaction between the cells. Like CD4 or CDS molecules, some of the accessory molecules may also function as signal transducers for T cell activation.

The accessory molecules don't interact with MHC- antigen complex. The binding of accessory molecules between T cell and APC are independent of the binding between TCR with MHC-antigen complex.

Memory T Cells

A remarkable feature of the acquired immune system is its memory of antigens that have previously entered into the body. The immune responses induced during the first entry of antigen into the host are called primary immune responses. During primary immune response, T and B cells are activated against the particular antigen. The activation of T and B cells and the development of effective immune responses against the antigen take 5 to 7 days during the first entry of the antigen.

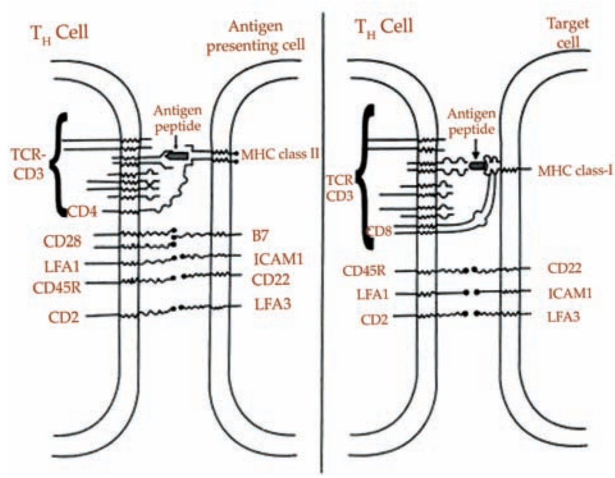


Figure 12: Schematic diagram of bindings between various surface molecules of T_H cell and APC and between T_c cell and target cell.

The bindings between surface molecules strengthen the interaction between the cells and leads to signal transduction and activation of T_H cell or T_C cell

But during second and subsequent entry of similar antigen, the immune system immediately identifies the antigen and mounts early and effective immune responses (referred to as secondary immune responses). When compared to the responses during the first exposure, the responses during subsequent exposures are early and vigorous. The immune system remembers every antigen that has entered into the body (like a police officer remembering a thief whom he captured once).

Virgin T cells released from thymus are in a resting state and they don't divide. If antigens do not activate the virgin T cells the virgin T cells die shortly after their release from thymus. On the contrary, if the virgin T cell is activated by its contact with the antigen, the T cell continues to live and divide many times. Some daughter cells become effector T cells while other daughter cells become memory T cells. The effector T cell functions are required for the immediate action against the antigen, which is already present in the host. Whereas the memory T cell functions are reserved for future encounters with the similar antigen, if the antigen happens to enter into the host again.

When the activating stimulus (antigen) is removed, the effector T cell activities subside over a period of several days.

Memory T cells have either long life or capable of self- renewal and they persist for years. Antigen specific memory CTLs has been detected in humans after 30 years of vaccination.

The virgin T cells express 205 to 220 kD isomers called CD45RA on their surface. Whereas the memory T cells express an 180kD isoform called CD45RO on their surface. Memory T cells also express high levels of adhesion molecules.

Helper T Cell Differentiation into T_H1 and T_H2 Cells:

In 1980s it was observed in mice that there were two types of helper T cells secreting two different sets of cytokines. One class referred to as T_H l produced cytokines that stimulated strong cellular immunity but weak antibody response. The other class referred to as produced the opposite effect; the cytokines secreted by T_H2 cells evoke strong antibody response but relatively weak cellular response.

It appears that T_H l and T_H2 cells are derived from common T_H cells. Such differentiation probably involves an intermediary stage called the T_H0 cell, which can secrete both IFNγ and IL-4. It is believed that the subsequent differentiation of T_H0 cells into T_Hl or T_H 2 depend upon the effects of other cytokines (such as IL-4 or IL-12) in the environment on the T_H0 cells.

The cytokines secreted by T_H l cells seems to play important roles in CMI responses, whereas, the cytokines produced by Tpj2 cells seems to play important role in humoral immune responses.

i. IL-2 and IFNγ produced by T_H1 cells enhance the microbial killing power of macrophages. The macrophages in turn, kill the intracellular bacteria.

ii. On the other hand, IL-4, IL-5 and IL-10 produced by T_H2 cells act mainly on B cells and induce antibody production, and antibody class switching. Thus, T_H2 cytokines act mainly against extracellular microbes through antibodies.

Leishmania is a genus of trypanosomes that are responsible for the disease leishmaniasis.

How T_H 0 Cell Differentiates into or T_H2 Cell?

The molecular events responsible for the differentiation of T_H0 cells into T_H1 or T_H2 cells are not known. However, it is believed that cytokines in the microenvironment of T_H0 cells are the principle factors, which determine the T_H0 cell differentiation into T_H1 or phenotypes (Figure 13).

i. In vitro and in vivo studies have shown that IL-4 induces the T_H0 cells to differentiate into T_H2 cells. But the source of IL-4 for the differentiation is not known. Mast cells may be the source of IL-4 for T_H0 cell differentiation.

ii. Differentiation of T_H0 cells into T_H1 cells needs IFNγ. The following events are suggested for the source of IFNγ:

Intracellular bacteria (such as **Leishmania** major, Mycobacterium leprae) stimulate macrophages and the stimulated macrophages secrete IL-12.

IL-12 acts on NK cells and the NK cells in turn secrete IFNγ.

IFNγ secreted by NK cells and IL-12 are believed to act upon T_H0 cells and leads to the differentiation of T_H0 cells into T_H1 cells.

Furthermore, when the T_H0 cells are differentiating into T_H1 cells, there is an associated inhibition of T_H2 cytokine secretion. Similarly, when T_H0 cells are differentiating into T_H2 cells, there is an associated inhibition of T_H1 cytokines secretion.

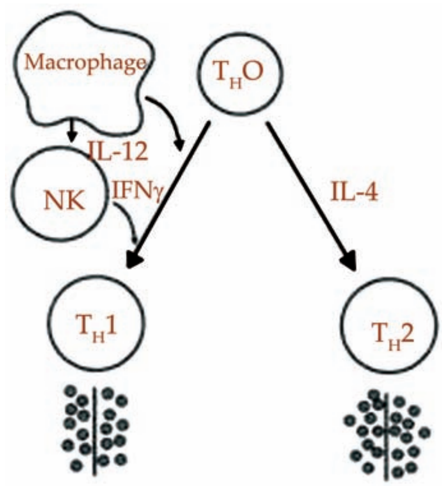


Figure 13: Differentiation of the T_H cell into T_H1 cell or T_H2 cell.

The microenvironment of T_H0 cell is believed to be responsible for the differentiation of T_H0 cell into T_H1 or T_H2 cell. The intracellular bacteria within the macrophage stimulate the macrophages to secrete IL-12. IL-12 acts on NK cell and the NK cell in turn secretes IFNγ. IFNγ in the microenvironment is responsible for differentiation of T_H0 cell into T_H1 cell. On the other hand, presence of IL-4 in the microenvironment leads to the differentiation of T_H0 cell into T_H2 cell

i. Thus IFNγ not only promotes cell differentiation, but it also prevents T_H1 cell development (by inhibiting IL-4 secretion).

ii. IL-4 not only promotes Th2 cell differentiation, but it also prevents T_H1 cell development (by inhibiting IL- 2 and IFNγ production).

This sort of polarization of immune responses towards T_H1 or T_H2 occurs especially in chronic parasitic infections.

Example 1:

T_H1 dominated immune response is seen in a mouse strain infected with Leishmania major. L. major is an intracellular parasite. L.major reside inside the macrophages and induce the macrophages to secrete IL- 12. The IL-12 promotes a T_H1 response against L. major. The lymphokines secreted by T_H1 cells in turn activate the macrophages to kill the intracellular parasite. In contrast, there are few mice strains, which can't kill L. major.

In these mouse strains, L. major infection leads to a T_H2 type of immune response. T_H2 response leads mainly to antibody production; but antibodies are ineffective against intracellular organisms. Since these mice strains don't develop a T_H1 response, macrophages are not activated (due to the absence of T_H1 cytokines). Consequently, the L. major multiplies and kill the mice. Therefore development of T_H1 response is essential for protection against L. major infection.

Example 2:

There are two main forms of leprosy (caused by the bacteria Mycobacterium leprae) called, tuberculoid leprosy (less aggressive form, wherein the infection is controlled by macrophages) and lepromatous leprosy (more severe form of leprosy, wherein the infection is uncontrolled). It is suggested that the promotion of $T_H$cells into T_H1 or may be responsible for the development of these two extreme forms of leprosy. Development of T_H1 response contains the infection and the person develops tuberculoid form of leprosy. Whereas development of response leads to an inability on the part of the macrophages to kill the bacteria; and this results in the dissemination of bacteria to many parts of the body and development of **lepromatous leprosy**.

Lepromatous leprosy is a form of leprosy characterized by pale macules in the skin.

Apart from chronic infections, T_H1 mediated responses are found in experimental autoimmune diseases. T_H1 responses are probably responsible for tissue damage in experimental autoimmune diseases.

i. T_H1 responses are implicated in inbred strain of NOD mice developing diabetes. There are evidences to suggest that induction of T_H2 responses in these mice may protect them from diabetes. Injection of IL-4 in NOD mice prevents or delays the onset of diabetes. T_H2 responses are found to be dominant in allergic diseases.

Compared to IFNγ producing T cell clones, higher proportion of IL-4 producing T cell clones were isolated from the peripheral blood of patients with atopic diseases of skin and lung. Cytokines IL-4 and IL-5 are believed to be responsible for the pathophysiology of these conditions because IL-4 and IL-5 produce increased IgE synthesis and increased eosinophil production, respectively.

Down Regulation of T cell Immune Responses:

Once the antigen is eliminated, the continued function of effector T cells is no longer beneficial to the host.

The mechanism of termination of T cell function is not fully known. CTLA-4 is a T cell surface molecule. It is believed that CTLA-4 acts as an important negative regulator of T cell function.

It was earlier explained that B7 molecule on APC binds to CD28 molecule on T_H cell and this binding acts as an important co-stimulatory signal for T_H activation. However, B7 molecule on APC can also bind to another T_H molecule called CTLA-4. But, B7 binding with CTLA- 4 on T_H cell causes down regulation of T_H cell activation.

CD28 is expressed by resting T_H cell, whereas the CTLA-4 is absent on resting T_H cell. CTLA-4 is expressed on an activated T cell. During an immune response against antigen, initially, the T_H cell is activated by the binding of CD28 (on T cell) with B7 (on APC). CD28 binding with B7 acts as an important co-stimulatory signal for the activation of T_H cell.

After the T_H cell is activated, CTLA-4 molecules appear on the activated T_H cell.

If the CTLA-4 molecules on activated T_H cell happen to bind to B7 molecules (on APC), negative signals are sent into the T_H cell, leading to the down regulation of T_H cell activation. Hence it is suggested that CTLA-4 acts as a regulatory molecule of activated T_H cell (Figure 14).

T Cells With γ/δ Chains of ICR

Most of the T cells in the circulation express α and β chains in their TCRs. But a small subset (less than 5 %) of mature T cells does not express α/β chains in their TCR. Instead they have different amino add chains designated γ and δ. The physiologic roles of γ/δ cells is uncertain. Certain γ/δ T cells recognize nonpeptide antigens derived from mycobacteria in vitro, and substantial increase in the number of these cells have been observed in patients with tuberculosis and other mycobacterial infections.

γ and δ T cell population appears to be a major population in skin, intestinal epithelium, and respiratory tract epithelium. The selective localization of γ/δ T cells in these sites may be related to their role in protection against microbes entering through these sites.

Anergy

B7 molecules are constitutively expressed on dendritic cells. But macrophages and B cells express B7 molecules after their activation. The co-stimulatory signal (between CD28 with B7) is essential for the activation and consequent proliferation and differentiation into effector T cells and memory T cells.

In the absence of co- stimulatory signal (CD28 and B7) the T cell doesn't proliferate in spite of TCR and MHC-antigen complex binding. Such a non-responsive state of T cell is referred to as anergy. IL-2 is essential for T cell proliferation. Lack of co-stimulatory signal results in very low IL-2 production and consequently the T cell proliferation does not occur.

(A) Resting T_H cell expresses CD28 molecules on its surface. Binding of CD28 (on resting T_H cell) with B7 (on APC) acts as an important co stimulatory signal for the activation of T_H cell, and (B) The activated T_H cell expresses molecules called CTLA- 4 on its surface. Binding between CTLA-4 (on activated T_H cell) with B7 molecule (on APC) is believed to send negative signal into the T_H cell, leading to the down regulation of $T_H$cell activation

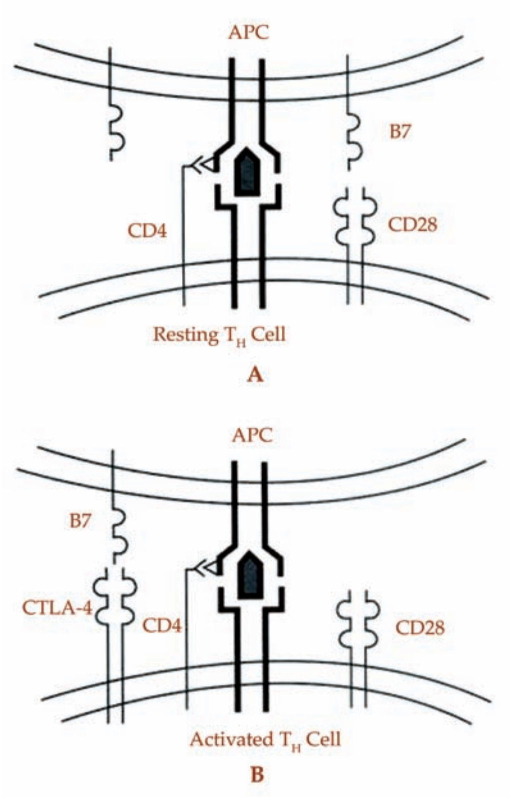


Figure 14: A and B: Down regulation of activated T_H cell.

Suppressor T Lymphocytes:

Apart from helper and cytotoxic subpopulations of T cells, it is proposed that another subpopulation of T cells called suppressor T cell population also exists. Suppressor T cells are suggested to suppress the humoral and CMI responses. However, it is uncertain whether suppressor T cells really contribute a separate functional subpopulation of T cells.

T Lymphocyte Development in Thymus:

The term T cell maturation is used to denote the events within the thymus that lead to the co-ordinated expression of ICRs, co-receptors, growth factor receptors, and adhesion molecules on T cells. These events occur through interactions of T cells with thymic cells. Cytokines, especially IL-7 and thymic hormones are implicated in T cell maturation. The entire mechanisms behind the T cell maturation guided by thymic cells are not known.

Bone marrow releases progenitor T cells into the circulation. The progenitor T cells released from the bone marrow are not mature T cells. Further maturation of T cells occurs in an organ called thymus, situated in the superior mediastinum. The progenitor T cells released form bone marrow into the circulation migrates to the thymus. T cells in the thymus are also referred to as **thymocytes**.

Thymocytes are hematopoietic progenitor cells present in the thymus

Thymus is covered by a fibrous capsule from which fibrous bands (trabeculae) penetrate and divide the parenchyma of the thymus into a number of lobules. Histologically each lobule has two distinct regions, the cortex or peripheral region and medulla or central region. The cortex is further divided into an outermost (or subcapsular) cortex and inner (or deeper) cortex. The anatomic divisions mentioned above correspond to functionally distinct microenvironments, which support specific phases of thymocyte maturation.

The thymic epithelial cells in the cortex of the thymus have long (about 25 μm) cytoplasmic processes and hence they are known as dendritic epithelial cells. The dendritic epithelial cells interact with the thymocytes and guide the thymocyte differentiation into mature T cells. The thymocyte-dendritic epithelial cell interaction results in the formation of cell complexes called lymphoepithelial complexes. The lymphoepithelial complexes are also called as nurse cells. The thymic nurse cells are composed of a dendritic epithelial cell that has internalized 20 to 40 thymocytes by emperipolesis.

During its stay in the thymus, T cell receptor (TCR) gene rearrangement occurs in the thymocyte.

There are two main purposes behind the TCR-gene rearrangement:

1. TCR (transcribed by rearranged TCR gene) of a T cell should bind to self-MHC molecules [because the TCR recognizes antigen presented in association with self- MHC molecule only]. Differentiating thymocytes capable of binding to self-MHC molecules are allowed to live by a process known as positive selection of thymocytes.

2. The TCRs shouldn't bind to self-peptides of the host. If the TCR binds to a self-peptide, the host tissue itself will be destroyed [a condition known as autoimmunity]. Differentiating thymocyte whose TCR has a high-affinity for self-MHC molecule is eliminated through a process known as negative selection of thymocytes.

Thymus is divided into three anatomical regions, the subcapsular region, the cortical region, and the medullary region. The progenitor T cells from bone marrow enter the thymus and migrate to the subcapsular region. The T cell development starts in the subcapsular region. As the thymocytes differentiate, they move from subcapsular region to cortex region, and then to medullary region.

The progenitor T cells released from the bone marrow are immature T cells. The progenitor T cells do not express CD4, CD8, or TCR molecules on their surface. During their stay in thymus, the thymocytes progress through a series of differentiation stages.

Since the progenitor T cells entering the thymus lack CD4 and CD8 molecules they are called double-negative ($CD4^+CD8^-$) thymocytes. The double-negative T cells differentiate and begin to express both CD4 and CD8 molecules on their surface. The thymocytes at this stage expressing both CD4 and CD8 molecules are called double-positive ($CD4^+CD8^+$) thymocytes. The double-positive thymocytes also express α and β chains of TCRs.

Positive Selection of Thymocytes

The T cell can bind to antigen only when the antigen is presented by self-MHC molecule on APC (self-MHC restriction). During their stay in the thymus, TCR-gene rearrangement occurs in the thymocytes. If the TCR-gene rearrangement in a thymocyte results in the formation of TCR, which can bind to self-MHC molecule, such thymocyte is allowed to progress further.\ Whereas, a thymocyte whose TCR is unable to bind to self-MHC molecule is eliminated (because such a thymocyte cannot bind to antigen presented by the APC, and hence is of no use to the host). The thymus allows the progression of thymocytes whose TCRs are capable of binding to self-MHC molecules by a process known as positive selection of thymocytes.

1. A double-positive thymocyte whose TCR binds to self- MHC class 1 molecule on the thymic epithelial cell receives a maturation signal and a survival signal; and the cell undergoes a positive selection. Consequently, the cell stops expressing CD4 molecules and express only CDS molecules. The cell becomes a single- positive ($CD8^+$) thymocyte.
2. Another double-positive thymocyte whose TCR binds to self-MHC class II molecule on thymic epithelial cell receives a maturation signal and a survival signal; and the cell undergoes a positive selection. Consequently, the thymocyte stops expressing CDS molecules and express only CD4 molecules. The cell becomes a single-positive ($CD4^+$) thymocyte.

3. Double-positive thymocytes that's TCRs are unable to bind to either self-MHC class I molecule or self- MHC class II molecule don't receive any surviving signals and they die by apoptosis.

Negative Selection of Thymocytes

T cell differentiation should produce T cells, which should react with foreign antigens, but not self-antigens (If T cells capable of binding to self-antigens are released as mature T cells, they will react with self-antigens and destroy host cells). The purpose of deleting T cells capable of reacting with self-antigens is believed to be achieved through the negative selection of thymocytes.

The details of negative selection are not completely understood. In the medulla of thymus, the positively selected thymocytes interact with the self-MHC class I and class II molecules present on the surface of dendritic cells and macrophages. Some of the positively selected thymocytes have low-affinity TCRs for self-antigens presented by self-MHC molecules; while others have high-affinity TCRs for self-antigens presented by self- MHC molecules.

Thymocytes bearing high-affinity TCRs for self-antigens presented by self-MHC molecules die by apoptosis. Whereas, thymocytes that are capable of reacting with self-MHC molecule plus foreign antigen are allowed to differentiate further to attain maturity. Single-positive ($CD4^+CD8^-$ or $CD4^-CD8^+$) T cells are released into the circulation as mature T cells.

In spite of intense research, there are many questions yet to be answered with respect to the positive and negative selection of T lymphocytes in the thymus.

Superantigens and T cell Activation

Activation of T_H cell occurs when the antigen is presented in association with MHC class II molecule by the APC to the T_H cell. Usually, antigens cannot activate T_H cells unless the antigen is presented by the APCs. However, there are some antigens (such as bacterial toxins and retroviral proteins) that can activate T_H lymphocytes without being processed and presented by APCs and such antigens are called superantigens.

The super antigen is not processed and presented by the APC to the T_H cell. From outside the cells, the superantigen binds the MHC class II molecule of the APC and the p chain of TCR; and the superantigen acts as a 'clamp' between these two cells (Figure 15). This binding leads to the activation of T_H cell.

Upon exposure of the host to superantigens, enormous number T_H cells are activated as described above. Activation of enormous number of T_H cells results in sudden release of large amounts of cytokines from activated T_H cells. Sudden release of large amounts of cytokines is injurious to the host and causes many severe clinical symptoms (such as toxic shock syndrome or food poisoning by Staphylococcus aureus enterotoxin).

Toxic shock syndrome (TSS)

In 1980s, toxic shock syndrome (wherein, the patient develops sudden skin rash, fever, hypotension and even death) became epidemic among young, primarily white woman during menstruation. A strong correlation between TSS and recovery of Staphylococcus aureus from vaginal cultures of affected persons was found. Most of the isolated S. aureus produced a toxin called toxic shock syndrome toxin-1.

> Microscopically, in a Wright's stained peripheral blood smear, a normal lymphocyte has a large, dark-staining nucleus with little to no eosinophilic cytoplasm. In normal situations, the coarse, dense nucleus of a lymphocyte is approximately the size of a red blood cell (about 7 μm in diameter).

This toxin acts as a superantigen and activates massive number of T_H cells leading to sudden release of large amounts of cytokines. The sudden release of large amounts of cytokines is responsible for the symptoms. Epidemiologically, TSS was associated with the use of certain brands of hyperabsorbent tampons during menstruation. Public education and removal of such tampons from market has resulted in marked decrease in TSS incidence.

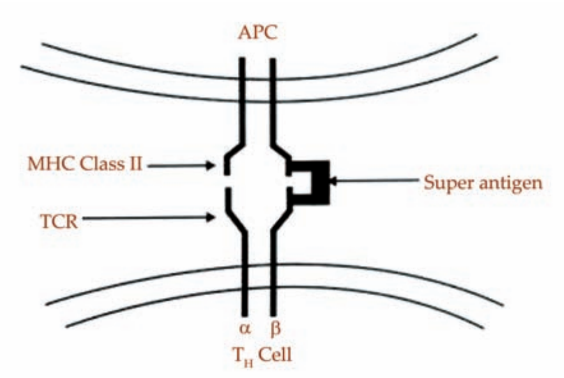
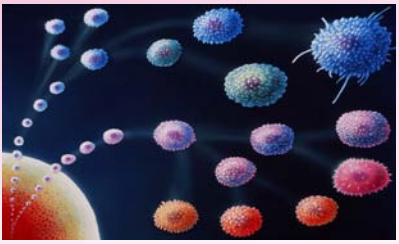


Figure 15: T_H cell activation by super antigen.

The super antigen is not processed and presented by the APC to the T_H cell. The super antigen lies outside the T_H cell and APC and binds these two ceils. Like a clamp, super antigen binds to the β chain of TCR and the MHC class II molecule on APC. This binding leads to the activation of T_H cell resulting in the release of large quantities of cytokines. The sudden release of large quantities of cytokines by numerous T_H cells is responsible for the clinical condition

Superantigens do not bind to the anti-genbinding site of Vβ chain of TCR, which is specific for a particular antigen only. But super antigens bind to β chain outside the variable region. Since super antigens bind outside the TCR-antigen binding cleft, any T_H cell expressing a particular Vβ sequence will be activated by a super- antigen. Hence a super antigen can bind to a significant percentage (about 5%) of the total T_H population in a host. Consequently massive amounts of cytokines are released leading to systemic toxicity.



CORE CONCEPT: LYMPHOCYTE FUNCTIONS

Lymphocytes are a component of complete blood count (CBC) tests that include a white blood cell differential, in which the levels of the major types of white blood cells are measured. Such tests are used to assist in the detection, diagnosis, and monitoring of various medical conditions.

Lymphocytes are significantly involved in maintaining the body's immune response mechanism through the development of humoral and cellular immunity.

- **Humoral immunity –** involves how B cells recognize and respond to antigens. These B cells transform themselves into plasma cells, which later produce antibodies to act upon any type of antigen.
- **Cellular immunity –** includes transplant rejection, which is a defense against intracellular organisms or against neoplasms. Cellular immunity uses T cells to get rid of antigens. The development and production of T cells depend on the thymus, which is why they are regarded as "T" cells.

Complete blood count (CBC) is a blood panel requested by a doctor or other medical professional that gives information about the cells in a patient's blood, such as the cell count for each cell type and the concentrations of various proteins and minerals

The majority of T cells can live longer with an average span of 4-5 years, but can also survive for as long as 20 years. They are also known for leaving and then re-entering the circulation many times during their entire life span.

It is difficult to differentiate B and T cells through blood film viewing since they are almost identical. However, they can be identified through immune cell markers.

Normal Count of White Blood Cells (WBCs) and Lymphocytes

Lymphocytes are one of the components counted in a **complete blood count** (CBC) test. A CBC test includes white blood cell differentials. The test is used to determine, diagnose, or monitor various medical conditions in a person. The lymphocyte count may vary for both adults and children. The lymphocyte level can be analyzed in the blood tests. The values can be seen in a CBC result.

An adult should have a normal WBC range of 4,500-11,000 WBC/mcL or 4.5-11.0 x 10^9 per liter, which is 1 percent of the total volume of blood in the body. When it comes to the lymphocyte count, it should be in the range of 800-5,000 lymphocytes/mcL or 0.8-5.0 x 10^9 per liter, which is 18-45 percent of the total white blood cells (WBCs) present in the body.

When the WBC count starts depleting to fewer than 2,500 WBC/mcL, it is considered as leukocytopenia (low WBC count). On the other hand, when it starts increasing to more than 30,000 WBC/mcL, then it is called as leukocytosis (high WBC count), which is again, not a good sign.

When it comes to the lymphocyte count, fewer than 800 lymphocytes/mcL is termed as lymphocytopenia (low lymphocyte count) and greater than 5,000 lymphocytes/mcL is regarded as lymphocytosis (high lymphocyte count) count in adults. In the case of children, the threshold varies per age. Children can go as high as 9,000 lymphocytes/mcL.

A percentage may also be given to you, which includes T-cells, B-cells, and natural killer (NK) cells. It should be between 15 and 40 percent of white blood cells.

It is important to remember that the normal values can vary from one laboratory to another.

Lymphocytosis (High Lymphocyte Count)

Causes

There are many causes that can lead to a variation in an abnormal lymphocyte count, which may become life-threatening if not treated on time. Infections, chronic inflammation caused by autoimmune disorders, and cancer of the lymphatic system or blood are some of the common reasons for a high lymphocyte count.

A high lymphocyte count is called as lymphocytosis. The lymphocyte count tends to increase when there is an infection in the body. In most cases, a high lymphocyte count shouldn't pose much of a threat since it is quite normal during and after suffering from an illness.

However, if there is a consistent high lymphocyte count, then it can be due to serious diseases that would need immediate medical attention. Such conditions include:

- **Hepatitis -** it causes inflammation of the liver. Symptoms include abdominal swelling, pain, fatigue, and yellowing of the skin and eyes.
- **Being HIV-positive or AIDS -** it damages the immune system. It can take many years for the person to develop AIDS from HIV. The body finds it difficult to fight against infections. The virus is transmitted by coming into contact with infected blood, having unprotected sex, from an infected mother to the baby, and through breastfeeding.
- **Viral infections -** can also increase the count of lymphocytes during viral infections such as influenza, measles, infectious mononucleosis (mono), and adenovirus infections.

- **Vasculitis -** in this condition, the blood vessels become inflamed, which causes changes in the walls of the blood vessels. The changes include scarring, weakening, thickening, or narrowing of the blood vessels. Vasculitis can be either acute or chronic.
- **Tuberculosis -** it is a bacterial infection that may cause lymphocytosis. It affects the lungs and may also affect the central nervous system, genitourinary system, lymphatic system, circulatory system, as well as the bones and joints. It is one of the most deadly infectious diseases.
- **Whooping cough -** respiratory illnesses like whooping cough and flu can increase the lymphocyte count. Symptoms of whooping cough include severe coughing, which is accompanied by a whooping noise. Medications of cough are usually not helpful. The doctor may prescribe antibiotics. Whooping cough can also be prevented through vaccinations. Flu causes respiratory distress, fever, and fatigue. In extreme cases, the condition can turn out to be deadly. Rest and increased intake of fluids are recommended. However, in some cases, antiviral medications are prescribed by the doctor.
- **Cytomegalovirus infection -** the infection spreads through infected body fluids such as breast milk, semen, blood, saliva, and urine. However, people are rarely affected by CMV. An individual's risk of contracting CMV increases if he or she has a weak immune system or is pregnant.
- **Acute lymphocytic leukemia (ALL) -** it is the cancer of the bone marrow and blood. In this disease, immature blood cells are produced quickly, hence, the term "acute". In ALL, the lymphocyte count goes high. Other symptoms include frequent nosebleeds or infections, fever, bone pain, bleeding gums, and swollen lymph nodes that lead to lumps.
- **Chronic lymphocytic leukemia (CLL) -** it is also a cancer of the bone marrow and blood. However, unlike ALL, this type of leukemiaslowly progresses, which is why it is referred to as "chronic". In this disease, the white blood cells are affected. Symptoms include enlarged but painless lymph nodes, weight loss, fatigue, and frequent infections.
- **Multiple myeloma -** it is a form of cancer that affects plasma cells. Some may experience symptoms such as a change in the RBC count, kidneys, the immune system, or the bones.
- **Crohn's disease -** it is an autoimmune disease, wherein the digestive tract becomes chronically inflammed. Symptoms include weight loss, **diarrhea**, and abdominal pain.

Symptoms

The person may or may not experience any symptoms. However, in some cases, the patient would have the following symptoms:

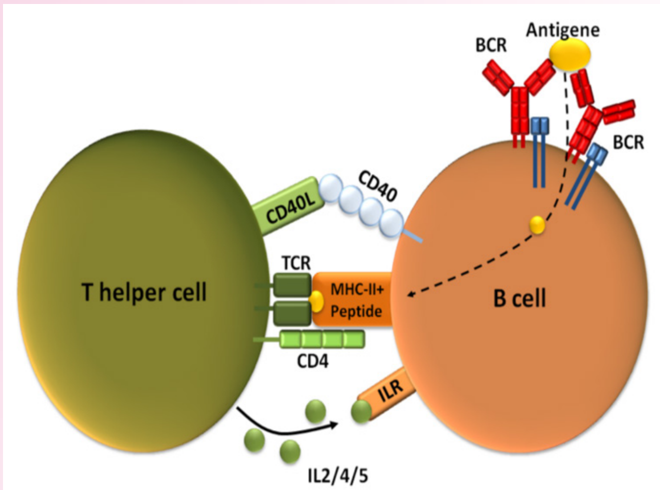
- Sore throat

- Sudden increase or decrease in body temperature along with fever and general exhaustion
- Fever
- Fatigue
- Night sweats
- Infection
- Loss of weight or appetite
- Enlargement of the spleen (may also cause stomach fullness)
- Irregular breathing
- Swollen lymph nodes in the neck, groin, or stomach
- External signs such as infection of the nose, relatively low energy levels, or redness of the oral mucous membrane
- Nausea, constipation, diarrhea, and vomiting
- Insomnia and general disorders of the nervous system
- Increased performance of the bone marrow tissues
- Symptoms of respiratory diseases

Diarrhea is the condition of having at least three loose or liquid bowel movements each day.

A CONCEPT OF T-INDEPENDENT ANTIGEN

A T-independent antigen refers to an antigen that can stimulate an immune response without the direct involvement of T cells. In the immune system, T cells and B cells play distinct roles in recognizing and responding to antigens. T cells are typically involved in cell-mediated immune responses, while B cells are responsible for producing antibodies in humoral immune responses.



Here's a breakdown of the concept of T-independent antigens and their significance:

1. B Cell Activation: B cells are a type of immune cell that can recognize antigens directly through their B cell receptors (BCRs) on the cell surface. When a B cell encounters an antigen that matches its BCR, it becomes activated.
2. T-Dependent vs. T-Independent Activation: B cell activation can occur in two main ways: T-dependent and T-independent. In T-dependent activation, B cells require help from T helper cells (CD4+ T cells) to become fully activated and generate a strong immune response. T-independent activation, on the other hand, occurs without the need for T cell assistance.
3. Types of T-Independent Antigens: T-independent antigens are often large molecules with repetitive epitopes, which are specific parts of antigens that B cells recognize. There are two types of T-independent antigens:

- Type 1 T-independent antigens: These antigens have a strong ability to cross-link multiple BCRs on a single B cell. This cross-linking provides sufficient signaling for B cell activation. Examples include bacterial lipopolysaccharides (LPS).

- Type 2 T-independent antigens: These antigens have repeating structures that can stimulate B cells but do not induce strong B cell activation on their own. They often require additional co-stimulatory signals or cytokines to achieve a robust immune response. Examples include polysaccharides found in bacterial capsules.

4. Response and Antibody Production: When B cells are activated by T-independent antigens, they can differentiate into plasma cells, which are specialized cells that produce antibodies. However, the antibody response to T-independent antigens may be limited compared to T-dependent responses. T-dependent responses typically result in the production of higher-affinity antibodies and the formation of immunological memory.
5. Role in Immune Defense: T-independent antigens are often associated with the initial immune response against certain bacterial infections. They can also play a role in the immune response to polysaccharide-rich pathogens. While T-independent responses are effective, they may not provide the same level of long-term protection as T-dependent responses.
6. Vaccine Design: Understanding T-independent antigens is important for vaccine design. Some vaccines include T-independent antigens to elicit rapid antibody responses, particularly in situations where a quick immune response is crucial.

SUMMARY

- Lymphocytes are a type of white blood cell that play a crucial role in the immune system. They are responsible for recognizing and attacking foreign substances, such as pathogens or antigens, and maintaining immune responses.
- Lymphocyte type of white blood cell (leukocyte) that is of fundamental importance in the immune system because lymphocytes are the cells that determine the specificity of the immune response to infectious microorganisms and other foreign substances.
- The bacteria are acted upon by the lysozymes in the macrophage and the microbial proteins are split into short antigenic peptides.
- B lymphocytes are produced from the hematopoietic stem cells in the bone marrow of adult. The mature B lymphocytes released into circulation from the bone marrow are in a resting or virgin state.
- The T cell receptor (TCR) of helper T cell binds to the MHC class II-antigen peptide complex on the B cell.
- Lag phase is the interval between the times of entry of antigen to the time of detection of antibodies against the antigen in blood.
- T lymphocytes develop from the hematopoietic stem cells in the bone marrow. The progenitor T cells released from the bone marrow into blood circulation are immature T cells.
- The thymic epithelial cells in the cortex of the thymus have long (about 25 μm) cytoplasmic processes and hence they are known as dendritic epithelial cells.
- A high lymphocyte count is called as lymphocytosis. The lymphocyte count tends to increase when there is an infection in the body.

MULTIPLE CHOICE QUESTIONS

1. **Where does the development of lymphocytes primarily occur?**
   a. Liver
   b. Spleen
   c. Bone marrow
   d. Kidneys
2. **Which type of stem cell gives rise to lymphocytes?**
   a. Erythropoietic stem cell
   b. Myeloid stem cell
   c. Hematopoietic stem cell
   d. Megakaryocytic stem cell
3. **B-cell development involves the rearrangement of genes encoding:**
   a. Immunoglobulins
   b. Interleukins
   c. Insulin receptors
   d Enzymes
4. **Positive and negative selection of T cells takes place in the:**
   a. Liver
   b. Spleen
   c. Thymus
   d. Lymph nodes
5. **Which stage of B-cell development is characterized by the presence of a pre-B cell receptor (pre-BCR)?**
   a. Early pro-B cell
   b. Late pro-B cell
   c. Immature B cell
   d. Mature B cell

Answers

1. (c) 2. (c) 3. (a) 4. (c) 5. (c)

REVIEW QUESTIONS

1. What are the functions of T cells and B cells?
2. Explain the activation of Cytotoxic T (Tc) Cells.
3. Explain the b cell development and function.
4. What does abnormal plasma cells mean?
5. Explain the difference between the primary and secondary immune responses.

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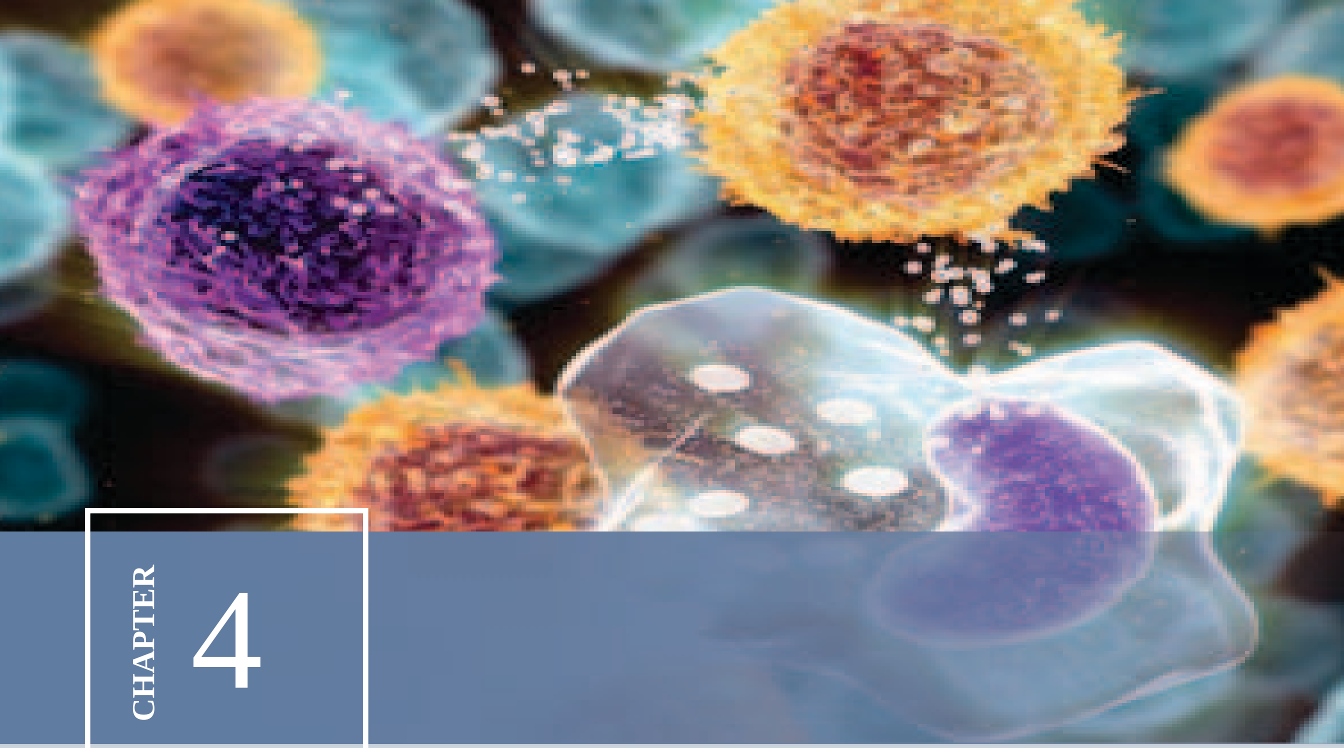
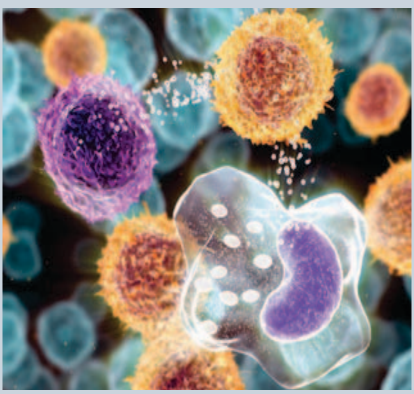


CHAPTER 4

CYTOKINES



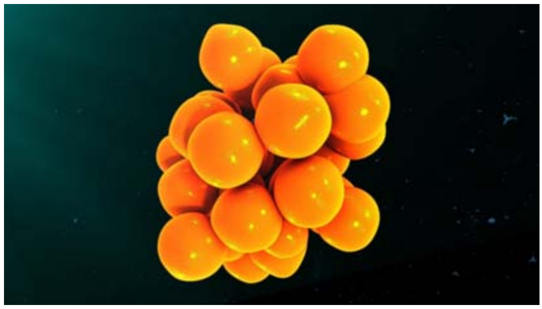
Cytokine is a general term used for small secreted proteins that are key modulators of inflammation. Cytokines are produced in response to invading pathogens to stimulate, recruit, and proliferate immune cells. Cytokines includes interleukins (IL), chemokines, interferons, and tumor necrosis factors (TNF).

After studying this chapter, you will understand the core concept of:

- Cytokines: meaning, characteristics and uses
- Mechanism of cell mediated immune response
- Mechanism of antibody medicated immune response

INTRODUCTION

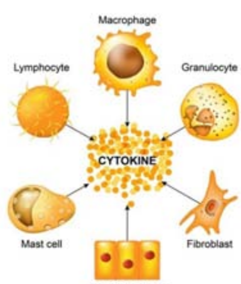
The term "cytokine" is derived from a combination of two Greek words - "cyto" meaning cell and "kinos" meaning movement. The most current terminology used to describe cytokines is "immunomodulating agents," or agents that modulate or alter the immune system response. Cytokines are small and membrane-bound protein-based cell signaling molecules that aid cell-to-cell communication in immune responses and stimulate the movement of cells towards sites of inflammation, infection, and trauma.



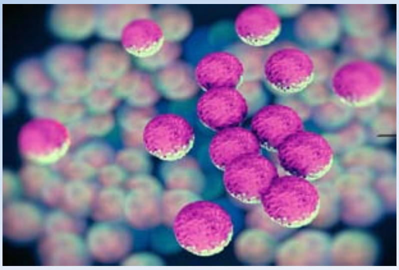
Cytokines transmit various signals for cell survival, proliferation, differentiation, and functional activity, circulate in picomolar concentrations, and may increase in magnitude almost a thousand-fold in response to an infection or inflammation.

Cytokines are crucial in controlling the growth and activity of the immune system cells and blood cells. They can be used to treat cancer and/or help prevent or manage chemotherapy side effects when injected, either subcutaneously, intramuscularly, or intravenously. They also help to boost anti-cancer activity by sending signals that can cause abnormal cell death and increase the longevity of normal cells.

Cytokines can exert systemic as well as local effects. A cytokine's actions may affect the same cell it was secreted from, other nearby cells or may act in a more endocrine manner and produce effects across the whole of the body, such as in the case of fever, for example. Cytokines have a large distribution of sources for their production, with nearly all cells that have a nucleus capable of producing interleukin 1 (IL-1), interleukin 6 (IL-6), and tumor necrosis factor-alpha (TNF-α), particularly endothelial cells, epithelial cells, and resident macrophages.



Cytokines are essential molecules that coordinate immune responses, influence cell behavior, and maintain tissue homeostasis. Their intricate network of interactions and functions makes them pivotal in both normal physiological processes and pathological conditions, and understanding their roles is crucial for advancing our knowledge of immunology and medical research.

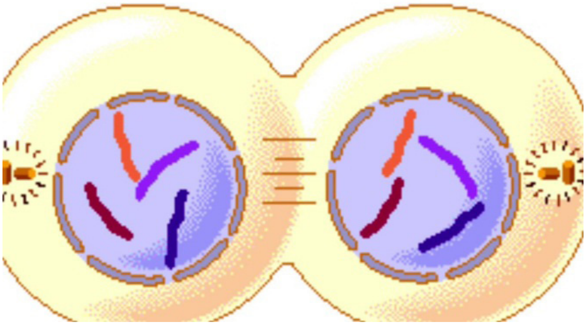


CORE CONCEPT: CYTOKINES: MEANING, CHARACTERISTICS AND USES

Cytokines are cell signaling molecules that aid cell to cell communication in immune responses and stimulate the movement of cells towards sites of inflammation, infection and trauma.

Cytokines are the proteins secreted by the cells of immune system that control the immune responses by interaction between the neighboring cells. In other words Cytokines are the signaling molecules like hormones and are the end products of interaction among immune cells. Cytokines like other signaling molecules tend to bind the specific receptors on the target cells but the structure of cytokines and their receptors is very different. Once the cytokines bind to their receptors, transcription factors are produced as a result of changes in the cell behavior by the process called as **signal transduction**. Transcription factors stimulate the selected genes for transcription which then secrete new cytokines or signaling molecules.

Signal transduction is the transmission of molecular signals from a cell's exterior to its interior.



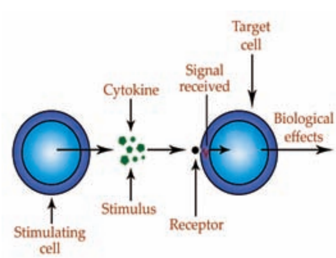
Cytokine, any of a group of small, short-lived proteins that are released by one cell to regulate the function of another cell, thereby serving as intercellular chemical messengers.

Cytokines effect changes in cellular behavior that are important in a number of physiological processes, including reproduction, growth and development, and injury repair. However, they are probably best known for the roles they play in the immune system's defense against disease-causing organisms. As part of the immune response, cytokines exert their influence over various white blood cells (leukocytes), including lymphocytes, granulocytes, monocytes, and macrophages. Cytokines produced by leukocytes are sometimes called interleukins, while those produced by lymphocytes may be referred to as lymphocytes. Cytokines typically are not stored within the cell but instead are synthesized "on demand," often in response to another cytokine. Once secreted, the cytokine binds to a specific protein molecule, called a receptor, on the surface of the target cell, an event that triggers a signaling cascade inside that cell. The signal ultimately reaches the nucleus, where the effects of the cytokine are manifested in changes in gene transcription and protein expression—i.e., genes, which code for proteins, may be turned on or off, and protein production may be stimulated or inhibited. Many different cytokines have been identified, and their activities, at least in part, are known. In some cases, one cytokine can interact with a variety of different cell types and elicit different responses from each cell. In other cases, different cytokines can elicit the same response from a cell. Some cytokines are known to induce or augment the activities of other cytokines, and sometimes their interactions occur through a cascading effect; however, the regulation of and cooperation between these various chemical signals still remain unclear in many cases. The classification of cytokines is problematic because so much remains to be learned about them, but they can be divided into five categories: interleukins, interferons, colony-stimulating factors, tumor necrosis factors, and growth factors.

Because cytokines are known to play a role in many disease processes, they have the potential to be used in treating a variety of disorders. For example, clinicians monitor levels of cytokines in the blood to assess the progression and activity of certain inflammatory states, such as septic shock. Measuring cytokine production also is useful in determining an individual's immunocompetence, or ability to fight off infection. Cytokines are used as therapeutic agents in treating persons with cancer and immunodeficiency disorders and those undergoing organ transplantation. Cytokines in conjunction with certain vaccines can enhance the vaccines' effectiveness.

Types of Cytokines

Depending on their function, cytokines are classified into the following types:



Chemokines

Chemokines are primarily involved in stimulating white blood cells, particularly leukocytes, in response to the detection of bacteria, fungi, and viruses. Their main job is to send a signal to draw the white blood cells at the infected area. Chemokines alert the immune system about the site of infection, which is targeted by the infection-fighting cells to kill the pathogens.

Colony-stimulating Factors (CSF)

CSFs assist the body to increase the WBC count. They signal the white blood cells to multiply in order to defend the body of infections. CSFs are of two types: granulocyte colony-stimulating factors (G-CSFs) that signal bone marrow to make more number of neutrophils, the most prominent type of WBCs, while granulocyte-macrophage colony-stimulating factors (GM-CSF) stimulate the bone marrow to increase production of two types of WBCs that include neutrophils and macrophages.

Interferons

Interferons do the job of activating natural killer (NK) cells, primarily to clear viral infections. NK cells are also expert at destroying tumor cells and considered as the primary mechanism to defend the immune system against cancer.

Interleukins

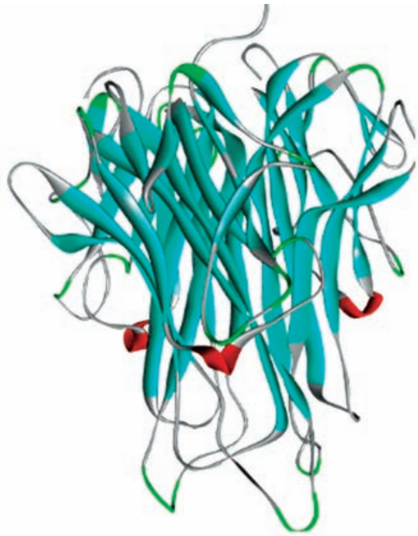
Interleukins play a key role in generating inflammatory response. They signal proliferation of T lymphocytes, which helps to contain the infection. Over 36 different interleukins have been identified that help regulate the immune response. Interleukins are named numerically starting from interleukin-1 (IL-1) to interleukin-36 (IL-36).

Transforming Growth Factors (TGFs)

They are a group of proteins that are primarily involved in generation of healthy cells. TGF-alpha, a type of transforming growth factor is secreted by macrophages brain cells. It helps in generation of epithelial tissue that lines the internal cavities of the body. TGF-beta, apart from cellular growth, are also involved in cellular differentiation, a process by which one cell type changes to another.

Tumor Necrosis Factor (TNF)

TNF are a group of proinflammatory cytokines secreted by macrophages. These cell signaling proteins play a key role in systemic inflammation. They also induce fever in response to an infection and are also involved in combating carcinogenesis and viral proliferation.

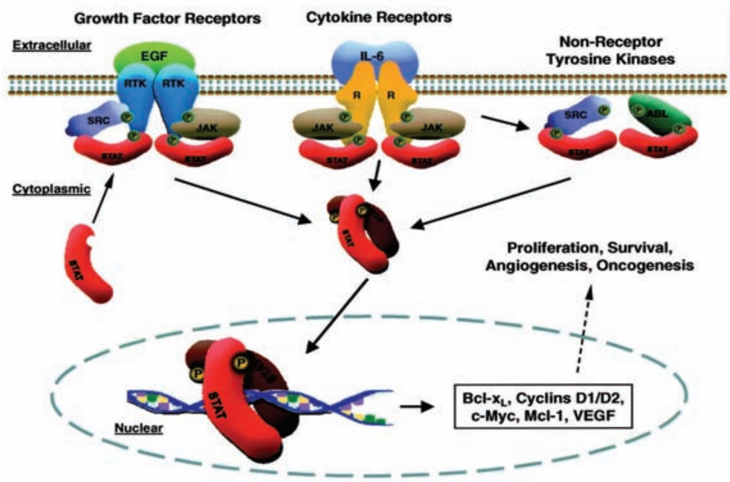


Depending on the type of cell involved in the production of cytokines, they are classified into two major types; monokines and lymphokines. The former are secreted only by monocytes, while the later are released by lymphocytes. Both monocytes and lymphocytes are a type of WBCs that help in regulating immunity. Cytokines can also be classified as pro-inflammatory and anti-inflammatory. While the pro-inflammatory ones induce inflammation in response to tissue injury, the function of anti-inflammatory cytokines {(IL-4), (IL-10), and (IL-13)} is exactly the opposite. Their purpose is to lower the inflammatory response. Chronic **inflammation**, such as in rheumatoid arthritis, is harmful to the health, hence to mitigate this inflammatory action, anti-inflammatory cytokines are released into the body.

Inflammation is part of the body's immune response. Infections, wounds, and any damage to tissue would not be able to heal without an inflammatory response.

Cytokines, Growth Factors and Hormones

Cytokines, growth factors (GF), and hormones are all chemical messengers that mediate intercellular communication. The regulation of cellular and nuclear functions by cytokines, growth factors, and peptide or protein hormones is initiated through the activation of cell surface receptors (Rc). All receptors have two main components: 1) a ligand-binding domain that ensures ligand specificity and 2) an effector domain that initiates the generation of the biological response upon ligand binding. The activated receptor may then interact with other cellular components to complete the signal transduction process. Many growth factors bind to receptors that are linked through G-proteins to membrane-bound phospholipase C (PLC). Activation of PLC cleaves phosphatidylinositol 4, 5-bisphosphate (PIP2) to form diacylglycerols (DAG) and D-myo-inositol-1, 4, 5-triphosphate (IP_3). IP_3 regulates intracellular $Ca^{2+}$by binding to the IP_3 receptor on the endoplasmic reticulum (ER) and stimulating Ca^{2+} release from the ER.



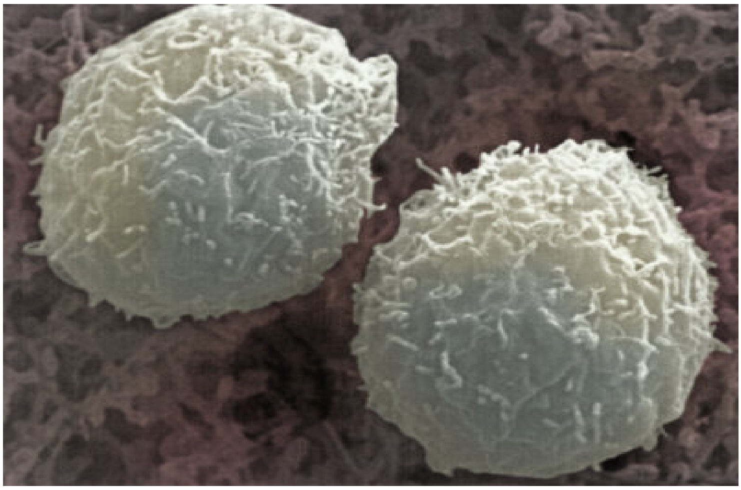
Free intracellular Ca^{2+} can bind to calmodulin, and this Ca^{2+}-calmodulin complex, in the presence of cyclic-AMP (cAMP), activates protein kinase A (PKA) by binding to the regulatory subunit of the enzyme. DAG binds to and activates protein kinase C (PKC). Other hormone receptors may be linked through G-proteins to adenylate cyclase (AC) instead of PLC.

Activation of AC increases the cellular levels of cAMP and, in the presence of the Ca^{2+}-calmodulin complex, will activate PKA. Additionally, some growth factor and cytokine receptors are protein tyrosine kinases (PTK) that are directly activated by ligand-receptor interaction.

Activation of any of the protein kinases, PKA, PKC, or PTK, catalyzes the phosphorylation of other proteins within the cell. Enzymes that are activated or inhibited by phosphorylation may mediate functional processes within the cell, while others may be one step in a protein kinase cascade that regulates nuclear events. Steroid hormones (i.e. estrogen, glucocorticoids), thyroid hormone, vitamin D_3, and retinoids are all small lipophilic molecules that easily penetrate both the cellular and nuclear membranes to enter the nucleus where they bind to their respective receptors that are ligand-dependent transcription factors. These ligand-receptor complexes bind to specific DNA response elements in the promoter region and regulate gene expression.

Differences between Enzymes and Hormones

Enzymes are the biological catalyst which speed up the rate of biochemical reactions without undergoing any changes. Hormones are molecules, usually a peptide (e.g.: insulin) or steroid (e.g.: estrogen) that is produced in one part of an organisms and triggers a specific cellular reactions in target tissues and organs some distance away.

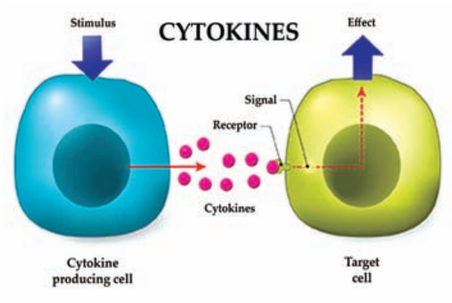


S.N.	Enzymes	Hormones
1.	Mostly enzymes perform reactions at the place of origin i.e. in cells where they are produced.	Hormones perform activity at some distance away from the site of origin.
2.	Enzymes are biological catalyst. They catalyze the biological reactions.	Hormones are not catalyst. They simply initiate biochemical reactions.
3.	All enzymes are generally proteins. There are some exceptions like ribozymes (RNA with catalytic activity).	The hormones may be polypeptides, terpenoids, steroids, phenolics compounds or amines.
4.	Enzymes are not translocate from one part to another part of cell.	Most of the hormones show polar translocation.
5.	As enzymes are catalyst, at the end of reaction they remain unchanged and can be reutilized.	As hormones are not catalyst, they participate in biological reaction and their chemical composition is changed and cannot be reutilized as such.
6.	They are macromolecules with higher molecular weight.	They have only low molecular weight.
7.	They are non-diffusible through cell membrane.	They are diffusible through cell membrane.
8.	They either act intracellularly or carried by some ducts to another site.	Generally carried by blood to a target organ.
9.	It increases the rate of metabolic physiological processes.	They may be either excitatory or inhibitory in their action.
10.	They catalyze reversible reactions.	Hormone controlled reactions are not reversible.
11.	Reaction rate increases with increase in their concentration up to a limit.	Deficiency or overproduction of hormone causes metabolic disorders or diseases.
12.	They act quickly.	Some hormones are quick acting, while some are slow acting with a lag period.
13.	They are not used in metabolic functions.	They are used up in metabolic functions.
14.	They cannot regulate morphogenesis.	Generally regulate morphogenesis, especially secondary sex character.
15.	Examples: • Oxidoreductases • Transferases • Hydrolases	Examples: • Insulin, • Glucagon, • T3, T4,

Nomenclature of Cytokines

Remembering cytokines and their functions can be challenging due to their diversity and complexity.

Most of the cytokines are named according to the Interleukin nomenclature subcommittee of the international union of immunological societies. Although the definition of cytokines is quite broad, but together they can be classified as lymphokines, interleukins, interferons, chemokines etc. depending on their function, cell of secretion, or target of action. The name interleukin was coined initially for those cytokines that mediate signaling between lymphocytes and leukocytes. In other words leukocytes were thought to be the principal target for interleukins but now this definition is no longer in function and the name interleukin is given to any new cytokine discovered routinely. Cytokines are produced mainly in response to viral infection or in response to immune attack. Interferons are the cytokines that interfere with viral RNA and protein synthesis to prevent viral replication.



Types of interferon – Interferons are mainly divided into two types.

- Type I Interferons --- Interferon-α (IFN- α) and interferon-β (IFN- β).
- Type II interferon --- Interferon-γ

Tumor necrosis factors (TNFs) can kill tumor cells and colony stimulating factors assist in regulation of stem cell activities. Chemokines play role in leukocyte recruitment and circulation.

Table 1: Different cytokines and their examples

Cytokine types	Examples
Lymphokines	MAF (Macrophage activating factor)
Interleukins	IL-7, IL-13, IL1-7
Interferons	IFN-α, IFN –β, IFN-γ, IFN-ω, IFN -
Tumor necrosis factors	TNF- α (Cachectin), TNF – β (lymphotoxin)
Colony stimulating factors	GSF (Granulocyte colony stimulating factor)
Polypeptide growth factors	EGF (Epidermal growth factor)
α-Chemokines	PF-4 (Platelet factor 4)
β-Chemokines	MCP-1 (Monocyte chemoattractant protein 1)
Transforming growth factors	TGF- α, TGF - β
Stress proteins	Heat shock proteins
RANTES	

General Characteristics of Cytokines

Different cytokines are structurally diverse, thus they share several properties:

i. **Cytokines bind to specific receptors on target cell membrane:**

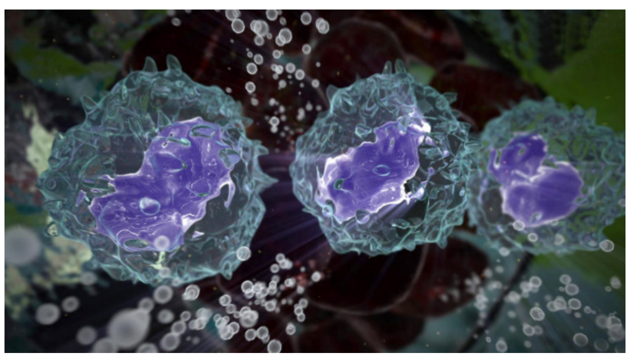
All cytokines initiate their action by binding to specific membrane receptors on target cells. Binding with receptors triggers the signal transduction pathways that ultimately alter gene expression in the target cells. Cytokines and their receptors show very high affinity for each other, with dissociation constant ranging from 10^{-10} to 10^{-12}M. As a consequence, only very little quantities of cytokine are required to elicit a biological effect.

Cytokines may be "good" when stimulating the immune system to fight a foreign pathogen or attack tumors. Other "good" cytokine effects include reduction of an immune response eg interferon β reduction of neuron inflammation in patients with multiple sclerosis.

ii. **Cytokine secretion is a brief, self- limited event:**

The synthesis of cytokines is transient, i.e., their synthesis is initiated by new gene **transcription** as a result of cellular activation. Such transcriptional activation is transient as the mRNA encoding most cytokines are unstable. Once synthesized, cytokines are rapidly secreted but never remain stored as preformed molecules.

Transcription is the first step of gene expression, in which a particular segment of DNA is copied into RNA by the enzyme RNA polymerase.



iii. **The action of cytokines is pleiotropic and redundant:**

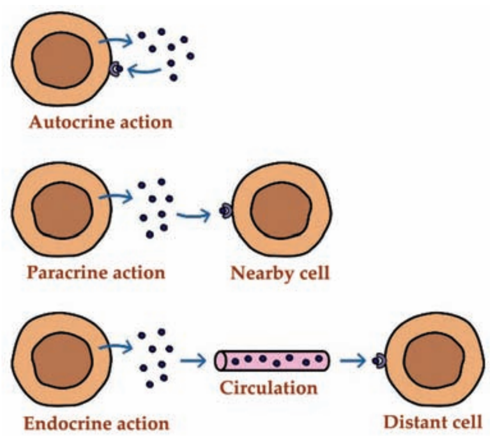
A given cytokine may show different biological effects on different target cells – it is the pleiotropic action of cytokines. On the other hand, redundancy refers to the property of multiple cytokines having the same functional effects. Because of this property, mutation of one cytokine gene may not have functional consequences, as other cytokine may compensate its effect.

iv. **Cytokines show synergy, antagonism and cascade induction:**

Cytokine synergism occurs when the combined effect of two cytokines on cellular activity is greater than that of the additive effects of the individual cytokine. In some cases, cytokines show antagonism that is, the effect of one cytokine inhibits or offset the effects of another cytokine. Cascade induction occurs when the action of one cytokine on a target cell induces that cell to produce one or more other cytokines which in turn may induce other target cells to produce other cytokines.

v. **Cytokines exert autocrine, paracrine and endocrine action:**

A particular cytokine may act on the same cell that secreted it, exerting autocrine action; it may act on a target cell in close proximity to the producer cell, exerting its paracrine action. In some cases, it may bind to target cells in distant part of the body, exerting its endocrine effect.



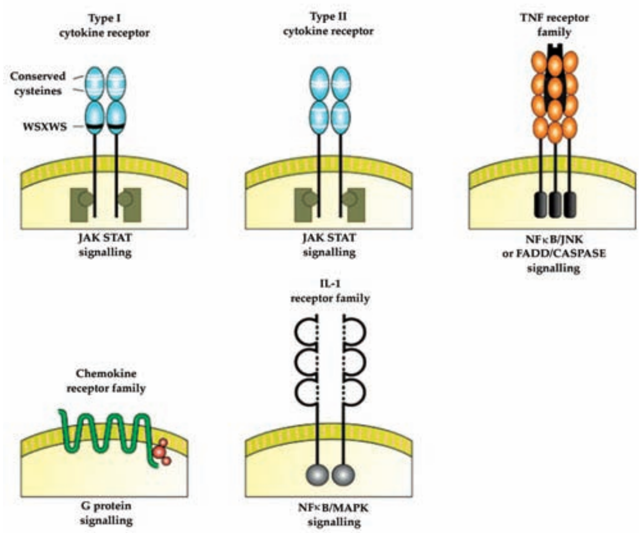
Cytokine Receptors

The cytokine receptors are trans membrane proteins with extracellular portions that bind with cytokines and cytoplasmic portions that are responsible for initiating intracellular signaling pathways. On the basis of homologies among the extracellular cytokine binding domains, cytokine receptors are classified into five families:

- Immunoglobulin super family receptors
- Class I cytokine receptor family (also known as haematoprotein receptor family)
- Class II cytokine receptor family (also known as the interferon receptor family)
- TNF-receptor family
- Chemokine receptor family.

Functional Categories of Cytokines

Cytokines may be classified into three main functional categories based on their principal biologic actions.



Cytokines acting as mediators and regulators of innate immunity

Different cytokines that regulate and mediate innate immunity are produced by mononuclear phagocytes in response to microbial agents such as lipopolysaccharide of bacteria, double stranded-RNA of virus etc. These cytokines act mainly on endothelial cells and leucocytes to stimulate early inflammatory reactions to microbes, and some functions to control these responses. Examples of such cytokines, their sources and biological actions are tabulated in Table 2.

Table 2: Cytokines of innate immunity

Cytokines of Innate Immunity

Cytokine	Principal cell source	Principal cellular targets & biologic effect
TNF	Macrophage, T cell	Endothelial cell: activation Neutrophil: activation Hypothalamus: fever Many cell types: apoptosis
IL-1	Macrophage, endothelial cell, epithelial cell	Endothelial cells: activation
IL-12	Macrophages Dendritic cells	NK cells & T cells: IFN-γ synthesis, increased cytolytic activity
IFN-γ	NK cells, T cells	Activation of macrophages
Type I IFNs	IFN-α: Macrophages IFN-β: Fibroblasts	All cells: antiviral activity, increased MHC I expression NK cells: activation

Cytokines acting as mediators and regulators of adaptive immunity

These cytokines are mainly produced by T lymphocytes in response to specific recognition of foreign antigens. Many of such cytokines regulate the growth and differentiation of various lymphocytes. Thus, they stimulate the activation phase of T-cell dependent immune responses. Other cytokines may recruit, activate and regulate specialized effector cells like mononuclear phagocytes, neutrophils and eosinophils to eliminate antigens in the effector phase of adaptive immune responses.

Cytokines acting as stimulators of haematopoiesis

Cytokines stimulating haemotopoiesis are produced by bone marrow stromal cells, leucocytes and other cells. These cause stimulation of growth and differentiation of immature leucocytes (Figure 1).

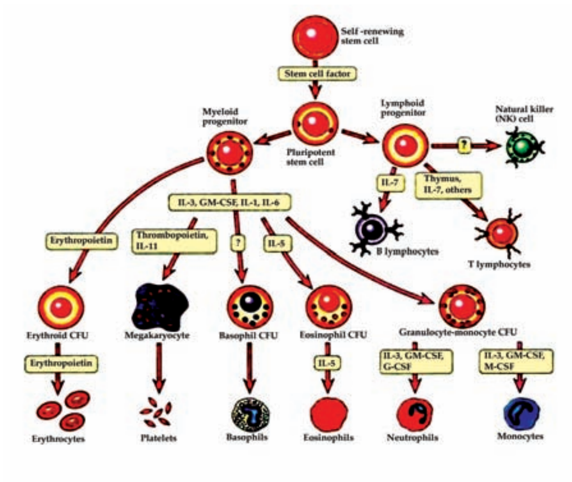


Figure 1: Roles of cytokines in haematopoiesis. Different cytokines stimulate the growth and maturation of different lineages of blood cells. CFU, colony-fonning unit; CSF, colony-stimulating factor.

Table 3: Haematopoietic cytokines

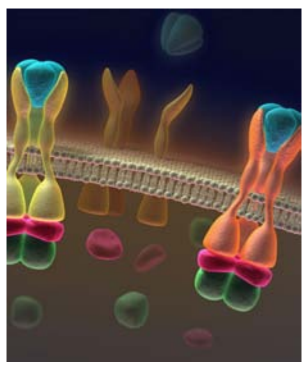
Haematopoietic growth factor	Sites of production	Main functions
Erythropoietin	Kidney, liver	Erythrocyte production
G-CSF	Endothelial cells, fibroblasts, macrophages	Neutrophil production
Thrombopoietin	Liver, kidney	Platelet production
M-CSF	Fibroblasts, endothelial cells, macrophages	Macrophage and osteoclast production
SCF/c-*kit* ligand	Bone marrow stromal cells, constitutively	Stem cell, progenitor cells survival/ division; mast cell differentiation
Flt-3 ligand	Fibroblasts, endothelial cells	Early progenitor cell expansion; pre-B cells
GM-CSF	T cells (T_H1 and T_H2), macrophages, mast cells	Macrophage, granulocyte production; dendritic cell maturation and activation
IL-3	T cells (T_H1 and T_H2), macrophages	Stem cells and myeloid progenitor cell growth; mast cells
IL-5	Activated helper T cells - T_H2 response only	Eosinophil production murine B-cell growth
IL-6	Activated T cells monocytes, fibroblasts, endothelial cells	Progenitor cell stimulation; platelet production; immunoglobulin production in B cells
IL-11	As above	As LIF
IL-7		T-cell survival

In general, the cytokines of innate and adaptive immunities are produced by different cell populations and act on different target cells. A comparative account of these cytokines is represented in Table 3. However, these distinctions are not absolute because same cytokines may be produced during both innate and adaptive immune reactions. Moreover, different cytokines produced during such reactions may have overlapping actions.

Cell is the smallest unit of life. Cells are often called the "building blocks of life". The study of cells is called cell biology.

Signal Transduction and Pathways

During the signal transduction the first step is the binding of cytokines to its receptors. Once the binding occurs, the receptor transmits a signal to the **cell** to alter its behavior. This phenomenon involving conversion of an extracellular signal into a series of intracellular events is called signal transduction. As Cell signaling needs to be precise and quick, enzyme cascade serve this function by the amplification of the responses rapidly. The important features of signal transduction include binding of cytokine to a cell surface receptor, stimulation of a transducer protein by the receptor, secondary activation of various enzymes, production of new transcription factors and ultimately gene stimulation leading to changed **cell** behavior.



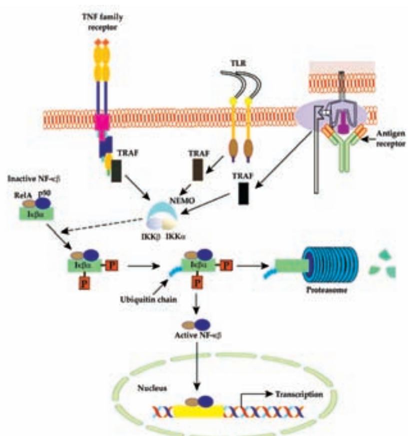
Although there are many pathways of signal transduction but three pathways are of utmost significance to the immune system.

- The NF-kB pathway
- The NF-AT pathway
- The JAK/STAT pathway

The NF-kβ pathway

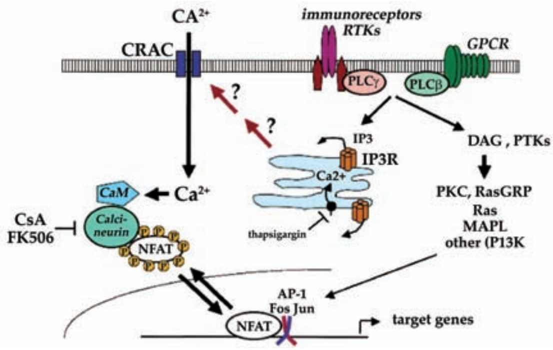
NF-kβ has its origins in the family of five transcription factors that play a crucial role in inflammation and immunity. There are more than 150 genes that get expressed on NF-kβ stimulation. There are three major pathways for NF-kβ activation. These are the classical pathway, alternative pathway and the third pathway that is stimulated by DNA damaging drugs and Ultraviolet light. The classical pathway involves proinflammatory signaling and gets activated by inflammatory cytokines, TLRs and antigen receptors (figure 2). IKK (Ikβ) complex is the central regulator of NF-kβ and all the signals induced by various stimuli focus on the same complex IKK. IKK complex is rich with kinase activity and on activation IKK complex phosphorylates Ikβ. Consequently Ikβ dissociates from the NF-kβ and the newly formed Ikβ becomes ubiquinated and is destroyed by proteasomes. Dissociation between Ikβ and NF-kβ liberates NF-kβ and it enters the nucleus to activate kβ motif containing promoters on DNA. This stimulates many genes and cytokines IL-1β, IL-6, IL-18, IL-33, TNF-α, GM-CSF, and IL-4. The second pathway or alternative pathway is required for lymphocyte development and is stimulated by a subset of TNF receptors. The third pathway does not require any IKK activation.

- IKK- Ikβ kinase
- Ikβ- Inhibitor of NF-kβ
- NEMO- NF-κB essential modulator
- TRAF- Tumor necrosis factor receptor-associated factor



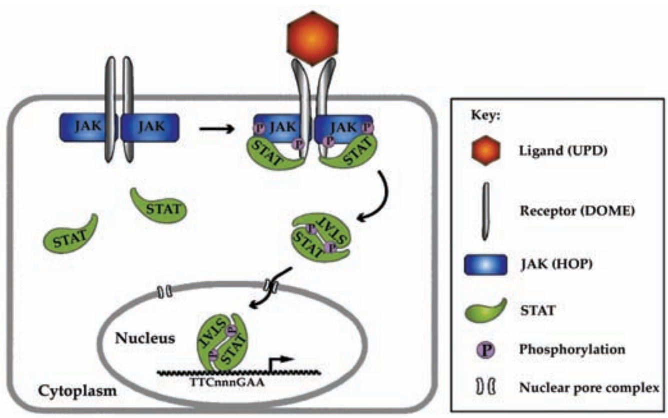
The NF-AT Pathway

The response generated to a T- cell is transmitted from the T cell receptor to a signal transducing complex called CD3 which is followed by clustering of CD3 chains in lipid rafts. Immunoreceptor tyrosine-based, activation motifs (ITAMs) are the specific amino acid sequences in the cytoplasmic domains of CD3 proteins. ITAMs later on become binding sites for several tyrosine kinases on activation. Zeta associated protein – 70 (ZAP-70) is one of the TKs and on phosphorylation can stimulate three signaling pathways. In the first pathway activation of calcineurin, a phosphatase is required and it ends up by generating NF-kβ and NF-AT. In the second pathway, ZAP-70 stimulates a protein kinase C which is required for activation of NF-kβ. The third pathway activates ras, a GTP-binding protein, fos and jun.



The JAK-STAT pathway

More than 40 cytokines use the JAK-STAT pathway. The stimulated JAK (Janus kinase) molecules phosphorylate tyrosine residues on one of several STAT proteins (signal transducers and activators of transcription). Phosphorylated STAT proteins then dimerize and dissociate from JAK and get shifted to the nucleus where they regulate the expression of target genes.



Cytokine-Related Diseases

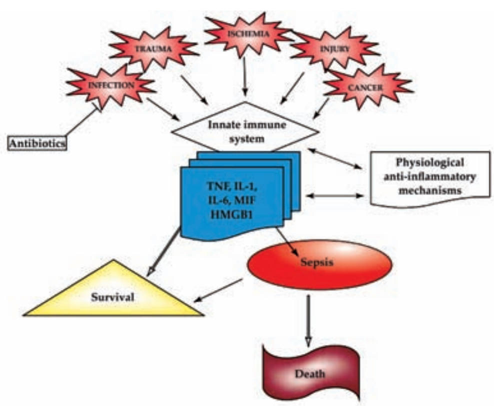
Defects in the complex regulatory networks controlling the expression of cytokines and cytokine receptors may lead to various abnormal conditions. Both over-expression and under-expression of cytokines and their receptors may result in several diseases.

- **Bacterial septic shock:** When bacterial cell-wall cytotoxins stimulate macrophages to overproduce IL-1 and TNF-α, bacterial septic shock may develop. The symptoms of bacterial septic shock, which is very often fatal include a drop in blood pressure, fever, diarrhoea and blood clotting in various organs. Higher levels of TNF-α were found in patients who died of meningitis than in those who recovered.
- **Bacterial toxic shock:** Many microorganisms are known to produce toxins that act as super-antigens which, unlike the antigens, are not internalized, processed and presented by antigen presenting cells. The super-antigens can

activate large number of T cells to produce excessive amount of cytokines. These elevated cytokines, in turn, can induce systemic reactions that include fever, widespread blood clotting and shock.

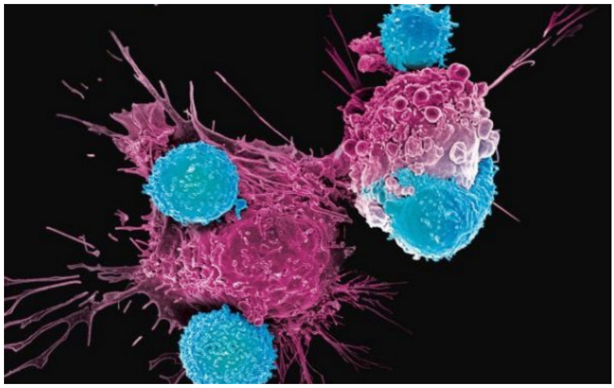
- **Lymphoid and myeloid cancers:** Development of some types of cancer is associated with the abnormal production of cytokines. For example, excessive high levels of IL-6 are secreted by cardiac myxoma cells, myeloma and plasma cytoma cells and cervical and bladder cancer cells. In myeloma cells, IL-6 acts in an autocrine manner to stimulate cell proliferation. Monoclonal antibodies to IL-6 if added to in vitro cultures of myeloma cells, their growth was found to be inhibited.
- **Chagas' disease:** Chagas' disease is caused by the protozoan parasite Trypanosoma cruzi and is characterized by severe immuno suppression. It has been found that in presence of T. cruzi, T cells show a dramatic reduction in the expression of a subunit of IL-2 receptors, resulting in their inactivation of T cells with many antigens. These, in turn, lead to **immunosuppression** in patients with Chagas' disease.

Immunosuppression is a reduction of the activation or efficacy of the immune system. Some portions of the immune system itself have immunosuppressive effects on other parts of the immune system, and immunosuppression may occur as an adverse reaction to treatment of other conditions.



Therapeutic Uses of Cytokines and their Receptors

From the above discussion it appeared that most cytokines are powerful mediators of innate and adaptive immunities. Now a days, a number of cytokines and soluble cytokine receptors have been purified and cloned. Many of such cytokines notably, Interferons and colony-stimulating factors, such as GM-CSF have proven to be therapeutically very useful.



Interferons

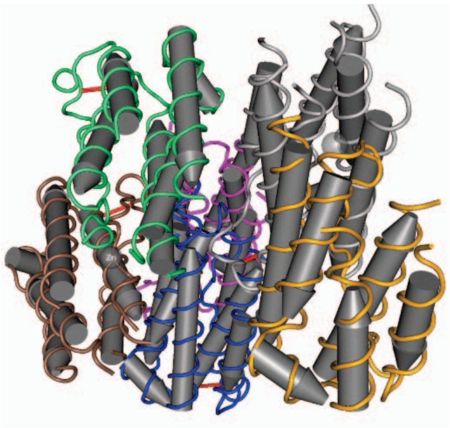
Different types of interferons have antiviral activity and various other effects including the capacity to induce cell differentiation, to inhibit proliferation by some cell types, to inhibit angiogenesis and to function in various immunoregulatory roles.

Some examples include:

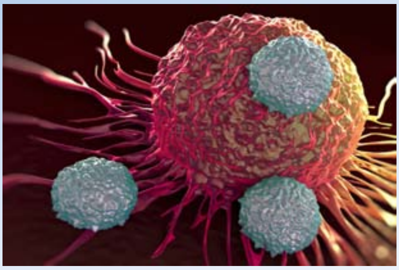
- INF-α (trade name Roferon and Intron- A) has been used for the treatment of hepatitis C, hepatitis B and many types of cancer. Chronic myelogenous leukemia, Kaposi's sarcoma, non-Hodgkins lymphoma, cutaneous T-cell lymphoma and multiple myeloma etc. respond well to treatment with IFN-α.
- INF-β. In the autoimmune neurologic disease multiple sclerosis (MS), a progressive neurologic dysfunction occurs. Treatment of that patients with IFN-p provides longer period of remission and reduces the severity of relapses.
- IFN-y has found to be effective for the treatment of a rare hereditary disease, chronic granulomatous disease (CGD) where the patients' phagocytic cells are seriously impaired to kill the ingested microbes. Therapy of CGD patients with IFN-y significantly reduces the incident of infections.

> Interferons are proteins that inhibit viruses from replicating. If a cell gets invaded by a virus, it releases interferons. This signals other cells to put up their shields so the virus does not spread.

The use of interferons in clinical practice is likely to expand, particularly as more is learned about their effects in combination with other therapeutic agents.



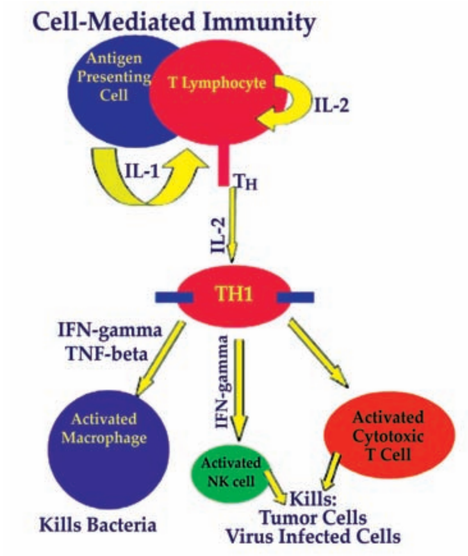
Although, a number of factors are likely to raise difficulties in adapting cytokines for safe and effective routine medical use. These include the need to maintain appropriate dose levels, to formulate effective combination of cytokines with other drugs or cytokines, and in some cases, to control cytokines side effects or toxicity. However, some cytokine-related therapies that either decrease or increase the immune response still offer promise of reducing graft rejection, treating certain cancers and immunodeficiency diseases, and reducing allergic reactions.



CORE CONCEPT: MECHANISM OF CELL MEDIATED IMMUNE RESPONSE

The immune response is the way in which your body recognizes and defends itself against bacteria, viruses, and other substances that are foreign and harmful.

The recruitment of the leukocyte and its activation is largely governed by the subset of CD4+ cells. Each leukocyte is specified for certain job and the cooperation of leukocyte to the lymphocytes is an important connection between innate and adaptive immune response. In general Th1 cells activate macrophages, Th2 activates eosinophils, and Th17 activates neutrophils. The Th1 cells activate the macrophage mediated phagocytosis, Th2 stimulates the production of IgE in response to a helminthic infection, and Th17 activates neutrophils to extracellular microbes including bacteria and fungi. Cytotoxic T lymphocytes (CTLs) kill the microbes which escape the phagocytosis and replicate in the cytoplasm.



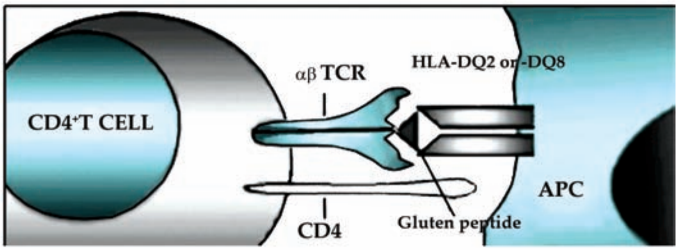
Migration of Lymphocyte at the Site of Infection

Majority of the leukocyte and lymphocyte are migrated towards the site of inflammation and infection, respectively in order to nullify the effect. Once the naïve lymphocyte encounters an antigen presented by the MHC molecule over the antigen presenting cells, they differentiate and migrate as an effector cells through the blood vessels to different parts of the body. The migration of lymphocyte is an interesting phenomenon orchestrated by different cytokines and specific receptors present over the surface of the lymphocytes. Naïve T lymphocytes express L-selectin and chemokine receptor CCR7 which help them to adhere to the lymph node and its surrounding tissues. After antigen stimulation the naïve T lymphocyte decrease the expression of L-selectin and CCR7 and increase the expression of sphingosine 1-phosphate receptors (S1PR1), which help their migration into the blood vessels. Th1 cells produces large amount of ligands that binds to E and P-selectin which recognize selectins secreted at the site of bacterial infection and trigger a strong innate immune response. Th2 cells express different chemokine receptors such as CCR3, CCR4 and CCR8 which recognize specific Chemokines secreted at the site of helminthic infection. Similarly, Th17 migration to the site of fungal and bacterial infection is modulated by CCR6 and other cytokines. Usually the subset of T lymphocytes is directed to the site of action with the help of different chemokines and their respective ligands. Similarly memory cells express different integrins that binds to the fibronectins over the extracellular matrices and help in their migration.

Functions of CD4+ Helper Cells

CD4+T cells recognize peptides presented on MHC class II molecules, which are found on antigen presenting cells (APCs). As a whole, they play a major role in instigating and shaping adaptive immune responses. The function of CD4+ cells are divided into four distinct stages.

- Leukocyte recruitment- T lymphocytes produces many cytokines that helps in the recruitment of neutrophils, eosinophils and monocytes at the site of infection or injury.
- Leukocyte activation- Leukocytes gets activated by the binding of CD40L on lymphocyte to the CD40 express over the antigen presenting cells.
- Expansion- The activation response gets expanded by the secretion of cytokines by the T lymphocytes.
- Control- The response of the immune system is controlled by some inhibitory cytokines such as IL-10.



Function of Th1 cells

Naïve T cells after encountering an MHC loaded peptide over the antigen presenting cells differentiates into Th1 cells. Th1 cells secretes interferon-γ (IFN-γ) which is a key cytokine involve in its functioning. IFN-γ acts on the activated macrophages and increases its phagocytosis efficiency and microbial killing. In addition IFN-γ also stimulates the B cells to secrete the IgG antibody that helps in opsonization and activation of complement pathway. Occasionally they also produce tumor necrosis factor that activate the neutrophils and promotes inflammation.

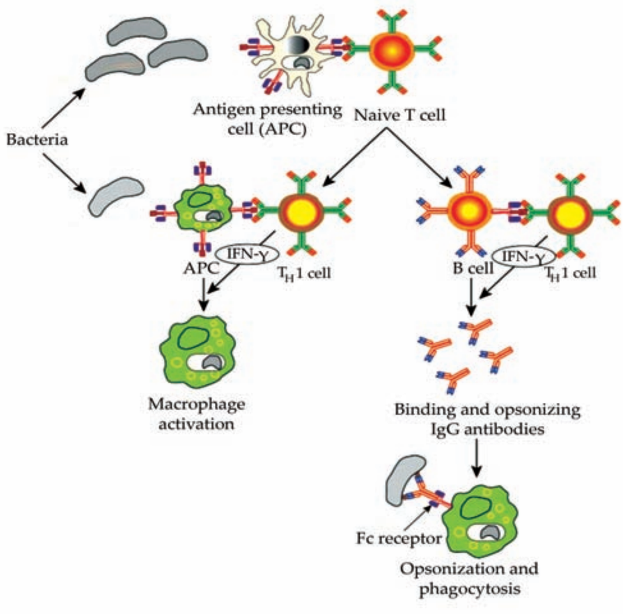


Figure 2: Functions of Th1 cells.

Function of Th2 cells

Naïve T cells after encountering an MHC loaded peptide (helminths) over the antigen presenting cells differentiates into Th2 cells. Th2 secretes IL-4, IL-13, and IL-5. IL-4 activates the B cells to secrete IgG antibody which can act on the parasites. In addition, IL-4 also stimulates the production of IgE antibody that helps in degranulation of mast cells. In terms of physiological aspect IL-4 along with IL-13 increases the mucus secretion and peristalsis of the intestinal tract. Moreover IL-4 and IL-13 act synergistically to activate macrophages which helps in fibrosis and tissue repair. Increase in the secretion and motion of the intestine is another way to combat the harmful effect of microbes entering into the digestive tract. IL-5 activates the eosinophil that helps in clearing the helminthic **infection**.

Infection is the invasion of an organism's body tissues by disease-causing agents, their multiplication, and the reaction of host tissues to the infectious agents and the toxins they produce.

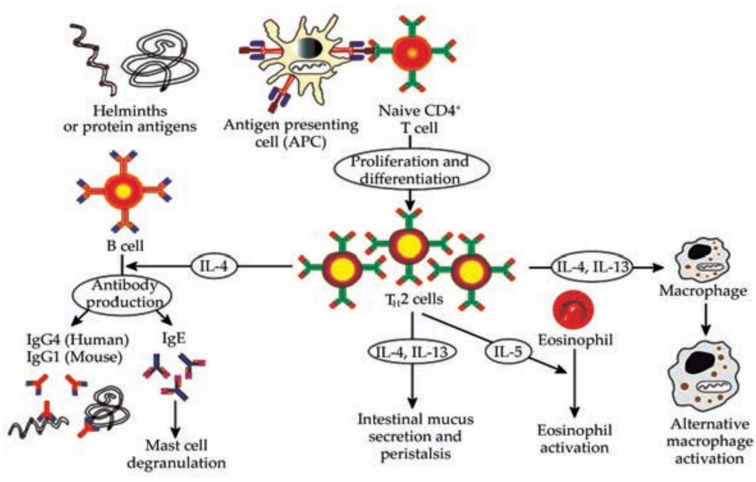


Figure 3: Functions of Th2 cells.

Function of Th17 Cells

After differentiation of naïve T cells into Th17 effector cells, they start secreting IL-17 and IL-22. IL-17 is a unique cytokine that do not have any homologue in the body. IL-17 stimulates the production of chemokines, TNF, IL-1, IL-6, and granulocyte-colony stimulating factors (G-CSF) which take part in inflammatory reactions. In addition, it is also involved in the production of defensins from the epithelial surface which in turn acts as a barrier for the invading pathogens. IL-22 is a relatively new interleukin and its function is not well established. However IL-22 is known to maintain the epithelial integrity, increase barrier function and involve in the repair activity following inflammation.

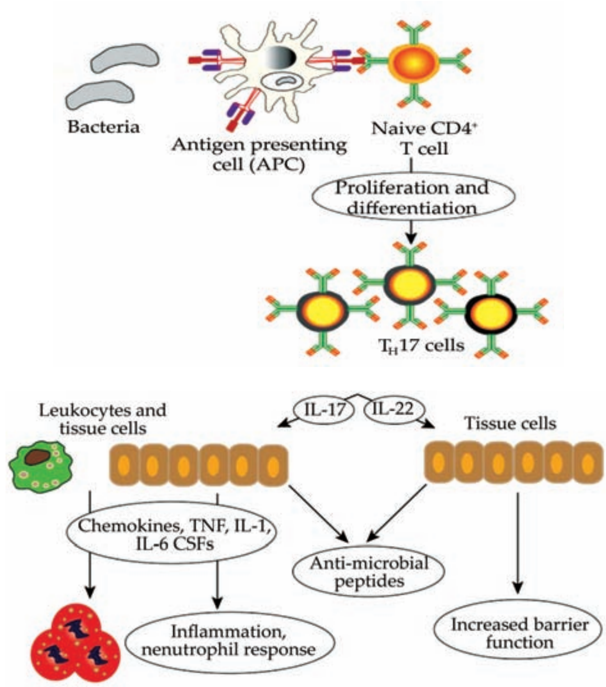
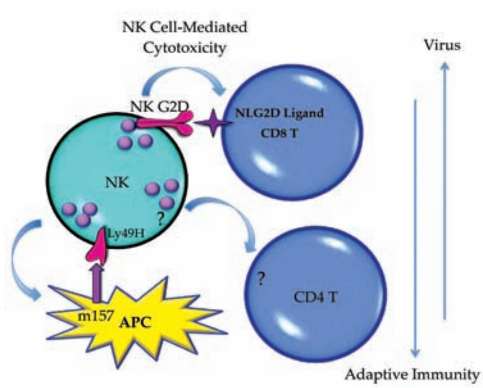


Figure 4: Functions of Th17 cells.

Function of CD8+ cytotoxic T lymphocyte

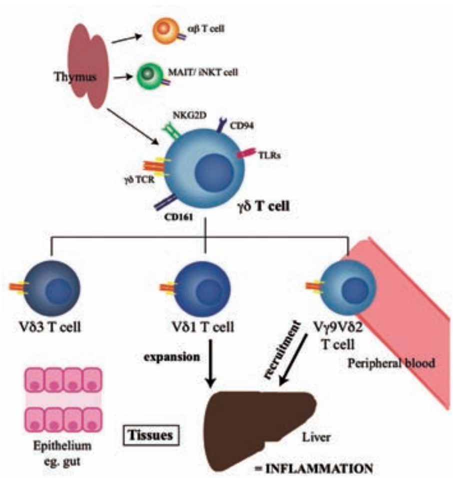
Cytotoxic T lymphocytes (CTLs) get activated after antigenic recognition over the MHC in the antigen presenting cells. The costimulatory molecules and their close association with other receptors results in the formation of immunological synapse (ICAM-1 to LFA1). Activated CTLs secrete cytotoxic granules such as granzyme and perforin. Perforin is homologue to complement protein-9 and is known to produce holes and perforation in the target cell. Perforin polymerize to form an aqueous pore in the target cells through which granzyme enters the target cells. Granzyme is serine proteases which is responsible for the cleavage and activation of caspase in the cells leading to the apoptosis of the target cells. In addition, activated CTLs also secrete granules such as serglycin which are known to assemble the granzyme and perforin over the surface of target cells.



CTLs also kill the target cells in granule independent fashion by Fas Ligand mediated pathway. The Fas ligand binds to the death receptors that are expressed in many cell types and activates the caspase mediated apoptosis pathway. The infection of intracellular microbes to host needs participation of CTLs because it will eliminate the reservoir of infection by killing the infected cell. Hepatitis B and C virus infection to a host activates the CTLs in order to kill the liver cells to eliminate the reservoir.

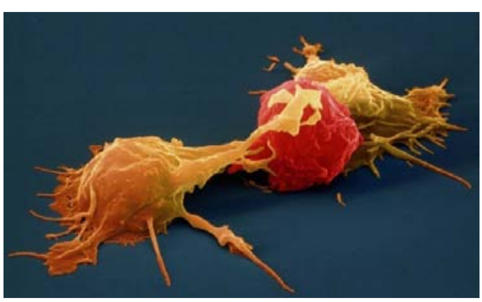
γδ T Cells

These are clonally distinct cells containing heterodimer of γ and δ chains, which are homologous to α and β chains in αβ T cells. Usually less than 5% of total circulating lymphocytes are composed of γδ T cell receptor. The γδ T cells present in the epithelial surface of bowel are called as intraepithelial lymphocytes. The γδ T cells do not recognize the MHC loaded peptide and are MHC independent. They require protein and non-proteinous antigen that do not require processing by MHC molecules.



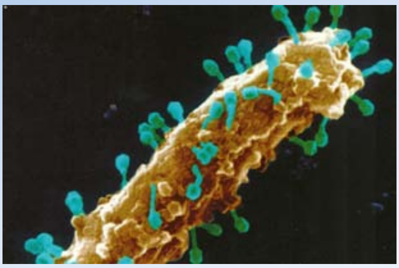
Natural Killer Cells

T cells population that express the CD56 marker over their surface are called natural killer or NK cells. NK cells recognizes the lipid bound to CD1 receptor (MHC class I homologue). NK cell secretes IL-4 and interferon-γ after activation and help in the production of antibody by B cells. NK cell plays an important role in the protection against mycobacterium infection.



Points to remember

- Cell mediated immunity is a type of adaptive immune response that is stimulated by microbes and can be transferred to naïve animal by T cells.
- Differentiation and migration of lymphocytes are orchestrated by different cytokines.
- Both CD4+ and CD8+ cells are responsible for cell mediated immunity
- CD4+ cells are comprised of Th1, Th2, and Th17 subsets and have different effector functions.
- CD8+ cells are converted into cytotoxic T lymphocyte upon differentiation and help in eradication of infection by killing the target cells.



CORE CONCEPT: MECHANISM OF ANTIBODY MEDICATED IMMUNE RESPONSE

The Immune response is the body's response caused by its immune system being activated by antigens. The immune response can include immunity to pathogenic microorganisms and its products, as well as autoimmunity to self-antigens, allergies, and graft rejections.

The humoral immunity is the other essential arm of adaptive immune response which is mainly governed by the antibodies. The humoral immune response is important against extracellular microbes, microbial toxins, fungi, and viruses. Any defect in the production of antibody leads to severe complications in immune compromised individuals. Current day **vaccine** strategies are mostly directed to increase the antibody production against specific pathogens.

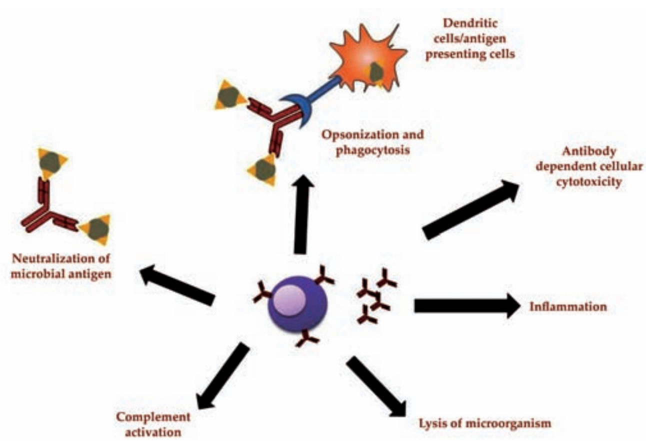


Figure 5: Overview of effector functions of antibody.

Neutralization of Microorganisms and their Toxins

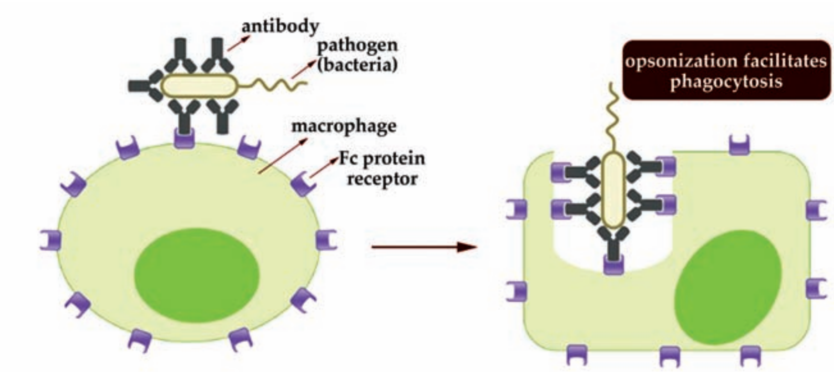
Most of the microorganisms attack and enter the host body after binding to a specific cell receptor present over the host cell. The specific structures present over the surface of bacteria, viruses, parasite or fungi are responsible for their attachment to the cell receptors. When the microorganism and its toxin enters or are produced inside the body the host immune system produces the antibody specific to bind their surface molecules.

Omega-3 and omega-6 fatty acids are regarded as precursors for the production of new cytokines, so it is advisable to consume food sources rich in these fatty acids.

The binding of specific antibodies to these molecules in the microbes and toxin makes it inaccessible to the host cell and hence neutralized. The host cell produces antibodies against the surface glycoprotein hemagglutinin in influenza virus infection in order to inhibit their binding to the sialic acid receptor present on the respiratory epithelium. Similarly, tetanus toxin binds to the cellular receptors over the neurotransmitter junction leading to paralysis and lockjaw condition. The antibody produced in Clostridium tetani infection neutralizes the toxins and inhibits its binding to receptors present in the motor end plates. Neutralization of the microbes and their toxins only requires antigen binding region of the antibody, i.e Fab or F $(ab)_2$ fragments. Most of the neutralizing antibodies in the blood are of IgG and IgA types in the mucosal surface.

Opsonization and Phagocytosis

Coating of microbes by the IgG type antibody helps in their phagocytosis by macrophages and neutrophils, the phenomenon is called opsonization. The Fc receptor of the IgG containing microbes binds to the neutrophils and participates in their intracellular degradation and killing. In addition to IgG molecules, complement protein C3b also helps in the opsonization and phagocytosis of the microbes by binding to leukocytes.



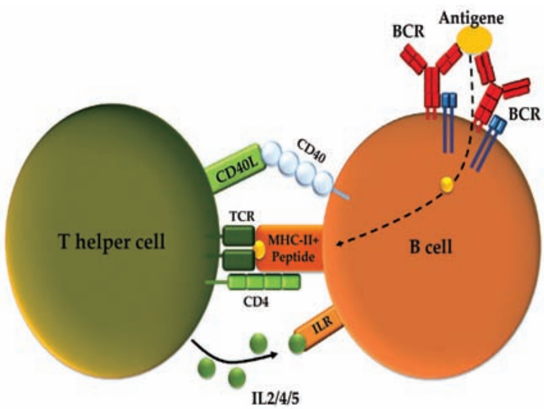
The substance that helps in opsonization and phagocytosis including antibodies and complements are called as opsonins.

Table 4: Functions of different antibody

Types	Function
IgG	Neutralization of toxins and microbes. Opsonization of antigens Activation of classical pathway of complement system Antibody dependent cell mediated cytotoxicity Transfer through placenta to neonates Negative feedback to B cell activation
IgA	Neutralization over the mucosal surface Activation of lectin and alternate pathway of complement system
IgM	Activation of classical pathway of complement system Antigen receptor on the naïve B cell

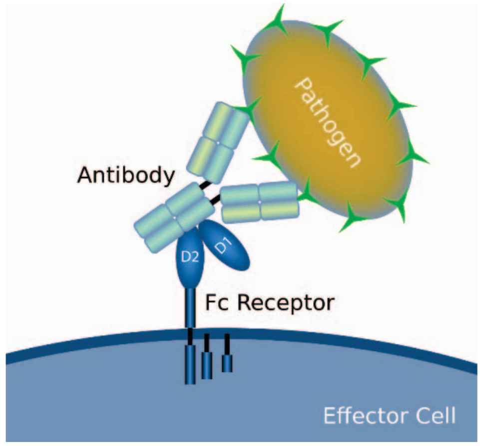
Antibody Dependent Cell-mediated Cytotoxicity

Many subclasses of IgG antibodies bind to infected cells and are recognized by the Fc receptors of NK cell. NK cell contains Fc receptor called FcγRIII (CD16) that binds to the antibody coated infected cells. Many therapeutic monoclonal antibodies against tumor cells are working on the same principle by coating the tumor cells which facilitates their killing by NK cells.



Antibody Dependent Killing of Helminths

Helminths are larger in size and cannot be processed by phagocytic cells of the immune system. Antibodies like IgE, IgG, and IgA coats the surface of helminths which can bind to the Fc receptor present over the eosinophils. Eosinophils contain granules in their cytoplasm which are responsible for the secretion of microbicidal major basic proteins. Binding of the antibody coated helminths to Fc receptor degranulates the eosinophils and secretion of major basic protein which is responsible for their killing. Mast cells also take part in the clearance of helminths from the body. Mast cell releases many cytokines and chemokines which attracts and helps in the degranulation of eosinophils and in turns secretion of major basic protein.

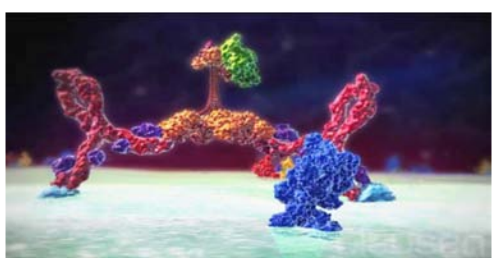


The Complement System

The complement system consists of different serum proteins that works through different pathways and interacts with each other in order to kill the invading microbes. The complement system is activated by the microbial antigen and the antibody produced against them by the host immune system. The proteins of the complement system attaches sequentially over the surface of microbes and help in their lysis. The complement system comprises of classical, alternate, and lectin pathways. The classical pathway is activated by the antibody binding with an antigen, alternate pathway activates in absence of antibody, and lectin pathway activates in response of lectin that binds to mannose.

Degranulation is a cellular process that releases antimicrobial cytotoxic or other molecules from secretory vesicles called granules found inside some cells. It is used by several different cells involved in the immune system, including granulocytes and mast cells.

The final step in all these three pathways leads to the formation of membrane attack complex. These complexes are around 100 Å in diameter and forms, a channel that allows the free flow of water and ion into the microbial cells. The movement of water and ions inside the cells increases the osmotic pressure of the cells and its rupture. The pore formed by complement proteins are similar to what we observe in case of NK cells, where instead of water and ions, perforin and granzyme enters and kills the cell.



Function of Complement System

- Promotes phagocytosis of the microbes by opsonization.
- Stimulates inflammation by **degranulation** of mast cells.
- Lysis of microbes by forming a membrane attack complex.
- Solubilization of those complexes that are unwanted for the body and helps in their clearance.

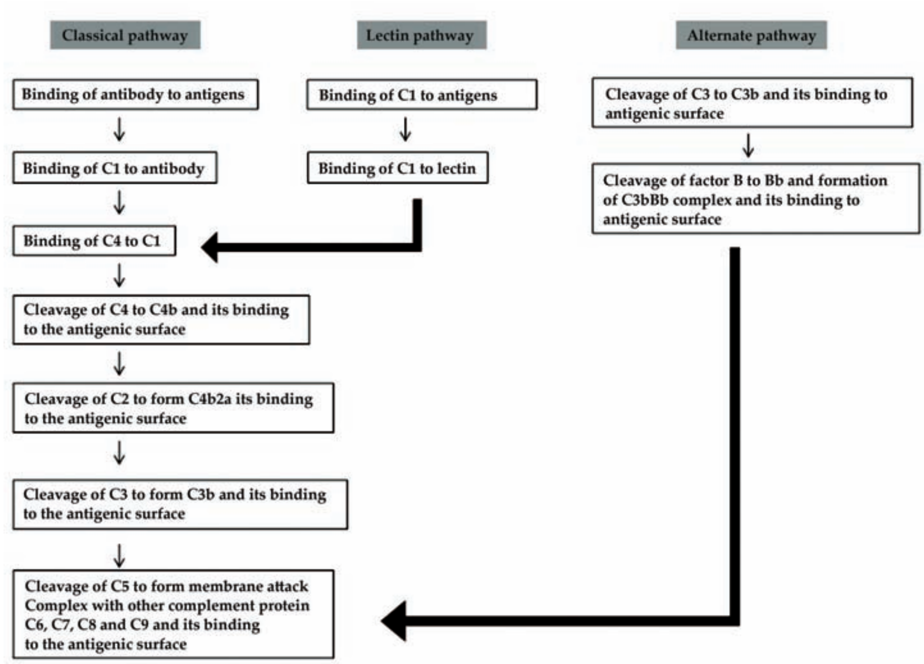


Figure 6: Different complement pathways.

Role of Complement in Disease

The complement system plays a critical role in inflammation and defense against some bacterial infections. Complement may also be activated during reactions against incompatible blood transfusions, and during the damaging immune responses that accompany autoimmune disease. Deficiencies of individual complement components or inhibitors of the system can lead to a variety of diseases (Table 5), which gives some indication of their role in protection against disease.

Table 5: Diseases associated with complement deficiencies

Complement Deficiency	Disease
C3 and Factor B	Severe bacterial infections
C3b-INA, C6 and C8	Severe Neisseria infections
Deficiencies of early C components C1, C4, C2.	Systemic lupus erythematous (SLE), glomerulonephritis and polymyositis
C1-inhibitor	Hereditary angioedema

Neonatal Immunity

The immune system of neonates is relatively under-developed and relies on maternal antibodies transmitted in utero to modify and control the severity of neonatal diseases. These antibodies are of the class IgG, which can be transported across the placenta. This is an active, selective process that involves intracellular pathways and is specifically mediated by the neonatal Fc receptor FcRn. Fc receptors are expressed by the trophoblastic cell layer which surrounds the developing foetus. Once the receptor has bound IgG it is transported to the foetal circulation. The body of neonate is sterile and passive transfer of antibody takes place by placenta before birth and through milk immediately after birth. The passive transfer of antibodies to neonates is important in providing immunity during the early phase of life when the immune system is not fully functional. Transfer of antibody through the gut in early life phase is called transcytosis. The maternal IgG is transferred through placenta and IgG and IgA through milk.

Transfer of antibodies begins at 17 weeks of gestation and increases as gestation advances. Maximal transfer occurs from the third trimester onwards. By 33 weeks of gestation maternal and foetal IgG are at equivalent levels and by 40 weeks of gestation the foetal IgG concentration is higher than maternal IgG. Factors that may reduce transplacental antibody transfer include:

- pathogen insult that affect placental integrity, such as HIV or Malaria;
- very high concentrations of maternal IgG which cause antibodies to compete for the receptor and
- The age of the foetus.

Premature infants born before 28 weeks gestation have very low levels of maternal antibodies and prematurity is also responsible for impaired cellular immunity. Having impaired humoral and cellular responses results in a "physiological immunodeficiency" which leaves premature infants vulnerable to both viral and bacterial pathogens.

Features of the Neonatal Immune System

- ***Pattern recognition***: Neonatal responses to pathogen associated molecular patterns (PAMPs) are reduced compared to adults. However pattern recognition receptor (PRR)-expression levels are similar. It appears that the molecules that transduce the signal (for example interferon response factor 3 – IRF3) have reduced function. This leads to reduced production of key inflammatory mediators, for example interleukin-12 (IL-12) and interferon-α (IFNα). PRR function increases over time, and the increase in capacity occurs in proportion to time since birth rather than 'gestational' age, suggesting that it is controlled by exposure to the environment and removal of maternal influence.

- ***T-cell response*:** There is a well-documented skewing of the neonatal T-cell response towards T helper 2 (Th2). This is associated with the reduction in IL-12 and IFNα production by neonatal antigen-presenting cells (APC). This may have an effect on the immune response to antigens seen in early life – possibly inducing an allergic type response.
- ***B-cell response*:** Antibody production in early life is reduced. In particular the antibody response to polysaccharide antigens is reduced. This is a particular problem with regards to bacterial infections, to which newborn children are highly susceptible. This failure to produce antibody is associated with several factors including reduced T cell help, fewer follicular dendritic cells and germinal centers and reduced signaling through the CD40 ligand family members.

Downstream effects of the neonatal immune response

The immaturity of the neonatal immune response has an effect on three important areas:

- *Increased susceptibility to infection*. The recognition of infectious agents is reduced in early life and it is therefore easier for a pathogen to invade the host. Neonates are also less experienced so have no immune memory against infection.
- *Decreased vaccine efficacy*. In a similar fashion to infection, reduced recognition of vaccine antigens as foreign means that induction of protective memory responses to vaccines are reduced. There is also an effect of maternally derived antibody which may mask key epitopes of the vaccine.
- *Development of asthma and allergy*. The Th2 skewing of the T cell response is hypothesized to drive the development of allergic responses to antigen in early life.

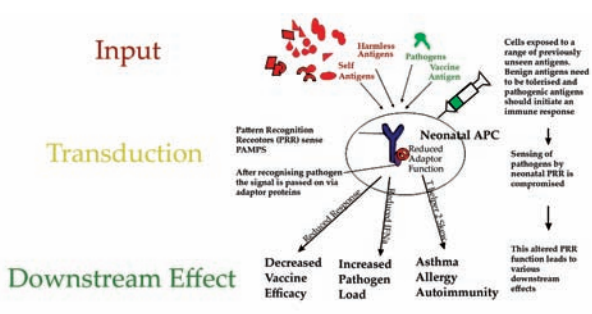
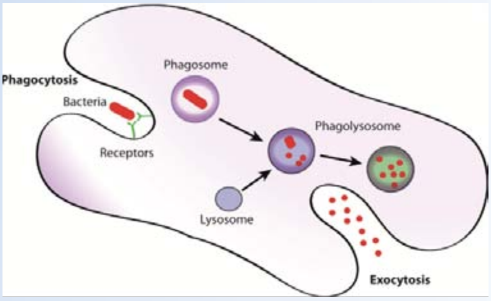


Figure 7: Downstream effects of reduced antigen presenting cell function in early life.

The neonatal immune system is exposed to a large number of previously unseen antigens. The majority of these antigens are benign and therefore should be tolerated but some are dangerous and therefore should induce an immune response. The neonatal antigen presenting cells (APC) have reduced recognition of antigens regardless of source (self, benign, pathogenic, vaccine). This is mediated at the level of pattern recognition receptors (PRR) or their adaptor molecules. This leads to reduced immune responses to these antigens and has an impact on vaccine efficacy, disease susceptibility and, possibly by skewing responses to Th2, the development of asthma and allergy.

A CONCEPT OF PHAGOCYTOSIS

Phagocytosis is a vital process by which certain cells in the immune system, known as phagocytes, engulf and digest foreign particles, microorganisms, cellular debris, and other materials. It is a key mechanism of the innate immune response, which serves as the body's first line of defense against infections and foreign invaders.



Phagocytosis plays a crucial role in identifying and eliminating harmful substances to maintain health and protect the body from pathogens. Here's an in-depth explanation of the concept of phagocytosis:

1. Phagocytes: Phagocytes are specialized immune cells that are capable of carrying out phagocytosis. The two primary types of phagocytes are neutrophils and macrophages. Neutrophils are the most abundant type of white blood cells and are usually the first to arrive at the site of infection. Macrophages are larger phagocytic cells that are found in tissues throughout the body.
2. Recognition and Attachment: The process of phagocytosis begins with the recognition of foreign particles or microorganisms by receptors on the surface of phagocytes. These receptors can recognize specific molecular patterns associated with pathogens, such as bacterial cell wall components. Once recognized, the phagocyte attaches to the target through receptor-ligand interactions.
3. Engulfment: After attachment, the phagocyte extends pseudopodia (temporary projections of the cell membrane) to surround the target particle. The pseudopodia fuse together, enclosing the particle within a vesicle called a phagosome. The phagosome is now internalized within the phagocyte.
4. Phagosome Maturation: The phagosome undergoes a series of maturation stages as it moves deeper into the phagocyte's cytoplasm. It fuses with lysosomes, which contain enzymes capable of breaking down the ingested material.
5. Digestion: Within the phagolysosome (the fused phagosome and lysosome), the engulfed material is exposed to a cocktail of

digestive enzymes. These enzymes break down the particle into smaller fragments, which can then be utilized or expelled from the cell.

6. Elimination of Debris: Once digestion is complete, the residual waste products are either excreted from the cell or stored within lysosomes until they can be safely expelled.
7. Inflammatory Response: Phagocytosis is often associated with inflammation. Phagocytes release chemical signals, such as cytokines, that attract other immune cells to the site of infection or injury. This results in increased blood flow, recruitment of immune cells, and activation of the immune response.
8. Antigen Presentation: Macrophages also play a role in antigen presentation, a process by which they display fragments of ingested pathogens (antigens) on their cell surfaces. This presentation helps activate specific immune responses by T cells.
9. Role in Immune Defense: Phagocytosis is a crucial defense mechanism against various pathogens, including bacteria, fungi, viruses, and cellular debris. It contributes to the rapid removal of potential threats before they can establish infections.
10. Professional Phagocytes: Neutrophils and macrophages are professional phagocytes, meaning they are specifically adapted for phagocytosis and constitute a significant part of the immune system's initial response to infections.

In summary, phagocytosis is a fundamental process of the innate immune system that enables phagocytes to recognize, engulf, and digest foreign particles and microorganisms. It is a critical component of immune defense, helping to prevent infections and maintain the body's health and homeostasis.

SUMMARY

- Cytokines are crucial in controlling the growth and activity of the immune system cells and blood cells. They can be used to treat cancer and/or help prevent or manage chemotherapy side effects when injected, either subcutaneously, intramuscularly, or intravenously.
- Cytokines are the proteins secreted by the cells of immune system that control the immune responses by interaction between the neighboring cells.
- Chemokines are primarily involved in stimulating white blood cells, particularly leukocytes, in response to the detection of bacteria, fungi, and viruses.
- Cytokine synergism occurs when the combined effect of two cytokines on cellular activity is greater than that of the additive effects of the individual cytokine.
- The cytokine receptors are trans membrane proteins with extracellular portions that bind with cytokines and cytoplasmic portions that are responsible for initiating intracellular signaling pathways.
- The humoral immunity is the other essential arm of adaptive immune response which is mainly governed by the antibodies.
- Coating of microbes by the IgG type antibody helps in their phagocytosis by macrophages and neutrophils, the phenomenon is called opsonization.
- The immune system of neonates is relatively under-developed and relies on maternal antibodies transmitted in utero to modify and control the severity of neonatal diseases.

MULTIPLE CHOICE QUESTIONS

1. **Which of the following is a pro-inflammatory cytokine?**
    a. IL-10
    b. IFN-alpha
    c. IL-1
    d. TGF-beta
2. **Which cytokine is known for its role in promoting T cell proliferation and activation?**
    a. IL-6
    b. IL-2
    c. TNF-alpha
    d. IFN-gamma
3. **Interferons are cytokines primarily involved in:**
    a. Regulating hematopoiesis
    b. Promoting cell growth
    c. Enhancing phagocytosis
    d. Antiviral responses
4. **Chemokines are cytokines that:**
    a. Enhance antibody production
    b. Promote cell division
    c. Attract immune cells to specific locations
    d. Regulate inflammation
5. **Which cytokine is important for regulating both humoral and cell-mediated immune responses?**
    a. IL-10
    b. IL-4
    c. IL-12
    d. IL-17

Answers

1. (c) 2. (b) 3. (d) 4. (c) 5. (c)

REVIEW QUESTIONS

1. What do you understand by cytokines?
2. Define the different types of cytokines.
3. Differentiate between enzymes and hormones.
4. Write the general characteristics of cytokines.
5. Describe the functional categories of cytokines.
6. Explain the signal transduction and pathways.
7. Elaborate the therapeutic uses of cytokines and their receptors.
8. What is the neutralization of microorganisms and their toxins?
9. Discuss about complement system.
10. Write the downstream effects of the neonatal immune response.

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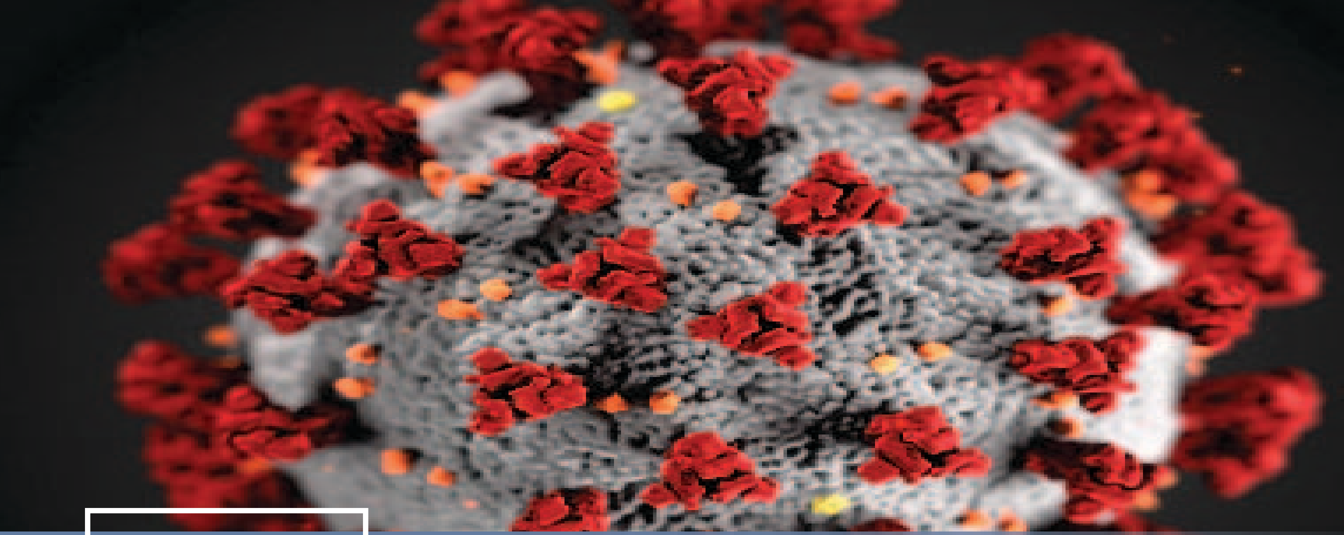
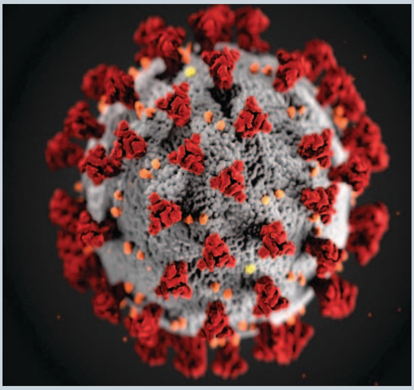


CHAPTER 5

IMMUNITY TO PATHOGENS



When pathogens do enter the body, the innate immune system responds with inflammation, pathogen engulfment, and secretion of immune factors and proteins.

It has been known for some time that the response to infection by the adaptive immune system, which includes T and B cells, includes a mix of immune boosters and immune suppressors that rein in the immune system, ensuring it doesn’t overreact and cause more damage than the invading pathogen.

After studying this chapter, you will understand the core concept of:

- Immunity to Microbes
- Transplant Immunology
- Tumor Immunology

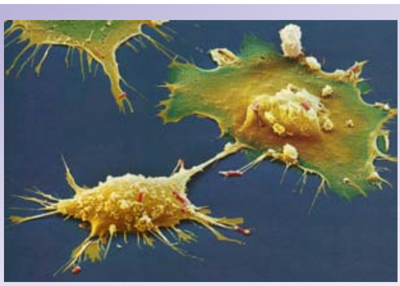
INTRODUCTION

The human body is constantly exposed to a wide array of microorganisms, or pathogens, that have the potential to cause diseases. To combat these threats, the immune system has evolved intricate defense mechanisms that collectively work to protect the body from infections. Immunity to pathogens refers to the ability of the immune system to recognize, respond to, and neutralize these foreign invaders while sparing the body's own healthy cells.

Immunity to pathogens is a dynamic and complex process involving both innate and adaptive immune responses. These responses work together to detect, target, and eliminate invading microorganisms, thereby safeguarding the body from infections. The interplay between the immune system and pathogens is a fascinating area of study with significant implications for public health, medicine, and the development of vaccines and treatments.

Our immune system contains several different types of cells whose job is to monitor conditions throughout the body and respond to signs of damage. Second, the chaperones have to eliminate the cause of the disruption (the party crashers). Our immune system has cells that eliminate pathogens or produce proteins that eliminate pathogens. Third, the chaperones can remember the identities of the party crashers which allows them to more easily find the party crashers if they return. After removing a pathogen, our immune system can remember the identity of the pathogen so that if it returns, our immune system can remove the pathogen before it makes us sick.

- Antibodies from B-cells
- Cytotoxic T-lymphocytes (CD8+)
- Helper T-cells (CD4+) to stimulate and regulate B-cells and CTL



CORE CONCEPT: IMMUNITY TO MICROBES

The development of an infectious disease in an individual involves complex interactions between the microbe and the host.

The key events during infection include: entry of the microbe, invasion and colonization of host tissues, evasion of host immunity, tissue injury or functional impairment.

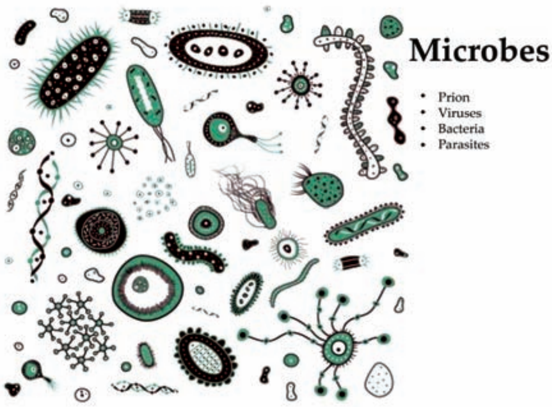
Microbial infections are best prevented by both innate and adaptive immune responses. The innate immune system takes care of early defense while the adaptive immune system offers a longer and potential response. In addition, adaptive immune responses are more specific and confer protection from repeated attacks by producing memory cells. As microbes differ a lot in their host attacking regime, their removal from the affected patient requires efficient effector systems. Adaptive immunity is developed in such a way that it permits the affected host to respond favorably to different types of **microbes**.

Microbes can be multi-cellular or single-celled organisms and include bacteria, protozoa, and some fungi and algae.

The result of many microbial infections is decided by the balance between microbial schemes for withstanding immunity and the host immune responses. Microbes have adapted several ways to combat the immune response.

Immunity against microbes performs almost similar to other defense mechanisms. Although it is essential for host survival but sometimes it may cause damage to the host tissue itself.

Some microbes especially viruses have the potential to be latent. In such cases the host immune response does not allow the microbe to spread but the microbes survive in the latent form, i.e. infection may prevail under specific conditions like stress etc.



Immunity to Bacteria

Bacterial infections cause a variety of immune responses from the host depending on the specific bacterium involved. Understanding the virulence of bacteria provides insight into the medical approaches that can be used to treat infection and prevent disease.

Immunity to Extracellular Bacteria

Extracellular bacteria are those that multiply and reside outside the host cell. These bacteria mainly affect the cells in two ways. They either attack by causing inflammation and tissue damage or by producing toxins.

Innate Immunity to Extracellular Bacteria

Innate immunity to extracellular bacteria essentially involves three processes.

Stimulation of Phagocytes

Phagocytes takes the help of surface receptors and Fc receptors to identify extracellular bacteria and its opsonization with the help of antibodies, respectively. Most of these receptors are associated with promotion of phagocytic activity and microbicidal activity.

Induction of Inflammatory Response

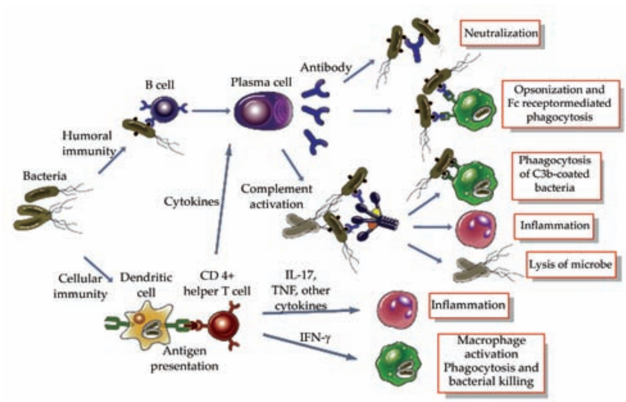
Antigen presenting cells like dendritic cells in addition to phagocytes are stimulated by microbes and these cells secrete cytokines which are responsible for causing leukocyte infiltration at the site of inflammation.

Activation of Complement System

Both gram positive and gram negative bacteria stimulate alternative pathway of complement system and mannose expressing bacteria stimulate lectin pathway of complement system by binding to mannose binding lectin.

Adaptive Immunity to Extracellular Bacteria

The immunity that plays major role against extracellular bacteria is the humoral or antibody mediated immunity as it prevents the infection by neutralizing the toxins. Usually polysaccharide antigens are prototypic thymus-independent antigens and humoral immunity is the basic line of defense against polysaccharide-rich encapsulated bacteria. The antibodies in such cases defend the body by neutralization, opsonization, phagocytosis and stimulation of complement system. Extracellular bacteria also stimulate the production of CD4+ helper T cells which induces inflammation and phagocytic activity. Besides this, these antigens may cause some mutational disorders and also the affected individual may have reduced immune response towards microbial infections.



Some Important Definitions

- ***Septic shock-*** It is the after effect of some gram positive and gram negative bacterial infections represented by intravascular coagulation.
- ***Superantigen*** - Some bacterial toxins or antigens stimulate non-specific activation of T-cells leading to polyclonal T-cell activation. These antigens are called **superantigens**.

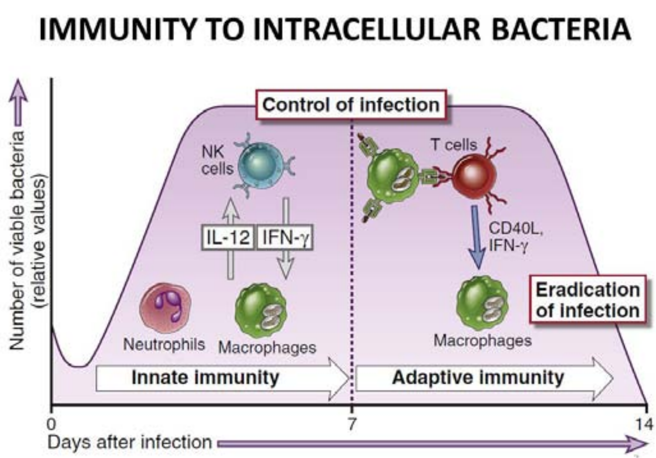
Superantigens are a class of antigens that cause non-specific activation of T-cells resulting in polyclonal T cell activation and massive cytokine release.

Immune Evasion by Extracellular Bacteria

- Polysaccharide antigens or encapsulated bacteria are more lethal as compared to a strain devoid of capsule because they resist phagocytosis.
- Capsulated bacteria inhibit alternate pathway of complement system due to the presence of sialic acid.
- One more way of evading immune response by extracellular bacteria is due to the genetic edition of surface antigens. E.g. surface antigen of some specific bacteria is contained in their pili. Pili contain a protein antigen called "pilin" and this pilin undergoes gene variation. Pili are the structures of bacteria responsible for bacterial adhesion to host cells.

Immunity to Intracellular Bacteria

Some intracellular bacteria like pathogenic or facultative are able to multiply within the phagocytes, so their elimination from the patients requires modified strategies.

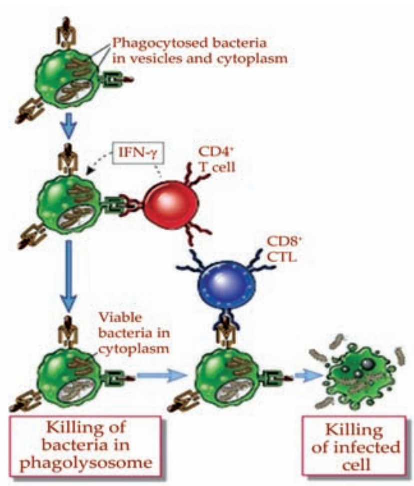


Innate Immunity to Intracellular Bacteria

Phagocytes and natural killer cells provide innate immunity to the intracellular bacteria. However some bacteria survive and multiply easily in the phagocytes, the phagocytes need to be stimulated by the secretions of these bacteria in order to clear the infection. The secretions from these bacteria are recognized by TLRs and cytoplasmic proteins of the NOD-like receptor (NLR) family so that they stimulate the phagocytes to degrade the invading bacteria. In addition to the intracellular bacteria, activated natural killer cells produce IFN-γ, which consecutively stimulates macrophages and cytokines. Although innate immunity provides protection from most of the bacteria but some intracellular bacteria like *Listeria monocytogenes* need cell mediated immunity in order to be eliminated from the body.

Adaptive Immunity to Intracellular Bacteria

T cell-mediated immunity plays a significant role in providing protection against intracellular bacteria. CD4+ T-cells and CD8+ cytotoxic T lymphocytes are the two major forms of cell mediated immunity that participate in phagocytosis or killing of infected cells, respectively. Both the, CD4+ T-cells and CD8+ cytotoxic T lymphocytes work together to provide protection against the intracellular bacteria. Granulomatous inflammation acts as a marker for most of the infections due to intracellular bacteria, which occurs because of T-cell and macrophage stimulation. Macrophage stimulation that occurs as an antigenic response towards intracellular microbes is sometimes able to cause tissue damage.



The response shown by different patients towards the intracellular microbes decides the development of the disease and its consequence. One neat example of such type of response is shown by leprosy patients. Leprosy is a disorder caused by ***Mycobacterium leprae*** and it exists in two forms, the lepromatous and tuberculoid form. Lepromatous form is characterized by feeble cell-mediated immune response and high specific antibody titer while the tuberculoid form shows low specific antibody titer but very strong cell-mediated immune response. Although the reasons attributed to this type of response are still speculated and not yet verified, one of the factors that are given significance is regarding varied pattern of cytokine production and T-cell differentiation in patients.

Dodging of Immune System by Intracellular Bacteria

Intracellular bacteria tend to dodge the immune system in many ways comprising evading into the cytosol or preventing phagolysosome fusion and by overpowering the reactive oxygen species by their microbicidal activity. These bacteria have the potential to cause chronic infections because they can survive the phagocyte mediated elimination and thrive for years in the body and may show reversion of the disease.

Immunity to Fungi

Mycoses is another term for fungal infections. Fungal infections are as important as other microbial infections but the lack of animal models for mycoses makes it less informed. Neutrophils and macrophages serve as the outstanding mediators of innate immunity against fungal infections. Neutrophils release lysosomal enzymes and fungicidal substances like reactive oxygen species, which phagocytose fungi to kill them within the cell. Cell-mediated immunity is effective mechanism of adaptive immunity against fungal infections. It functions by preventing the spread of fungi to other tissues. Certain Fungi also evoke antibody response of protective value.

Immunity to Viruses

Immune responses towards viruses either function by blocking the infection or by getting rid of infected cells. Type I interferons participate in innate immunity while neutralizing antibodies take part in the **adaptive immunity**.

Adaptive immunity is the protection of a host organism from a pathogen or toxin. It is mediated by B cells and T cells, and is characterized by immunological memory.

Innate Immunity to Viruses

Type I interferons inhibit the infection while the killing of infected cells is mediated by Natural killer (NK) cells. Type I interferons prevent viral replication by triggering an “antiviral state”. NK cells are significant in early stages of the infection because in later stages adaptive immune responses progress. NK cells kill the infected cells and also identify infected cells where the virus has shut off class I MHC expression as an evading mechanism from CTLs. The importance of evading mechanism lies in the fact that the liberation of NK cells from a normal state of inhibition occurs only when MHC class I expression is turned out and not active.

Adaptive Immunity to Viruses

High affinity antibodies produce adaptive immune response against viral infections by preventing virus binding to the host cells, and by CTLs which bring out elimination of infected cells by killing them. CTLs like CD8+ T-cells identify viral peptides by class I MHC molecule. Further virus infected cell is phagocytosed by the antigen presenting cells such as dendritic cells. Dendritic cells process the viral antigen and present it to naïve CD8+ T-cells. Some of the CD8+ T-cells replicate massively to kill the infected cells. In some cases the virus persists in the infected individual without active replication leading to latent infection. CTLs may lead to tissue injury even if the infectious virus is not dangerous to the body.

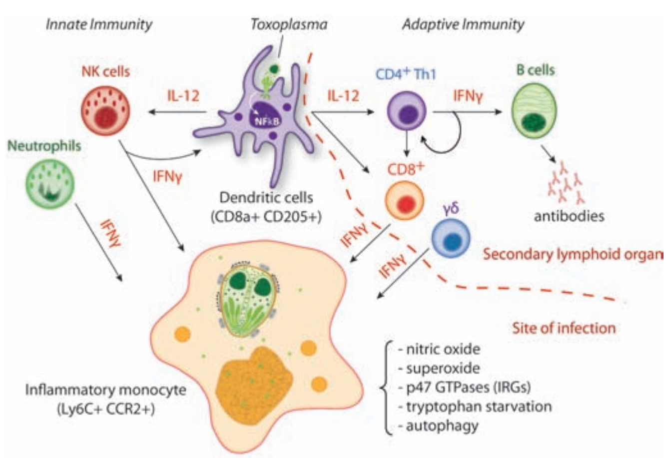
How Viruses Deceive Immune System?

Viruses have adopted numerous strategies for escaping the immune system.

- Viruses can change their surface antigens to avoid immune response. Generally surface glycoproteins containing T-cell epitopes undergo changes by point mutation or reassortment of genes especially in RNA viruses.
- Some viruses escape the immune surveillance by inhibiting the antigen presentation process and by inactivating the immunocompetent cells.
- Suppression of immunosuppressive molecules is also one of the strategies adopted by viruses.

Immunity to Parasites

Parasitic infections are mostly the infections caused by protozoa, ectoparasites and helminths. The parasitic infections are mostly chronic because of weak innate immunity. Besides weak immunity, parasites have a knack of evading host immune response very easily.



Innate Immunity to Parasites

Phagocytosis is the main innate immune response to parasitic infections but many parasites are able to escape the immune system. E.g. some helminths have thick teguments that enable them to evade the cytocidal mechanism of neutrophils and macrophages. Very few parasites have the potential to activate alternate pathway of complement system but the parasites that recoup from infected patients acquire resistance to complement mediated lysis.

Adaptive Immunity to Parasites

Parasites exhibit diverse adaptive immune response. Cell mediated immunity is the principal defense mechanism against parasitic infections.

Stimulation of macrophages by Th1 cell derived cytokines is especially directed by cell mediated immunity to neutralize the antigens. Helminths are removed by IgE antibody and eosinophil-mediated killing as well as other leukocytes.

Immune Evasion by Parasites

Parasites have varied mechanism to evade immune response due to some modifications.

- Parasites escape the immune system by preventing host immune response and by discounting their immunogenicity.
- They evade immune mechanism by bringing antigenic variation including changes in surface antigens by masking or shedding their antigens.
- One more way to evade **immune response** is by developing resistance to immune effector mechanisms.

Immune response is the body's response caused by its immune system being activated by antigens.

Table 1. Mechanisms of immune evasion by parasites

Mechanisms	Parasite
Antigenic variation	Trypanosomes, Plasmodium
Prevention of host immune responses	Filaria, trypanosomes
Shedding masking antigen	Entamoeba
Acquired resistance to Complement	Schistosomes

Approach for Vaccine Development

Vaccination is an effective strategy for restraining infections. Remarkably potent vaccines are those that are successful in provoking high-affinity antibodies and memory cells. Most of the vaccines developed, function by inducing humoral immune response in the host.

Attenuated and Inactivated Bacterial and Viral Vaccines

Vaccines containing intact nonpathogenic microbes are made after attenuating the virus or by killing the microbes taking care of their immunogenicity. Attenuated viral vaccines prove beneficial because they evoke effective innate and adaptive immune responses. Live attenuated bacterial vaccines used nowadays offer protection but for small duration while live attenuated viral vaccines elicit good response and long lasting immunity. Viral vaccines perform better because of their adaption in cell culture. However live viral vaccines always face a potent risk of reversion to virulence and hence safety is the main concern for such viruses. To minimize such risks inactivated vaccines are used such as influenza vaccine.

Purified Antigen (Subunit) Vaccines

Subunit vaccines are those that contain a purified antigen and also needs to be given along with an immunogenic enhancer called an adjuvant. Construction of subunit vaccine involves isolation of a specific protein from a virus or bacteria before administration. Diphtheria and tetanus are the best examples of subunit vaccines.

Synthetic Antigen Vaccines

This concept involves identification of an epitope or an antigen and designing it in the laboratory to be used in future as a vaccine.

Live Viral Vaccines Involving Recombinant Viruses

Recombinant viruses involve the insertion of genes that encode an antigen into a noncytopathic virus, which provide immunity following its introduction to a susceptible host. They offer protection by eliciting both innate and adaptive immune response. However in some cases safety is the issue.

DNA Vaccines

This founds the basis of most fundamental work being done in current times. DNA vaccine strategy involves inoculation of a plasmid containing complementary DNA encoding a protein antigen. DNA vaccines elicit both humoral and cell mediated response even without any adjuvant administration but their effectiveness needs more experimentation and verification.

Adjuvants

Adjuvants are the immunogenic substances that do not provide immunity by themselves but enhance the immunogenicity of the vaccines to which they are administered with.

Passive Immunization

Passive immunization is another way of providing protective immunity. This can be done by transfer of specific antibodies and are useful in cases like snake bite. Passive immunity does not offer long lasting memory because it is short lived and does not generate memory.

Passive immunity is a form of immunity which occurs when antibodies are transferred from one person to another individual, or when antibodies of animal origin are introduced to a human. This type of immunity is short acting, and is typically seen in cases where a patient needs immediate protection from something and he or she cannot form antibodies quickly enough independently.

In natural passive immunity, antibodies are passed from a mother to a child. Antibodies can be transferred through the placenta, or transmitted through the colostrum, a liquid which is produced in the breasts for a baby's first meal. The antibodies transmitted through the colostrum and placenta generally only last for several weeks, which is long enough to allow the baby to start to build up its own **immune system** and to make its own antibodies.

Immune system is a host defense system comprising many biological structures and processes within an organism that protects against disease.

Artificial passive immunity involves the introduction of antibodies through means such as injection. *For example*, in the treatment of some diseases, patients may be given a serum derived from patients who have recovered to help them fight the disease. This practice is sometimes seen when people are dealing with an outbreak of a new or extremely virulent disease for which no known treatment is available.

Prophylactic treatments of antibodies are sometimes given when people are exposed to diseases like rabies, botulism, tetanus, and diphtheria. By giving a patient passive immunity, a doctor can help the patient recover from a disease which the patient's body is unable to fight. This type of immunity is not the same thing as vaccination, a process in which small amounts of antigens are introduced to the body to encourage it to form its own antibodies so that it can fight those antigens in the future.

When the body makes its own antigens, as seen when someone is vaccinated, this is known as active immunity. Active immunity lasts much longer than passive immunity, sometimes persisting over a lifetime to keep someone from getting sick, in marked contrast with the temporary state of passive immunity.

Inducing active immunity is often preferred, when possible, because it will support the patient's health in the long term. However, artificially creating active immunity is very complicated, as it involved the controlled introduction of antigens in the body, and this can be dangerous for some patients, as seen when so-called "live" vaccines cause outbreaks.

Table 2. Vaccine development strategies

Type of vaccine	Examples
Live attenuated or killed bacteria	Bacillus Calmette-Guerin, Cholera
Viral vectors	Clinical trials of HIV antigens in Canarypox vector
Live attenuated viruses	Polio, rabies
Subunit vaccines	Tetanus toxoid, diphtheria toxoid
Conjugate vaccines	*Haemophilus influenzae*
Synthetic vaccines	Hepatitis
DNA vaccines	Clinical trials ongoing for several infections

CORE CONCEPT: TRANSPLANT IMMUNOLOGY

Transplantation is the process of moving cells, tissues or organs from one site to another for the purpose of replacing or repairing damaged or diseased organs and tissues. It saves thousands of lives each year. However, the immune system poses a significant barrier to successful organ transplantation when tissues/organs are transferred from one individual to another.

Transplant immunology is the study of immune responses to transplantation of organs and other donor material, for the purpose of preventing rejection and increasing transplant success. This field includes researchers working in lab settings on transplant medicine, as well as doctors who interact directly with patients and perform transplants. People with a variety of medical and scientific backgrounds can pursue careers in transplant immunology, with many research centers using interdisciplinary teams for research.

The risk of rejection is one of the key obstacles with transplanting donor tissue and other materials. If the recipient's immune system identifies the transplanted material as foreign and dangerous, it will attack, degrading the material. This can lead to death or severe illness for the recipient. Transplant immunologists study how the immune system responds to transplants with the goal of finding a way to interrupt it and encourage the body to accept the donor tissue.

Specific cells in our bodies engulf pathogens, digest those pathogens and present small pieces of those pathogens on their cell membrane.

One aspect of transplant immunology involves screening donors and recipients to identify good matches. In addition to matching by blood type, researchers can also check for a number of other characteristics, searching for things like proteins in the donor material that the recipient's body will attack. Matching can take time if someone has an unusual blood type or other characteristics. By matching donors and recipients as precisely as possible, people can reduce the risk of rejection.

Another area of interest involves the development of anti-rejection drugs. Many of these work by suppressing immune responses so the recipient's body cannot start to attack the donor material. Developing drugs to suppress rejection without interfering with the immune system's ability to fight off actual infections is of interest to many researchers. Researchers develop drug regimens, study potential candidates for anti-rejection drugs, and follow up on patients to see how well they tolerate specific drugs.

Transplant immunology is also a topic of interest for people developing artificial transplant materials, such as heart valves made from plastics and metals, grafts cultured in a lab using donor cells, or grafts of animal origin. The body usually responds to foreign objects by rejecting them, and developing effective transplants requires understanding how and where the body tags foreign materials to see if it is possible to develop artificial transplant materials that will not trigger a rejection response.

People interested in transplant immunology careers will need to go to medical school or pursue graduate work in the sciences with a focus on immunology and related topics. Education for people in these fields can last 12 years or more. Transplantation is a method of treating the patient with malfunctioned organ or tissue with the healthy one. The organ to be taken is called graft from a healthy individual referred as donor. The individual who receives the graft is called recipient. The transplantation is called orthotropic if the graft is used for an identical anatomical position and heterotropic if used in a different anatomical location. The blood from one individual can be transferred to another individual of same blood group and the process is called as transfusion. The method of transplantation can turn into rejection if the graft belongs to a genetically different individual. The phenomenon of rejection is a large and one of the challenging areas of research in the field of immunology. The immunology behind a tissue or cell rejection is due to the adaptive immune response.

An inbred strain of animal is genetically identical because of the homozygous nature of all the genes except the sex chromosome.

A graft transplanted from one part of body to other of a same individual is called **autologous graft.**

A graft transplanted from one individual to a genetically identical individual is called **syngeneic graft.**

A graft transplanted from one individual to a genetically different individual is called **allogeneic graft (allograft).**

A graft transplanted between individuals of two different species is called **xenogeneic graft (xenograft).**

The substances recognized as foreign by the host immune system in an allograft are called **alloantigens** and the lymphocytes and antibody against those are called **alloreactive.** Similarly, for xenograft the substances are called **xenoantigens** and counteracting immunity is called **xenoreactive.**

Immunity to Allograft

Polymorphic genes called histocompatibility genes are responsible for the recognition of foreign transplant cells, and are very diverse among the individuals of same species. Some of the basic rules of transplantation are as follows

- Cells and organ transplanted between genetically identical individuals are not rejected.
- Cells and organ transplanted between two genetically non identical or different individuals are always rejected.
- The offspring of two different inbred animals usually do not reject the organ or tissue from either of the parents.
- The parents of an offspring belonging to two different inbred strains usually reject the graft taken from the offspring.

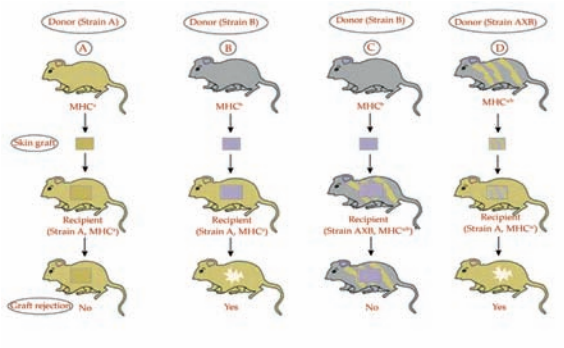


Figure 1. Genetics of graft rejection in mice.

The molecules responsible for rejection of transplants are well known to be major histocompatibility complex (MHC). The alloantigen from a foreign donor is presented to the T cell by MHC molecules. The alloantigens are presented by typically two ways, direct and indirect.

In **direct presentation** , MHC molecules from the donor is directly presented to host T cells to elicit the cell mediated immune response without the involvement of host antigen presenting cells or MHC molecule.

In **indirect presentation**, the alloantigens are captured and processed by host antigen presenting cells and presented to T cells to elicit the immune response.

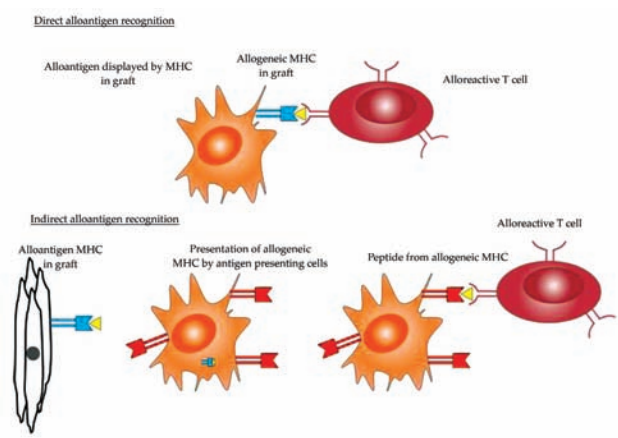


Figure 2. Direct and indirect recognition of allogeneic antigens.

Activation of Alloreactive Lymphocytes

The alloantigens stimulate the B and T cell response similar to that of a protein antigen. Alloantigens can be recognized by T cells either by direct or indirect way and trigger a T cell immune response. Naïve lymphocyte migrates to lymph node after sensitization and differentiates into an effector cells that migrate back to the graft in order to induce rejection. Usually many of the T lymphocytes responding to alloantigens are memory T cells. In addition to alloantigen recognition by T lymphocyte, costimulatory molecules such as B7-1 also activate proliferation of lymphocytes. Similar to the processing of a protein antigen, the alloantigens that are processed by MHC class I can activate the CD8+ T cell response and those processed by MHC class II can activate CD4+ specific immune response.

Inhibition of Allograft Rejection

The rejection of the allograft in a recipient having fully functional immune system is inevitable. It has been a challenging task to prevent the rejection of allograft in a recipient. An important area of research in the field of immunology is to avoid the rejection of an allograft and allow it to survive without any immune reactivity.

Many strategies have been adopted to avoid the rejection

- Immunosuppressive drugs can be used to prevent graft rejection. Cyclosporine can inhibit the transcription of genes responsible for T cell activation and IL-2 secretion. Rapamycin inhibits the growth factor responsible for T cell activation.
- Some antimetabolites that have the potential to kill the activating T lymphocyte can be used to prevent the alloreactive response.
- **Inhibition of costimulatory** molecule is another strategy to prevent the rejection of the allograft.
- **Anti-inflammatory** drugs such as **corticosteroids** are frequently used to create an immunosuppressive condition in an individual and allow the survival of graft.

Corticosteroids are a class of steroid hormones that are produced in the adrenal cortex of vertebrates, as well as the synthetic analogues of these hormones.

Xenogeneic Transplantation

Transplantation of an organ from a different species has been a great interest for the scientist working on different human disease models. Transplantation of organs from pigs to human is an interesting outcome of the transplantation immunology. Pigs are preferred species for transplantation in human as compared to other species because of the anatomical compatibilities. However, the presence of natural antibodies such as IgM to the xenograft causes a hyperacute reaction in the recipients. The natural antibodies are mostly directed towards the carbohydrate present over the cell surface. Xenograft can also be rejected by T cell mediated immune responses similar to what is observed for the allograft.

Blood Transfusion

Blood transfusion is also a kind of transplantation in which blood is transferred from one individual to other having same blood group. The concept of blood grouping was put forth by Karl Landsteiner, who divided the blood into A, B, and O based on the presence of specific antigens. A, B, and O are the carbohydrate moiety present over the surface of blood cells. An individual having blood group A contains type "A" antigen and antibody against type B, so it will show rejection against blood group B. Same is true for blood group B antigen, which contains antibody against A. The blood group AB contains antigen against both A and B and do not produce antibody against any antigen hence the person of blood group AB can take blood from any individual hence called as universal recipient. The blood group O contains no antigen over its surface and produces antibody against both A and B hence can give the blood to any blood group individual and is called as universal donor. Individual having blood group AB can give blood to only AB individuals while blood group O individuals can only accept blood from an O group individual.

Table 3. Blood group antigens

	Blood group A	Blood group B	Blood group AB	Blood group O
Antigens on RBC	A	B	Both A and B	None
Antibody Against	B	A	None	Both A and B
Accept blood from	A and O	B and O	A, B, AB and O	Only O
Donate blood to	A	B	AB	A, B, AB and O

Other Blood Group Antigens

The ABO blood group may be modified by the enzymes involved in the modification of the surface glycoproteins.

The enzyme glycosyltransferase plays an important role in the terminal carbohydrate addition in the blood group antigens. The modification in which a different enzyme fucosyltransferase incorporates a fucosyl group at any other terminal position of the blood group antigen, results in the formation of **Lewis antigen.** Lewis antigen has a capacity to bind with E and P type selectin, making it a useful tool for immunologist.

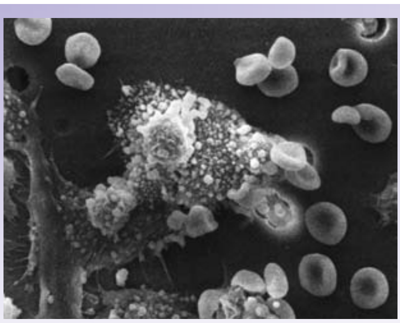
Another important blood group antigen is called Rh factor, named after Rhesus monkey from which it was first identified.

The person can be Rh + or Rh - based on the presence or absence of Rh factor in the blood cells. The Rh - mother carrying an Rh + child can be sensitized by the Rh + factor circulating through the placenta in her blood. Since Rh factor will act as an antigen in mother's body and she will produce an IgG antibody against Rh factor. Any subsequent pregnancy can lead to abortion because of the destruction of the blood cells by antibodies against Rh factor; such condition is called erythroblastosis fetalis.

Graft Versus Host Diseases

Graft versus host disease (GVHD) occurs when a host is unable to reject the graft due to some form of immunosuppression. GVHD is usually caused by the antibody against MHC molecules of the host. GVHD is initiated by the grafted T cells that recognize the host alloantigens and mount an immune response in the form of effector T cell. The condition may be acute or chronic based on the severity of mounted immune response. Acute may be fatal while chronic GVHD leads to dysfunction of the affected organs. Natural killer cells, cytokines, and cytotoxic T lymphocytes are the major players of GVHD.

Each B-cell and T-cell expresses a receptor on its surface that binds to a unique biological molecule which is usually a peptide from a protein but can be nucleic acid or carbohydrate.



CORE CONCEPT: TUMOR IMMUNOLOGY

Tumor immunology describes the interaction between cells of the immune system with tumor cells. Understanding these interactions is important for the development of new therapies for cancer treatment.

Another important role of the immune system is to identify and eliminate tumors. This is called immune surveillance. The *transformed cells* of tumors express antigens that are not found on normal cells. To the immune system, these antigens appear foreign, and their presence causes immune cells to attack the transformed tumor cells. The antigens expressed by tumors have several sources; some are derived from oncogenic viruses like human papillomavirus, which causes cervical cancer, while others are the organisms own proteins that occur at low levels in normal cells but reach high levels in tumor cells. One example is an enzyme called tyrosinase that, when expressed at high levels, transforms certain skin cells (e.g. melanocytes) into tumors called melanomas. A third possible source of tumor antigens are proteins normally important for regulating cell growth and survival, that commonly mutate into cancer inducing molecules called oncogenes.

A combination of hypoxia in the tumor and a cytokine produced by macrophages induces tumor cells to decrease production of a protein that blocks metastasis and thereby assists spread of cancer cells.

The main response of the immune system to tumors is to destroy the abnormal cells using killer T cells, sometimes with the assistance of helper T cells. Tumor antigens are presented on MHC class I molecules in a similar way to viral antigens. This allows killer T cells to recognize the tumor cell as abnormal. NK cells also kill tumorous cells in a similar way, especially if the tumor cells have fewer MHC class I molecules on their surface than normal; this is a common phenomenon with tumors. Sometimes antibodies are generated against tumor cells allowing for their destruction by the complement system. Clearly, some tumors evade the immune system and go on to become cancers. Tumor cells often have a reduced number of MHC class I molecules on their surface, thus avoiding detection by killer T cells. Some tumor cells also release products that inhibit the immune response; for example by secreting the cytokine TGF-β, which suppresses the activity of macrophages and lymphocytes. In addition, immunological tolerance may develop against tumor antigens, so the immune system no longer attacks the tumor cells. Paradoxically, macrophages can promote tumor growth when tumor cells send out cytokines that attract macrophages, which then generate cytokines and growth factors such as tumor-necrosis factor alpha that nurture tumor development or promote stem-cell-like plasticity.

Immune Surveillance

The concept of immune surveillance was given by Macfarlane Burnet in 1950's. According to his definition of Immune surveillance physiologic function of the immune system is to identify cancerous or precancerous cells and remove them from the body before they cause any harm.

General features of tumor immunity

- One of the most important features is that tumors elicit specific adaptive immune responses. This can be proved due to the presence of T lymphocytes, natural killer cells and macrophages in the surroundings of the tumor cells.
- Tumor immunity is not sufficient to stop the growth of tumors. This study pattern undoubtedly raises questions about the concept of Immune surveillance.
- Outside source can be used to stimulate immune system to prevent and destroy the tumor cells.

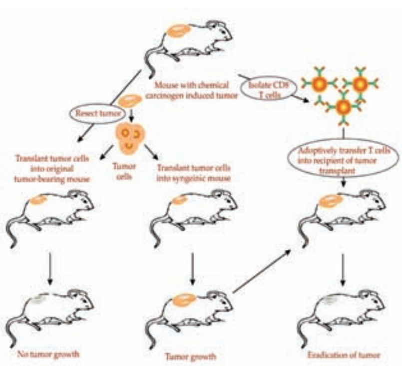


Figure 3. Schematic representation of tumor immunity. Mice treated surgically for chemical induced carcinogen gets immune to tumor growth following the injection of same tumor cells.

Similarly, transfer of CD8+ cells to a recipient mice having tumor transplant rejects the tumor growth. However transplantation of same into a syngeneic mouse turned into a tumor mass.

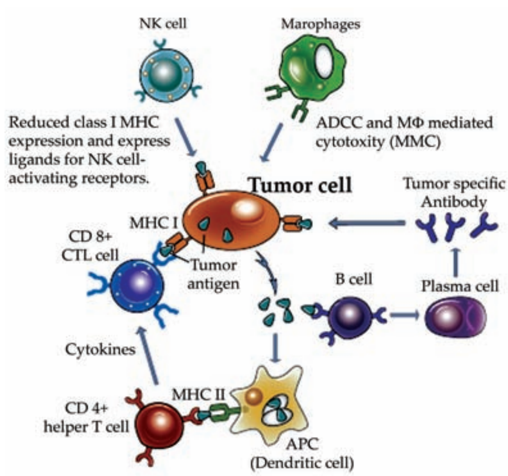
Tumor Antigens

These are the antigens produced in tumor cells. Tumor antigens are classified based on how they express. Oncogenes and mutated tumor suppressor genes are identified as tumor antigens.

- **Tumor specific antigens –** are the ones that do not express on normal cells but only on tumor cells.
- **Tumor associated antigens** -are the ones that express on tumor as well as normal cells.
- **Antigens of oncogenic viruses** - Oncogenic viruses such as Epstein – Barr virus, human papillomavirus and papovaviruses are associated with certain types of cancers in humans and animals. The end products of these oncogenic viruses act as tumor antigens and induce immunogenic response. This concept of immune response against virus induced cancers has paved way for the synthesis of vaccines against the tumor causing viruses.
- **Oncofoetal antigens** - These are the proteins that are highly expressed in cancer cells as well as in foetus undergoing development but are absent in the adult cell.
- **Tissue specific differentiation antigens** - These are tissue specific molecules expressed only on normal cells of origin and are not expressed on cells from other tissues.

Immune Response to Tumors

Tumor Immune response mostly occurs in two forms either by innate immune response or by adaptive immune response.



Innate Immune Response to Tumors

Natural Killer Cells (NK Cells)

Around 15% of mammalian blood lymphocytes are composed of NK cells. NK cells can be activated by interferons from virus infected cells or by IL-12 from activated macrophages. NK cells are large, granular, and non-phagocytic cells that are derived from bone marrow. NK cells can kill certain tumor cell lines and are quite effective in eliminating the cells that diminish class I MHC expression. Studies also indicate that patients with deficiency of NK cells are more likely to suffer from EBV- associated lymphomas. NK cells express CD56 and CD16 antigen receptors over their surface. Activation of NK cells by antigen antibody reaction through CD16 kills the target cells.

Macrophages

Macrophages can prevent the spread of cancer based on their activation state. Activated macrophages can kill transformed cells more efficiently than the normal cell.M1 cells especially treat the tumor cells like an infectious organism and produces cytokine tumor necrosis factor (TNF) to kill the tumor but M2 macrophages on the other hand are associated with **tumor progression**.

Tumor progression is the third and last phase in tumor development. This phase is characterized by increased growth speed and invasiveness of the tumor cells.

Adaptive Immune Response to Tumors

T-lymphocytes

The basis of adaptive tumor immunity is to destroy tumor cells by CD8+ CTLs. Functioning of CD8+ cell requires cross presentation of the tumor antigen by the dendritic cells. Although CD8+ CTLs have a substantial role to play in killing the transformed clones but not much is known about the efficacy of CD4+ helper T –cells in tumor immunity.

Antibodies

These are known to kill tumors either by stimulating antibody-dependent cell mediated cytotoxicity or by the activation of complement system. Even though there is some immune response by antibodies in which NK cells mediate the destruction of tumor cells but antibody response towards tumors is not quite effective in most of the tumor cases.

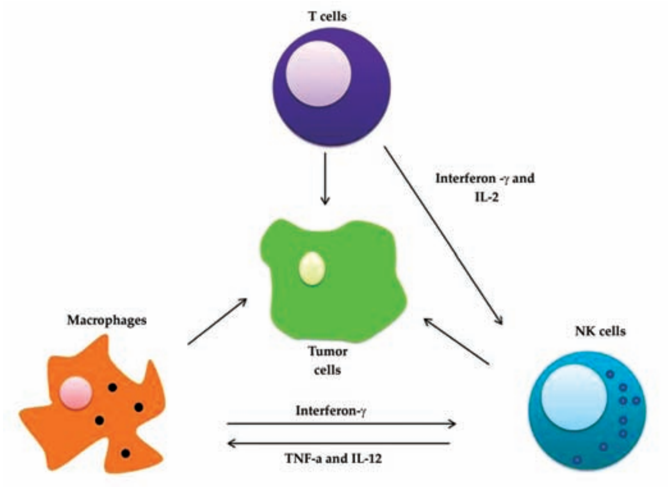


Figure 4. Immune response to tumor.

Evasion of Immune Response by Tumors

Immune evasion by tumors can be divided into intrinsic or extrinsic mechanisms based on how these mechanisms are mediated.

Intrinsic Mechanism of Immune Evasion by Tumor Cells

- Tumor cells are highly prone to mutation due to increased mitotic rate and hence the antigen expression may change at times and within a particular period tumor cells may lack or be deficit of the antigens that generate immune response.
- Immune system may not have access to the tumor cells because of antigen masking effect of **glycocalyx** molecules. Glycocalyx molecules help in hiding the tumor cells from the immune system by getting expressed in higher amounts in tumor cells.
- Tumor cells may prevent the immune response by associating with molecules that destroy the immune system.
- Some of the released products from the tumor cells may prevent the immune response. e.g TGF-β inhibits lymphocytes and macrophages.
- Tumor cells do not express class II MHC molecules so they may not elicit effective T cell immune response.

Extrinsic Cellular Suppression of Anti-tumor Immunity

- M2 macrophages are associated with progression of tumor growth and thus these macrophages may suppress the T- cell response to the tumor cells.
- T- Cell immunity can be lowered by the presence of regulatory T cells.
- Myeloid – derived suppressor cells (MDSCs) are detrimental to T cell response and may recruit regulatory cells to suppress the immune response.

Immunotherapy for Tumors

Immunotherapy has a potential role in transformed cells because it is less harmful than other known treatments for cancer (chemotherapy, radiotherapy and surgery). Some of the roles that immunotherapy can play are mentioned below

- Treatment of patient with cytokines and costimulators may help to increase the immunity against tumor.
- Tumor antigens can be used as a vaccine for the infected individuals.
- Tumor immunity can be augmented by blocking the inhibitory pathways.
- Stimulation of immune response can be done by local administration of polyclonal activators of lymphocytes.
- Another way of inducing immune response is to transfer antibodies and other immune effectors passively.
- Adoptive cellular immunotherapy can be approached which involves transfer of cultured immune cells in the host body.
- In some patients transfer of hematopoietic stem cell transplants combined with alloreactive T cells help in elimination of tumor.
- Monoclonal antibodies which are tumor specific may prove useful in some cases.

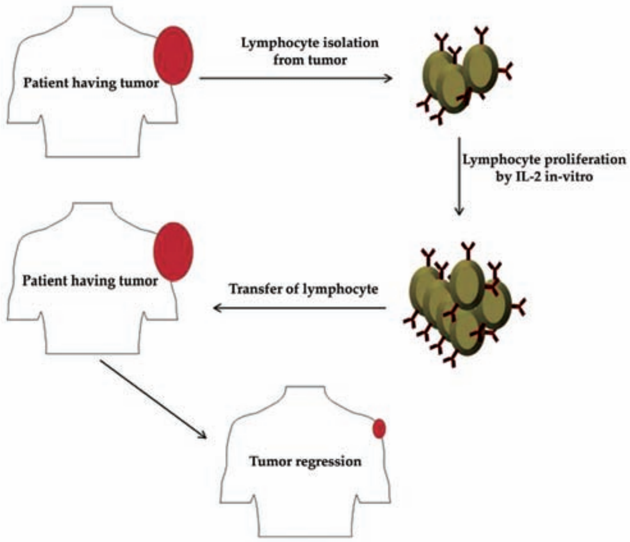
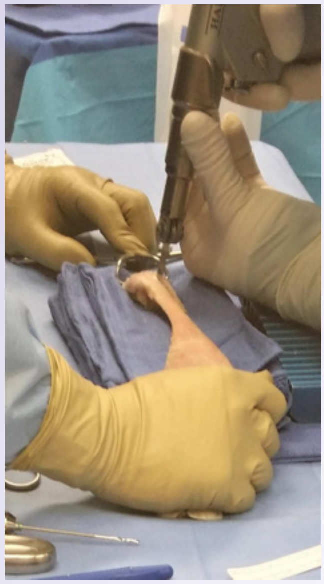


Figure 5. Adoptive transfer of antibodies in immunotherapy for cancer.

A CONCEPT OF ALLOGRAFT

An allograft is a medical term used to describe a type of organ or tissue transplantation in which the donor and recipient are genetically distinct individuals of the same species. In other words, an allograft involves the transfer of biological material, such as organs, tissues, or cells, from one person (the donor) to another person (the recipient) who is not genetically identical but belongs to the same species. Allografts are commonly performed in various medical contexts, including organ transplantation, tissue replacement, and cellular therapy.



Key points about allografts:

1. Compatibility and Immunology: Since the donor and recipient are not genetically identical, there is a potential for the recipient's immune system to recognize the transplanted tissue or organ as foreign and mount an immune response. This immune response can lead to graft rejection, where the recipient's immune system attacks and damages the transplanted tissue.
2. Immunosuppression: To prevent graft rejection, recipients of allografts typically receive immunosuppressive medications. These drugs suppress the immune system's ability to mount an aggressive response against the transplanted tissue, increasing the chances of graft survival. Immunosuppression carries risks, including increased susceptibility to infections and other complications.
3. Types of Allografts: Allografts can involve various types of tissues or organs, including kidneys, livers, hearts, lungs, pancreas, corneas, skin, bone, and blood vessels. Depending on the organ or tissue being

transplanted, different surgical techniques and immunosuppressive regimens may be used.

4. Matching and Compatibility: While it is not necessary for the donor and recipient to be genetically identical for an allograft, certain human leukocyte antigen (HLA) matching is important to reduce the risk of graft rejection. HLA molecules play a critical role in the immune system's recognition of self and non-self, and closer HLA matching between donor and recipient improves graft survival.
5. Success and Challenges: The success of allograft transplantation depends on various factors, including the type of graft, the donor-recipient relationship, the recipient's overall health, and the effectiveness of immunosuppression. While allograft transplantation has saved countless lives and improved the quality of life for many recipients, challenges such as graft rejection and long-term immunosuppression remain significant issues.
6. Advancements: Advances in medical technology, surgical techniques, and immunosuppressive therapies have contributed to improved outcomes for allograft recipients. Research continues to focus on reducing the risks associated with graft rejection and improving long-term graft survival.
7. Ethical Considerations: The availability of suitable donors and the ethical considerations surrounding organ and tissue donation raise important questions about equitable access, informed consent, and the ethical implications of allograft transplantation.

SUMMARY

- Immunity to pathogens is a dynamic and complex process involving both innate and adaptive immune responses. These responses work together to detect, target, and eliminate invading microorganisms, thereby safeguarding the body from infections.
- Immunity against microbes performs almost similar to other defense mechanisms. Although it is essential for host survival but sometimes it may cause damage to the host tissue itself.
- Phagocytes takes the help of surface receptors and Fc receptors to identify extracellular bacteria and its opsonization with the help of antibodies, respectively.
- Intracellular bacteria tend to dodge the immune system in many ways comprising evading into the cytosol or preventing phagolysosome fusion and by overpowering the reactive oxygen species by their microbicidal activity.
- Parasitic infections are mostly the infections caused by protozoa, ectoparasites and helminths. The parasitic infections are mostly chronic because of weak innate immunity.
- Transplant immunology is also a topic of interest for people developing artificial transplant materials, such as heart valves made from plastics and metals, grafts cultured in a lab using donor cells, or grafts of animal origin.
- Polymorphic genes called histocompatibility genes are responsible for the recognition of foreign transplant cells, and are very diverse among the individuals of same species.
- Graft versus host disease (GVHD) occurs when a host is unable to reject the graft due to some form of immunosuppression. GVHD is usually caused by the antibody against MHC molecules of the host. GVHD

MULTIPLE CHOICE QUESTIONS

1. **Which component of the immune system is responsible for the body's initial response to pathogens?**
   a. Antibodies
   b. Memory cells
   c. Innate immunity
   d. Cell-mediated immunity
2. **What is the primary role of antibodies in the immune response?**
   a. Destroy infected cells
   b. Produce cytokines
   c. Bind to and neutralize pathogens
   d. Enhance inflammation
3. **Which type of immunity involves B cells and T cells recognizing specific antigens on pathogens?**
   a. Innate immunity
   b. Active immunity
   c. Adaptive immunity
   d. Passive immunity
4. **What is the purpose of vaccination in building immunity to pathogens?**
   a. To directly kill pathogens in the body
   b. To introduce weakened or harmless antigens to stimulate immune memory
   c. To create inflammation at the site of infection
   d. To replace damaged immune cells
5. **Which immune component is responsible for remembering past pathogen encounters and facilitating a faster response upon re-exposure?**
   a. Phagocytes
   b. Cytokines
   c. Memory cells
   d. Antibodies

Answers

1. (c)   2. (c)   3. (c)   4. (b)   5. (c)

REVIEW QUESTIONS

1. How many process involved in innate immunity to extracellular bacteria
2. How does viruses deceive immune system? Describe.
3. What is passive immunity? Explain.
4. What is transplant immunology? Discuss.
5. Explain about the inhibition of allograft rejection.
6. What do you understand by blood transfusion? Describe.
7. What is tumor immunology? Describe the general features of tumor immunity.
8. Write the short notes on the following:
   a. T-lymphocytes
   b. Antibodies.

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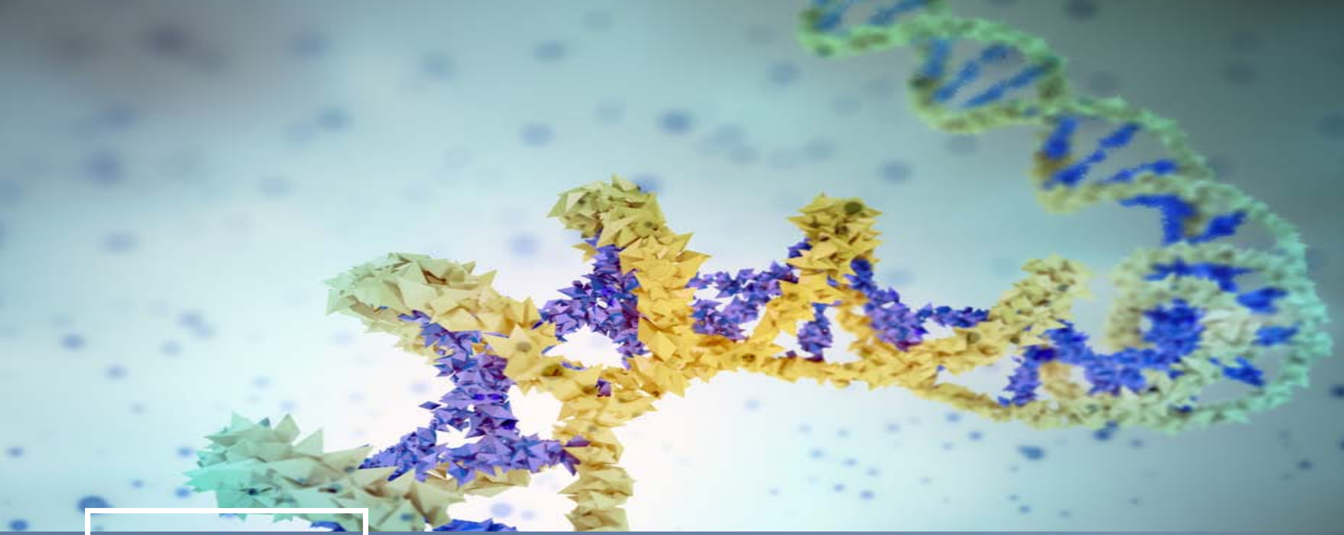
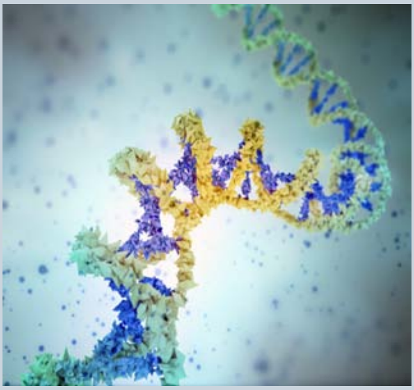


CHAPTER 6

FUNCTIONALITY OF TRANSPLANTATION IMMUNOLOGY



Transplantation is the process that involves the transfer of organs, tissues, or cells from one site to another. Transplantation typically takes place across individuals with the aim of repairing or replacing damaged organs or tissues. Owing to the rapid advancements in modern medical science, thousands of lives are saved each year by successful transplants.

After studying this chapter, you will understand the core concept of:

- Transplantation: Meaning, Types and Mechanism
- Grafts and Tissue Transplantation
- Organ Transplantation
- Organ donation: Special Legal and Ethical Problems
- Rejection

INTRODUCTION

Transplantation is the process of moving cells, tissues, or organs, from one site to another, either within the same person or between a donor and a recipient. If an organ system fails, or becomes damaged as a consequence of disease or injury, it can be replaced with a healthy organ or tissue from a donor. Organ transplantation is a major operation and is only offered when all other treatment options have failed. Consequently, it is often a life-saving intervention.

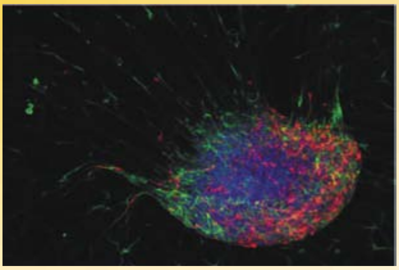
Transplantation immunology is a specialized field within immunology that focuses on understanding the immune responses that occur when tissues or organs from one individual (the donor) are transferred to another individual (the recipient). The success of transplantation depends on a delicate balance between the immune system's protective functions and its potential to recognize and reject foreign tissue.

The immune system plays a critical role in transplantation. The complex mechanisms of immunity, which under normal circumstances work to identify foreign microbes and direct the immune system to destroy them, pose a significant barrier to successful transplantation. Rejection of a transplant occurs in instances where the immune system identifies the transplant as foreign, triggering a response that will ultimately destroy the transplanted organ or tissue.

The intensity of the immune response against the organ or tissue, also commonly referred to as the graft, will depend on the type of graft being transplanted and the genetic disparity between the donor and recipient. To reduce the possibility of rejection, the donor and recipient are carefully matched for immune compatibility prior to transplantation. However, the small pool of eligible donors can make it difficult to find a donor-recipient match and there will always be a degree of rejection against the graft. A critical undersupply of donated organs means that waiting lists for transplants are extremely long. Patients needing a kidney transplantation, for example, wait on average 944 days (more than two and a half years) for a life-saving transplant.

The field of transplantation has undergone a dramatic expansion. Sophisticated surgical procedures have enabled the transplantation of almost any type of vital organ or tissue as an effective therapeutic modality. Since most transplants are done from genetically different donors, we must consider the immunological response of the recipient to the transplantation antigens presented by the donor graft. In general, this immune response is designed to cause rejection of the transplanted tissue and both cellular (or lymphocyte-mediated) and humoral (antibody-mediated) mechanisms are involved. However, under certain circumstances, a state of immunological tolerance develops which permits long-term normal function of a transplanted organ. This overview is intended as an introduction to the field of transplantation immunology. Since most transplants are performed between different members of the same species, we will focus on the basic mechanisms of allogeneic immune responses in transplant immunity.

Transplant immunity results after exposure of a recipient to different histocompatibility antigens. These transplantation antigens have been grouped as major (i.e. HLA and ABO blood groups) and minor histocompatibility antigens (although the latter have been documented in mice and other animals, little is known about them in humans). HLA antigens are controlled by a series of genes on chromosome 6, referred to as the human Major Histocompatibility Complex MHC). These genes have been classified into major categories. HLA-A and HLA-B are examples of Class I genes and HLA-DR and HLA-DQ represent Class II genes. These genes are highly polymorphic. For instance, HLA-A has more than 20 alleles and HLA-B has more than 50 alleles. Therefore, it is highly unlikely that two unrelated persons have the same HLA type. Both HLA class I and class II alloantigens can induce transplant immunity at humoral (antibody) and cellular (T lymphocyte) immune levels. ABO incompatibility is important only in antibody-mediated injury of the graft.Since transplant immunity leads to rejection, it seems best that the patient and donor are matched for histocompatibility antigens. Although this is possible for genetically identical twins (syngeneic grafts) and to a lesser extent between HLA- and ABO-identical siblings, the vast majority of transplants come from incompatible unrelated (cadaveric) donors. Immunosuppressive drugs are given to control transplant immunity so that rejection can be prevented or controlled and that the recipient develops a long-term acceptance (or tolerance) of the transplant. Rejection can be mediated by antibodies, lymphocytes or both and can manifest itself in different ways: hyperacute rejection (during the early post-transplant period), acute rejection (may occur at any time) and chronic rejection (a slowly developing process causing a progressive decline in graft function). Transplantation immunology represents a critical intersection between medical advances, immunological research, and patient care. By understanding the intricacies of immune responses to transplanted tissues, scientists and clinicians aim to improve the success rates of transplantation, enhance patient outcomes, and pave the way for novel therapeutic approaches.



CORE CONCEPT: TRANSPLANTATION: MEANING, TYPES AND MECHANISM

Transplantation is the transfer (engraftment) of human cells, tissues or organs from a donor to a recipient with the aim of restoring function(s) in the body. When transplantation is performed between different species, e.g. animal to human, it is named xenotransplantation.

The term "transplantation" means removing something from one location and introducing it in another location, hence in immunology the term transplantation is used to refer the procedure involved in the replacement of a nonfunctional or damaged organ or tissue with a good, and functional organ or tissue to safe guard the life of an individual.

The implanted tissue or organ is known as “graft”. Organs that can be transplanted are heart, kidneys, liver, lungs, pancreas, penis, eyes and intestine.

Tissues include bones, tendons, cornea, heart valves, veins and skin. The advancement in surgical techniques introduced the idea of transplantation of required organs or tissues. It is now possible to transfer a variety of tissues and organs from a healthy donor to an ailing recipient or from one site to another in the same individual (eg. skin grafting in burn cases) through sugery.

The **immune system** is a host defense system comprising many biological structures and processes within an organism that protects against disease.

Even though development in surgical techniques allowed clinicians to perform transplantation of organs or tissues, certain barriers such as availability of donors, expenditure for the clinical procedure and implant rejection reactions etc. made it a difficult and cumbersome procedure.

Since the **immune system** has been evolved to protect its own self from the attack of foreign bodies, it holds the power of identifying its own cells from foreign cells.

Cells express a set of membrane bound antigens called auto antigens. The auto antigens act as identification marks for the immune system to identify its own cells from foreign cells.

Immune system identifies the graft through the antigens present on their cells and reacts in a similar way as it reacts to a pathogen.

Hence the unique power of immune system is responsible for graft rejection. But immunological rejection is exempted for genetically identical or allograft (Own tissue taken from some other region), because the antigens present on the graft tissue are identical with its own antigens and immune system fails to identify them as foreign cells to mount any action.

First systematic study of transplantation was reported in 1908 by Alexis Carrel. In 1935 a Russian surgeon attempted the first human kidney transplant, but mismatch of blood group between donor and recipient induced a rejection reaction.

The first successful human kidney transplant between identical twins was succeeded in 1954. Many failures in transplantation procedure insist the importance of knowledge in immunobilogy.

Discovery of human blood groups was one of key event that helped to understand the cause for transplant rejection. Now it is understood, that matching of donor and recipients blood groups is essential for the survival of tissue grafts, because the vascular endothelium express blood group antigens. Matching of blood group antigens other than A, B, AB, O is not critical for transplantation, because they are expressed only on red Blood cells.

Study of Mouse skin graft rejection further improved the knowledge of transplantation immunology, and it helped a lot in understanding major his to compatibility antigens.

As recognition of foreign cells by recipient T cells is associated with MHC antigens of foreign cells, knowledge about MHC is essential to understand transplantation reactions.

You may also participate in sports. However, you must take enough precaution to prevent injuries to your kidney. Proper padding and protection around your lower abdominal region and back is of utmost necessity.

Transplantation is a useful procedure in surgical repair or replacement of diseased tissues or organs. It has been well-established that an animal accepts grafts of its own tissue or that of an identical twin, while a graft from another animal of the same species is rejected. The immune response induced by the transplantation (HLA) antigens present in all mammalian cells is the reason for the rejection of exogenous grafts.

The first record of plastic surgery is found in Sushrut Sanhita (800 BC). The first Indian surgeon Sushruta reconstructed a severed nose of a patient with the patient's own skin flap.

Transplantation may be fresh organs or stored organs in orthotropic grafts, the transplant is applied in anastomosed normal site (e.g. skin graft), whereas heterotopic grafts are placed in anatomically abnormal sites (e.g.. thyroid transplanted in subcutaneous tissue).

Types of Transplant

- Auto-graft (Autogenic graft). It is a tissue of one site engrafted to another site in the same individual.
- Isograft (Syngenetic graft). It is a graft placed in another individual of the same genetic constitution (e.g. monozygous twins or members of inbred strains).
- Allograft (Homograft or allogeneic graft). It is a graft transfer between two genetically different members of the same species.
- Xeno graft (Xenogeneic graft). Grafts between members of different species are called xeno-graft. It was formerly known as heterograft.

Auto-graft and isograft are usually accepted and survives causing a minimum inflammatory reaction. Allografts and xenografts usually undergo necrosis and are rejected due to genetic and antigenic incompatibility.

Allograft Reaction:

Rejection of the graft by recipient is called allograft reaction. The most successful organ transplant is that of kidney. Other organ transplants such as bone marrow has been tried but with little success

Mechanism of Transplant Rejection

Changes observed in human renal allograft rejection is almost similar to that observed in mice:

- The graft becomes vascularized within a few days and appears to be accepted initially.
- Between 3-9 days, there is increasing infiltration of the graft by lymphocytes and monocytes with marked reduction in circulation. At this stage, very few plasma cells invade the graft.
- Between 10-11 days, the necrosis begins to be visible to the naked eye.
- On 12th or 13th day, the graft sloughs off completely.

An **antibody** (Ab), also known as an immunoglobulin (Ig), is a large, Y-shaped protein produced mainly by plasma cells that is used by the immune system to neutralize pathogens such as pathogenic bacteria and viruses.

Cell mediated reaction is almost responsible for the rejection. Rejection is brought by helper T cells which activate cytotoxic T cells, macrophages and B cells when second allograft from the same donor is applied on a sensitized recipient, it will be rejected in 5-6 days. In this accelerated rejection of second graft second set reaction, antibodies play an important role along with cell-mediated immunity.

Antibodies are formed abundantly from the 11th day onward of transplantation. Thus a graft is rejected either by sensitized T cells or by circulating antibodies. **Antibody** induces platelet aggregation or type II hypersensitivity reaction (ADCC), Antibody Dependent cell mediated cytotoxicity.

Types and mechanisms of rejection

1. **Hyperacute rejection**: **Pre-formed anti-donor antibodies** bind to graft endothelium immediately after transplantation (thrombosis, ischemic damage, and rapid graft failure)
2. **Acute cellular rejection**: **T cells destroy graft parenchyma** (and vessels) by cytotoxicity and inflammatory reactions
3. **Acute humoral rejection**: **Antibodies damage graft vasculature**
4. **Chronic rejection**: Dominated by arteriosclerosis, T cell reaction and secretion of cytokines induces **proliferation of vascular smooth muscle** cells, associated with **parenchymal fibrosis**

Tissue Typing and Matching in Transplantation

ABO compatibility is essential in all tissue transplantation. Survival of an allograft depends on HLA compatibility. HLA antigens are expressed on the surface of leucocytes (especially lymphoid cells). A full HLA type of an individual contains two haplotypes (haplotype—one strand of chromosome pair), inherited from each parent.

There is reasonable opportunity for genetic matching of siblings of a patient, 25% chance for both haplotypes and 50% chance of one haplotype. With many HLA antigens discovered, there is possibility of thousand halo-types. There is a remote chance that two random individuals will have completely identical haplotypes. The best HLA compatible donors are selected from the family members.

Leucocytes Grouping

The HLA antigens of class I type on leucocytes are identified by sera obtained from:

- Multiparous women who usually possess antibodies to HLA antigens of their husbands
- Recipient of multiple blood transfusion
- Volunteers who are repeatedly skin grafted
- Monoclonal antibodies prepared against HLA antigens.

Cytotoxic Test:

A purified suspension of blood lymphocytes of donor is mixed with the recipient's serum and a panel of standard sera for HLA antigens in presence of complement. When lymphocytes carry the appropriate antigens, the cytotoxic antiserum will combine with the lymphocytes (target cell).

The antigen-antibody complex will activate the complement and damage the cell membrane and the permeability of cell membrane is increased and the same is detected by dye (trypan blue) uptake into the cell, i.e. the dead cells are stained by the dye.

Mixed Leucocyte Culture Test

The test detects class II antigens of HLA complex and cell mediated immunity. Lymphocytes from both donors and recipients are cultured together (co-cultivation). The test is based on the principle that T-lymphocytes—when exposed to incompatible HLA antigen—undergo blastoid transformation, take up thymidine and divide.

Lymphocyte Transfer Test

Peripheral blood lymphocytes of recipient is injected intradermally into several prospective donors. Delayed hypersensitivity reaction at the injection site is indicative of immune response of recipient leucocytes against the donor tissue. Persons showing no response against recipient's leucocytes is chosen as donor.

Detection of Preformed Antibody

Donor's lymphocytes are matched with recipient's serum. If preformed antibodies are present in patient's serum, the lymphocytes of the donor undergo lysis.

Prevention of Graft Rejection in Transplantation

Immunosuppression

When HLA typing of the recipient and donor is well-matched, a state of immunosuppression is produced in the recipient so that the transplant tissue survives for a long time. By irradiation, corticosteroids and anti-lymphocytic serum (ALS) the immunological reactivity may be non-specifically depressed. Such patients are susceptible to infections and prone to development of lymphomas.

Transplantation in Anatomically Protected Sites

There are certain anatomically protected sites where allografts are permitted to survive, e.g. cornea, cartilage and testicle grafting. The avascularity of cornea limits the entry of lymphocytes into the graft.

Immunological Enhancement

The circulating antibody produced against graft antigens can—under circumstances—protect the graft from cell mediated immune response. This phenomenon is known as "Immunological enhancement". If the recipient animal is previously immunized with one or more injections of tissue from the prospective donor and transplant applied subsequently, the transplant survives for longer duration.

Following transplantation of kidney in rats by this method, the graft maintains normal function for a year or more. The immunological enhancement phenomenon can be passively transferred from immunized animal to a normal animal. In certain human kidney transplantations, the method has been used with some success.

CORE CONCEPT: GRAFTS AND TISSUE TRANSPLANTATION

Transplantation is the most regulated field in medicine and requires a detailed knowledge of the clinical as well as the non-clinical issues of a program to succeed in a highly competitive field. Organ and Tissue Transplantation is a series of seven volumes that will go over the science, the administrative and regulatory issues making a contemporary transplant program successful.

Transplants of animal tissue have figured prominently in mythology since the legend of the creation of Eve from one of Adam's ribs. Historical accounts of surgical tissue grafting as part of the cure of patients date back to the early Hindu surgeons who, about the beginning of the 6th century BCE, developed techniques for reconstructing noses from skin flaps taken from the patient's arm. This method was introduced into Western medicine by Italian surgeon Gaspare Tagliacozzi in the 16th century. The flap was left attached to the arm for two to three weeks until new blood vessels had grown into it from the nose remnant. The flap was then severed and the arm freed from the reconstructed nose.

It was found that extremely thin pieces of skin could be cut free and would obtain enough nourishment from the serum in the graft bed to stay alive while new blood vessels were being formed. This free grafting of skin, together with the flap techniques already mentioned, constituted the main therapeutic devices of the plastic surgeon in the correction of various types of defects. Skilled manipulations of such grafts produced surprising improvements in the appearance of those born with malformed faces and in the disfigurements resulting from severe burns. Cornea, which structurally is a modified form of transparent skin, can also be free grafted, and corneal grafts have restored sight to countless blind eyes.

Blood transfusion can be regarded as a form of tissue graft. The blood-forming tissues—bone marrow cells—can also be transplanted. If these cells are injected into the bloodstream, they home to the marrow cavities and can become established as a vital lifesaving graft in patients suffering from defective marrow.

The chief distinguishing feature of organ and limb grafts is that the tissues of the organ or limb can survive only if blood vessels are rapidly joined (anastomosed) to blood vessels of the recipient. This provides the graft with a blood supply before it dies from lack of oxygen and nourishment and from the accumulation of poisonous waste products. As can be seen from the examples cited, living-tissue grafts may be performed for a variety of reasons. Skin grafts can save life in severe burns, can improve function by correcting deformity, or can improve appearances in a cosmetic sense, with valuable psychological benefits. Organ grafts can supply a missing function and save life in cases of fatal disease of vital organs, such as the kidney.

A tissue removed from one part of the body and transplanted to another site in the same individual is called an autograft. Autografts cannot be rejected. Similarly, grafts between identical twins or highly inbred animals—isografts—are accepted by the recipients indefinitely.

Grafts from a donor to a recipient of the same species—allografts or homografts—are usually rejected unless special efforts are made to prevent this. Grafts between individuals of different species—xenografts or heterografts—are usually destroyed very quickly by the recipient.

Tissue or organ grafts may be transplanted to their normal situation in the recipient and are then known as orthotopic—for example, skin to the surface of the body. Alternatively, they may be transplanted to an abnormal situation and are then called heterotopic—for example, kidneys are usually grafted into the lower part of the abdomen instead of into the loin (the back between the ribs and the pelvis), as this is more convenient. If an extra organ is grafted, it is called auxiliary or accessory—for example, a heterotopic liver graft may be inserted without removal of the recipient's own liver.

Grafts are usually performed for long-term effects. Occasionally, the limited acceptance of a skin allograft may be lifesaving by preventing loss of fluid and protein from extensive burned surface in severely ill patients.

The graft also provides a bacteria-proof covering so that infection cannot occur. When the allograft is removed or rejected, the patient may be sufficiently recovered to receive permanent autografts.

Certain tissues, including bone, cartilage, tendons, fascia, arteries, and heart valves, can be implanted even if their cells are dead at the time of implantation or will be rejected shortly thereafter. These are structural implants rather than true grafts or transplants. They are more akin to the stick to which a rose is attached for support—although their support is essential, their function does not depend on biological processes. In fact, xenografts or inert manufactured devices may often be equally suitable substitutes.

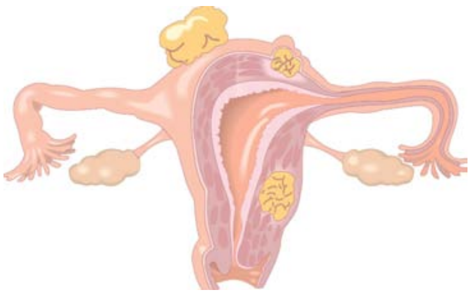
Tissue Transplants

Skin

Most skin grafting is with autografts; the special indication for skin allografts in severely burned patients has been mentioned. Skin allografts seem to be rejected more aggressively than any other tissue, and there are many experimental situations in which skin grafts between two inbred strains of animal fail, although kidney grafts between the same strains survive indefinitely. With autografts, the donor skin is limited to what the patient has available. If allografts were not rejected, skin from cadavers could be used for coverage of burned areas without the need for subsequent autografting.

Flaps

Flaps as used by Tagliacozzi are particularly valuable if adipose (fat) tissue as well as skin has been lost. This is because flaps are thicker than skin grafts; they consist of the full thickness of skin, with fascia and fat, or may also contain muscle. The procedure of raising a flap and keeping the donor site adjacent to the recipient bed can be complicated and uncomfortable for the patient. The cosmetic results are good, and the fat and other tissues under the skin contained in the flap can be used to cover exposed bone or to allow movement in a contracted joint or to fashion a new nose.



Full-thickness free-skin grafts

Full-thickness free-skin grafts, which include dermis and **epidermis**, are the maximum thickness that can survive without a blood supply; they are therefore in some danger of failure to survive. These grafts produce good cosmetic appearances and are especially useful on the face. The main defect of a full-thickness free-skin graft is that, unless it is very small, the donor site from which it comes becomes a defect that needs to be closed in its own right and May itself need skin grafting.

The **epidermis** is the outer layer of the three layers that make up the skin, the inner layers being the dermis and hypodermis.

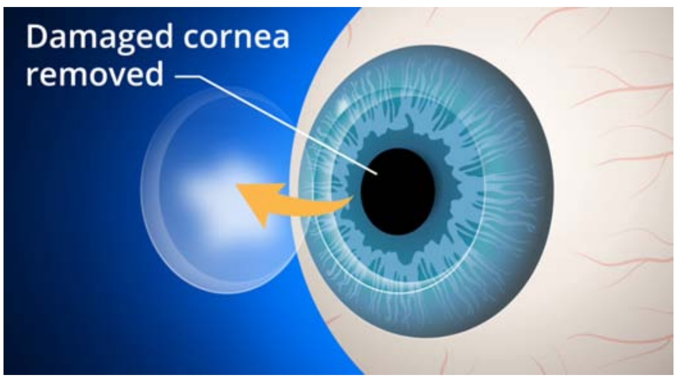
Split or partial-thickness skin grafts

Split, or partial-thickness, skin grafts are by far the most commonly used grafts in plastic surgery. Superficial slices of skin the thickness of tissue paper are cut with a hand or mechanical razor. The graft, which contains living cells, is so thin that it usually gains adequate nourishment directly from the raw surface to which it is applied, and the risk of failure to take (that is, to survive in the new location) is therefore much less than with full-thickness grafts. Another major advantage is that the donor site is not badly damaged. It is tender for only two or three weeks, and it resembles a superficial graze both in appearance and in the fact that healing takes place from the deep layer of the skin left behind. Split skin grafts can be taken quickly from large areas to cover big defects. They tend to have an abnormal shiny reddish appearance that is not as satisfactory cosmetically as the other types of skin graft.

Other tissue transplants

Cornea

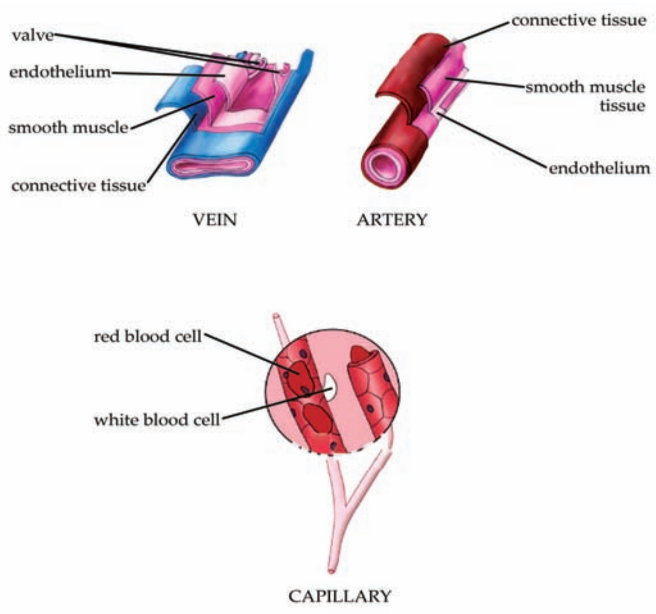
There are certain forms of blindness in which the eye is entirely normal apart from opacity of the front window, or cornea. The opacity may be the result of disease or injury, but, if the clouded cornea is removed and replaced by a corneal transplant, normal vision can result. Since cells of the cornea remain viable for some 12 hours after death, a cornea can be grafted if it is removed within that period. Cooling will slow the process of deterioration, although the sooner the section of cornea is transplanted the better. The graft bed to which a cornea is transplanted has no blood supply. Nourishment comes directly by diffusion from the tissues. Because most rejection factors are carried in the bloodstream, the lack of blood vessels permits most corneal allografts to survive indefinitely without rejection. Rejection can occur if, as sometimes happens, blood vessels grow into the graft.



Blood Vessels

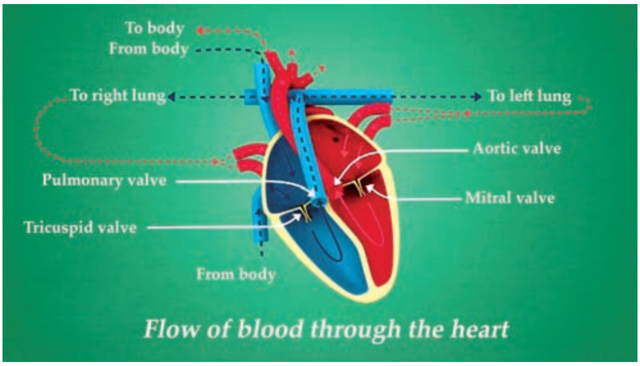
By far the most satisfactory blood-vessel transplant is an autograft, similar in principle to skin autografts. Blood-vessel grafts are frequently used to bypass arteries that have become blocked or dangerously narrowed by fatty deposits, a condition caused by degenerative atherosclerosis (hardening of the arteries). Such atherosclerotic deposits in the coronary and carotid arteries are responsible, respectively, for most heart attacks and strokes. If atherosclerosis affects the main artery of the leg, the result is first pain in the calves and then gangrene of the foot, necessitating amputation of the leg. If dealt with early, the effects of the arterial blockage can often be overcome by removing a nonessential superficial vein from the leg, reversing it so that the valves will not obstruct blood flow, and then joining this graft to the affected artery above and below the block—thus bypassing the obstruction. Grafting for coronary artery bypass has become one of the most common surgical operations in developed countries.

Vein or arterial allografts are far less successful. In time the walls tend to degenerate, and the vessels either dilate, with the danger of bursting, or become obstructed.



Heart valves

Valvular diseases of the heart can be dangerous, since both a blocked valve and a valve that allows blood to leak backward create a strain on the heart that can lead to heart failure. If the valve is seriously damaged, it can be replaced with a xenograft valve or a manufactured mechanical valve. Neither is ideal. Xenograft valves have a normal central blood flow, but after a few years they may become rigid and cease to function. Plastic valves—usually of the ball-valve or trapdoor types—force blood to flow around the surface of the ball or trapdoor flap, and this tends to damage red blood cells and cause anemia. Synthetic heart valves require ongoing anticoagulation therapy.



Bone

When fractures fail to unite, autografts of bone can be extremely valuable in helping the bone to heal. Bone allografts can be used for similar purposes, but they are not as satisfactory, since the bone cells are either dead when grafted or are rejected. Thus, the graft is merely a structural scaffold that, although useful as such, cannot partake actively in healing.

Fascia

Fascia, sheets of strong connective tissue that surround muscle bundles, may be used as autografts to repair hernias. The principle of use is like that for skin.

Nerves

Nerves outside the brain and spinal cord can regenerate if damaged. If the delicate sheaths containing the nerves are cut, however, as must happen if a nerve is partially or completely severed, regeneration may not be possible. Even if regeneration occurs it is unlikely to be complete, since most nerves are mixed motor and sensory paths and there is no control ensuring that regenerating fibres take the correct path. Thus, there will always be some fibers that end in the wrong destination and are therefore unable to function. Defective nerve regeneration is the main reason why limb grafts usually are unsatisfactory. A mechanical artificial limb is likely to be of more value to the patient.

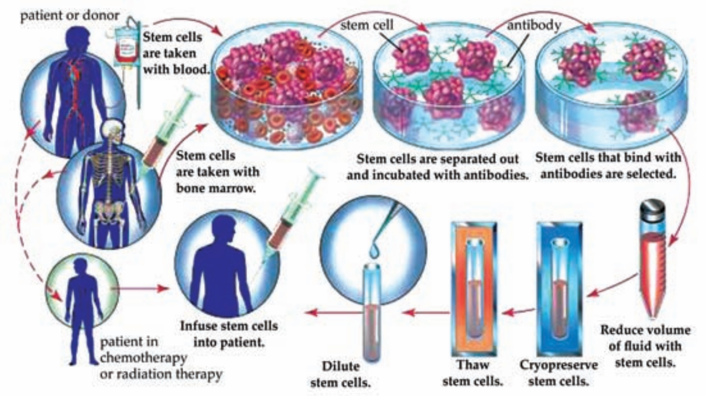
Blood

Blood transfusion has been one of the most important factors in the development of modern surgery. There are many lifesaving surgical procedures that are possible only because the blood loss inevitable in the operation can be made up by transfusion. Blood transfusion is of value in saving life following major injury, bleeding ulcers, childbirth, and many other conditions involving dangerous loss of blood. Purified blood components can be transfused to treat specific defects; for example, platelets are used to correct a low platelet count, and clotting factor VIII is given to counteract the clotting defect in classic **hemophilia**.

Haemophilia, also spelled hemophilia, is a mostly inherited genetic disorder that impairs the body's ability to make blood clots, a process needed to stop bleeding.

Bone marrow

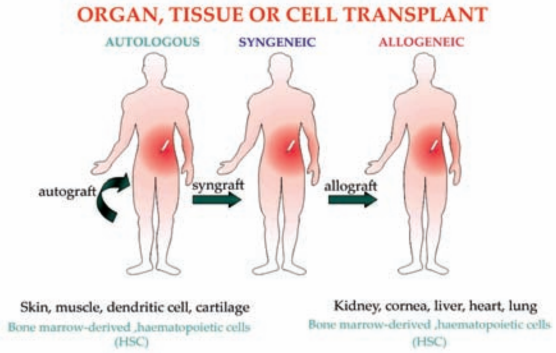
Diseases in which the bone marrow is defective, such as aplastic anemia, may be treated by marrow grafting. Some forms of leukemia can be cured by destroying the patient's bone marrow—the site of the cancerous cells—with drugs and irradiation. Marrow grafting is then necessary to rescue the patient. There is a tendency for the patient to reject the allografted marrow, and there is an additional hazard because immune cells in a marrow graft can react against the patient's tissues, causing serious and sometimes fatal graft-versus-host disease. To avoid these complications, special immunosuppressive treatment is given. The use of monoclonal antibodies (see below Monoclonal antibodies) to selectively remove harmful lymphocytes from the donor marrow has produced encouraging results in preventing graft-versus-host disease.



CORE CONCEPT: ORGAN TRANSPLANTATION

Organ transplantation is a medical procedure in which an organ is removed from one body and placed in the body of a recipient, to replace a damaged or missing organ. The donor and recipient may be at the same location, or organs may be transported from a donor site to another location.

Transplant, also called graft or organ transplant, in medicine, a section of tissue or a complete organ that is removed from its original natural site and transferred to a new position in the same person or in a separate individual. The term, like the synonym graft, was borrowed from horticulture. Both words imply that success will result in a healthy and flourishing graft or transplant, which will gain its nourishment from its new environment.



Definition of Terms Used In Transplantation

You may not be familiar with the terms used in transplantation science. These are explained below

Autograft: cell tissue taken from one site of body and transplanted to the damaged site. Best example is skin grafting in a burns case.

Isograft: it is the term used when tissue/organ taken from one member of the species is transplanted in another member of the same genetic makeup (syngeneic) of the same species. Examples are liver, kidney transplantation done between identical twins.

Allograft: it is the term used when tissue/organ taken from one member of the species is transplanted in another member of the same species. There will be genetic differences between recipient and donor.

Xenograft: Tissue or organ taken from one species is transplanted in another species.

Donor: Person who donates the organ or tissue.

Recipient: Person who receives the tissue or organ from the donor.

MHC: Major histocompatibility complex.

HLA: Histocompatibility locus antigens

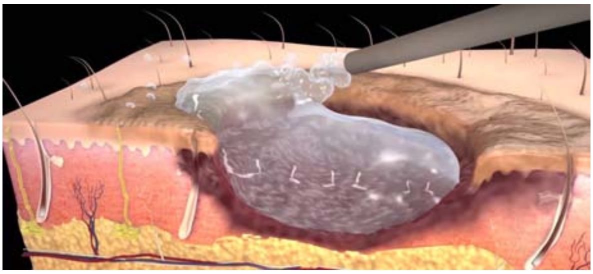
Classification of Grafts

Grafts are classified according to the site, species and type of donor involved in transplantation. Briefly it is described below.

According to species

- Autologous grafts also known as autografts:
  - o Grafts transplanted from one part of the body to another in the same individual
- Syngeneic grafts also known as isografts:
  - o Grafts transplanted between two genetically identical individuals of the same species, e.g. between identical twins

- Allogeneic grafts also known as allografts:
  - o Grafts transplanted between two genetically different individuals of the same species e.g. kidney from son to father, etc.
- Xenogeneic grafts also known as xenografts:
  - o Grafts transplanted between individuals e.g. organ from an animal species transplanted in another species.



Immunologic Basis of Allograft Rejection

In earlier days when there was no understanding about transplantation, one of the major problems in transplantation was the rejection of the graft. Gradually after studies it was found that the grafts rejection is a kind of specific immune response to the antigens in the grafted organ/tissue. The phenomenon was specific and imparted immune memory in the recipient of graft if the graft from the same donor was transplanted again the recipient will reject the graft faster compared to first time.

This was due to the immune memory of the first graft.

Grafts rejection can be of two types:

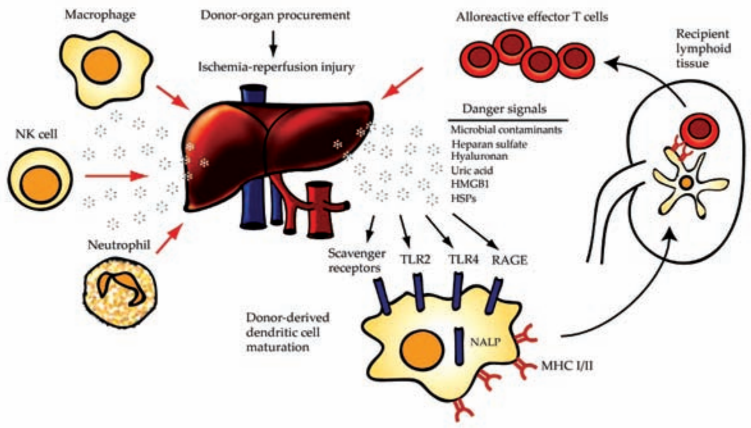
- First set rejection
- Second set rejection

First set rejection

In initial set of experiments in experimental animals the skin graft was transplanted from one animal to another animal of the same species. The graft appeared to be accepted initially as it was vascularized and was normal morphologically and functionally. However it was seen that by fourth day the graft was invaded by lymphocytes, it was inflamed, blood vessels were occluded by thrombi, and ischemia (decreased vascularity) and necrosis (death of graft tissue) set in and graft became shrinked looking like a scab and sloughed off by 10th day. This is first set rejection.

Second set rejection

A second graft transplanted from same donor in the same recipient results in accelerated rejection. The same set of events as described above happens but at an accelerated rate and graft is rejected by 6th day.



Immunologic Basis of Graft Rejection

Transplantation antigens are the antigens which need to be taken into consideration for both the donor and the recipient to ensure successful transplantation. These include Major histocompatibility antigens (MHC molecules), Minor histocompatibility antigens (HLA antigens) and other alloantigens (ABO blood group, and some tissue specific antigens, etc.). Mismatch of these antigens in donor and recipient is a strong factor for graft rejection.

Mechanism of allograft rejection

- Cell-mediated Immunity: Recipient's T cell-mediated cellular immune response against alloantigens on grafts (recognize the allogenetic MHC molecules as foreign) lead to graft rejection.
- Humoral Immunity: Complements activation, antibody dependent cell cytotoxicity, opsonization, etc. This is brought about by enhancing and blocking antibody which results in hyperacute rejection of graft.
- Role of NK cells: NK cells get activated and help in graft rejection.

Effector Mechanisms of Allograft Rejection

- Allograft rejection can be of two types:
- Host versus graft reaction (HVGR)
- Graft versus host reaction/disease (GVHR/GVHD)

Host versus graft reaction (HVGR)

The host rejects the graft as seen in conventional organ transplantation. This may be due to mismatch of MHC and other alloantigens. HVGR can be Hyperacute rejection (occurs within minutes of transplantation), acute rejection (occurs within days to 2 weeks after transplantation, 80-90% of cases occur within 1 month) and chronic rejection (develops months or years after acute rejection reactions have subsided). Antibody against ABO blood type antigen; antibody against VEC antigen and antibody against HLA antigen, either preformed or formed after the graft due to mismatch/ partial match is the basic cause of rejection of graft.

Graft versus host reaction/disease (GVHR/GVHD)

Here the graft is immunologically more powerful and so kind of starts reaction against the host. Examples are bone marrow transplantation and immune cells transplantation. In graft versus host disease (GVHD) the immunocompetent cells in the graft react against the minor histocompatibility antigens and other antigens which can damage the host.

Prevention and Treatment of Allograft Rejection

Detailed laboratory work is extremely important that will ensure that the donor and recipient match genetically and this will help to minimize the graft rejection. The tests include tissue typing. The other modalities to minimize graft rejection are imunosuppressive therapy and induction of immune tolerance.

Tissue typing includes

- ABO and Rh blood typing
- Crossmatching (Preformed antibodies)
- HLA typing
  - o HLA-A and HLA-B
  - o HLA-DR

> A person who had to undergo a kidney removal surgery (nephrectomy); such a surgery is conducted when a person has an infected kidney or diseased, malfunctioning/cancerous growths on the kidney.

Immunosuppressive therapy

Various drugs are given to suppress the immune response of the host. These include:

- Cyclosporine
- Azathioprine, Cyclophosphamide
- Ab against T cell surface molecules
- Anti-inflammatory agents like corticosteroids, etc.

Induction of immune tolerance by:

- inhibition of T cell activation by injecting soluble MHC molecules in the recipient
- administration of anti-IL2R mAb
- administration of Th2 cytokines like anti-TNF-a, anti-IL-2?anti-IFN-g mAb

Xenotransplantation

The demand for transplantation of organs damaged due to various diseases is increasing day by day. The availability of organs is a big issue. Various scientists are trying to transplant organs from animal species in the humans (experimental stage). Such grafts are called xenografts. So xenotransplantation is defined as the transplantation of organ from one species to another species e. g. pig heart in a human being and so on. So far there has been no success.

The kidney

The surgery of kidney transplantation is straightforward, and the patient can be kept fit by dialysis with an artificial kidney before and after the operation. The kidney was the first organ to be transplanted successfully in humans, and experience is now considerable. Effective methods of preventing graft rejection have been available since the 1960s.

Fatal kidney disease is relatively common in young people. When there is deterioration of kidney function, eventually, despite all conventional treatment, the patient becomes extremely weak and anemic. Fluid collects in the tissues, producing swelling, known as edema, because the kidneys cannot remove excess water. Fluid in the lungs may cause difficulty in breathing and puts an excessive strain on the heart, which may already be suffering from the effects of high blood pressure as a result of kidney failure.

Waste products that cannot be removed from the body can cause inflammation of the coverings of the heart and the linings of the stomach and colon. As a result, there may be pain in the chest, inflammation of the stomach leading to distressing vomiting, and diarrhea from the colitis. The nerves running to the limbs may be damaged, resulting in paralysis. Treatment with the artificial kidney followed by kidney grafting can eliminate all these symptoms and has a good chance of permitting the dying person to return to a normal existence. Unfortunately, in most countries only a minority of patients receive this treatment because of a shortage of donor kidneys. Artificial kidney treatment lasting about three to four hours, two to three times a week, removes all the features of kidney failure in one to two months. The patient then is able to leave the hospital and can be assessed as to suitability for a transplant. As has been mentioned, the kidney graft is heterotopic. The diseased kidneys are left in place, unless their continued presence is likely to impair the patient's health after a successful graft.

Transplantation and Postoperative care

The patient may receive a kidney from a live donor or a dead one. Cadaver kidneys may not function immediately after transplantation, and further treatment with the artificial kidney may be required for two to three weeks while damage in the transplanted kidney is repaired. The patient is given drugs that depress immune responses and prevent the graft from being rejected. Immediately after the operation, for the first week or two, every effort is made to keep the patient from contact with bacteria that might cause infection. The patient is usually nursed in a separate room, and doctors and nurses entering the room take care to wear masks and wash their hands before touching the patient. The air of the room is purified by filtration. Close relatives are allowed to visit the patient, but they are required to take the same precautions. When stitches have been removed, the patient is encouraged to get up as much as possible and to be active, but, in the first four months after the operation, careful surveillance is necessary to make sure that the patient is not rejecting the graft or developing an infection. The patient may be discharged from the hospital within a few weeks of the operation, but frequent return visits are necessary for medical examination and biochemical estimations of the blood constituents, to determine the state of function of the graft, and to make sure that the drugs are not causing side effects. Each patient requires a carefully adjusted dose of the immunosuppressive drugs that prevent transplant rejection.

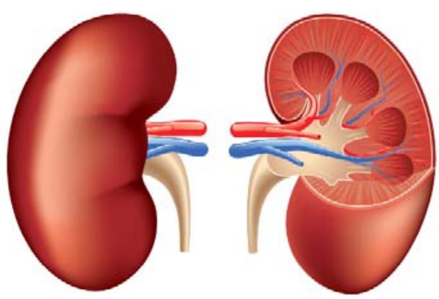
Rejection is caused by the immune system identifying the transplant as foreign, triggering a response that will ultimately destroy the transplanted organ or tissue. Long term survival of the transplant can be maintained by manipulating the immune system to reduce the risk of rejection.

Once the dosage of immunosuppressive drugs is stabilized, patients are encouraged to go back to a normal existence and return to work. The only restrictions are that they must continue to take their drugs and make frequent visits to the outpatient department for surveillance. Patients can return even to heavy work, such as driving a bulldozer, but more often a relatively light job is preferable. Women can bear children after a transplant, and men can become fathers. The course of events is not always so happy, unfortunately. If the patient rejects the kidney or develops a serious infection, it may be necessary to remove the graft and stop administration of the immunosuppressive drugs. The patient must then return to regular maintenance treatment with an artificial kidney but may receive a second or even a third graft.

Data on kidney transplant results

In kidney grafts involving identical twins, in which case rejection is not a problem, recipients have survived more than 25 years. A number of patients who have received kidneys from unrelated cadaver donors have survived more than 20 years, demonstrating that in some patient's rejection can be controlled with standard immunosuppressive drugs. There has been a gradual improvement in the overall results of kidney transplants. The patient mortality has declined to around 10 percent per year, death usually being due to infection associated with immunosuppressive treatment; to complications of dialysis in patients whose kidneys have failed; or to other facets of kidney disease, such as high blood pressure and coronary artery disease. Recipients also face an increased risk of developing malignant growths, particularly lymphomas (malignant diseases of the **lymphatic system**). One cause of this may be related to the effects of immunosuppressive treatment. Kidney graft survival has improved since the introduction of the immunosuppressive agent cyclosporine (also called cyclosporin A), and many centres have achieved a one-year survival rate of about 96 percent and a five-year rate of about 80 percent for patients with a functioning kidney graft from an unrelated cadaver donor. One-year survival rates of 90 to 95 percent have been attained for kidney grafts from well-matched living donors. Thus, the patient who develops permanent kidney failure has a reasonable chance of successful treatment from a combination of dialysis and kidney transplantation. Those fortunate enough to receive a well-functioning kidney can expect complete rehabilitation.

The **lymphatic system** is part of the vascular system and an important part of the immune system, comprising a network of lymphatic vessels that carry a clear fluid called lymph (from Latin, lympha meaning "water") directionally towards the heart.



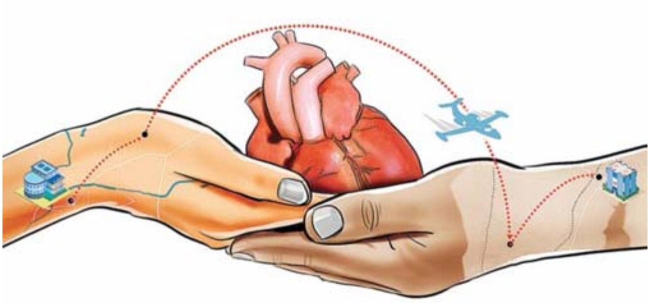
The heart

The heart is a pump with a built-in power supply; it has a delicate regulatory mechanism that permits it to perform efficiently under a wide range of demands. During moments of fear or intense exercise, the heart rate increases greatly, and the contractions become more forceful. Cessation of the heartbeat has been, throughout the ages, the cardinal sign of death. Thus, it is perhaps not so surprising that there was an intense public interest when the first attempts were made at transplanting a human heart. The objectives of heart transplantation, nevertheless, are the same as those of other organ grafts.

Donor and recipient are carefully matched prior to transplantation to minimise the risk of rejection. They are matched based on their blood group, tissue typing, and how the recipient's blood serum reacts to donor cells.

One of the most important advances in surgery after World War II was in direct operations on the heart. Heart valves are repaired or replaced with artificial valves, and techniques have been developed so that the heart can be stopped and its function taken over temporarily by an electrical pump. If, however, the muscle of the heart is destroyed, as occurs in certain diseases, the only operation that can cure the patient is to replace the heart with a graft or possibly an artificial heart. Blockage of the coronary arteries and certain other heart muscle diseases can kill the patient because the muscle of the heart cannot contract properly. A patient with one of these diseases who is close to dying is, therefore, a possible recipient for a heart transplant.

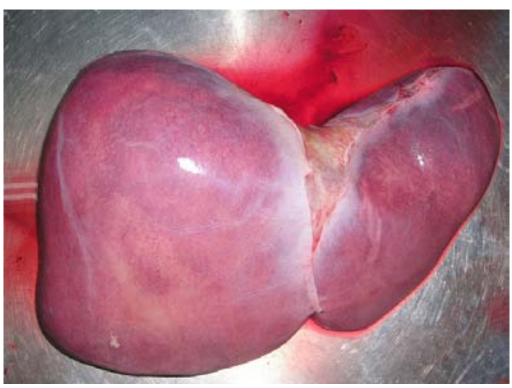
A group of American investigators perfected the technique of heart transplantation in the late 1950s. They showed that a transplanted dog's heart could provide the animal with a normal circulation until the heart was rejected. The features of rejection of the heart are similar to those of the kidney. The cells that produce immune reactions, the lymphocytes, migrate into the muscle cells of the heart, damage it, and also block the coronary arteries, depriving the heart of its own circulation. Some of the lymphocytes (i.e., B cells) also secrete antibodies that are toxic. In most experiments it was more difficult to prevent rejection of the heart than of the kidney. Despite this, rejection was prevented for long periods in animals. Based on this experimental work, the next step was to transplant a human heart into a patient dying of incurable heart disease. This step was taken in 1967 by a surgical team in Cape Town, South Africa.



In the years immediately following the first transplant, numerous heart allografts were performed at medical centres throughout the world. Unfortunately, many recipients succumbed to rejection of the transplanted organ. Furthermore, the heart is more sensitive to lack of blood than the kidneys are; it must be removed from the donor more quickly and can be preserved without damage for only a short period of time. Because of these difficulties—particularly the problem of rejection—the number of heart transplants performed worldwide dropped considerably after the initial excitement abated. Steady advances in detecting and treating rejection were made throughout the 1970s, however, and the introduction of the immunosuppressant cyclosporine in the 1980s brought even further improvements in the long-term survival rates for heart graft recipients. Interest in the procedure revived, and numerous heart transplants were performed. A number of patients have lived five or more years after the operation, and heart grafting has become an accepted therapy for otherwise incurable heart disease.

The liver

The liver is a complicated organ that produces clotting factors and many other vital substances in the blood and that removes many wastes and toxic by-products from the circulation. It is, in effect, a chemical factory. The two categories of fatal liver disease that may be treated by liver grafting are nonmalignant destructive diseases of the liver cells—for example, cirrhosis—and primary liver cancer affecting either the main liver cells or the bile ducts. The liver is extremely sensitive to lack of blood supply and must be cooled within 15 minutes of the death of the donor. The operation can be difficult, since the liver is rather large and of complex structure. Both its removal from the cadaver and its grafting into the recipient are major surgical operations. The operation is more difficult in humans than in animals—particularly, the removal of the diseased liver from the recipient. This may be much enlarged and adherent to surrounding structures so that its removal may result in serious bleeding. Once transplanted, the liver must function immediately or the patient will die. There is no treatment available that is comparable to the use of the artificial kidney for kidney disease.

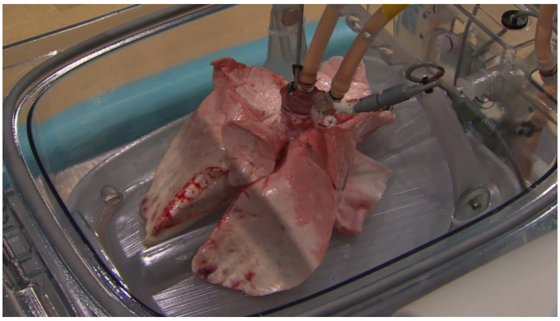


If the liver functions well immediately after transplantation, the rest of the management is similar to that followed in kidney operations, and the same drugs are given. Many early liver transplantation operations failed, but an increasing number have successfully restored dying patients to normal existence. Children do especially well following liver transplantation. The commonest fatal liver disease in childhood is a congenital deficiency of bile ducts called biliary atresia. Several centres have obtained an 80 to 90 percent one-year survival in children after liver grafting, although up to 25 percent of these patients may require retransplantation because of failure of the first graft.

The lung

Chronic fatal disease of the lung is common, but the progress of the disease is usually slow, and the patient may be ill for a long time. When the lung eventually fails, the patient is likely to be unfit for a general anesthetic and an operation. The function of the lung is to allow exchange of gases between the blood and the air. The gas passes through an extremely fine **membrane** lining the air spaces. This exposure to air makes the lungs susceptible to infection, more so than any other organs that have been grafted. It is consequently not surprising that infection has caused failure of many lung transplants. Even a mild rejection reaction can severely damage the gas-exchange membrane, and the patient may die before the rejection is reversed. The actual ventilation of the lungs by rhythmic breathing is a complicated movement controlled by nerves connecting the brain to the lungs and to the muscles that produce the breathing. Cutting the nerves can interfere with the rhythmicity of breathing, and this may be an important cause of the difficulties of successfully transplanting both lungs. Nevertheless, these difficulties have been overcome. If only one lung is transplanted, however, the patient's own diseased lung may interfere with the function of the graft by robbing it of air and directing too much blood into the graft.

A **membrane** is a selective barrier; it allows some things to pass through but stops others. Such things may be molecules, ions, or other small particles.



The heart and lungs

The technique of transplanting the heart and both lungs as a functioning unit was developed in animal experiments at Stanford Medical Center in California. Despite the technical feasibility of the operation, rejection could not be controlled by conventional immunosuppression. With the availability of cyclosporine, researchers were able to obtain long-term survivors with combined heart–lung transplants in primate species. Applications to human patients have been remarkably successful. Approximately two-thirds of the patients who initially received transplants at Stanford survived. Other centres subsequently adopted this form of treatment for patients with severe lung fibrosis and failure of the right side of the heart, which pumps blood into the lungs. Unfortunately, many organ donors have been maintained on ventilators, a process that frequently leads to lung infections; as a consequence, the availability of donor heart–lung units is quite limited.

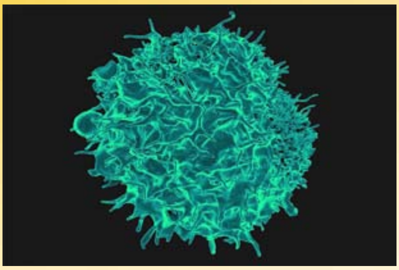
The pancreas

The pancreas consists of two kinds of tissues: endocrine and exocrine. The latter produces pancreatic juice, a combination of digestive enzymes that empty via a duct into the small intestine. The endocrine tissue of the pancreas—the islets of Langerhans—secrete the hormones insulin and glucagon into the bloodstream. These hormones are vital to the regulation of carbohydrate metabolism and exert wide-ranging effects on the growth and maintenance of body tissues. Insufficient insulin production results in type 1 diabetes mellitus, a disease that is fatal without daily injections of insulin. Even with insulin therapy, many diabetics suffer kidney failure and blindness due to the disease's effects on the small blood vessels. A pancreas transplant can potentially prevent the progression of these complications.

Much effort has been devoted to removing the islets of Langerhans from the pancreas with a view to grafting the separated islets or even the isolated insulin-producing beta cells. Unfortunately, it is very difficult to obtain sufficient islets from the fibrotic human pancreas, and it appears that isolated islets are highly susceptible to rejection. A number of clinical attempts at islet grafting have been made.

Transplanting the vascularized pancreas has been more encouraging. It is customary to graft the body and tail of the pancreas; that is, half the pancreas is transplanted, using the splenic artery and vein for vascular anastomosis. One of the difficulties with this procedure has been dealing with the digestive juice produced by the transplanted pancreas. A further complicating factor has been the fact that corticosteroids—frequently used for immunosuppression in transplant patients—aggravate diabetes. The availability of cyclosporine permitted the avoidance of corticosteroids. Later, triple-drug immunosuppression, using cyclosporine, tacrolimus, and mycophenolate mofetil, was introduced.

Pancreas transplantation is particularly attractive when a patient with diabetic kidney failure can receive a kidney and pancreas graft from the same donor. A technique with encouraging early results has been to insert the pancreas graft very close to the patient's own pancreas in the so-called paratopic position. This allows drainage of insulin directly into the liver, while the pancreatic juice is diverted into the stomach, where the digestive enzymes are inhibited by stomach acid. It is certainly most gratifying to patients who have been undergoing regular dialysis and taking insulin to be free from both these onerous treatments and to be permitted to eat and drink without restriction. The five-year functional survival rate for pancreas transplant is about 90 percent. It is of interest that the vascularized pancreas probably is less susceptible to rejection than the kidney.



CORE CONCEPT: ORGAN DONATION: SPECIAL LEGAL AND ETHICAL PROBLEMS

Organ donation is giving an organ to help someone who needs a transplant. Transplants can save or greatly enhance the lives of other people. But this relies on donors and their families agreeing to donate their organ.

Legal aspects

In countries with established transplant programs, organ transplantation is highly regulated. Of particular concern is organ donation, with legal, medical, and social issues surrounding the procurement of organs, without compensation, for transplantation? Many of those issues are overcome by organ registries, in which individuals choose to become organ donors.

Through such registries, donors can indicate which organs they are willing to donate upon death. Whether a person is a registered organ donor can then be indicated on a personal identification card (e.g., a driver's license), authorizing organ procurement once the individual is deceased. In the absence of legal consent via registration as an organ donor, organ procurement representatives are required to consult with next of kin for authorization to obtain organs from the deceased person.

Ethical Considerations

Defining death

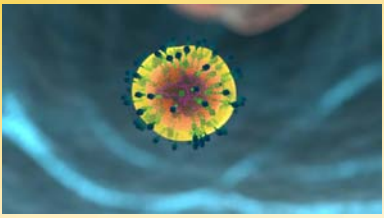
Transplantation raises important ethical considerations concerning the diagnosis of death of potential donors, and, particularly, how far resuscitation should be continued. Every effort must be made to restore the heartbeat to someone who has experienced sudden cardiac arrest or to restore breathing to someone who cannot breathe. Artificial respiration and massage of the heart, the standard methods of resuscitation, are continued until it is clear that the brain is dead. Most physicians consider that beyond this point efforts at resuscitation are useless.

In many countries, the question of how to diagnose brain death—that is, irreversible destruction of the brain—has been debated by neurologists and other medical specialists. Most of these experts agree that when the brainstem is destroyed, there can be no recovery. The brainstem controls the vital function of breathing and the reflexes of the eyes and ears, and it transmits all information between the brain and the rest of the body. Most countries have established strict guidelines for how brainstem death is to be diagnosed and what cases are to be excluded—for example, patients who have been poisoned, have been given drugs, or have developed **hypothermia**. The neurological signs of brainstem death must be elicited by a trained clinician who is not concerned directly with the transplant operation. These signs are recertified after an interval, and, if there is the slightest doubt, further reverifications are made until the criteria are unequivocally met. The guidelines are not seriously disputed, and there has never been a recovery in a case that fulfilled the criteria of brainstem death.

Hypothermia is a medical emergency that occurs when your body loses heat faster than it can produce heat, causing a dangerously low body temperature.

Shortage of donors

Another area of ethical concern is the dilemma posed by the shortage of donor organs. Advances in immunosuppressive therapy have put increasing pressure on the supply of donor organs, and medical personnel sometimes find themselves having to determine who among the potential recipients should receive a lifesaving graft. Furthermore, there is a danger of commercial interests becoming involved with people willing to sell their organs for personal gain, and there is definite risk of illegal organ trafficking, in which organs are procured from unwilling donors and then sold to facilities that offer transplant services.



CORE CONCEPT: REJECTION

Transplant rejection is a process in which a transplant recipient's immune system attacks the transplanted organ or tissue. Transplant rejection can be lessened by determining the molecular similitude between donor and recipient and by use of immunosuppressant drugs after transplant.

Humans possess complex defense mechanisms against bacteria, viruses, and other foreign materials that enter the body. These mechanisms, which collectively make up the immune system, cannot, unfortunately, differentiate between disease-causing microorganisms and the cells of a lifesaving transplant. Both are perceived as foreign, and both are subject to attack by the immune system. This immune reaction leads to rejection, the greatest problem in successful tissue and organ grafting.

Immune responses

In order to understand why rejection occurs and how it may be prevented, it is necessary to know something of the operations of the immune system. The key cells of the immune system are the white blood cells known as lymphocytes. These are of two basic types: T lymphocytes (T cells) and B lymphocytes (B cells). These cells have the capacity to distinguish "self" substances from such "nonself" substances as microorganisms and foreign tissue cells. Substances that provoke an immune reaction are recognized by the presence of certain molecules, called antigens, on their surface.

T lymphocytes are responsible for cell-mediated immunity, so named because the T cells themselves latch onto the antigens of the invader and then initiate reactions that lead to the destruction of the nonself matter. B lymphocytes, on the other hand, do not directly attack invaders. Rather, they produce antibodies, proteins that are capable of initiating reactions that weaken or destroy the foreign substance. The overall immune reaction is exceedingly complex, with T lymphocytes, B lymphocytes, macrophages (scavenger cells), and various circulating chemicals waging a coordinated assault on the invader.

Transplant rejection is generally caused by cell-mediated responses. The process usually occurs over days or months, as the T lymphocytes stimulate the infiltration and destruction of the graft. The transplant may be saved if the cell-mediated reactions can be suppressed. Antibody attack of transplanted tissues is most apparent when the recipient has preexisting antibodies against the antigens of the donor.

Selection of Donor and Tissue Matching

The factors that provoke graft rejection are called transplantation, or histocompatibility, antigens. If donor and recipient have the same antigens, as do identical twins, there can be no rejection. All cells in the body have transplantation antigens except the red blood cells, which carry their own system of blood-group (ABO) antigens. The main human transplantation antigens—called the major histocompatibility complex, or the HLA (human leukocyte antigens) system—are governed by genes on the sixth chromosome. HLA antigens are divided into two groups: class I antigens, which are the target of an effector rejection response; and class II antigens, which are the initiators of the rejection reaction. Class II antigens are not found in all tissues, although class I antigens are. Certain macrophagelike tissue cells—called dendritic cells because of their finger-like processes—have a high expression of class II antigens. There has been much interest in trying to remove such cells from an organ graft, so that the rejection reaction will not be initiated.

Tissue typing involves the identification of an individual's HLA antigens. Lymphocytes are used for typing. It is important also that the red blood cells be grouped, since red-cell-group antigens are present in other tissues and can cause graft rejection. Although transplantation antigens are numerous and complicated, the principles of tissue typing are the same as for red-cell grouping. The lymphocytes being typed are mixed with a typing reagent, a serum that contains antibodies to certain HLA antigens. If the lymphocytes carry HLA antigens for which the reagent has antibodies, the lymphocytes agglutinate (clump together) or die. If the lymphocytes of both the recipient and the potential donor are killed by a given serum, then, as far as that typing serum is concerned, the individuals have antigens in common. If neither donor nor recipient lymphocytes are affected, then donor and recipient lack antigens in common. If the donor lymphocytes are killed but not those of the recipient, then an antigen is present in the donor and is missing from the recipient. Thus, by testing their lymphocytes against a spectrum of typing sera, it is possible to determine how closely the recipient and donor match in HLA antigens. As a final precaution before grafting, a direct crossmatch is performed between the recipient's serum and donor lymphocytes. A positive cross match usually contraindicates the donor–recipient transplant under consideration.

There is now considerable knowledge concerning the inheritance of transplantation antigens, but, even so, tissue typing is not sufficiently advanced to give an accurate prediction of the outcome of a graft in an individual case, particularly when the donor and recipient are not related to one another. In accordance with Mendelian laws of inheritance, a person obtains one of a pair of chromosomes from each parent. Therefore, a parent-to-child transplant will always be half-matched for transplantation antigens. Siblings have a one-in-four chance of a complete match of the HLA antigens, a one-in-four chance of no match, and a one-in-two chance of a half-match.

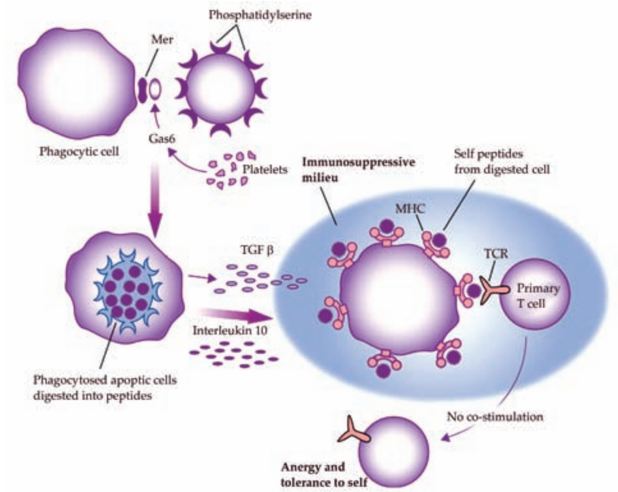
The blood transfusion effect

Following a blood transfusion, some patients become sensitized to the transplantation antigens of the donor, so it was expected that prior blood transfusion could only harm the recipient's prospects for a successful organ graft. Careful analysis of results, however, showed the contrary. Although a great deal of effort has been expended to determine the mechanisms involved, researchers still do not know how the immune system is modified by prior blood transfusions. Most centres now give blood transfusions before transplantation, though some patients do develop HLA antibodies against a wide spectrum of the population and therefore become very difficult to transplant. This pool of highly sensitized patients is getting larger throughout the world, not only from blood transfusions but also from patients who have rejected kidney grafts and are back on dialysis and from women who have had multiple pregnancies.

A special application of the blood transfusion effect involves repeated small blood transfusions from a potential donor who is a close relative of the patient. If sensitization does not occur, subsequent kidney graft results are excellent. Some patients, however, develop a positive cross match to donor lymphocytes and cannot receive a graft from that donor. This, combined with the increasing number of patients who readily develop HLA antibodies, has resulted in a decline in the use of donor-specific blood transfusion.

Immunosuppression

The aim of transplantation research is to allow the recipient to accept the graft permanently with no unpleasant side effects. With current drugs that are used for this purpose, after some months the dosage can often be reduced and sometimes even stopped without the graft's being rejected. In such a case, the patient is no longer as susceptible to infections. There would appear to be adaptation of the recipient toward the graft and the graft toward the recipient. The adaptation is probably akin to desensitization, a process used sometimes to cure patients suffering from certain types of allergies by giving them repeated exposure to small doses of the allergen to which they are sensitive.



Azathioprine

Azathioprine is one of the most widely used immunosuppressive agents; it also has been used to treat leukemia. It can be given by mouth, but the dose must be carefully adjusted so that the blood-cell-forming tissues in the bone marrow are not damaged, which could lead to infections and bleeding. The white blood cell and platelet counts need to be determined frequently to make sure that azathioprine is not being given in too large a dose. It is an extremely valuable drug and has been the basis of most immunosuppressive regimens in patients with organ grafts. At first, high doses are given, but eventually the doses may be reduced. Even years after transplantation, small doses of azathioprine may still be needed to maintain coexistence between graft and host.

Corticosteroids

Cortisone and its relatives, prednisone and prednisolone, are very useful in patients with organ grafts. They can be given by mouth, but, although not damaging to the blood-forming cells, they do predispose the body to infection, cause stunted growth in children, and have other injurious effects. Persons receiving these substances may develop complexion problems with swollen faces and may tend to gain weight and become diabetic, and their bones may become brittle. Few recipients of organ transplants, however, can do without corticosteroids, particularly during an active rejection crisis.

Antilymphocyte and antithymocyte globulins

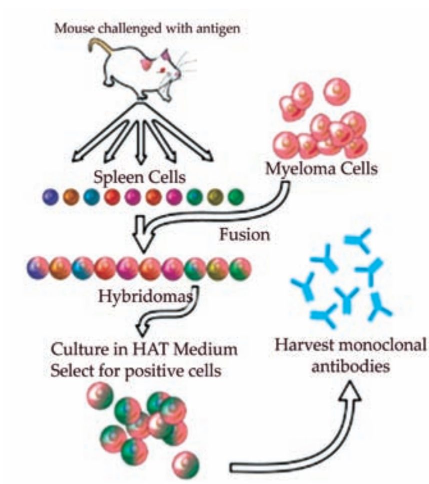
If rabbits receive repeated injections of mouse lymphocytes, they become immunized and develop antibodies against the mouse cells. The serum from the rabbits' blood can be injected into mice and will often prevent them from rejecting grafts, both from other mice and even, sometimes, from other species. Such antilymphocyte serums can be produced between a variety of species, but in higher mammals, particularly humans, it has been difficult to obtain a powerful immunosuppressive serum without side effects of toxicity.

The activity of the antilymphocyte serum lies in its gamma globulin, which contains the antibody proteins. Antilymphocyte globulin is used in humans but contains many proteins that are ineffective and may be harmful. It can be added to immunosuppressant and steroid treatment regimens, and it is extremely useful in treating rejection crisis in kidney graft recipients who have not responded to corticosteroids. The horse has usually been used to produce antilymphocyte serum for the treatment of human patients, but some persons are sensitive to horse proteins and become extremely ill when treated with horse serum. Such patients may, however, be successfully treated with rabbit antilymphocyte globulin.

Antithymocyte globulin is a similar antibody treatment but is produced in animals, usually rabbits or horses, through inoculation with human thymocytes. Antithymocyte globulin triggers a significant reduction in levels of circulating T lymphocytes. It commonly is used to induce immune suppression for kidney transplantation, helping prevent graft rejection.

Monoclonal antibodies

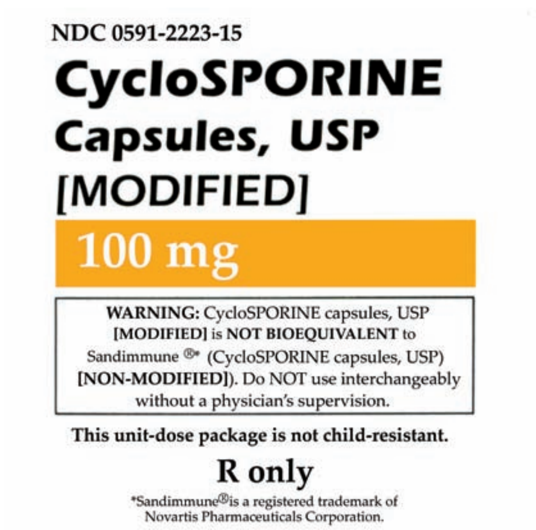
An important development in antibody production followed the discovery that an antibody-forming lymphocyte can be fused with a cancerous bone marrow cell. The resulting hybrid cell produces the antibody specified by its lymphocyte progenitor, while from the cancer cell it obtains the characteristic of multiplying indefinitely in laboratory cultures. The culturing of the hybrid yields a clone of cells that produce one specific antibody—a “monoclonal” antibody. Such agents are exclusively specific in action and there is no theoretical limit to the number of antibodies that can be produced by different hybrid cell lines. Monoclonal antibodies can be regarded as highly specific antilymphocyte globulin without many of the unwanted materials that are present in the ordinary polyclonal antilymphocyte serum described above. Some monoclonal antibodies have been produced that are effective as immunosuppressive agents in humans.



Cyclosporine

Cyclosporine was found as a natural product of an earth fungus by researchers at the pharmaceutical company Sandoz Laboratories. It is a stable cyclic peptide with powerful immunosuppressive activity affecting especially the T lymphocytes. Cyclosporine was found to prevent organ graft rejection in a number of animal species. When the drug was used in humans, the expected immunosuppressive effect was again observed. It has been used in recipients of all types of organ grafts with improved immunosuppressive results. Cyclosporine can be toxic to the human kidney, however, and may cause permanent renal damage. The drug also increases the growth of hair on the face and body. It is a difficult drug to use because, being fat soluble, its absorption is variable and each patient needs to be individually studied to ensure that the dosage is adequate but not excessive.

It is clear that none of the agents so far used to prevent rejection is ideal. No one would use such dangerous agents except as a last resort in a desperate situation. This, unfortunately, is the exact plight of a person in need of a vital organ transplant. Immunosuppression is, however, much more effective and less dangerous than it used to be, and advances with chemical derivatives, in particular monoclonal antibodies and nontoxic analogues of cyclosporine, have brought significant improvements in immunosuppressive therapy.



Organ and Tissue Banks

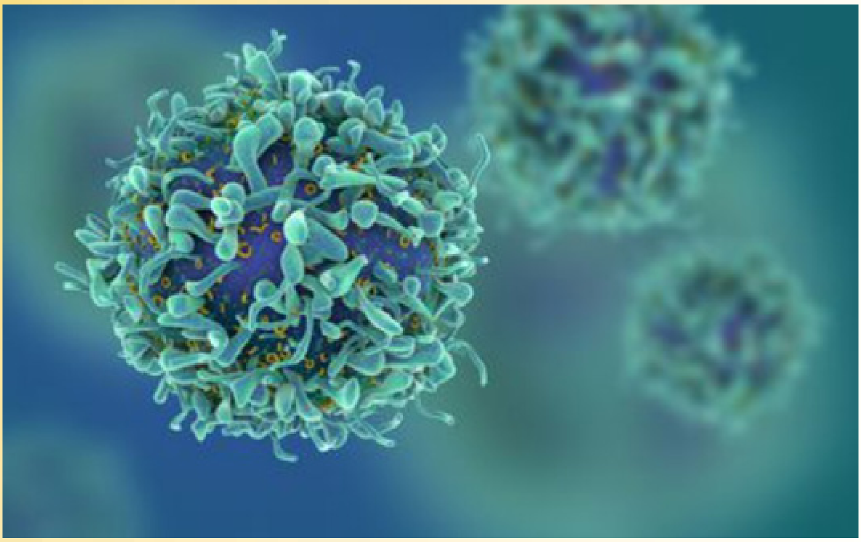
Without a blood supply, organs deteriorate rapidly. Cooling can slow down the process but cannot stop it. Organs differ in their susceptibility to damage. At body temperature, irreversible destruction of the brain occurs after more than 3 to 5 minutes; of the heart, liver, pancreas, and lung, after 10 to 30 minutes; of the kidney, after 50 to 100 minutes; and of the skin and cornea, after 6 to 12 hours. The shorter the time the organ is deprived of its blood supply, the better. Although the cornea can be removed for grafting at relative leisure, every minute is of vital importance for a liver transplant. When a kidney is removed from a living donor, it is not necessary to use elaborate preservation techniques. The operations on the donor and recipient are performed at the same time, and the recipient is prepared to receive the graft by the time that the donor organ is removed. Cadaver kidneys are removed as soon as possible after the donor's death, preferably within an hour. Cool solutions are infused into the blood vessels of the kidney, which is then kept at 4 °C (39 °F) in a refrigerator or surrounded by ice in a vacuum flask. At the same time, the recipient is prepared for operation. Kidneys can be conserved in this simple way for 24 to 48 hours with little deterioration, and during this time they can be moved for long distances. For a kidney to be preserved from 48 to 72 hours, a complicated machine is required to provide artificial circulation. Cool oxygenated physiological solutions with the same osmotic pressure as blood are passed through the blood vessels of the kidney. The imperfections of the machinery mean that there is a slow deterioration of the organ that does not occur normally in the body. To keep a kidney undamaged for longer than 72 hours is difficult. **Blood cells**, spermatozoa, and certain other dissociated tissue cells can be frozen to subzero temperatures and kept alive indefinitely. Special preserving fluids will prevent cell destruction by ice crystals, but these fluids have damaging effects if introduced into whole organs such as the kidney.

A **blood cell**, also called a haematopoietic cell, hemocyte, or hematocyte, is a cell produced through hematopoiesis and found mainly in the blood.

For the heart, lung, liver, and pancreas, grafting is performed as quickly as possible, preferably within eight hours for the liver and pancreas, four hours for the heart, and two hours for the combined heart-and-lung graft. Much research will be necessary before it is possible to keep organs banked in the way that blood can be stored.

A CONCEPT OF IMMUNOSUPPRESSION

Immunosuppression refers to the intentional suppression or modulation of the immune system's activity. This is typically done through medications or treatments to prevent or reduce the immune response. Immunosuppression has various medical applications, including preventing organ transplant rejection, treating autoimmune diseases, and managing inflammatory conditions. While immunosuppression can be beneficial in certain situations, it also comes with potential risks and considerations.



Key points about immunosuppression:

1. Purpose and Goals: The primary purpose of immunosuppression is to reduce the activity of the immune system. This can be desirable in cases where the immune system's response is harmful, such as in organ transplantation, where the immune system might reject the transplanted organ, or in autoimmune diseases, where the immune system attacks the body's own tissues.
2. Transplantation: Immunosuppressive drugs are commonly used after organ transplantation to prevent the recipient's immune system from recognizing the transplanted organ as foreign and attacking it. These drugs help promote the acceptance and long-term survival of the transplanted organ.
3. Autoimmune Diseases: In conditions like rheumatoid arthritis, lupus, and multiple sclerosis, where the immune system mistakenly attacks healthy tissues, immunosuppressive therapies can help reduce inflammation and tissue damage.

4. Inflammatory Conditions: Immunosuppression is also used to manage severe inflammatory conditions, such as certain forms of inflammatory bowel disease (IBD) or severe asthma, where excessive immune responses can lead to tissue damage.
5. Immunosuppressive Drugs: Medications used for immunosuppression include corticosteroids, calcineurin inhibitors, mTOR inhibitors, and monoclonal antibodies that target specific immune cells or molecules. These drugs can be administered orally, intravenously, or through other routes.
6. Risks and Side Effects: Immunosuppression weakens the body's ability to fight infections. Patients on immunosuppressive therapy are more susceptible to bacterial, viral, and fungal infections. Long-term use of these medications can also increase the risk of certain cancers, metabolic disorders, and other complications.
7. Individualized Treatment: The dosage and duration of immunosuppressive therapy vary depending on the specific medical condition, patient's response, and potential side effects. Treatment plans are often tailored to the individual patient's needs.
8. Monitoring and Care: Patients on immunosuppressive therapy require careful monitoring by healthcare professionals to assess the effectiveness of treatment, manage potential side effects, and adjust medications as needed.
9. Balancing Act: Finding the right balance between suppressing the immune response and maintaining immune function is essential. Too much immunosuppression can lead to increased susceptibility to infections, while too little can result in rejection of transplanted organs or exacerbation of autoimmune diseases.

SUMMARY

- Transplantation is the process of moving cells, tissues, or organs, from one site to another, either within the same person or between a donor and a recipient.
- Matching of blood group antigens other than A, B, AB, O is not critical for transplantation, because they are expressed only on red Blood cells.
- Transplantation may be fresh organs or stored organs in orthotropic grafts, the transplant is applied in anastomosed normal site (e.g. skin graft), whereas heterotopic grafts are placed in anatomically abnormal sites (e.g.. thyroid transplanted in subcutaneous tissue).
- A tissue removed from one part of the body and transplanted to another site in the same individual is called an autograft.
- The surgery of kidney transplantation is straightforward, and the patient can be kept fit by dialysis with an artificial kidney before and after the operation.
- Cadaver kidneys may not function immediately after transplantation, and further treatment with the artificial kidney may be required for two to three weeks while damage in the transplanted kidney is repaired.
- In kidney grafts involving identical twins, in which case rejection is not a problem, recipients have survived more than 25 years.
- Chronic fatal disease of the lung is common, but the progress of the disease is usually slow, and the patient may be ill for a long time.
- The pancreas consists of two kinds of tissues: endocrine and exocrine. The latter produces pancreatic juice, a combination of digestive enzymes that empty via a duct into the small intestine. The endocrine tissue of the pancreas—the islets of Langerhans—secrete the hormones insulin and glucagon into the bloodstream.

MULTIPLE CHOICE QUESTIONS

1. **What is the primary concern of transplantation immunology?**
   a. Creating artificial organs
   b. Matching blood types
   c. Ensuring donor-recipient histocompatibility
   d. Using stem cells for tissue regeneration
2. **Graft rejection refers to:**
   a. The integration of a transplanted organ
   b. The success of a transplant procedure
   c. The immune system's attack on transplanted tissue
   d. The transplantation of tissues between closely related individuals
3. **Which type of graft rejection occurs within minutes to hours after transplantation?**
   a. Hyperacute rejection
   b. Acute rejection
   c. Chronic rejection
   d.Delayed-type hypersensitivity
4. **Immunosuppressive medications are used in transplantation to:**
   a. Enhance the immune response
   b. Induce graft-versus-host disease (GVHD)
   c. Dampen the immune response to prevent graft rejection
   d. Stimulate antibody production
5. **The process of crossmatch testing in transplantation involves:**
   a. Determining the patient's blood type
   b. Testing the compatibility of donor and recipient cells
   c. Assessing the physical fitness of the recipient
   d. Measuring the level of immunosuppression required

Answers

1. (c) 2. (c) 3. (a) 4. (c) 5. (b)

REVIEW QUESTIONS

1. What is transplantation immunology?
2. Discuss about tissue typing and matching in transplantation.
3. Explain the organ transplantation.
4. Define the antilymphocyte and antithymocyte globulins.
5. Focus on monoclonal antibodies.
6. What is cyclosporine?
7. Explain the blood transfusion effect.

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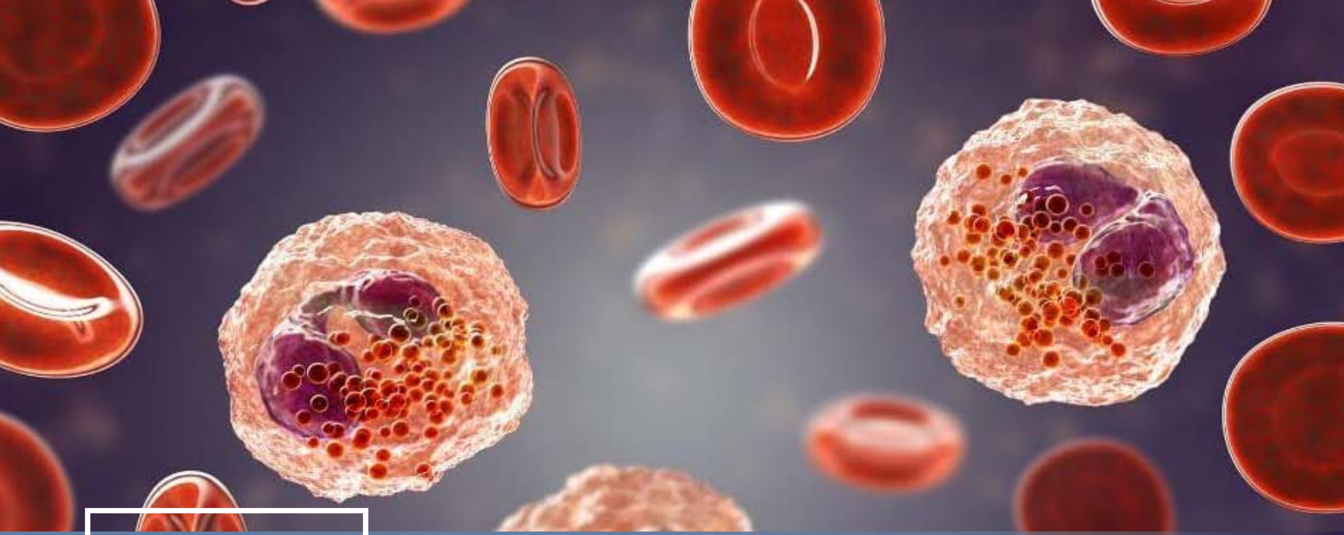
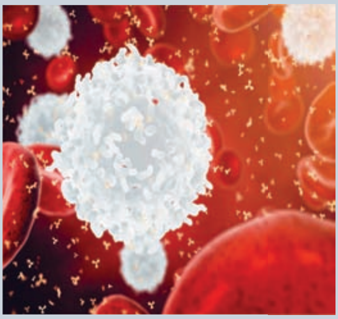


CHAPTER 7

IMMUNE SYSTEM DISORDERS AND DISEASES



The concept of immune system disorders and diseases revolves around the disruptions, dysfunctions, and malfunctions that can occur within the immune system, leading to a wide range of health conditions. The immune system, which is responsible for defending the body against pathogens and maintaining its overall health, can sometimes fail to operate properly, resulting in various disorders and diseases. Here's an overview of the key concepts within this field:

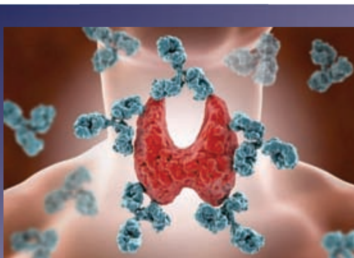
After studying this chapter, you will understand the core concept of:

- Autoimmunity and autoimmune diseases.
- Immunodeficiency disorders.
- Hypersensitivity reactions (Type I to Type IV).
- Immunology in infectious diseases.

INTRODUCTION

The immune system is a complex network of cells, tissues, and organs that work together to defend the body against harmful pathogens, such as bacteria, viruses, fungi, and parasites. It plays a critical role in maintaining overall health by identifying and eliminating foreign invaders while also recognizing and removing abnormal or damaged cells within the body.

However, in some cases, the immune system may malfunction, leading to immune system disorders and diseases. These conditions can arise from various factors, including genetic predisposition, environmental triggers, and autoimmune responses. Immune system disorders can affect different parts of the immune system, leading to a wide range of symptoms and health consequences.



CORE CONCEPT: AUTOIMMUNITY AND AUTOIMMUNE DISEASES

Autoimmunity is a condition in which the immune system mistakenly attacks and damages the body's own healthy tissues and cells, considering them as foreign invaders. Normally, the immune system is designed to recognize and eliminate harmful pathogens, such as bacteria and viruses, while preserving the body's own cells and tissues. However, in cases of autoimmunity, this self-tolerance breaks down, leading to the development of autoimmune diseases.

Autoimmunity is the system of immune responses of an organism against its own healthy cells, tissues and other normal body constituents.

Normally the function of immune system in our body is to recognize foreign elements and to destroy these before they could harm us either by humoral immune response (specific antibody formation) or cell mediated immune response by activation and clonal expansion of T cells. Thus the immune system defends the body against infections and certain other diseases by identifying, attacking, and destroying germs and other foreign substances. Sometimes the immune system makes a mistake and starts attacking the body's own tissues or organs. This is called **autoimmunity**. There are many autoimmune diseases one example being type 1 diabetes in which the Islets cells (produce Insulin) in the pancreas are destroyed by the immune system.

An autoimmune disease is a case of mistaken identity; it is an abnormal condition in which the body reacts against constituents of its own tissues. The result may be simple hypersensitivity reaction and or autoimmune disease when the body begins attacking its own healthy tissues. We can say it is a case of mistaken identity resulting in failure of the immune system to differentiate between self and non self. About 5 % to 7 % of adults suffer from autoimmune diseases and two thirds of these are females. Somehow left handed people are more prone the reason for this is not known.

This failure to differentiate between self and non self may result due to some extraneous environmental factors like some viral infections and exposure to some mutagenic agents; can be due to the breakdown and failure of immune regulation and due to some aberration in the genes. Whatever the reason the result is autoimmune disease which may involve a particular organ when it is called an organ specific disease (e.g. Addison's disease involving Adrenal glands) or it may involve particular cells/tissues all over the body when it is called non-organ specific or disseminated disease (e.g. Rheumatoid arthritis).

Definition

Autoimmunity refers to the body's development of intolerance of the antigens on its own cells i. e. there is an immune response to one's own tissue antigens. This type of body response results in a disease state characterized by a specific antibody or cell-mediated immune response against the body's own tissues (auto antigens). So, we can say that autoimmunity is the breakdown of mechanisms responsible for self tolerance and induction of an immune response against components of self.

The immunological mechanism of the body is dependent on two major factors: (1) the inactivation and rejection of foreign substances and (2) the ability to differentiate between the body's own antigens ('self') and foreign ('non self'). It is not yet known exactly what causes the body to fail to recognize self proteins as its own and to react to them as if they were foreign resulting in autoimmunity and may be autoimmune diseases. Prominent examples include celiac disease, diabetes mellitus type 1 (IDDM), sarcoidosis, systemic lupus erythematosus and many others.

Autoimmune Disease States

The autoimmune diseases can be divided into systemic, localized and haemolytic disorders, depending on tissue/cells affected and the clinico-pathologic features

Systemic Autoimmune Diseases

These diseases are associated with auto antibodies to antigens which are not tissue specific. One example can be polymyositis, here the tissue involved are muscles, however the auto antibodies are found against the auto antigens which are often ubiquitous "t-RNA synthetases". Another example is rheumatoid arthritis (RA). There is symmetric poly arthritis with muscle wasting and may be associated with myositis, and vasculitis, etc. The specific marker (auto antibody) found in blood in these patients is Rheumatoid Factor (RF) which is usually 19 s IgM. RF is an antibody against Fc fragment of immunoglobulins. Other systemic autoimmune diseases are polyarteritis nodosa, systemic lupus erythematosus and Sjogren's syndrome as shown in the figure.

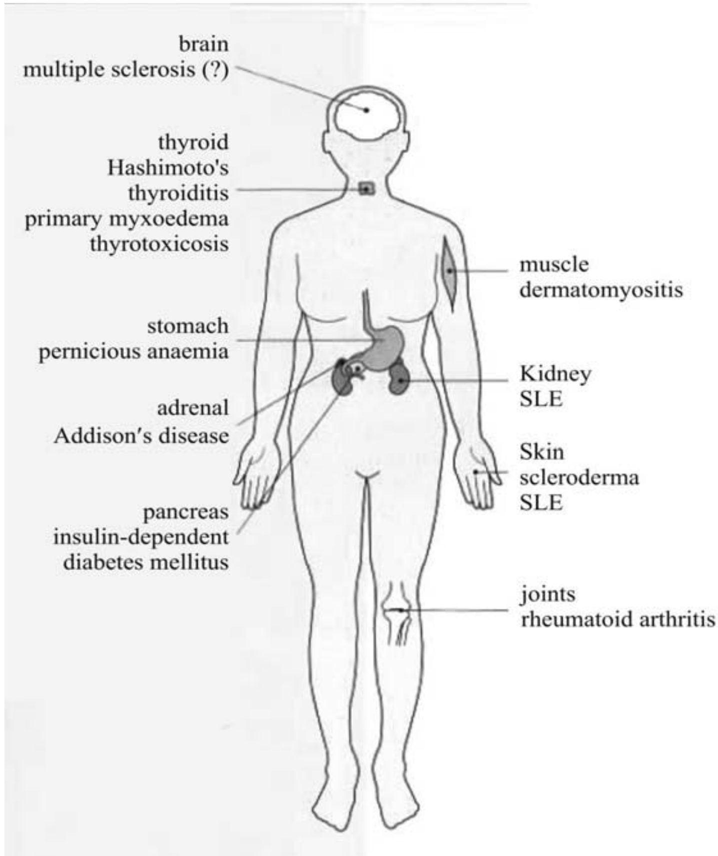


Figure 1. Two types of autoimmune disease.

Hyperthyroidism is a condition in which the thyroid gland makes too much thyroid hormone.

So, let us recap, in case of systemic autoimmune diseases the incriminating antigens and the autoimmunity are distributed in many tissues. The systemic diseases are:

- – Rheumatoid arthritis
- – Systemic lupus erythematosus (SLE)
- – Scleroderma
- – Primary Sjogrens's syndrome
- – polymyositis

Organ Specific or Localized Autoimmune Diseases

As the name indicates in these cases the autoimmunity involves a particular organ. One best studied organ is thyroid and examples are Hashimoto's disease which affects thyroid gland causing lymphadenoid goitre and the other is Graves disease causing thyrotoxicosis. Anti thyroglobulin antibodies

If you suspect you have an immune system disorder or disease, consult a healthcare provider. Proper diagnosis and treatment are essential for managing these conditions effectively.

are produced in both these cases and these can be shown in sera of the patients by various tests. However, the pathology in the two is different and so are the resulting symptoms. In Hashimoto' goitre there is hypothyroidism and in Graves disease there is **hyperthyroidism**. Another example is Addison's disease in which the adrenal glands are affected. There is lymphocytic infiltration of the adrenal glands and production of antibodies directed against zona glomerulosa. Other diseases include autoimmune disease of eyes, brain, skin and many others.

In organ specific autoimmune diseases the implicated antigens and the autoimmunity are restricted to specific organs in the body

- – Type I diabetes
- – Goodpasture's syndrome
- – Multiple sclerosis
- – Grave's disease
- – Hashimoto' thyroiditis
- – Myasthenia gravis

Haemolytic Autoimmune Diseases

Auto antibodies are formed against RBCs leading to autoimmune haemolytic anaemia; auto antibodies may form against platelets resulting in autoimmune thrombocytopaenia; and formation of anti leucocyte antibodies resulting in autoimmune leucopaenia and so on.

Causes of Autoimmune Diseases

Let us understand why normally immune response does not occur against our own tissue antigens. This is due to "Tolerance to self antigens" which is acquired by various mechanisms. Failure in immune recognition of self and injury of self tissues (autoimmunity) results from a loss of self tolerance. Let us discuss the mechanisms of self tolerance and how it is broken down to result in autoimmunity.

Mechanisms of Self Tolerance

One mechanism is that the clones of lymphocytes which act against self antigens are deleted. This is the "clonal deletion" theory. Clonal deletion is mediated by ubiquitous self antigens. The second is inactivation of developing lymphocytes so our immune system becomes self tolerant, no activation of immunity against self antigens as specific lymphocytes are either deleted or inactivated. Clonal inactivation can be mediated by tissue-specific antigens.

Peripheral T cell tolerance mechanisms:

These are explained below:

- Immunological Ignorance: Very few self proteins contain peptides that are presented by a given MHC molecule at a level sufficient for T cell activation, Autoreactive T cells are present but not normally activated.
- Suppressor or regulatory T cells: mediate active suppression of autoreactive cells
- Immunologically privileged sites: no lymphatic drainage or non-vascularized areas; presence of immunosuppressive factors.

Peripheral B cell tolerance mechanisms

- Contact with soluble antigens: this leads to downregulation of surface IgM, and so there is inhibition of signaling resulting in anergic (non reactive) cells and so no immune response occurs against these soluble antigens, there is tolerance to these antigens.
- Another mechanism of self tolerance is the Fas-mediated apoptosis (programmed cell death) of anergic B cell following secondary encounter with CD4 T cell

Mechanisms of breakdown of self tolerance resulting in autoimmunity:

There may be loss of “Self Tolerance” germinal centers due to some genetic aberration or some environmental factors.

MHC (Major Histocompatibility complex) molecules normally should present the peptide (antigen) at optimal level to T cells for immune response to occur. The level of autoantigenic peptide presented is determined by polymorphic residues in MHC molecules that govern the affinity of peptide binding.

If the peptide is presented at levels too low to engage effector T cells, there will be tolerance to such self antigens and if peptide is presented at very high levels then there is clonal deletion or anergy (tolerance).

Only a few peptides can act as autoantigens so there are a relatively few autoimmune syndromes.

Individuals with a particular autoimmune disease tend to recognize the same antigens with the same MHC.

Autoimmunity (breakdown of tolerance) arises most frequently to tissuespecific antigens with only certain MHC molecules that present the peptide at an intermediate level recognized by T cells without inducing tolerance.

Gene mutation/s leads to one or more autoimmune diseases. Autoimmune diseases are associated with particular MHC genotypes. Sometimes there may be more than one gene involved and the result is complex autoimmune diseases.

Environmental factors which can cause breakdown of self tolerance include pathogens (bacterial, viral and others), drugs, hormones, and toxins. These are just a few ways that can trigger autoimmunity.

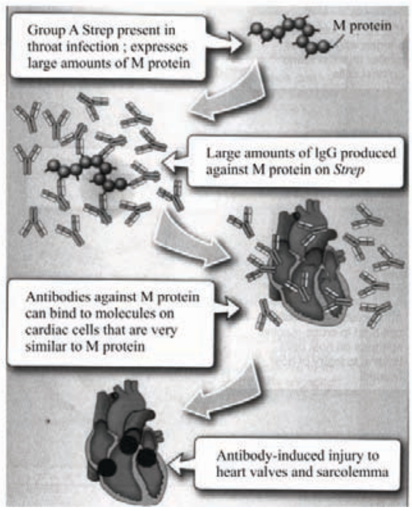
- Drugs
  - Drug induced lupus
- Toxins
  - Example: Toxic Oil Syndrome
- Hormones
  - Example: Endometriosis and preeclampsia are both thought to be autoimmune in nature

Complement deficiency: deficiencies in the classical complement pathway renders patients more likely to develop immune complex diseases like:

- SLE
- RA

Molecular mimicry

Rheumatic fever is a classic example of molecular mimicry as shown in picture below.



Let us enumerate causes of autoimmunity

- Lymphocytes abnormalities and polyclonal lymphocytes activation
- Cytokine dysregulation due to changes in lymphocytes
- Failure of central tolerance and peripheral tolerance
- Overcome of Ag released from hidden location
- Antigen modified/ generated by molecular changes in the body
- Molecular mimicry
- Alteration in Ag processing
- Infection
- Genetic factors.

Immunopathology of Autoimmune Diseases

Immune response is a physiological reaction which occurs within an organism in the context of inflammation for the purpose of defending against exogenous factors.

These are autoimmune diseases, so the pathology is immune response to self antigens as explained above. Antibody mediated **immune response** by formation of antibody to antigens on cell surface or in tissue matrix. Examples are glomerulonephritis (Ab to basement membrane); rheumatic myocarditis (molecular mimicry as described above), etc. Another mechanism is immune complex mediated, Type III immune response to self antigen Examples are Rheumatoid arthritis. The third mechanism is cell mediated (Type IV immune response) immune response against brain componants, example is experimental auto immune encephalomyelitis.

So you see type II, III and Type IV immune responses are involved in immunopathology of autoimmune diseases.

LE Cell

The LE cell was so termed because of its exclusive presence in the bone marrow of 25 patients with confirmed or suspected SLE at the Mayo Clinic. LE cells are polymorphonuclear cells which have phagocytosed the nuclear material of other cells. LE cells are not usually found in peripheral blood. However, sometimes LE cell phenomenon could form in the buffy coat of peripheral blood after a period of]. LE cells are common in bone marrow but have also been found in synovial fluid, cerebrospinal fluid and pericardial and pleural effusions from patients with SLE.

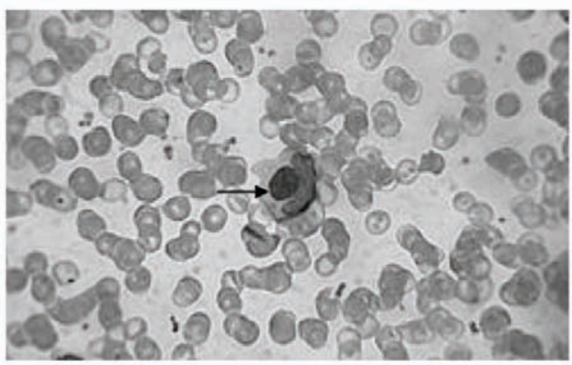


Figure 2: LE cells are formed in response to an auto immune factor present in plasma in cases of SLE.

Diagnosis of Autoimmune Diseases

It is very difficult to exactly diagnose autoimmune diseases. Clinical presentation and physical examination are done to make a probable diagnosis. Laboratory tests performed are of general nature and in case of organ specific disease specific tests can be performed.

General tests:

- C Reactive Protein
- Autoantibody titers (anti DNA, anti phospholipids, etc)
- Presence of Rheumatoid Factor
- Erythrocyte sedimentation rate

Disease specific tests:

- Neurological exam – MS
- Fasting glucose – Diabetes

Treatment of Autoimmunity

The general principle is to somehow stop the immune response to self antigens. This can be achieved by the following methods:

- Immunosuppression (e.g., prednisone, cyclosporin A);
- Removal of thymus (some Myasthenia Gravis patients);
- Plasmapheresis (to remove Antigen+Antibody complexes);
- T-cell vaccination (to activate suppressing T cells so that immune response to self antigens is suppressed??);
- Block MHC with similar peptide, so antigen processing is stopped;
- anti-CD4 monoclonal Antibody to inhibit immune response;
- anti-IL 2R monoclonal Antibody to inhibit immune response.

CORE CONCEPT: IMMUNODEFICIENCY DISORDERS

Immunodeficiency disorders, also known as primary immunodeficiency disorders (PIDD), are a group of genetic conditions characterized by a weakened or absent immune system. In individuals with immunodeficiency disorders, the immune system's ability to defend against infections and diseases is compromised, making them more susceptible to recurrent or severe infections. These disorders can affect both the innate immune system (the body's first line of defense) and the adaptive immune system (which produces specific responses to pathogens).

There are more than 400 known types of primary immunodeficiency disorders, each with its own specific genetic mutation and pattern of immune dysfunction.

Immunodeficiency disorders represent defects in the immune system that result in weakened or dysregulated immune defense. These deficiencies may occur as primary diseases or secondary conditions.

Take proactive steps to prevent infections. Practice good hand hygiene, avoid close contact with sick individuals, and follow recommended vaccination schedules.

Primary Immunodeficiencies

PIDs constitute inherent defects in immunity, most of which arise from inborn deviations in the genetic code. More than 300 PIDs have been identified. These conditions have been placed into 9 categories corresponding with their clinical and immunologic phenotypes by an international group of experts who evaluate these diseases every 2 years (Table 1). Although PIDs were previously believed to represent rare conditions, they are now known to affect 1 of every 1,200 to 2,000 individuals, with growing prevalence due to increased recognition. The magnitude of this disease frequency, therefore, mandates that all healthcare providers recognize the associated clinical infectious and noninfectious characteristics, testing options, and management recommendations for children with PIDs. And because so many PIDs present in childhood, and several are now captured by newborn screening, their understanding by pediatricians remains especially important.

Table 1. Primary Immunodeficiency Categories

1. Immunodeficiencies affecting cellular and humoral immunity
2. Combined immunodeficiencies with associated or syndromic features
3. Predominantly antibody deficiencies
4. Diseases of immune dysregulation
5. Congenital phagocytic defects
6. Defects in intrinsic and innate immunity
7. Autoinflammatory disorders
8. Complement deficiencies
9. Phenocopies of PIDs

Infectious Signs and Symptoms of PID

Infections serve as the key hallmark warning signs for the presence of PID. (4) In general, children with PIDs develop infections more frequently, more severely, and more atypically. Although noninfectious characteristics of PIDs can result from inappropriate immune function and, the link between PID and infection remains essential and must be regarded as a prompt to consider PID in pediatric practice. Each of the 9 PID categories conveys a different risk for certain infectious susceptibilities in accordance with the molecular mechanism for the impaired immunity. Offending organisms encompass the entire spectrum of pathogens, including bacteria, viruses, fungi, and parasites. Thus, awareness of infections associated with defective humoral immunity, T-cell function, phagocytic capacity, complement activity, and other innate immunity becomes essential.

Defects in humoral immunity that cause PID result in impaired antibody production. Infants receive maternal immunoglobulin (Ig) G transplacentally. Because the halflife of IgG is approximately 21 days in the circulation, children have protection from these maternal antibodies until approximately 3 to 4 months of life. At that time, characteristic infections tend to ensue, although some antibody deficiencies are known to present at older ages. Because antibodies serve as critical elements for protection of the sinopulmonary tract, affected individuals typically develop recurrent otitis media, sinusitis, and pneumonia. Infections are by no means limited to these sites, however, and can result in sepsis, meningitis, osteomyelitis, or chronic diarrhea. For example, X-linked agammaglobulinemia is characterized by a lack of B cells due to mutations in BTK located at Xq22.1. Absence of B cells results in failure to produce IgG, IgA, IgM, and IgE. In affected boys, disappearance of maternal antibodies is associated with the appearance of recurrent sinopulmonary tract infections, classically by polysaccharideencapsulated bacteria, such as **Streptococcus pneumoniae** and Haemophilus influenzae. The lack of antibodies also results in susceptibility to infections by mycoplasmal and pseudomonal organisms, Salmonella species, Campylobacter jejuni, rotavirus, and Giardia lamblia.

Streptococcus pneumoniae is a Gram-positive bacterium that is responsible for the majority of community-acquired pneumonia.

T-cell deficiency produces combined humoral and cellular immune defects. Because of this extensive effect on immune function, affected individuals demonstrate susceptibility to opportunistic fungal, viral, mycobacterial, and parasitic infections in addition to the infections observed in antibody deficiencies. Classic pathogens include candidal species, Pneumocystis jirovecii, Mycobacteria species (including bacille Calmette-Guérin), and Cryptosporidium parvum. Viral infections surpass the ordinary and are notably severe, disseminated, or persistent. Severe combined immunodeficiency disease (SCID) serves as a classic example of primary T-cell deficiency. The diagnosis is established by either 1) the absence or a very low number of T cells (<300 CD_3+ T cells/mm^3) with no or very low T-cell function (<10% of the lower limit of normal) as determined by response to phytohemagglutinin or 2) the presence of T cells of maternal origin. Mutations in at least 16 genes lead to SCID, and most forms are caused by disrupted T-cell development secondary to abrogated T-cell receptor formation or cytokine signaling (Table 2). The most common type of SCID in the United States results from mutations in IL2RG at Xq13.1 and is, thus, found in boys. SCID represents a true pediatric emergency, and mortality from infections is certain if affected infants are not treated appropriately. Life-threatening infections are caused by respiratory syncytial virus, rhinovirus, parainfluenza virus, adenovirus, cytomegalovirus, and even attenuated vaccine strains of measles virus, varicella virus, rotavirus, and oral poliovirus. Invasive bacterial and fungal infections also occur, the former mostly resulting from impaired humoral immunity secondary to the defects in T-cell immunity.

Table 2. Causes of Severe Combined Immunodeficiency Disease

ASSOCIATED GENE	LYMPHOCYTE PHENOTYPE	DEFECTIVE MECHANISM
IL2RG	$T^-B^+NK^-$	Signaling for IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21
JAK3	$T^-B^+NK^-$	Same as *IL2RG*
RAG1	$T^-B^-NK^+$	Formation of T- and B-cell receptors
RAG2	$T^-B^-NK^+$	Same as *RAG1*
DCLRE1C	$T^-B^-NK^+$	Same as *RAG1*
PRKDC	$T^-B^-NK^+$	Same as *RAG1*
LIG4	$T^-B^-NK^+$	Same as *RAG1*
NHEJ1	$T^-B^-NK^+$	Same as *RAG1*
IL7R	$T^-B^+NK^+$	Signaling for IL-7
PTPRC	$T^-B^+NK^+$	Receptor signaling and T-cell activation
CD3D	$T^-B^+NK^+$	T-cell receptor signaling
CD3E	$T^-B^+NK^+$	Same as *CD3D*
CD247	$T^-B^+NK^+$	Same as *CD3D*
CORO1A	$T^-B^+NK^+$	Same as *CD3D*
AK2	$T^-B^-NK^-$	Stem cell maturation
ADA	$T^-B^-NK^-$	Lymphocyte survival

Because neutrophils play a significant role in the clearance of bacterial and fungal pathogens, PIDs due to defective phagocytic capacity are marked by infections caused by these organisms. Phagocytic impairment can result from inadequate neutrophil quantity or function. Classic infections in children with neutrophil deficiency include skin or solid organ abscesses, cellulitis, pneumonia, sepsis, and severe gingivitis or periodontitis. Other signs of a phagocytic defect include omphalitis, poor wound healing, and aphthous stomatitis. Affected individuals have an increased risk of life-threatening infections from Staphylococcus aureus and Pseudomonas aeruginosa. As a key illustration, chronic granulomatous disease (CGD) serves as an example of PID caused by impaired neutrophil function. This condition arises from defects in components of nicotinamide adenine dinucleotide phosphate oxidase, which produces the oxidative burst needed for neutrophils to kill phagocytized organisms. The most common form of CGD results from mutations in CYBB, which is located at Xp21.1-p11.4. Thus, most children with CGD are boys. CGD can be recognized by susceptibility to catalase-positive organisms, such as S aureus, Serratia marcescens, Klebsiella species, Burkholderia cepacia, Aspergillus species, and Nocardia species Infections caused by unusual pathogens, especially methylotrophic microorganisms capable of using single-carbon compounds as their sole source of energy, should also raise concern for the presence of CGD. In children with CGD who have received bacille Calmette-Guérin immunization, infection by the bacterium itself should be considered. Characteristic sites of various infections in CGD include the skin, lungs, lymph nodes, liver, spleen, central nervous system, and bones.

Children with defects in primary complement deficiencies can be recognized by infections caused by encapsulated bacteria. Complement deficiency, whether of the early or terminal classical complement pathway components, should be considered in children who have invasive meningococcal disease, such as sepsis or meningitis from Neisseria meningitidis. Individuals who have primary early classical complement pathway protein (C1, C4, C2, and C3) deficiencies further demonstrate susceptibility to other encapsulated bacteria, such as S pneumoniae and H influenzae.

Finally, children with PIDs due to other defects in innate immunity exhibit susceptibility to a variety of pathogens, depending on the critical defense mechanism affected. For example, natural killer cell deficiency should be suspected in individuals with recurrent, unusual, or severe herpesviral infections. Children with defects in the interferong–interleukin-12 signaling pathway, on the other hand, develop atypical mycobacterial infections. Perturbation of other key innate immune signaling pathways leads to chronic mucocutaneous candidiasis, which should be suspected in children with thrush that is unusually recurrent or unresponsive to nystatin. Finally, defects in toll-like receptor (TLR) signaling pathways offer an excellent example of the diversity of infectious susceptibilities affected by various innate immunity disorders. Humans express 10 TLRs, each of which performs an inherent danger-sensing function by recognizing specific intracellular or extracellular bacterial, fungal, or viral components. TLR3 deficiency leads to susceptibility to herpes simplex virus encephalitis. (16) IRAK-4 and MyD88, however, serve as adaptor molecules that mediate signaling downstream from all the other TLRs (except partly TLR4). Deficiencies of either of these proteins result instead in vulnerability to invasive infections (sepsis, meningitis, arthritis, and osteomyelitis, among others) that are caused by pyogenic bacteria, such as S aureus, S pneumoniae, and H influenzae, rather than by viruses or fungi.

Noninfectious Signs and Symptoms of PID

Improved understanding of PIDs and enhanced diagnostic capabilities, especially genetic testing, have provided greater awareness of the need to appreciate noninfectious manifestations, that can serve as early or initial warning signs for the presence of underlying PID. The pediatrician should keep in mind that inherently defective immunity not only results in an inability to defend against the external environment (ie, infections) but also produces impaired ability to navigate the internal environment, leading to autoimmune and autoinflammatory conditions. In some children, these conditions will appear before infections ensue and can engage nearly any organ system. Affected children may, therefore, first come to the attention of other specialist providers. Presence of any of these features in a pediatric patient, especially with early or very early onset, should prompt consideration of additional evaluation for underlying PID.

These noninfectious manifestations of PIDs are reviewed in the following accompanying tables and figures. Of importance, the lists of defective PID genes provided in the tables and figures are by no means comprehensive. Instead, they are intended to underscore the degree to which concern for PID is deserved.

Gastrointestinal disorders can reveal the presence of underlying PID (Table 3). Autoimmune enteropathy can be observed when children lack regulatory T cells. The autoimmune disease in other PIDs, on the other hand, leads to **inflammatory bowel disease**. In several PIDs, dysregulated inflammation rather than autoimmunity results in inflammatory bowel disease. Finally, the presence of intestinal inflammation with bowel atresias is nearly uniquely associated with TTC7A deficiency. Thus, children with chronic enteropathy, inflammatory bowel disease, or multiple intestinal atresias should all be considered for evaluation for underlying PID.

Inflammatory bowel disease is a term that describes disorders involving long-standing (chronic) inflammation of tissues in your digestive tract.

A variety of dermatologic conditions, many of which relate to underlying dysregulated immunity in the skin, can signal the existence of PID (Fig 3). Eczema is a presenting feature of several PIDs, most notably hyper-IgE syndrome and Wiskott-Aldrich syndrome. Neutrophilic pustular dermatosis, or Sweet syndrome, can be observed in proteasome-associated autoinflammatory syndromes. Ectodermal dysplasia, on the other hand, can suggest PID due to a defect in nuclear factor kappa-light-chain-enhancer of activated B cells signaling or store-operated calcium entry. Cutaneous granulomas have been associated with CGD, ataxia-telangiectasia, and Blau syndrome in addition to several other PIDs. Abnormal hair can also serve as a sign of PID. Hypopigmented hair is observed in ChédiakHigashi, Griscelli, and Hermansky-Pudlak syndromes, and thin or sparse hair is present in cartilage-hair hypoplasia and dyskeratosis congenita. Brittle hair due to trichorrhexis invaginata, so-called bamboo hair, is essentially pathognomic for Comèl-Netherton syndrome. Meanwhile, the complete absence of hair (alopecia totalis) should engender concern for NFKB2 or FOXN1 deficiency. Familial cold autoinflammatory syndrome is marked by a phenotype of cold-induced urticaria (as characterized by a wheal and flare response to an ice cube placed on the skin) and is caused by mutations in

Table 3. Gastrointestinal Manifestations of PIDs

NEARLY 40 PID GENES ARE ASSOCIATED WITH INFLAMMATORY BOWEL DISEASE OR NONINFECTIOUS ENTEROPATHY
ADAM17, AIRE, ARPC1B, BACH2, CD40LG, CD55, CTLA4, CYBA, CYBB, DKC1, DOCK8, FOXP3, IKBKG, IL10, IL10RA, IL10RB, IL2RA, ITCH, LRBA, MALT1, MVK, NCF1, NCF2, NCF4, NEIL3, NFAT5, NLRC4, NOD2, PIK3CD, PLCG2, STAT1, STAT3, STXBP2, TNFAIP3, TTC37, TTC7A, WAS, XIAP, ZBTB24

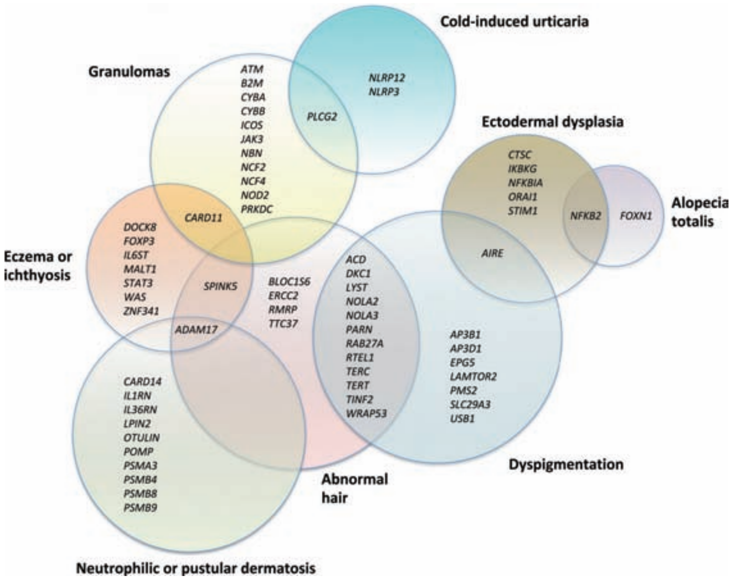


Figure 3. Dermatologic manifestations of primary immunodeficiencies.

$NLRP_3$ or $NLRP_{12}$. A different form of cold-induced urticaria that produces a negative ice cube test but significant wheal responses to evaporative cooling occurs in PLCg2- associated antibody deficiency and immune dysregulation. Finally, abnormal skin pigmentation can be found in several PIDs, including dyskeratosis congenita (reticular hyperpigmentation) and Chédiak-Higashi, Griscelli, or Hermansky-Pudlak syndromes (hypomelanosis or albinism).

Because PIDs are caused by deficient and dysregulated immunity, it comes as little surprise that some affected individuals will present with autoimmune disease as the chief or even sole manifestation (Fig 4). Noninfectious arthritis, including juvenile idiopathic arthritis, may be a sign of a hyperinflammatory PID condition. Autoimmune cytopenias are classically associated with impaired immunologic tolerance, whether through defective negative selection of T cells in the thymus or decreased regulatory T-cell inhibition of autoreactive T cells. They can also present in individuals with autoimmune lymphoproliferative syndrome (ALPS). Vasculitis serves as a feature of many of the periodic fever syndromes and several complement deficiencies. In terms of periodic fever syndromes, the known associated features underscore the need to consider further evaluation in a child who develops fevers of regular chronicity, arthritis, or vasculitis, and any other signs of inflammation. Most of these syndromes result from hyperactivation of inflammasomes, which are protein complexes critical for inducing inflammatory responses, and can be treated with targeted biological modifiers. Vasculitic disease is also known to occur in several newly described PIDs. Thus, it seems likely that novel PIDs will continue to be discovered as causes of vasculitis in children. Finally, evaluation for PIDs should be considered for children who present with systemic lupus erythematosus (SLE) or SLElike disease. SLE is known to be associated with deficiencies of almost all components of the classical complement pathway (except C9). Individuals with CGD and ALPS can also present with SLE-like features. It remains important to note that variants of the disease known as hemophagocytic lymphohistiocytosis (HLH) can present with an inflammatory syndrome known as macrophage activation syndrome. This condition must be recognized appropriately because it is treated more like HLH than other inflammatory conditions that the symptoms might mimic.

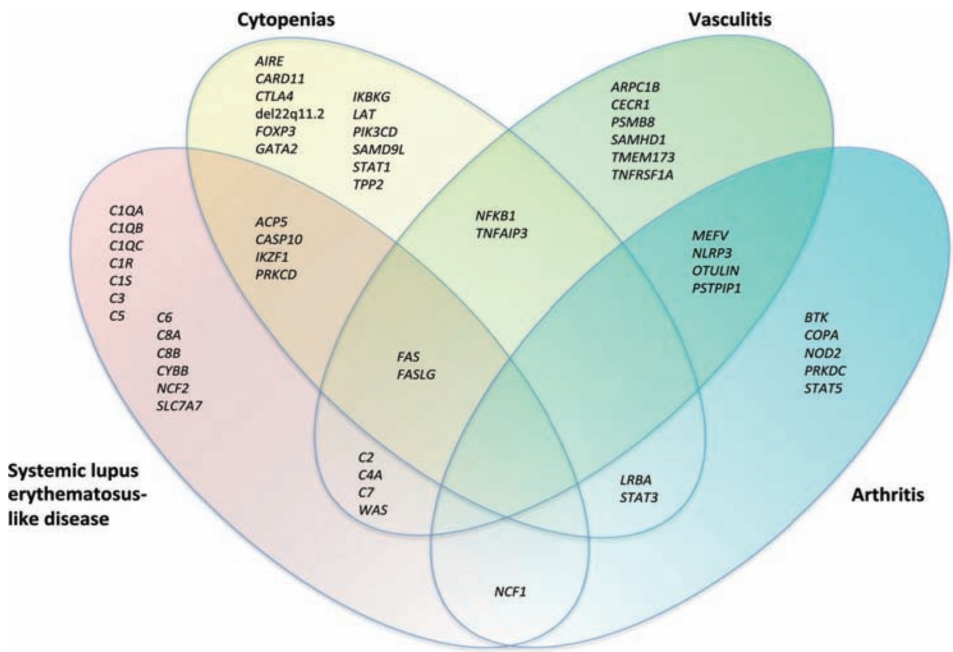


Figure 4. Autoimmune manifestations of primary immunodeficiencies.

Screening for PIDs should be strongly considered in children with early-onset endocrinopathies (Fig 5). Most of these endocrine disorders are autoimmune, resulting from the disrupted tolerance mechanisms inherent to the associated PIDs. Severe or difficult-to-control type 1 diabetes can indicate regulatory T-cell dysfunction. Primary adrenal insufficiency is an archetypal feature of AIRE, MCM_4, and $NFKB_2$ deficiencies. Hypoparathyroidism is observed in AIRE deficiency and DiGeorge anomaly. In fact, the association between neonatal hypocalcemia and DiGeorge anomaly remains well appreciated. Clinicians must realize that the absence of 22q11.2 hemizygosity does not exclude the diagnosis of DiGeorge anomaly, as nearly half of affected individuals may not have the characteristic deletion. Children who present with any features (eg, hypocalcemia or hypoparathyroidism, congenital cardiac disease, or developmental or neuroanatomical defects) characteristic of DiGeorge anomaly should, therefore, be evaluated for potential thymic dysfunction. Growth hormone deficiency represents a classic characteristic of $STAT_5B$ deficiency but is additionally known to occur in patients with $NFKB_2$ deficiency and ataxia-telangiectasia. Finally, thyroid disease seems underappreciated as a manifestation of PID but can provide a clue to the existence of underlying PIDs.

Two important hematologic/oncologic conditions should raise concern for underlying PIDs (Table 4). The PID mechanisms that lead to these 2 disorders include impaired control of immunologic activation and dysregulated proliferation of immune cells. First, lymphoproliferative disease suggests an inherent defect in immune function. For example, in ALPS, abnormal proliferation of immune cells occurs due to defective apoptosis. Meanwhile, several other PIDs are characterized by lymphoproliferative disease associated with Epstein-Barr virus infection. Lymphoid hyperplasia is observed in autosomal recessive hyper-IgM syndromes. Second, pediatric patients who develop HLH should be screened for PIDs if they do not have defects in any of the familial HLH genes. PIDs associated with HLH include Chédiak-Higashi, Griscelli, and Hermansky-Pudlak syndromes; CGD; periodic fever syndromes; and a variety of other conditions that affect cytotoxic T-cell or natural killer cell function.

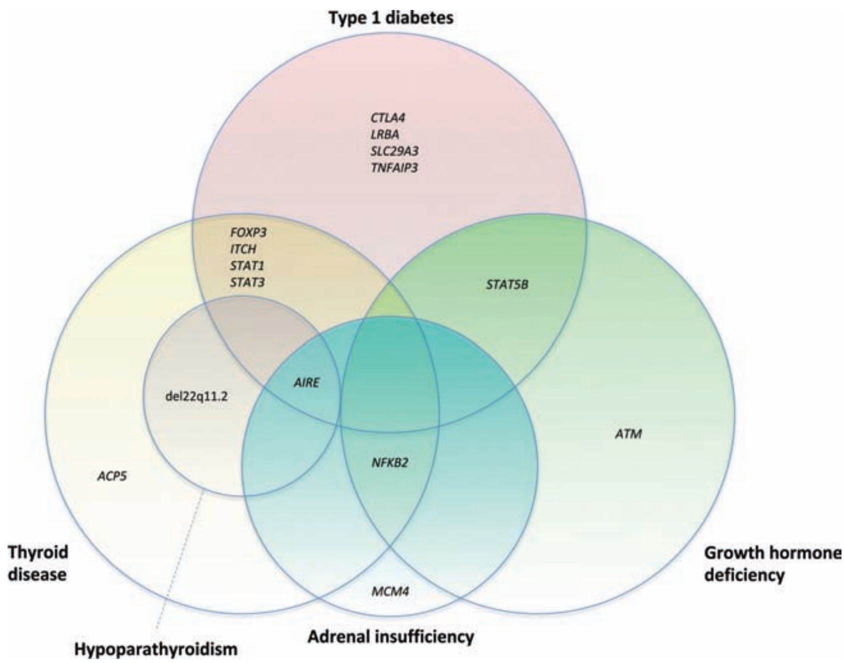


Figure 5. Endocrinologic manifestations of primary immunodeficiencies.

Several PIDs are characterized by neurologic signs (Fig 6). These abnormalities occur most typically due to roles that the damaged genes would normally serve outside of the immune system. That said, the precise mechanism for ataxia and hearing loss remains unclear in many conditions and is often not directly related to the defective immune function in the associated PID. Ataxia is a typical feature of ataxia-telangiectasia, Chédiak-Higashi syndrome, and Hoyeraal-Hreidarsson syndrome, among others. Hearing loss, on the other hand, is a fundamental attribute of Carnevale-Mingarelli-Malpuech-Michels syndrome, MuckleWells syndrome, H syndrome, and other PIDs. Next, developmental delay is clearly associated with a variety of PIDs, but providers often neglect to evaluate for defective immunity due to the attention given to the neurologic deficits. This association emphasizes the critical intersection between neurologic development and activity and immune function, especially in terms of preservation of DNA integrity. Doublestranded break DNA repair seems to play an essential role in cognitive development, and children with PIDs caused by a defect in this repair mechanism present with developmental delay. As another example, developmental delay is observed in immunodeficiency–centromeric instability– facial anomalies syndrome, which is caused by aberrant DNA methylation and chromatin structure.

Table 4. Hematologic/Oncologic Manifestations of PIDs

CONDITION	ASSOCIATED PID GENES
Lymphoid hyperplasia or lymphoproliferative disease	*AICDA, CASP8, CASP10, CD27, CD70, CECR1, CTLA4, FAAP24, FAS, FASLG, ITK, LAT, LRBA, MAGT1, MVK, PIK3CD, PIK3R1, PRKCD, RASGRP1, RBCK1, SH2D1A, SLC29A3, STAT3, STAT5B, STK4, TPP2, UNG, XIAP*
Hemophagocytic lymphohistiocytosis	*AP3B1, ATM, BTK, CARMIL2, CD27, CD3E, CYBA, CYBB,* del22q11.2, *DKC1, FAS, IKBKG, IL2RG, IL7R, ITK, LYST, MAGT1, MEFV, NCF1, NLRC4, ORAI1, PIK3CD, PRF1, RAB27A, RAG1, RAG2, SH2D1A, STAT1, STAT2, STAT3, STX11, STXBP2, TNFRSF1A, UNC13D, WAS, XIAP*

Other examples of developmental delay and PID caused by inappropriate processing of nucleic acids include ADAR, CHD7, SMARCAL1, and TRNT1 deficiencies. Aside from defects in nucleic acid handling, glycosylation is also known to play a vital role in immunologic and neurologic processes. As a result, a variety of congenital disorders of glycosylation represent PIDs that manifest as developmental delay. Last, unidentified DiGeorge anomaly should always be considered in a child with neurocognitive deficits, behavioral issues, or various psychiatric disorders. Thus, clinicians should be aware of these neurologic associations and consider immunologic evaluation accordingly.

Several rare noninfectious pulmonologic conditions can suggest the presence of underlying PIDs (Table 5). Interstitial lung disease has been reported in several PIDs. Pulmonary alveolar proteinosis is most commonly associated with CSF2RA deficiency but can also be observed in patients with other PIDs. Pulmonary capillaritis and hemorrhage should raise concern for COPA syndrome. Finally, eosinophilic pneumonia represents a key feature of lung disease, immunodeficiency, and chromosome breakage syndrome caused by $NSMCE_3$ deficiency.

Finally, clinicians should be aware of 2 additional noninfectious manifestations of PIDs (Table 6). First, allergies, which are often severe or directed toward numerous agents, can serve as the presenting feature of several PIDs. In many of these conditions, the atopy is accompanied by markedly elevated serum IgE levels. Second, skeletal dysplasia serves as a classic feature of spondyloenchondrodysplasia with immune dysregulation, cartilage-hair hypoplasia, and Schimke immuno-osseous dysplasia but can denote the presence of other PIDs as well.

When to Test for PID

The decision of when to test for or refer a child for evaluation of a suspected PID may be affected by several factors.

First, a history of recurrent, severe, or unusual infections or any of the noninfectious issues discussed previously herein merits immunologic evaluation. The definition of recurrent remains imprecise. It may not be unusual (17% incidence rate) for a 1-year-old child to have had 3 or more episodes of otitis media, and almost half of the children will have had at least 3 occurrences of otitis media by 3 years of age. One large study suggests that healthy children younger than 3 years may have 11 respiratory tract infections per year, and healthy children aged 3 to 5 years may have 8 respiratory tract infections annually. Thus, clinical judgment remains necessary. As a general rule of thumb, it is the clinical practice of the authors to advise pediatricians to consider PID in their patients who distinguish themselves from the general patient population regarding the occurrence of infections. Statistically speaking, a pediatric practice of 10,000 patients will have 5 to 10 cases of PID overall. Pediatricians should, therefore, be mindful of patients who exist as outliers from an infectious standpoint so that they will be sure to capture PID cases within their practice.

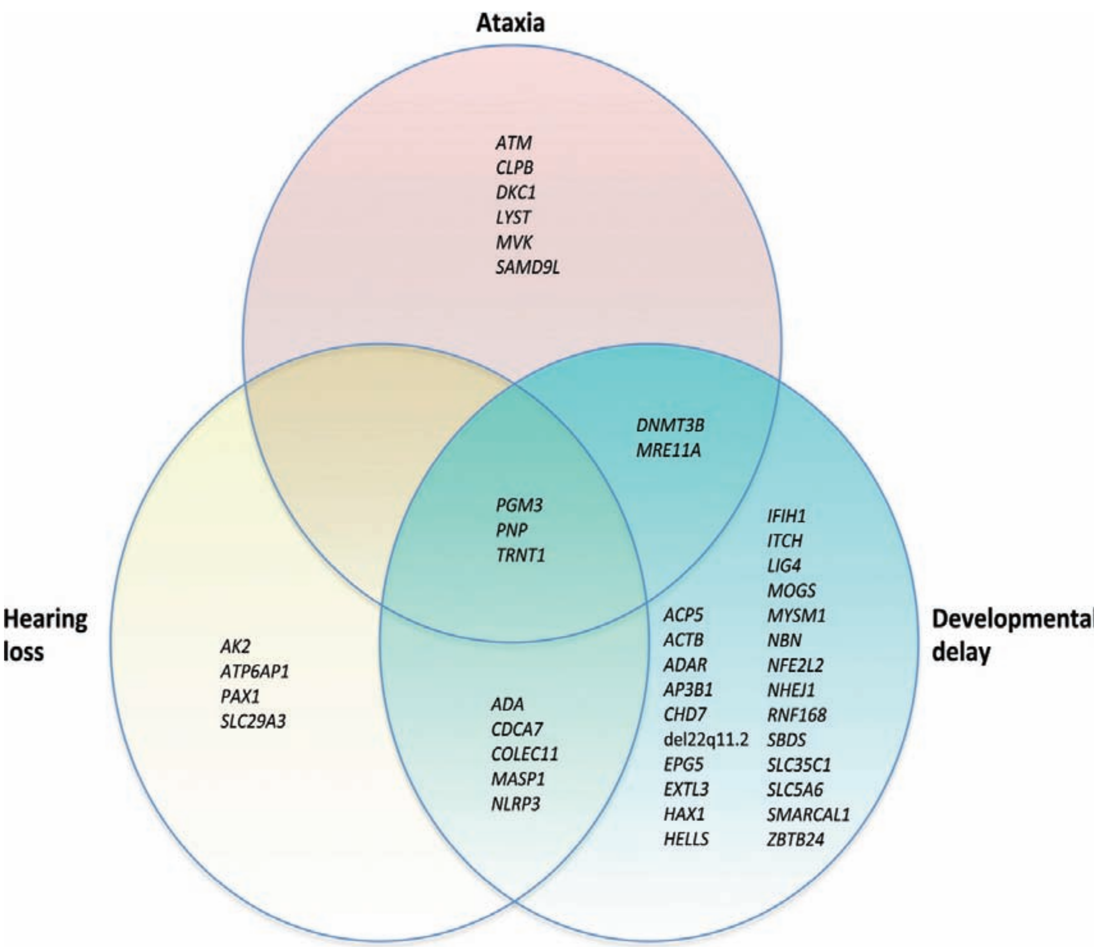


Figure 6. Neurologic manifestations of primary immunodeficiencies.

Table 5. Pulmonologic Manifestations of PIDs

CONDITION	ASSOCIATED PID GENES
Interstitial lung disease	*COPA, ITCH, STAT3, TMEM173, TNFAIP3, XIAP*
Pulmonary alveolar proteinosis	*CD40LG, CSF2RA, GATA2, SLC7A7*
Capillaritis and hemorrhage	*COPA*
Eosinophilic pneumonia	*NSMCE3*

Second, the family history may encourage immunologic testing. Underlying PID should be considered if family members have had recurrent or unusual infections, especially maternal uncles or other males. Other suspicious elements of the family history include consanguinity and early childhood mortality from infections or unexplained causes.

Table 6. Other Noninfectious Manifestations of PIDs

CONDITION	ASSOCIATED PID GENES
Allergies	*CARD11, CARMIL2, CTLA4, DOCK8, FOXP3, PGM3, SPINK5, WAS, WIPF1*
Skeletal dysplasia	*ACP5, ADA, ALG12, EXTL3, NBAS, PGM3, RNU4ATAC, RMRP, SMARCAL1*

Third, investigation for PID may be warranted in children with physical examination findings consistent with any of the noninfectious presentations mentioned. Children with failure to thrive should also raise suspicion for underlying PID. Furthermore, a child who demonstrates absent or minimal lymphatic tissue should unquestionably be assessed for an underlying PID.

Finally, infants with abnormal newborn screening tests for SCID must receive immunologic evaluation. As of the end of 2017, all but 2 states (Louisiana and Indiana) have implemented active SCID newborn screening programs. The screening test is designed to detect absent or low numbers of T-cell receptor excision circles in blood spots from Guthrie cards. Thus, an abnormal newborn screening test may signify the presence of T-cell deficiency.

Laboratory Testing for PID

Several tests are recommended for initial evaluation for suspected PID. First, because defects in humoral immunity predominate, quantitative measurement of serum immunoglobulin levels and functional assessment of specific antibody responses provide excellent screening tests for PID. Specific antibody function should be tested by determining antibody production toward immunizations. These functional activities represent the most standardized, most uniform, and best-studied humoral immune responses. As a more advanced diagnostic intervention, children older than 2 years can be additionally tested for humoral responses to the polysaccharide antigens in the 23-serotype pneumococcal polyvalent vaccine. Overall, these investigations are performed by obtaining baseline antibody titers, administering the relevant vaccine, and measuring the postimmunization titers (eg, approximately 3 weeks after immunization for tetanus toxoid; both 4 to 8 weeks and 6 months after immunization for the 23-serotype pneumococcal polyvalent vaccine). Next, a complete blood cell count with a manual differential count can be used to screen for quantitative Tcell deficiency through the absolute lymphocyte count (ALC). In a newborn, an ALC less than 2,500/mL (<2.5 10^9/L) suggests that T cells have not developed fully and that they may even be absent in whole or in part. From a PID perspective, it is essential for pediatricians to focus on and calculate the ALC in routine blood cell counts. Any abnormalities should be followed by a T-cell subset analysis performed by flow cytometry to ensure the appropriate presence and distribution of the different subpopulations. Absence of a cardiothymic silhouette on chest radiographs in an infant should also raise concern for the absence of T cells. For functional assessment of T cells, in vivo delayed-type hypersensitivity testing by the intradermal injection of T-cell antigens such as tetanus toxoid and Candida albicans can be performed in the office setting, but the recommended option would consist of referral to an immunology specialist for appropriate in vitro testing. The advantage of the latter method comes from the fact that it is highly quantitative and directly comparable with reference standards. Third, a total serum hemolytic complement assay can be ordered to exclude deficiency of any of the components of the classic complement pathway. This test is highly sensitive to sample handling, and true deficiency is likely only if the total serum hemolytic complement is zero or near zero. Next, a potential diagnosis of CGD should be examined using a dihydrorhodamine-1,2,3 flow cytometry–based assay. Nitroblue tetrazolium reduction is no longer preferred due to its subjectivity and lack of sensitivity. Clinicians should be aware that individuals with myeloperoxidase and glucose-6-phosphate dehydrogenase deficiencies can have abnormal dihydrorhodamine-1,2,3 test results. Children whose clinical history may suggest cyclic neutropenia should be evaluated by serial complete blood cell counts with differential counts. This protocol involves obtaining counts 2 to 3 times weekly for 6 to 8 weeks. Finally, patients for whom ataxia-telangiectasia is suspected should be screened with a serum a-fetoprotein level, which is typically elevated in affected individuals.

Children with suspected PIDs should then be referred to a clinical immunology expert for further evaluation because many different subspecialized immune tests are available that help in the further delineation of the different PIDs. Such additional laboratory testing may be warranted, and genetic testing may be considered necessary in some individuals. Importantly, we are rapidly moving toward a practice environment in which broad genetic testing (such as whole exome sequencing) may actually precede formal assessment of a PID in an otherwise sick child. This approach has the potential to conserve financial resources in some cases but often leaves the geneticist and immunologist with the task of determining whether certain genetic findings represent an explanation for the clinical and immunologic presentation. Nonetheless, it is essential for the pediatrician to know that several different sequencing “panels” exist to allow for follow-up assessment of abnormal SCID newborn screening tests and can capture all known SCIDassociated genes with rapid turnaround. Although genetic tests are truly empowering and can be definitive, they do not replace quantitative and functional immune tests, which are still needed to understand the severity of a given gene aberration.

Management of Children with PID

Infectious complications should be minimized as much as possible, and strategies should be dictated by the susceptibilities associated with the immunologic mechanism that is impaired. These precautions are especially important because any infection in an individual with PID has the potential to become severe, and repeated chest infections in particular can increase the risk of bronchiectasis and irreversible lung damage. Trimethoprim-sulfamethoxazole prophylaxis is indicated for the prevention of Pneumocystis pneumonia and is also used for prophylaxis against staphylococcal and nocardial infections in CGD. Children with susceptibility to atypical mycobacterial infections should be placed on azithromycin prophylaxis, and acyclovir is recommended as a prophylactic agent in individuals who are susceptible to invasive herpes simplex virus infections. Fluconazole can be used for prophylaxis against candidal infections, although some patients with chronic mucocutaneous candidiasis will require voriconazole for complete prevention. In children with CGD, itraconazole is typically used for prophylaxis against Aspergillus and other fungal infections. Antifungal prophylaxis is especially important in these patients, as Aspergillus infection represents the most significant cause of mortality. Children who are susceptible to Cryptosporidium parvum infections, such as boys with X-linked hyper-IgM syndrome, should avoid water parks and tap water. Live attenuated immunizations should not be given to individuals with certain PIDs and should, therefore, be approved by an immunology specialist before administration. Live attenuated immunizations should be administered to family members and household contacts, however, to provide herd protection. For all children, universal precautions, such as good handwashing, should be practiced. They do not need to be “kept in a bubble” but should be excluded from contact with individuals who have infections and from environments highly enriched for pathogens, such as child care. Children with severe T-cell deficiencies require reverse isolation in clinic and hospital settings. If transfusions become necessary for these individuals, they should receive only irradiated, cytomegalovirus-negative blood products. In all children with PIDs, a low index of suspicion should be entertained for infections. Antimicrobial drug therapy should be given promptly and withdrawn cautiously. It is also important to note that many of these practices become unnecessary in patients who have achieved full immune reconstitution through definitive curative measures, such as hematopoietic stem cell transplant or gene therapy. Finally, more advanced treatment options should be pursued with input from an expert in PIDs. For example, IgG replacement is fully indicated as a therapy for PIDs that contain humoral deficiency as a component and should be used to treat such patients with PID accordingly. Certainly, a wide array of IgG replacement options exists, and best practice continues to evolve. Furthermore, as knowledge increases regarding the defective mechanisms responsible for impaired immunity for various PIDs, novel targeted biological therapies are being developed and tested to more specifically restore appropriate immune function in these individuals. Ultimately, the greatest opportunity for definitive treatment for most children with PIDs will come through continued improvements in hematopoietic stem cell transplant and increasing advances in gene therapy, as directed by an immunology specialist.

Secondary Immunodeficiencies

Secondary causes of immunodeficiency should always be excluded in children with suspected PIDs. A brief review of these etiologies follows.

Iatrogenic causes of immunodeficiency must be considered. Medications stand as well-established causes of IgA deficiency and include antiepileptic drugs, such as phenytoin, carbamazepine, valproic acid, and zonisamide. IgA deficiency has also been linked to use of nonsteroidal anti-inflammatory drugs. Other medications associated with hypogammaglobulinemia include anti-inflammatory compounds (eg, sulfasalazine, gold, penicillamine, and hydroxychloroquine). Many medications given for chemotherapy in the transplant setting or for treatment of malignancies, such as cyclosporine, tacrolimus, mycophenolate mofetil, cyclophosphamide, azathioprine, and 6-mercaptopurine, can cause hypogammaglobulinemia and secondary T-cell deficiencies. Corticosteroids also suppress T-cell function and impair wound healing, and they have been proposed to induce hypogammaglobulinemia if administered for extended periods. Targeted biological modifiers are being implemented in a variety of disease conditions, such as autoimmune or inflammatory diseases and solid organ transplant, and they have been demonstrated to cause secondary immunodeficiency. The primary mechanism consists of intentional disruption of the immunologic host defense processes that are causing the undesirable clinical condition. For example, individuals who are receiving anti– tumor necrosis factor α therapy for inflammatory bowel disease are well-known to have increased risk for Mycobacterium tuberculosis infection or reactivation. Several newer biologic therapies (eg, rituximab and anti-CD19 chimeric antigen receptor T cells) are designed to cause depletion of B cells and result in hypogammaglobulinemia. Finally, surgical injury to the thoracic duct can cause hypogammaglobulinemia and quantitative T-cell deficiency through loss of lymphatic contents into the pleural cavity.

Other mechanisms of immunoglobulin and lymphocyte losses must be excluded as well. Additional explanations for pulmonary losses include chylothorax or exudative pleural effusions from obstruction of lymphatic flow or perturbed intrathoracic pressure. Gastrointestinal losses can also represent a significant cause of secondary immunodeficiency. Hypogammaglobulinemia, particularly loss of IgG, and lymphopenia can be observed in children with intestinal lymphangiectasia. Other sources of gastrointestinal losses include inflammatory bowel disease and chronic diarrhea. Last, nephrotic syndrome can even lead to **IgG deficiency** in some patients through urinary losses.

IgG deficiency is a health problem in which your body doesn’t make enough immunoglobulin G (IgG).

Nutritional status cannot be underestimated as a secondary cause of immunodeficiency, and it is said to represent the most common cause of an immune deficiency. Severe malnutrition results in decreased production of secretory IgA by mucosal tissues and lower numbers of B cells. Thymic involution occurs, total T and CD4þ T-cell counts decrease, and T-cell proliferation to mitogens and antigens becomes diminished. The absence of specific nutrients can affect immunity in a variety of manners. For example, lack of protein results in overall leukopenia and diminished innate immune function. Zinc deficiency, on the other hand, results in impaired neutrophil chemotaxis and phagocytosis, natural killer cell cytotoxicity, and T-cell differentiation. Vitamin E deficiency has been reported to cause decreased T-cell function, and vitamin B6 deficiency is associated with lymphopenia, decreased antibody production, and lower T-cell proliferative responses to mitogens. Furthermore, vitamin C deficiency is known to result in susceptibility to infections, especially pneumonia, and poor wound healing through impaired neutrophil chemotaxis, phagocytosis, and killing activity. On the opposite end of the spectrum, obesity can impair neutrophil activity, lymphocyte proliferation, and T-cell development. Thus, nutritional factors must be assessed.

Physiologic and psychological stress can also cause secondary immunodeficiency. For example, critically ill postinjury patients have been shown to have impaired neutrophil phagocytosis and killing of S aureus. As a second example, brain injury induces interleukin-10–mediated suppression of immune function. In addition, psychosocial stressors can certainly affect children, whether at home, at school, or in other settings. For instance, children with depression can have decreased neutrophil phagocytosis and bactericidal activity toward S aureus. On the whole, few studies have examined the effect of psychological stressors on immune function in children, but lessons may be learned from stress in adults, which is associated with reduced secondary antibody responses and lymphocyte proliferative responses to mitogens among other immunologic changes.

Human immunodeficiency virus (HIV) infection remains the most well-studied cause of secondary immunodeficiency; a comprehensive discussion remains beyond the scope of this review. HIV infection should be excluded by antibody- and antigen-based assays combined with nucleic acid testing. For neonates, only virologic testing should be performed due to transmission of maternal antibodies. These nucleic acid–based tests should be performed at ages 14 to 21 days, 1 to 2 months, and 4 to 6 months.

Other factors should be investigated as potential secondary causes of immunodeficiency. Anatomical issues, such as eustachian tube dysfunction and congenital fistulas, can lead to recurrent infections. Atopy can produce adenoidal hypertrophy that results in recurrent otitis media or sinusitis. It can also contribute to asthma, which may be misdiagnosed as recurrent pneumonia or bronchitis. Furthermore, allergic sensitization can exacerbate eczema, which can predispose toward recurrent cutaneous infections. In terms of oncologic etiologies, multiple myeloma and chronic lymphocytic leukemia are known to produce secondary immunodeficiency, but these malignancies do not commonly occur in children. No large studies have been conducted to examine the prevalence of secondary immunodeficiency in pediatric malignancies, however. Other disease conditions, such as cystic fibrosis and diabetes mellitus, are known to favor the appearance of recurrent infections. SLE has been associated with IgA deficiency, but it remains uncertain whether the correlation is causal. Certainly, some effect on immune function is likely, as 1 of the 11 criteria used for diagnosis includes leukopenia or lymphopenia. Infections themselves can suppress immunity. A well-established example is provided by measles infection, which can produce profound impairment of B- and T-cell function. Finally, preterm infants are born with an immunocompromised state. Transfer of maternal antibodies begins during the second trimester but accelerates during the final trimester. Mature T cells can be detected at 15 to 16 weeks of gestation, but polyclonal repertoire diversity is not achieved until 22 to 26 weeks of gestation, and thymic output remains low, resulting in low T-cell receptor excision circle levels and abnormal newborn screening tests for SCID.

A hypersensitivity reaction is an extreme or unnecessary immune response that the body has to an antigen.

CORE CONCEPT: HYPERSENSITIVITY REACTIONS (TYPE I TO TYPE IV)

A hypersensitivity reaction happens when the immune system mistakenly identifies harmless substances as harmful. There are several types, depending on the class of substance that triggers it.

An antigen or allergen can refer to a toxic or foreign substance that causes an immune reaction. After detecting an antigen and perceiving it as a potential threat, the immune system mounts an immune response to dispose of it. The body can produce different types of hypersensitivity reactions, depending on the antigen a person has exposure to and how the body responds to it.

There are four different types of hypersensitivity reactions. Some evidence suggests a potential fifth type, but this may actually be a subset of type 2 hypersensitivity reactions.

Each type of hypersensitivity reaction is an extreme immune response to an antigen. Each type of reaction differs based on the type of antigen the body identifies, what type of immune response the body generates, and how quickly the body produces the response.

Some people may refer to hypersensitivity reactions as allergies, as these are a form of hypersensitivity. Although people use these terms interchangeably, an allergic reaction typically refers to the signs and symptoms a person may experience, while a hypersensitivity reaction describes the immunological process that occurs in the body.

Type 1 Hypersensitivity Reaction

Type 1 hypersensitivityTrusted Source causes an immediate responseTrusted Source and occurs after a person has exposure to an antigen. With this type of reaction, the body responds to an antigen by producing a specific type of antibody called IgE.

There are different components that can trigger type 1 hypersensitivity responses, including antigens that come from:

- food products, such as nuts, shellfish, and soy
- animal sources, such as cats, rats, or bee stings
- environmental sources, such as mold, latex, and dust
- allergic conditions, such as allergic rhinitis, allergic asthma, and conjunctivitis

There are two stages to type 1 hypersensitivity: the sensitization stage and the effect stage.

During the sensitization stage, the person encounters the antigen but does not experience any symptoms. During the effect stage, the person has exposure to the antigen again. As the body now recognizes the antigen, it is able to produce a response that results in the symptoms that people typically experience with an **allergic reaction**.

Allergic reaction occurs when your body attacks a foreign, and typically harmless, substance.

Some physical symptoms of type 1 hypersensitivity can include:

- rash
- flushing
- hives
- itching
- edema
- wheezing
- rhinitis
- stomach cramps

Responses can also cause:

- nausea and vomiting
- shortness of breath
- cardiac symptoms
- loss of consciousness

The first step a doctor may take to diagnose type 1 hypersensitivity is assessing the person's history, including taking information on signs and symptoms and reviewing their medical records. After this, they will conduct a physical examination in addition to blood and allergy tests to help identify which antigen caused the reaction.

There are different treatments for type 1 hypersensitivity, depending on the cause of the reaction and how the body responds. Some people may require emergency medical treatment with an immediate effect, whereas people with mild symptoms may require other medications. Also, people should try to avoid the allergen in the future.

Some treatment options may include:

- adrenaline, or epinephrine
- systemic glucocorticoids
- antihistamines

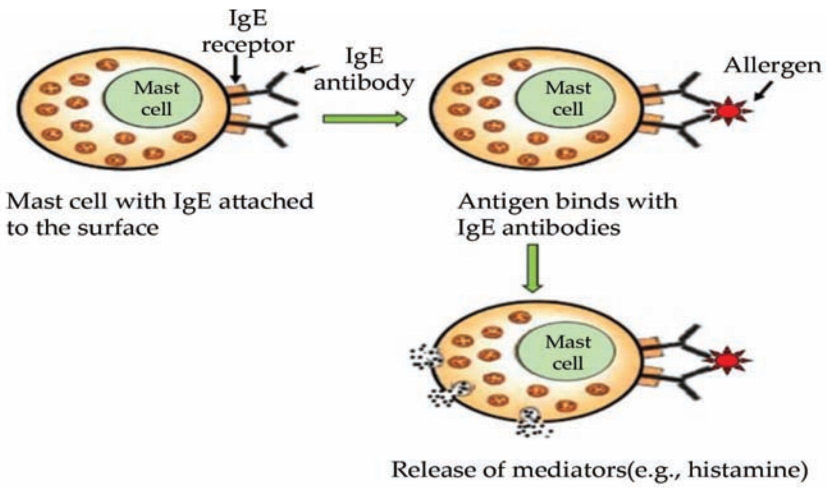


Figure 7: Type I hypersensitivity.

Type I hypersensitivity is the immediate hypersensitivity caused by IgE antibody and results anaphylaxis to insect venoms, drug and food. These allergic reactions are systematic or local due to the induction of IgE antibody to allergens. The type I hypersensitivity response is the product of an antigen cross link to membrane-bound IgE antibody of a basophil or mast cell. Histamine releases during the anaphylactic reaction and causes potential tissue damage inside the body (Figure 7).

Type 2 Hypersensitivity Reaction

Similar to type 1, type 2 hypersensitivityTrusted Source reactions also involve antibodies. In fact, type 2 and type 3 hypersensitivity both result from the same class of antibody, called IgG. The difference between them lies in the form of antigens that generate a response. Additionally, type 2 can also involve IgM antibodies.

Type 2 hypersensitivity causes cytotoxic reactions, meaning that healthy cells die as they respond to the antigens. This can cause long-term damage to cells and tissues, resulting in conditions such as:

- the blood disorder immune thrombocytopenia if there are not enough platelets
- autoimmune hemolytic anemia if the red blood cells burst
- autoimmune neutropenia if the body destroys neutrophils
- autoimmune conditions such as Graves' disease

Common causes of type 2 hypersensitivity reactions include drugs such as:

- penicillin
- thiazides
- cephalosporins
- methyldopa

There are different subsets of type 2 hypersensitivity, depending on the trigger and the response. Methods of diagnosis vary according to these subsets, as a doctor must be cautious to avoid provoking further damage. Diagnosis may involve direct immunofluorescence to help identify causative antibodies.

Treatment for type 2 hypersensitivity typically involves immunosuppressants to prevent the action of unusual antibodies. Treatment options may include:

- systemic glucocorticoids
- cyclophosphamide and cyclosporin agents
- intravenous immunoglobulin infusions
- plasmapheresis

Type II hypersensitivity responses are initiated by the toxic properties of antibody attached with antigens on the external of cells. The antibodies can activate complement dependent lysis which leads to tissue damage. In a cytotoxic reaction, the antibody responds right to the antigen which is attached to the cell membrane to activate cell lysis by complement induction. These antigens could be "self " i.e.

autoimmune reactions or "non-self ". Cytotoxic reactions are intermediated by IgM and IgG. One of the best examples of cytotoxic reactions is the Rh-incompatibility of a newborn. Other examples are blood transfusion reactions, Good pasture's syndrome and autoimmune (Figure 8).

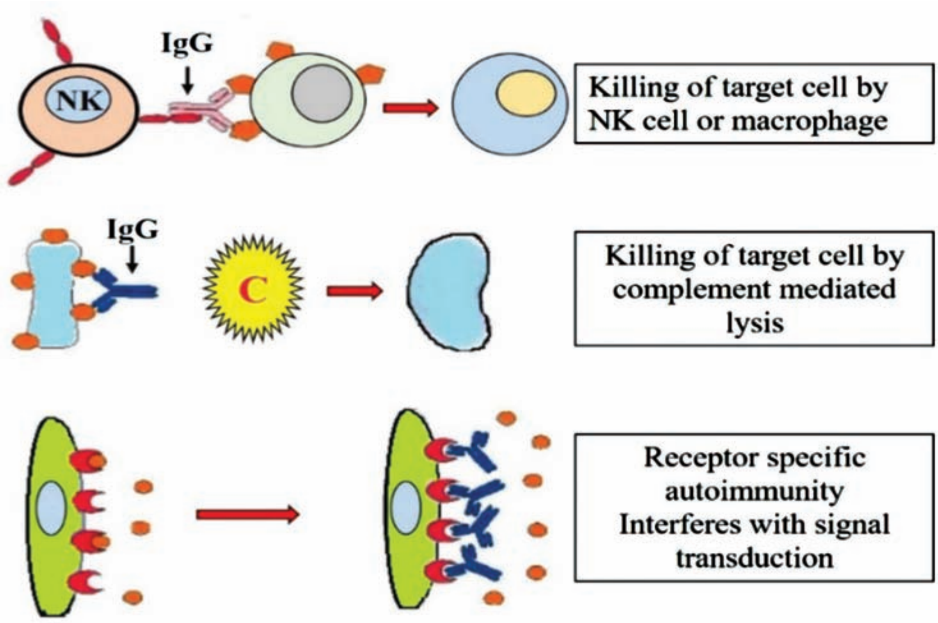


Figure 8: Type II hypersensitivity.

Type 3 Hypersensitivity Reaction

In type 3 hypersensitivityTrusted Source, antigens and antibodies form complexes in the skin, blood vessels, joints, and kidney tissues. These complexes cause a series of reactions that lead to tissue damage.

Causes of a type 3 hypersensitivity reaction can include:

- drugs that contain proteins from different organisms, such as antivenins
- the drug infliximab, which people use to manage autoimmune conditions
- animal sources, such as insect stings or tick bites

Type 3 hypersensitivity reactions can lead to:

- serum sickness
- lupus
- rheumatoid arthritis
- small-vessel vasculitis
- Henoch-Schönlein purpura

When diagnosing type 3 hypersensitivity, a doctor may look at a person's clinical history, perform a physical exam, and conduct various assessments, including tests of blood and urine samples, biopsies, and imaging scans.

There are many treatment options available, depending on the severity and presentation of the hypersensitivity response. Typically, treatment involves controlling the underlying condition. This may involve immunosuppression with systemic glucocorticoids and disease-modifying drugs.

Type III hypersensitivity is facilitated by the formation of antigen antibody complexes. IgG and IgM bind antigen, deposit antigenantibody (immune) complexes. This complex stimulates complement, which effects in PMN chemotaxis and initiation. PMNs then discharge tissue damaging enzymes to the cell. One of the most common examples of type III hypersensitivity response in human body is serum sickness (Figure 9).

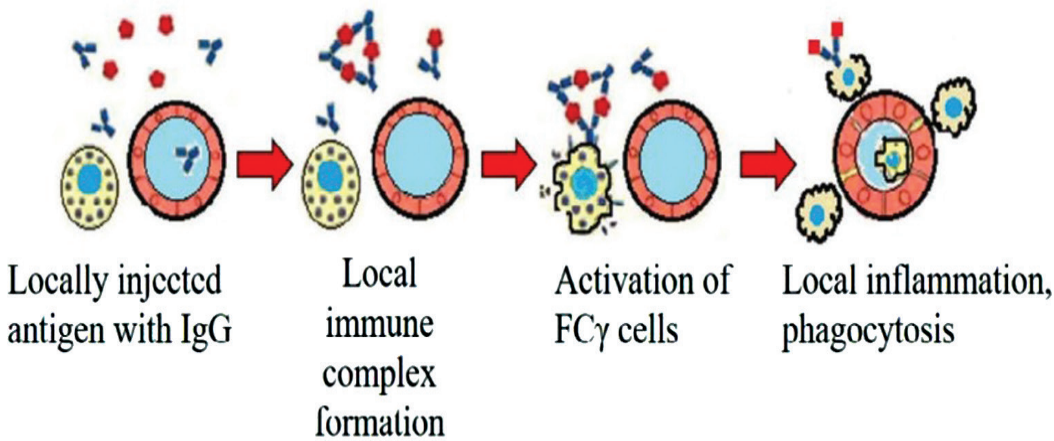


Figure 9: Type III hypersensitivity.

Type 4 Hypersensitivity Reaction

Unlike the other types, type 4 hypersensitivityTrusted Source reactions are cell-mediated.

Instead of antibodies, white blood cells called T cells control type 4 hypersensitivity reactions. Experts can further subdivide these reactions into type 4a, type 4b, type 4c, and type 4d based on the type of T cell involved and the reaction it produces.

This type also differs from the other three in that it causes a delayed reaction.

The three subsets of type 4 hypersensitivity are as follows:

- contact dermatitis
- tuberculin-type hypersensitivity
- granulomatous-type hypersensitivity

Some common causes of type 4 hypersensitivity reactions include exposure to poison ivy, certain metals, and drugs such as antibiotics or anticonvulsants.

Diagnosing contact dermatitis is often achievable using a skin biopsy and skin patch tests. A doctor is likely to use a chest X-ray when diagnosing tuberculin-type hypersensitivity. Granulomatous-type hypersensitivity is more challenging to diagnose, and a doctor may consider using any of the following to make an assessment:

- X-ray
- lymph node biopsy
- enzyme analysis
- salivary gland analysis

Treatment varies from case to case. With contact dermatitis, for example, a doctor may prescribe topical steroids. However, with tuberculin-type hypersensitivity, the doctor will use a normal procedure for tuberculosis. Common treatments for tuberculin-type hypersensitivity include:

- rifampin
- isoniazid
- pyrazinamide
- ethambutol

Treatment for granulomatous-type hypersensitivity can also include steroid therapy, but treatments vary depending on the condition that comes from the reaction. For example, a doctor may prescribe methotrexate as steroid treatment if a person presents with pulmonary sarcoidosis.

Delayed or type IV hypersensitivity was initially described by its period course in which the responses took 12-24 hours of time to progress and persevered for 2-3 days. Cell-mediated responses are introduced by T-lymphocytes and intermediated by effector T-cells and macrophages. This reaction contains the antigens attached to the surface of lymphocytes. The pre-sensitized lymphocytes can induce cytokines, which can damage cells. Many long-lasting diseases, including tuberculosis exhibits delayed type hypersensitivity (Figure 10).

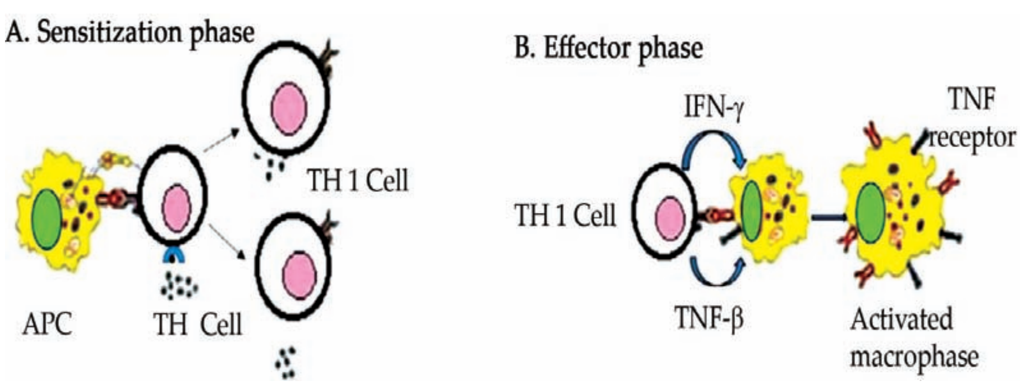
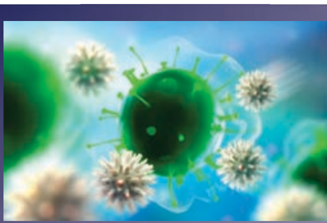


Figure 11: Type IV hypersensitivity.



CORE CONCEPT: IMMUNOLOGY IN INFECTIOUS DISEASES

Immunology in infectious diseases is a fascinating and critical field that delves into the intricate interactions between the human immune system and various pathogens. Infectious diseases are caused by microorganisms such as bacteria, viruses, fungi, and parasites, and understanding how our immune system responds to these invaders is crucial for preventing, treating, and managing infections. Immunology provides insights into the complex mechanisms that underlie our body's defense mechanisms, as well as the strategies employed by pathogens to evade these defenses.

At its core, immunology in infectious diseases explores how the immune system identifies, targets, and eliminates foreign invaders, while also maintaining a delicate balance to prevent unnecessary harm to the body's own tissues. This dynamic interplay involves a wide array of immune cells, molecules, receptors, and signaling pathways that work in harmony to mount effective responses against infections.

Immunology plays a central role in our understanding of infectious diseases and how the body defends itself against pathogens. Infectious diseases are caused by microorganisms such as bacteria, viruses, fungi, and parasites, and the immune system's response to these invaders is essential for preventing and controlling infections. Here's how immunology is involved in infectious diseases:

- Innate Immune Response:
  - The innate immune system is the body's first line of defense against pathogens. It includes physical barriers (skin, mucous membranes) and immune cells like neutrophils, macrophages, and dendritic cells.
  - Innate immune cells recognize common patterns on pathogens, called pathogen-associated molecular patterns (PAMPs), using pattern recognition receptors (PRRs).
  - Innate immunity triggers inflammation, recruits immune cells to the site of infection, and helps contain the infection until the adaptive immune response develops.
- Adaptive Immune Response:
  - The adaptive immune system provides specific and targeted responses to pathogens.
  - T cells and B cells are key players in adaptive immunity. T cells recognize infected cells and coordinate immune responses, while B cells produce antibodies that neutralize pathogens.
  - Memory cells generated during an adaptive immune response provide long-lasting immunity against future encounters with the same pathogen.

- Antigen Presentation:
  - Immune cells, particularly dendritic cells, present antigens (pieces of pathogens) to T cells, initiating a specific immune response.
  - Antigen presentation is crucial for activating T cells and B cells to recognize and respond to pathogens.
- Cytokines and Chemokines:
  - Immune cells communicate through signaling molecules called cytokines and chemokines.
  - Cytokines regulate immune responses, inflammation, and immune cell activity, while chemokines guide immune cells to the site of infection.
- Vaccination and Immunization:
  - Vaccines leverage the immune system's memory response to provide protection against infectious diseases.
  - Vaccines introduce harmless antigens from pathogens, stimulating a controlled immune response and generating memory cells.
- Immunopathology:
  - In some cases, the immune response can cause tissue damage and contribute to the severity of certain infectious diseases.
  - Immunopathology is evident in diseases like sepsis, where an excessive immune response can lead to organ dysfunction.
- Immune Evasion:
  - Pathogens often evolve strategies to evade the immune system, allowing them to establish chronic infections.
  - For example, some viruses can downregulate MHC molecules to evade T cell recognition.
- Host-Pathogen Interactions:
  - Understanding the interactions between host cells and pathogens helps identify targets for treatment and vaccine development.
  - Researchers study how pathogens attach to host cells, evade immune responses, and manipulate host machinery.

Immunology research has led to significant advances in our ability to diagnose, treat, and prevent infectious diseases. It has facilitated the development of vaccines, antiviral drugs, and immune-based therapies. Additionally, understanding immune responses to infections contributes to our knowledge of broader immune system functions and contributes to the development of novel treatment strategies for a wide range of diseases.

Innate Immune Response

The innate immune response is the first line of defense that the body mounts against invading pathogens, such as bacteria, viruses, fungi, and parasites. It is a rapid and nonspecific response that provides immediate protection, even before the adaptive immune response is fully activated. The innate immune system is a critical component of the body's overall defense mechanisms, and it serves as a crucial barrier against infections. Here's an overview of the key features and components of the innate immune response:

Key Features:

- **Rapid Response:** The innate immune response is quick to initiate, often within minutes to hours of encountering a pathogen.
- **Nonspecific:** It does not require prior exposure to the specific pathogen and responds to a wide range of pathogens in a similar manner.
- **Physical and Chemical Barriers:** Physical barriers, such as the skin and mucous membranes, prevent pathogens from entering the body. Chemical barriers, such as enzymes in saliva and stomach acid, can kill or inhibit the growth of pathogens.
- **Phagocytosis:** Phagocytic cells, including neutrophils and macrophages, engulf and destroy pathogens by surrounding them and digesting them.
- **Inflammation:** The innate response triggers inflammation at the site of infection. Inflammation increases blood flow, recruits immune cells to the site, and helps contain and eliminate the pathogen.
- **Complement System:** The complement system is a group of proteins that can enhance the immune response by promoting inflammation, opsonization (marking pathogens for destruction), and cell lysis.

Components of the Innate Immune Response:

- **Physical Barriers:** Skin, mucous membranes, and epithelial cells create a physical barrier that prevents pathogens from entering the body.
- **Phagocytic Cells:** Neutrophils, macrophages, and dendritic cells are specialized cells that engulf and destroy pathogens through phagocytosis.
- **Natural Killer (NK) Cells:** NK cells are lymphocytes that target infected or abnormal host cells, such as virus-infected cells or cancer cells.
- **Inflammatory Response:** Immune cells release signaling molecules called cytokines that promote inflammation. Inflammation increases blood flow, recruits immune cells, and creates an environment unfavorable for pathogens.
- **Acute-Phase Proteins:** These proteins are produced in response to inflammation and help regulate the immune response.

- **Pattern Recognition Receptors (PRRs):** These receptors on immune cells recognize common patterns on pathogens called pathogen-associated molecular patterns (PAMPs).
- **Toll-Like Receptors (TLRs):** A group of PRRs that recognize specific PAMPs and trigger immune responses.
- **Fever:** Elevated body temperature in response to infection helps inhibit pathogen growth and enhance immune responses.

The innate immune response sets the stage for the adaptive immune response, which follows later and provides targeted and long-lasting protection. Together, these two arms of the immune system work in harmony to defend the body against a wide range of pathogens and maintain overall health.

Adaptive Immune Response

The adaptive immune response is a highly specialized and specific defense mechanism that the body employs to target and eliminate specific pathogens. Unlike the innate immune response, which provides immediate but nonspecific protection, the adaptive immune response takes time to develop but offers long-lasting immunity and memory against particular pathogens. This response is characterized by its ability to "adapt" and tailor its actions to the unique features of each pathogen. Here's an overview of the key features and components of the adaptive immune response:

Key Features:

- **Specificity:** The adaptive immune response targets specific antigens (molecules on the surface of pathogens) in a highly specific manner.
- **Memory:** Upon encountering a pathogen for the first time, the adaptive immune system generates memory cells that "remember" the pathogen. This leads to a faster and more robust response upon subsequent encounters.
- **Slower Activation:** The adaptive response takes time to develop, typically days to weeks, as immune cells need to recognize the antigen, proliferate, and differentiate into effector cells.
- **Long-Lasting Protection:** The memory cells produced during the adaptive immune response provide long-lasting immunity against the same pathogen.

Components of the Adaptive Immune Response:

- **T Lymphocytes (T Cells):** T cells play a central role in coordinating the adaptive immune response. There are several types of T cells, including helper T cells, cytotoxic T cells, and regulatory T cells.
  - **Helper T Cells (CD4+ T Cells):** Helper T cells assist in activating other immune cells, such as B cells and cytotoxic T cells. They also promote the production of antibodies by B cells.

- **Cytotoxic T Cells (CD8+ T Cells):** Cytotoxic T cells directly attack infected host cells by recognizing and killing them.
- Regulatory T Cells (Tregs): Regulatory T cells help regulate and control the immune response, preventing excessive inflammation and autoimmunity.

- **B Lymphocytes (B Cells):** B cells produce antibodies, also known as immunoglobulins, that recognize and neutralize specific antigens. Antibodies can bind to pathogens, preventing their entry into cells or marking them for destruction by other immune cells.
- **Antibody-Mediated Immunity:** Antibody-mediated immunity, also called humoral immunity, involves the production of antibodies by B cells. These antibodies circulate in the bloodstream and neutralize pathogens, preventing infection.
- **Cell-Mediated Immunity:** Cell-mediated immunity involves the activation of T cells, particularly cytotoxic T cells, which directly attack infected host cells. This response is critical for clearing intracellular infections, such as those caused by viruses.
- **Major Histocompatibility Complex (MHC):** MHC molecules present antigens to T cells, allowing them to recognize and respond to specific pathogens.
- **Clonal Expansion:** Upon encountering an antigen, B cells and T cells undergo clonal expansion, where they proliferate to produce a larger population of effector cells capable of targeting the pathogen.
- **Immunological Memory:** Memory B cells and memory T cells are long-lived cells that "remember" specific antigens. They allow for a faster and more efficient response upon re-exposure to the same pathogen.

The adaptive immune response is a sophisticated and dynamic process that provides the body with a highly targeted defense against a wide range of pathogens. Its ability to generate memory ensures that the immune system can mount a swift and potent response upon encountering familiar pathogens, thereby preventing reinfections and contributing to overall immunity.

Antigen Presentation

Antigen presentation is a crucial process in the immune system that enables immune cells to communicate and initiate specific immune responses against pathogens. It involves the display of antigens—pieces of proteins from pathogens—on the surface of immune cells to be recognized by other immune cells. Antigen presentation plays a central role in both the adaptive immune response and the activation of various immune cells. Here's an overview of the process of antigen presentation:

Antigen-Presenting Cells (APCs): Antigen-presenting cells are specialized immune cells that capture, process, and present antigens to other immune cells. The most important APCs include:

- **Dendritic Cells:** Dendritic cells are professional APCs that capture antigens at the site of infection and migrate to lymph nodes to present antigens to T cells.
- **Macrophages:** Macrophages engulf and digest pathogens, and they present antigens to T cells to initiate immune responses.
- **B Cells:** B cells internalize antigens through their B cell receptors, process them, and present them to helper T cells.

Antigen Processing and Presentation:

- **Internalization:** APCs capture antigens by engulfing pathogens or taking up soluble antigens. These antigens are broken down into smaller peptides within the APC.
- **Antigen Processing:** Proteins from the pathogen are degraded into short peptides by proteases within the APC's compartments, such as endosomes and lysosomes.
- **Binding to Major Histocompatibility Complex (MHC) Molecules:**
  - MHC class I molecules present peptides from intracellular pathogens to cytotoxic T cells (CD8+ T cells).
  - MHC class II molecules present peptides from extracellular pathogens to helper T cells (CD4+ T cells).
- **Peptide-MHC Complex Formation:** Processed peptides bind to MHC molecules, forming a peptide-MHC complex.
- **Surface Expression:** The peptide-MHC complex is transported to the cell surface and presented to T cells.

Recognition by T Cells:

- **T Cell Receptor (TCR) Recognition:** T cells have T cell receptors on their surface that recognize specific antigens presented on the surface of APCs. The TCR binds to the peptide-MHC complex.
- **Co-Stimulation:** Co-stimulatory molecules on the APC and the T cell interact to ensure proper activation of the T cell.

Immune Activation:

- **Helper T Cell Activation:** Helper T cells recognize antigens presented by MHC class II molecules on APCs. This activation triggers helper T cells to secrete cytokines that assist in the activation of other immune cells, such as B cells and cytotoxic T cells.
- **Cytotoxic T Cell Activation:** Cytotoxic T cells recognize antigens presented by MHC class I molecules on infected cells. Activated cytotoxic T cells directly kill infected cells.

Antigen presentation is a critical step in initiating specific immune responses and coordinating immune reactions. It allows immune cells to communicate, activate, and orchestrate the appropriate defense mechanisms against a wide range of pathogens.

Cytokines and Chemokines

Cytokines and chemokines are signaling molecules that play essential roles in the immune system by mediating communication between immune cells and coordinating immune responses. These small proteins are produced by various cells, including immune cells, and act as messengers to regulate immune cell development, activation, migration, and interactions. Cytokines and chemokines play critical roles in inflammation, immune defense, tissue repair, and the overall balance of the immune system. Here's an overview of cytokines and chemokines:

Cytokines: Cytokines are a diverse group of proteins that facilitate communication between immune cells and other cells in the body. They can have both pro-inflammatory and anti-inflammatory effects, and their balance is crucial for maintaining a healthy immune response. Some key types of cytokines include:

- **Interleukins (IL):** These cytokines regulate communication between leukocytes (white blood cells) and are involved in various immune responses. For example, IL-1 and IL-6 contribute to inflammation, while IL-2 stimulates T cell proliferation.
- **Tumor Necrosis Factor (TNF):** TNF is involved in inflammation and immune responses. It plays a role in fighting infections but can also contribute to inflammatory diseases.
- **Interferons (IFN):** IFNs are essential for antiviral defense and modulating immune responses. They inhibit viral replication and activate immune cells.
- **Chemokines:** While chemokines are also a type of cytokine, they specifically regulate the migration of immune cells to sites of infection or inflammation.

Chemokines: Chemokines are a subset of cytokines that act as chemoattractants, guiding immune cells to specific locations within the body. They help orchestrate the movement of immune cells to sites of infection, injury, or inflammation. Chemokines are critical for processes such as:

- **Recruitment of Immune Cells:** Chemokines attract immune cells, such as neutrophils and macrophages, to sites of infection or tissue damage.
- **Cellular Trafficking:** Chemokines help immune cells move within tissues and navigate through the bloodstream to reach target areas.
- **Inflammation:** Chemokines are central to the initiation and resolution of inflammation by controlling the migration of immune cells and promoting their activation.

Roles in Immune Responses:

- **Inflammation:** Cytokines, including pro-inflammatory cytokines, initiate and sustain the inflammatory response, recruiting immune cells to the site of infection or injury.
- **Immune Cell Activation:** Cytokines activate immune cells, such as T cells and B cells, by promoting their proliferation, differentiation, and function.
- **Antiviral Defense:** Interferons are crucial for limiting viral replication and promoting an antiviral state in infected cells.
- **Immunomodulation:** Cytokines help regulate the balance between different immune responses, preventing excessive inflammation and autoimmunity.
- **Tissue Repair:** Some cytokines aid in tissue repair and wound healing after infections or injuries.

Cytokines and chemokines form a complex network of communication that allows immune cells to work together effectively and mount appropriate responses against infections and other challenges. Dysregulation of cytokines and chemokines can lead to immune-related diseases and disorders. Understanding these signaling molecules is essential for advancing our knowledge of the immune system and developing therapies for immune-related conditions.

Vaccination and Immunization

Vaccination and immunization are essential strategies used to prevent the spread of infectious diseases and protect individuals and populations from potentially serious illnesses. They involve introducing harmless or weakened forms of pathogens or their components into the body to stimulate the immune system and create immunity without causing the actual disease. Vaccination and immunization have been among the most successful public health interventions in history, saving millions of lives and reducing the burden of infectious diseases worldwide.

Key Concepts:

- **Vaccine:** A vaccine is a biological preparation that contains antigens from a specific pathogen (bacteria, virus, or other microorganism) or parts of it. Vaccines stimulate the immune system to recognize and remember the pathogen, creating an immune response without causing illness.
- **Immunization:** Immunization refers to the process of inducing immunity through vaccination. When a person receives a vaccine, their immune system responds by producing antibodies and memory cells that provide protection against future infections.

How Vaccines Work:

- **Antigen Exposure:** Vaccines contain harmless antigens from pathogens or components that mimic antigens. These antigens stimulate the immune system without causing disease.

- **Immune Response:** The immune system recognizes the antigens in the vaccine as foreign and mounts an immune response, including the production of antibodies and memory cells.
- **Memory Cells:** Memory B cells and memory T cells "remember" the antigens. If the person is later exposed to the actual pathogen, the immune system can quickly recognize and eliminate it.

Types of Vaccines:

- **Live Attenuated Vaccines:** These vaccines contain weakened forms of live pathogens. Examples include the measles, mumps, and rubella (MMR) vaccine.
- **Inactivated Vaccines:** These vaccines contain killed or inactivated pathogens or their components. Examples include the polio vaccine.
- **Subunit, Recombinant, and Conjugate Vaccines:** These vaccines use specific components of the pathogen, such as proteins or sugars, to trigger an immune response. Examples include the hepatitis B vaccine.
- **Viral Vector Vaccines:** These vaccines use a harmless virus (vector) to deliver genetic material from the pathogen, stimulating an immune response. Examples include some COVID-19 vaccines.
- **mRNA Vaccines:** These vaccines use a small piece of genetic material (mRNA) to instruct cells to produce a harmless part of the pathogen, triggering an immune response. Examples include some COVID-19 vaccines.

Benefits of Vaccination:

- **Prevents Disease:** Vaccines prevent serious and potentially life-threatening diseases, such as polio, measles, and influenza.
- **Herd Immunity:** High vaccination rates protect vulnerable individuals who cannot be vaccinated, such as those with weakened immune systems.
- **Reduced Disease Burden:** Vaccination has led to a significant reduction in the prevalence of many infectious diseases.
- **Cost-Effective:** Vaccination is a cost-effective public health intervention, reducing healthcare costs associated with treating diseases.

Vaccination schedules vary based on factors such as age, health status, and risk factors. It's important to follow recommended vaccination schedules to ensure optimal protection for individuals and communities. Vaccination remains a cornerstone of public health efforts to prevent and control the spread of infectious diseases.

Immunopathology

Immunopathology refers to the study of the immune system's response to disease, particularly focusing on situations where the immune system's normal functions become dysregulated or result in harmful effects. It encompasses the understanding of how the immune system can both contribute to and cause disease, leading to a range of immune-related disorders. Immunopathology plays a significant role in various medical fields, including immunology, pathology, and clinical medicine.

Key Concepts in Immunopathology:

- **Autoimmunity:** Immunopathology includes the study of autoimmune diseases, where the immune system mistakenly targets and attacks the body's own tissues. Examples of autoimmune diseases include rheumatoid arthritis, systemic lupus erythematosus (SLE), and multiple sclerosis (MS).
- **Hypersensitivity Reactions:** Immunopathology involves understanding hypersensitivity reactions, where the immune system overreacts to harmless substances (allergens), leading to allergies or other adverse responses.
- **Immune Deficiencies:** Immunopathology encompasses primary immunodeficiency disorders (genetic defects) and secondary immunodeficiencies (acquired deficiencies, e.g., due to HIV infection), where the immune system's weakened function leads to increased susceptibility to infections.
- **Immunologic Tolerance and Dysregulation:** Immunopathology explores how the immune system maintains tolerance to self-antigens and how breakdowns in this tolerance can lead to autoimmune diseases.
- **Inflammatory and Immune-Mediated Diseases:** Conditions marked by chronic inflammation and immune responses, such as inflammatory bowel disease (IBD) and psoriasis, fall under the realm of immunopathology.
- **Immunopathological Mechanisms:** Immunopathology delves into the mechanisms behind immune-mediated tissue damage, such as the role of immune complexes, cytokines, and cytotoxic T cells in causing harm.

Clinical Significance:

- **Diagnosis:** Immunopathology helps diagnose and classify various immune-related diseases by analyzing immune responses, antibody profiles, and cellular reactions.
- **Treatment:** Understanding immunopathology informs the development of targeted therapies, such as immunosuppressants for autoimmune diseases or immunomodulatory drugs to manage excessive immune responses.
- **Prognosis:** Immunopathology insights contribute to predicting disease progression and outcomes, guiding patient management.
- **Research and Drug Development:** Immunopathology research provides insights into disease mechanisms, leading to the development of novel

treatments and interventions.

- **Personalized Medicine:** Immunopathology supports personalized treatment approaches based on an individual's immune profile and disease characteristics.

Immunopathology is a multidisciplinary field that bridges immunology, pathology, and clinical medicine. Researchers and healthcare professionals in this field strive to unravel the complex interactions between the immune system and various diseases, with the ultimate goal of improving diagnosis, treatment, and management of immune-related disorders.

Immune Evasion

Immune evasion refers to the strategies that pathogens use to evade or manipulate the host's immune system in order to establish and maintain an infection. Pathogens have evolved various mechanisms to counteract immune responses and increase their chances of survival within the host. Immune evasion is a complex and dynamic process that involves interactions between pathogens and the host's immune cells and molecules.

Strategies of Immune Evasion:

- **Antigen Variation:** Some pathogens, particularly viruses and bacteria, can rapidly change the surface antigens they express. This makes it challenging for the immune system to recognize and target the pathogen effectively.
- **Mimicking Host Molecules:** Pathogens can produce molecules that resemble host molecules, effectively "hiding" from the immune system by not appearing foreign.
- **Suppression of Immune Cells:** Some pathogens can directly inhibit or destroy immune cells. For example, HIV targets and destroys CD4+ T cells, weakening the immune response.
- **Blocking Immune Signals:** Pathogens can interfere with immune signaling pathways, preventing the host from mounting an effective immune response.
- **Modulating Immune Responses:** Pathogens can manipulate the host's immune response, promoting anti-inflammatory responses to evade immune attack.
- **Inhibiting Complement Activation:** Pathogens can interfere with the activation of the complement system, which normally helps eliminate pathogens by promoting inflammation and cell lysis.
- **Inducing Tolerance:** Some pathogens can induce immune tolerance, making the host less responsive to the pathogen and allowing it to persist.
- **Intracellular Survival:** Pathogens that can live inside host cells are protected from immune surveillance. Examples include intracellular bacteria like Mycobacterium tuberculosis.

- **Immune Evasion Proteins:** Some pathogens produce specific proteins that directly target and interfere with immune cells or signaling pathways.

Examples of Immune Evasion:

- **HIV:** HIV infects and destroys CD4+ T cells, which are critical for immune responses. It also rapidly mutates its surface proteins, making it difficult for the immune system to generate an effective response.
- **Malaria:** The Plasmodium parasite that causes malaria changes the surface antigens on its red blood cell stages, evading immune recognition.
- **Herpesviruses:** Herpesviruses establish latent infections, hiding in nerve cells and periodically reactivating to cause disease.
- **Influenza Virus:** Influenza viruses frequently undergo genetic changes, allowing them to evade recognition by previously acquired immunity.
- **Mycobacterium tuberculosis:** This bacterium can evade immune responses by residing inside immune cells, preventing their destruction.
- **Cancer Cells:** Cancer cells can suppress immune responses, allowing them to evade immune recognition and destruction.

Understanding the mechanisms of immune evasion is critical for developing effective treatments and vaccines against infectious diseases. Researchers study these strategies to find ways to counteract them and develop interventions that can enhance the immune system's ability to recognize and eliminate pathogens.

Host-Pathogen Interactions

Host-pathogen interactions refer to the complex and dynamic relationships between microorganisms (pathogens) and their host organisms. These interactions are central to the outcome of infectious diseases and involve a wide range of molecular, cellular, and physiological processes. Host-pathogen interactions can influence the success of infection, the severity of disease, and the ability of the host's immune system to respond effectively. Understanding these interactions is crucial for developing strategies to prevent and treat infectious diseases.

Key Aspects of Host-Pathogen Interactions:

- **Attachment and Entry:** Pathogens often have specific mechanisms to attach to and enter host cells. This initial step is critical for establishing infection.
- **Immune Evasion:** Pathogens may use various strategies to evade the host's immune system, as discussed earlier. This can include antigen variation, inhibition of immune signaling, and interference with immune cell function.
- **Proliferation and Replication:** Pathogens replicate and multiply within host cells, often exploiting the host's resources for their own growth and survival.

- **Disease Progression:** The interactions between pathogens and host cells can lead to tissue damage, inflammation, and other factors contributing to the progression of disease symptoms.
- **Transmission:** Pathogens have evolved mechanisms to ensure their transmission to new hosts, often involving manipulation of host behavior, reproduction, or other factors.
- **Adaptation and Evolution:** Host-pathogen interactions can drive the evolution of both pathogens and hosts. Pathogens may evolve to better infect hosts, while hosts may develop genetic resistance or immunity.

Examples of Host-Pathogen Interactions:

- **Bacterial Infections:** Bacteria like Streptococcus pneumoniae can attach to host cells using surface proteins and evade immune detection by mimicking host molecules.
- **Viral Infections:** Viruses like the human immunodeficiency virus (HIV) target immune cells, replicate inside them, and evade immune surveillance.
- **Parasitic Infections:** Malaria parasites manipulate host red blood cells and evade immune recognition by frequently changing their surface antigens.
- **Fungal Infections:** Fungi like Candida albicans can evade immune responses and colonize host tissues.
- **Host-Microbiota Interactions:** The gut microbiota interacts with the host's immune system and influences its development and function.

Clinical Implications:

- **Treatment and Therapeutics:** Understanding host-pathogen interactions can inform the development of targeted therapies that disrupt these interactions and prevent infection.
- **Vaccine Development:** Vaccines aim to provoke immune responses that effectively counteract specific host-pathogen interactions, preventing infection or reducing disease severity.
- **Antibiotic Resistance:** Knowledge of host-pathogen interactions can help combat antibiotic resistance by identifying new targets for drug development.
- **Immune System Modulation:** Insights into host-pathogen interactions can guide efforts to modulate immune responses in cases of excessive inflammation or immune deficiency.

Studying host-pathogen interactions provides insights into the intricate strategies pathogens use to colonize and survive within hosts, as well as the host's mechanisms to recognize and eliminate these invaders. This knowledge is essential for advancing our understanding of infectious diseases and developing effective interventions to control and prevent them.

Immunology in Cancer

Immunology plays a significant role in understanding and treating cancer. The relationship between the immune system and cancer is complex, involving both the body's efforts to recognize and eliminate cancer cells (immune surveillance) and the ability of cancer cells to evade immune responses. This field, known as cancer immunology, explores ways to harness the immune system to target and treat cancer.

Key Aspects of Immunology in Cancer:

- **Immune Surveillance:** The immune system constantly monitors the body for abnormal or cancerous cells. Natural Killer (NK) cells and cytotoxic T cells are among the immune cells that can recognize and destroy cancer cells.
- **Tumor Antigens:** Tumors often express unique antigens that the immune system can recognize as foreign. These antigens can be targeted by immune cells for destruction.
- **Tumor Microenvironment:** The tumor microenvironment consists of various immune cells, blood vessels, and connective tissue surrounding the tumor. It can either promote or inhibit immune responses against the tumor.
- **Immune Evasion:** Cancer cells can develop mechanisms to evade immune detection and destruction. This includes downregulating expression of antigens and inhibiting immune cell function.
- **Immunotherapy:** Immunotherapy aims to enhance the body's immune response against cancer. It includes approaches such as checkpoint inhibitors (blocking molecules that inhibit immune responses), adoptive T cell therapy, and cancer vaccines.
- **Checkpoint Inhibitors:** Checkpoint inhibitors target molecules like PD-1 and CTLA-4, which cancer cells may exploit to suppress immune responses. Blocking these checkpoints can enhance the immune system's ability to recognize and attack cancer cells.
- **CAR-T Cell Therapy:** Chimeric Antigen Receptor (CAR) T cell therapy involves modifying a patient's T cells to express a receptor that targets specific cancer antigens. These modified T cells are then infused back into the patient to attack the cancer.
- **Cancer Vaccines:** Cancer vaccines stimulate the immune system to recognize and target cancer cells. Some vaccines use tumor antigens or genetic material to provoke an immune response.
- **Immune Checkpoint Proteins:** Some tumors overexpress immune checkpoint proteins, which inhibit immune responses. Targeting these proteins with antibodies can reactivate immune attacks against the tumor.

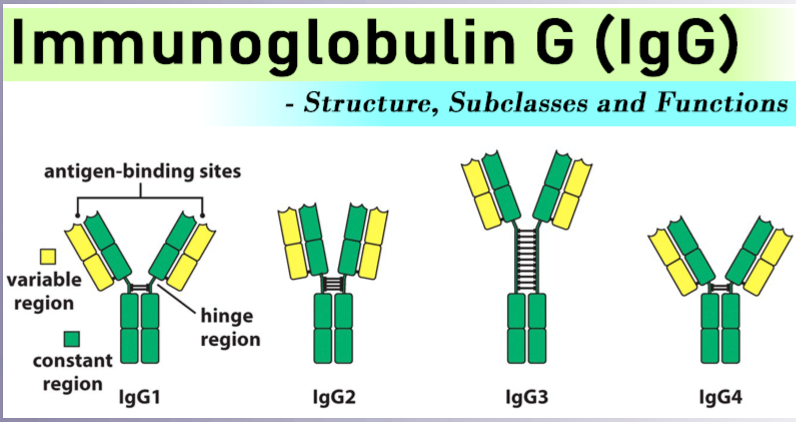
Clinical Applications:

- **Treatment:** Immunotherapy has revolutionized cancer treatment, leading to durable responses and extended survival in some cases. It is used in various cancer types, including melanoma, lung cancer, and certain leukemias.
- **Combination Therapies:** Combining immunotherapies with other treatments, such as chemotherapy or targeted therapies, can enhance their effectiveness.
- **Precision Medicine:** Identifying specific tumor antigens allows for personalized immunotherapy approaches tailored to a patient's unique cancer.
- **Monitoring Response:** Monitoring immune responses and changes in the tumor microenvironment can guide treatment decisions and predict patient outcomes.
- **Future Developments:** Ongoing research aims to improve the design of immunotherapies, understand resistance mechanisms, and develop new approaches for targeting cancer cells.

Immunology in cancer is a rapidly evolving field with significant implications for patient care. By harnessing the power of the immune system, researchers and clinicians are transforming the landscape of cancer treatment and offering new hope to patients with previously limited options.

A CONCEPT OF IMMUNOGLOBULIN G

Immunoglobulin G (IgG) is the most common type of antibody found in the human body. It represents approximately 75% of the total antibodies present in the serum, which is the clear part of the blood. IgG antibodies are produced and released by plasma B cells, a type of white blood cell.



- Antibodies, also known as immunoglobulins, are glycoproteins that play a crucial role in the immune system's defense against foreign substances and microorganisms. They have a Y-shaped structure, with two arms called paratopes that can bind to specific antigens, which are molecules present on the surface of pathogens.
- In humans, there are five different classes of antibodies: IgG, IgA, IgM, IgD, and IgE. Among these, IgG is the most abundant and versatile antibody class. It is found not only in the blood circulation but also in all other body fluids, including saliva, tears, and breast milk.
- One of the primary functions of IgG antibodies is to protect against bacterial and viral infections. They can neutralize pathogens by binding to their antigens and preventing them from entering or damaging the body's cells. IgG antibodies can also activate other immune cells, such as natural killer cells and macrophages, to destroy the pathogens directly through processes like Antibody-Dependent Cellular Cytotoxicity (ADCC) and Antibody-Dependent Cellular Phagocytosis (ADCP).
- Additionally, IgG antibodies play a crucial role in the immune system's complement system, which is a group of proteins that work together to eliminate pathogens. IgG antibodies can trigger a process called Complement-Mediated Cytotoxicity (CMC), where the complement proteins are activated to form a membrane attack complex that punctures and kills the targeted cells.
- The concentration of IgG in the serum is typically around 10 to 16 mg/mL, making it the major immunoglobulin in the extracellular spaces.

This high concentration reflects the important role of IgG in providing long-term immunity against infections. IgG antibodies can persist in the bloodstream for an extended period, providing a memory response to previously encountered pathogens. This memory response allows the immune system to mount a faster and more effective defense if the same pathogen is encountered again in the future.

Immunoglobulin G (IgG) is the most common antibody found in the human body. It plays a critical role in protecting against bacterial and viral infections by neutralizing pathogens, activating immune cells, and participating in the complement system. With its high concentration and ability to provide long-term immunity, IgG is a vital component of the immune system's defense mechanisms.

SUMMARY

- The immune system is a complex network of cells, tissues, and organs that work together to defend the body against harmful pathogens, such as bacteria, viruses, fungi, and parasites. It plays a critical role in maintaining overall health by identifying and eliminating foreign invaders while also recognizing and removing abnormal or damaged cells within the body.
- An autoimmune disease is a case of mistaken identity; it is an abnormal condition in which the body reacts against constituents of its own tissues. The result may be simple hypersensitivity reaction and or autoimmune disease when the body begins attacking its own healthy tissues
- MHC (Major Histocompatibility complex) molecules normally should present the peptide (antigen) at optimal level to T cells for immune response to occur. The level of autoantigenic peptide presented is determined by polymorphic residues in MHC molecules that govern the affinity of peptide binding.
- The LE cell was so termed because of its exclusive presence in the bone marrow of 25 patients with confirmed or suspected SLE at the Mayo Clinic. LE cells are polymorphonuclear cells which have phagocytosed the nuclear material of other cells.
- Immunodeficiency disorders represent defects in the immune system that result in weakened or dysregulated immune defense. These deficiencies may occur as primary diseases or secondary conditions.
- Gastrointestinal disorders can reveal the presence of underlying PID. Autoimmune enteropathy can be observed when children lack regulatory T cells. The autoimmune disease in other PIDs, on the other hand, leads to inflammatory bowel disease.
- Human immunodeficiency virus (HIV) infection remains the most well-studied cause of secondary immunodeficiency; a comprehensive discussion remains beyond the scope of this review. HIV infection should be excluded by antibody- and antigen-based assays combined with nucleic acid testing.
- A hypersensitivity reaction happens when the immune system mistakenly identifies harmless substances as harmful. There are several types, depending on the class of substance that triggers it.
- Immunology plays a central role in our understanding of infectious diseases and how the body defends itself against pathogens. Infectious diseases are caused by microorganisms such as bacteria, viruses, fungi, and parasites, and the immune system’s response to these invaders is essential for preventing and controlling infections.

MULTIPLE CHOICE QUESTIONS

1. Which of the following is an autoimmune disease?

a. Influenza

b. Tuberculosis

c Rheumatoid arthritis

d. Malaria

2. Immunodeficiency disorders are characterized by:

a. Hyperactive immune responses

b. Overproduction of antibodies

c. Impaired immune function

d. Autoimmune reactions

3. Which immune cell is primarily affected in HIV/AIDS?

a. B cells

b. Natural Killer (NK) cells

c. T cells

d. Macrophages

4. Allergic reactions are a result of:

a. Immune cells attacking healthy tissue

b. Immune suppression

c. Immune cells responding to harmless substances

d. Immune cells failing to respond to pathogens

5. Which type of hypersensitivity reaction involves antibodies of the IgE class?

a. Type I

b. Type II

c. Type III

d. Type IV

ANSWERS

1. (c) 2. (c) 3. (c) 4. (c) 5. (a)

REVIEW QUESTIONS

1. Differentiate between an autoimmune disease and an immunodeficiency disorder.
2. Describe the four major types of hypersensitivity reactions and provide an example of each.
3. What is the role of cytokines in the immune system and how do they contribute to immune responses?
4. Discuss the concept of immunotherapy in cancer treatment. Provide examples of different immunotherapy approaches.
5. What are primary immunodeficiency disorders and how do they differ from secondary immunodeficiencies?

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Core Concepts in Biology: Immunology

2nd Edition

The immune system is responsible for keeping our bodies healthy and protected against various diseases. It's the branch of biomedical science known as immunology that studies this intricate system of cells, tissues, and organs. This defense network works hard to safeguard us from harmful bacterial, viral, fungal, and parasitic pathogens.

Investigating the immune system's functionality and finding ways to utilize it to cure different kinds of illnesses is the fundamental mission of immunology. This field of study delves into the varied facets of the immune system, including white blood cells, cytokines, immune receptors, and antibodies. By diving into immunology, we learn about how our immune system detects and combats unfriendly substances, also called antigens. Discerning how the immune system distinguishes between self and other is another significant focus in immunology, as this ability guarantees that it specifically targets harmful pathogens while leaving healthy cells and tissues alone.

By delving into the field of immunology we can explore how our system detects and fights, against substances known as antigens. Understanding how our immune system differentiates between what belongs to our body and what doesn't also an aspect of immunology. This ability ensures that it specifically targets pathogens while leaving cells and tissues unharmed.

Immunology plays a role in vaccine production by studying how vaccines stimulate responses and protect us from specific diseases. It also focuses on researching conditions, allergies and immunodeficiencies to develop the effective treatments. Continuous research in this field leads to advancements in medicine, including the development of tools, therapies and preventive measures against immune related disorders, infectious diseases and cancer. Undoubtedly immunology drives innovation for a more resilient world.

The captivating field of immunology unravels the mysteries surrounding our bodies' defense against diseases and infections. By exploring core concepts, in immunology through explanations and comprehensive research readers can gain an understanding of the complex mechanisms within our immune system.

Immunology encompasses more, than combating pathogens; it also involves understanding the intricate equilibrium of immune responses and the risks associated with immune related disorders. Covering principles and cutting edge developments this book offers an examination of immunology making it a must read for students and individuals captivated by the captivating realm of our body protective mechanisms.

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Kaile Li is an accomplished associate professor in the Department of Immunology, known for his expertise in adaptive immunity and innate immunity. With a focus on understanding the intricate workings of the immune system, particularly its ability to adapt and respond to pathogens, his research delves into the mechanisms that drive immune responses at both the cellular and molecular levels. Through his work, Professor Li aims to contribute to the development of innovative treatments and therapies that harness the potential of the immune system to combat diseases and disorders. His dedication to advancing our knowledge in the field of immunology makes him a highly respected figure in the scientific community.

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